Service

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Service Operations and Processes

Service Operations and Processes includes service order management to plan and deliver services, service request management to process services requests originating from various sources, in-house service to manage services in-house at service centers, and case management to process issues that involve multiple processing steps.

[Service Order Management](#)

Service Order Management provides a variety of functions that allow you to set up end-to-end service processes. It is integrated with Service Contract Management as well as with logistics, inventory management, pricing, billing, and Finance.

[Service Request Management](#)

You use Service Request Management to monitor and fulfill requests for service by your customers. Service requests can be used internally by companies where a department delivers services as well as in external customer-facing scenarios. They enable you to measure and evaluate the quality of services provided.

[In-House Service](#)

In-House Service supports companies that offer maintenance and repair services for service objects (in-house service objects).

[Service with Advanced Execution](#)

Service with Advanced Execution is a process that is suitable for services which require extensive planning and execution. They generally have longer service times with a requirement to bill the time and service parts used for the service. The core characteristic of this process is the integration of the commercial aspects of Service with the planning and execution capabilities of Maintenance Management.

[Case Management](#)

Case Management enables you to consolidate, manage, and process information about a complex problem or issue in a central collection point, the case. Within a case, you can group diverse information, such as business partners, transactions, products, and documents. This information can reside in different physical systems.

Service Order Management

Service Order Management provides a variety of functions that allow you to set up end-to-end service processes. It is integrated with Service Contract Management as well as with logistics, inventory management, pricing, billing, and Finance.

Note

The following images contain links to documentation you might be most interested in. Click on each of the areas to learn more.

Service Orders

Please note that image maps are not interactive in PDF outputs.

Use service orders to provide short-term agreements and add planning

Service Quotations

Please note that image maps are not interactive in PDF outputs.

Use service quotations to provide binding offers and cost estimates to

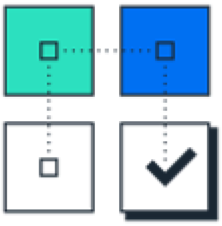
Service Order Templates

Please note that image maps are not interactive in PDF outputs.

Use service order templates to specify reusable sets of service-related data and

data for the provision of one-off services to your customers.	your customers and quickly create follow-up service orders.	minimize the time to create service orders or service quotations.

Service Confirmations



Please note that image maps are not interactive in PDF outputs.

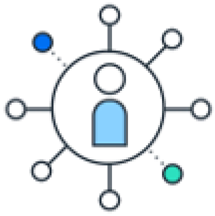
Use service confirmations to precisely record time and materials consumed to provide services.

Cross Functions

Please note that image maps are not interactive in PDF outputs.

Use functions that are available in more than one service transaction throughout Service Order Management.

Apps



Please note that image maps are not interactive in PDF outputs.

Use the apps in Service Order Management to support your service processes.

Interaction Center

The interaction center provides multichannel collaboration and communication tools to enable efficient and consistent customer service.

Interaction center (IC) agents can handle inbound and outbound service transactions using the phone, e-mail, fax, chat, and letter communication channels. They can process business transactions, such as service orders and enhance their productivity by using alerts and a knowledge search. All relevant account information is available to them in the IC, such as account data, service order status, and product-related information.

The IC provides agents with the tools they need for the processes that comprise their daily work. Generally, the IC supports the following processes:

- Communication process

  This business process involves the communication between an IC agent and a business partner (customer, employee, and so on) in real time, for example using the telephony channel. Since IC agents can also work asynchronously, processing e-mails, service requests, or workflow items from the agent inbox, the communication process is not necessarily performed at all times.

- Interaction process

  The interaction process is the central IC process, which often encompasses the communication process and the business transaction process, and is performed whenever an agent uses the IC to communicate or interact – either directly with a business partner or asynchronously by working on a business transaction belonging to a business partner.

  The interaction process starts in the interaction mode with either the identification and confirmation of the business partner or the selection of a business transaction, e-mail, workflow, or other item belonging to a business partner. The interaction process includes all steps that the agent takes to work on a customer request. The interaction process can optionally result in the creation of one or more business transactions.

- Business transaction process

Some simple customer or employee requests can certainly be resolved by providing a short answer or other information, not making use of a business transaction. However, frequently an interaction will result in one or more follow-on business transactions, such as a service request. In such cases, each business transaction is linked to the interaction record, providing a connection between the business process and the interaction process.

To start the interaction center, you use the `WUI` transaction and select a corresponding business role, such as **Service IC Agent**.

Related Information

Interaction Center WebClient
E-Mail Response Management System
Integrated Communication Interface

Interaction Center WebClient

Use

The interaction center (IC) WebClient is a thin-client, highly optimized desktop application for interaction center agents. It provides a comprehensive set of interaction center (IC) functions and supports various processes required to answer customer queries and work in corresponding business transactions.

Features

- Account Identification

  A customer search function that agents use to identify customers in the system. You can set up this function to identify customers even before agents accept contacts, based on automatic number identification (ANI) or the address of an incoming e-mail.

- Account Overview

  An overview that agents use to get complete information for an entire account, including individual account details, notes, roles, planned activities, and business partner relationships.

- Interaction Record

  A customer-focused overview of the current interaction and customer history.

- Interaction History

  A comprehensive interaction history that provides one view of a customer. Search and overview of business transactions for a customer. Agents can use various search criteria to find the business transactions they are looking for. Agents can record data both during and immediately after an interaction.

- Knowledge Article Search

  Interaction center agents can search for knowledge articles, which can include documents, links to Web sites, and screenshots, to answer customer queries.

- Agent Inbox

  A central worklist that allows agents to process incoming and outgoing mails, planned activities, and other work items.

- E-Mail

  An interface that agents use for processing incoming and outgoing e-mails, as well as completing unfinished e-mails.

- Chat

  An interface that agents use for processing incoming chat sessions.

- Scratch Pad

  An electronic notepad that agents use to record miscellaneous information during an interaction. Notes from the scratch pad can be imported into fields in the IC WebClient.

- Alerts

  Functions that combine system resources intelligently to guide agents during customer interactions.

- Queue Status

  A feature that agents use to quickly display the channels and queues they are assigned to, based upon configuration in the communication management software.

Specifics for Business Transactions in the Interaction Center

Use

This topic summarizes the specifics that apply to the processing of business transactions in the interaction center (IC).

Features

- The information from a confirmed account is automatically copied to any newly created business transaction.
- The business transactions that you use in the IC have a tiled layout. Alternatively, you can configure an overview layout for these business transactions in Customizing for the UI framework.

Settings

- Business transaction profile in IC

  You can make the following settings in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**:

  - Define which transaction types are available to the agent to create business transactions, and which transaction type is to be the default
  - Define whether dialog boxes are allowed and displayed automatically (automatic display is only supported by the interaction record)
  - Define the transaction type for transaction data and the additional business transactions separately

- Automatic routing

  Set up automatic routing. For information about the necessary settings in the rule modeler, see **Interaction Center WebClient** > **Business Transaction** > **Define Automatic Routing**.

Sales Transactions

Get an overview on how to create and change sales orders and sales quotations, display item proposals, as well as display and print sales transactions.

To use the sales transactions, you use the `WUI` transaction and select a corresponding role, such as Sales Professional.

Entering Sales Documents on the WebClient UI
Item Proposals in Sales Transactions
Searching for, Displaying and Printing Sales Transactions on the WebClient UI

Entering Sales Documents on the WebClient UI

Use

You use this function to create and change sales orders and sales quotations.

You carry out your presales activities such as Opportunities and Activities on the WebClient UI, and then, for example, process a rush order without having to exit.

This function is of particular use to sales employees who work mainly with the WebClient UI and therefore do not require the full range of functions offered by Sales Quotation and Order Management. For example, sales assistants can use this function to create simple sales orders.

Prerequisites

You can directly create and change a sales document on the WebClient UI once you have carried out the following tasks:

- You have assigned the relevant sales documents to a profile in Customizing under Service > Transactions > Settings for Sales Transactions on WebClient UI > Define Profiles for Sales Transactions.
- You have defined copy control of transaction types in Customizing for Service under Transactions > Settings for Sales Transactions on WebClient UI > Copy Control for Opportunities and Sales Transactions.
- If you want to use payment cards in sales orders, you have made settings in Customizing under Sales and Distribution > Billing > Payment Cards.

Features

When you maintain a sales document using the WebClient UI, the system enters data directly in SAP. This means that you can decide yourself which UI you use to process the sales order.

Creating Sales Quotations and Sales Orders as Follow-Up Transactions from an Opportunity

The sales representative or sales assistant has determined a potential sales opportunity, and has therefore created an opportunity. If the customer then requests a quotation, the sales employee can immediately create a sales quotation. To do so, the sales employee opens the opportunity that he or she created earlier, and creates the sales quotation as a follow-up transaction from the opportunity.

When you do so, the following data is copied to the sales quotation:

- Business partners
- Sales organization
- Distribution channel
- Division

- Sales office
- Sales group
- Product configuration
- Items
- Product
- Quantity
- Notes
- Unit

You can only display the following fields:

- Product
- Product ID
- Transfer
- Item number

You can navigate to the sales document using the transaction history in the follow-up activity.

Offering Alternative Items in Sales Quotations

In addition to the item requested by the customer, the sales employee can also offer an alternative item in the sales quotation. In order to offer the alternative item, when the sales employee copies the standard item and the alternative item from the opportunity to the sales quotation, the sales employee must make sure that the alternative item comes directly after (by item number) the standard item in the list of items.

Creating a Follow-Up Activity from a Sales Document

You can create a follow-up activity from a sales document. You can then navigate to the sales document using the transaction history of the follow-up activity. The system shows the follow-up activity in the following:

- Transaction history of sales orders and sales quotations
- Document flow shown in SAP GUI (transactions `VA02`, `VA22`, and `VA42`)

Correcting Errors

The system handles errors according to **Sales and Distribution** rules. This means that if errors occur, you cannot save the sales order. You can only continue processing the sales order after the errors have been corrected.

i Note

The assignment of digital payment cards is currently not supported on the WebClient UI.

Activities

Certain restrictions apply to this function. Therefore, it should only be used by sales employees who work with sales documents occasionally.

Example

You create a sales quotation as a follow-up transaction for an opportunity. The system copies data directly from the opportunity to the sales quotation.

Item Proposals in Sales Transactions

Use

- Item proposals are used as input help in sales orders and quotations.
- Item proposals are available on the WebClient UI for Service, including the Interaction Center (IC).

Prerequisites

You have set up the item proposals in sales transactions in Customizing by choosing **Service** > **Transactions** > **Settings for Sales Transactions on WebClient UI** > **Product Proposals**. Under the node **Past Orders**, you make the settings for item proposals from previous orders.

Features

Item proposals are displayed directly in the **Items** assignment block.

You can automatically display item proposals, or manually trigger the display. To manually trigger the display of proposals, set the indicator **Past Orders on Request** in the Customizing of product proposals, in the activity **Assign Method Schema to Transaction Type**.

Item proposals appear at the end of the item list as suggested items, that is without item numbers. You can transfer the suggested items as final order items, by entering an order amount. Only then can the system carry out pricing and an availability check for the items.

- **Item Proposals from Past Orders**

  The system proposes items from the previous sales transactions for this customer, that is products and order quantities.

  - You can define the transaction types and the number of transactions in a certain time frame that the system draws from to determine proposed items, for example the last 10 sales orders from the past three months.
  - You can define how the system calculates the proposed quantities: from the last order, or as an average quantity from the previously defined number of orders.
  - You can select the required transactions using a popup, or the system can immediately (after you have entered the customer data) display all products from the past transactions.

Searching for, Displaying and Printing Sales Transactions on the WebClient UI

Searching for and Displaying Sales Transactions on the WebClient UI

If you want to provide your customer with information on the current status of a sales order that has already been created, you can use the **Search** function to locate transactions quickly, and forward the necessary information to the customer immediately. In addition, this function allows you to enter changes requested by the customer quickly and easily in the sales transaction.

Displaying sales transactions related to the relevant account

You can display sales transactions, that is, sales quotations and sales orders related to the relevant account on the **Account** overview page, under the corresponding assignment blocks. This allows you to gain an overview of all existing sales transactions for a specific customer, for example, in preparation for a customer visit.

Sales representatives can also create new sales transactions, edit or delete existing sales transactions from these assignment blocks. When you create a new sales transaction from here, the **Sold-to Party** field is automatically filled with the account from the **Account** overview page. Once you save and return to the **Account** overview page, the newly created sales transaction appears in the relevant assignment block.

Data retrieval parameters are delivered for each of these assignment blocks, which enable you to filter which data you want to display on the corresponding assignment blocks. For example, you may only want to display sales orders with posting dates for last month.

The following data retrieval parameters are available for the corresponding assignment blocks:

- Quotations: **Posting Date, Transaction Type**, **Valid From/To**
- Sales Orders: **Posting Date, Transaction Type**

For more information on data retrieval parameters, see [Configurable Data Retrieval].

Printing Sales Transactions

You can print sales transactions, that is, quotations and sales orders from the WebClient UI. This allows sales representatives to preview, save or print sales transactions in order to check them, share them with their team, or file them to their documents. It is not intended for sales representatives to send transactions directly to their customers; this will continue to be done from the standard **Sales and Distribution** UI.

Once you have saved the sales transaction on the WebClient UI, you can use the **Preview Output** pushbutton for a print preview with a PDF viewer. You can also save locally or print out the preview using a PDF viewer. To print sales transactions, you can choose the output type set up in SAP **Sales and Distribution**, including PDF-based print forms, SAPscript and Smart Forms.

For more information, see PDF-Based Print Forms

Specifics for Interaction Center: Communication Management Software

Use

Communication management software (CMS) refers to software products that manage communication channels and handle individual contacts with customers through different channels, such as phone, e-mail, and chat. The Integrated Communication Interface (ICI) allows the CMS to communicate with the Interaction Center WebClient.

For more information about ICI, see [Integrated Communication Interface].

For information about the features that are available in the telephony channel, see [Communication Management Software].

For information about the features that are available in the chat and e-mail channels, see [Chat] and [E-Mail] in the Interaction Center WebClient documentation.

Prerequisites

- You have made the settings in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Communication Management Software Profiles**.
- Optionally, you have made the settings for checking the continued communication between the communication session, the CMS, and the Web browser in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Heartbeat Profile**.

  In this Customizing activity, you can make settings for the following checking methods:

  - Heartbeat without watchdog
  - Heartbeat and watchdog
  - CMS session availability

Business Process Push

Use

You can use this function to push business processes to external routers or communication management software (CMS) for distribution to interaction center (IC) agents based on their real-time availability. For example, the following business processes can be pushed:

- Business transactions, such as service orders
- Faxes and letters received through SAPconnect or the ArchiveLink scenario

The ID of the business process and a parameter that specifies the type of business process are transferred to the external router or CMS.

In addition, for each business transaction, you can define a set of additional parameters to represent, for example, status and priority. In order for the business process to be pushed and prioritized correctly in the CMS, you define rule policies to set the values for the parameters before these values are transferred to the CMS.

Prerequisites

- You have made the settings for business process push in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Business Process Push > Define Settings for Business Process Push**.
- You have assigned parameters to those agent inbox item types that you want to be pushed to external routers or CMS in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Business Process Push > Assign Parameters for Business Process Push**.
- You have defined rule policies to set the values for the parameters. For more information, see Rule Policies.
- You have made the settings for pushing faxes in Customizing for **Service** under **Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Define Receiving E-Mail/Fax Settings**.
- You have made the settings for pushing letters in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Business Process Push > Define Letter Settings**.

Features

Business transactions, such as service orders, are automatically transferred to an external router or CMS if the following conditions are met:

- The business transaction is saved and has ended, or is saved and escalated.
- The values of the parameters that have been defined in Customizing for the business transaction correspond to the values of the parameters that are set in the rule policies.

  For example, the business transaction must have status **In Process** and priority **High** to be pushed to an external router or CMS.

The business transaction is locked and the current employee responsible for the business transaction is removed. No changes to the business transaction and its status are possible until the transaction has been pushed to an IC agent.

When an IC agent is alerted that an inbound business transaction is routed, the business transaction ID and type are shown on the screen. The IC agent can either accept or reject the business transaction. The IC agent who accepts it is assigned as an employee responsible and can process that business transaction and change its status.

- Rule policies

  You define rule policies to set the values for the additional parameters for business transactions and cases before these values are transferred to the CMS. You define rule policies in the **Order Routing** and the **Intent Driven Interaction (IC WebClient)** contexts using attributes, conditions, and actions.

  ### Example

  You can create the following rule policy in the **Order Routing** context:

  Conditions:

  `If Priority Is Not Equal To Very High`

  Actions:

  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA`
  - `Send business process push parameter`

  You can create the following rule policy in the **Intent Driven Interaction (IC WebClient)** context:

  Conditions:

  `If Current Event Equals Order Was Saved`

  Actions:

  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = PRI`
  - `Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA`

  Conditions:

  `If Current Event Equals Interaction Ended`

  Actions:

  `Send business process push parameter`

- Pushing of faxes and letters received through SAPconnect or the ArchiveLink scenario

  If the Customizing settings for pushing faxes and letters have been made, faxes and letters received through SAPconnect or the ArchiveLink scenario are pushed to IC agents based on their real-time availability. The number at which the fax arrives is the indicator for pushing the fax.

An IC agent is alerted that an inbound fax or letter is pushed. The IC agent can then either accept or reject it. Once the IC agent accepts the fax or letter, the **Identify Account** page and the **Fax Preview** or **Letter Preview** is displayed.

ERMS E-Mail Push

Use

You can use this function to push e-mails to the communication management software (CMS) for distribution to an interaction center (IC) agent.

When e-mail is routed to an agent, the **Accept** and **Reject** context area buttons allow the agent to decide whether or not to accept the inbound message. If the agent accepts the e-mail, the system searches for the business partner based on the e-mail address. The agent can then use standard e-mail functions in the e-mail editor view.

Prerequisites

You have specified **Inbox** as the e-mail provider for the e-mail profile in Customizing for **Service** under **IC WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

Activities

- Create rules in the **ERMS** context to push e-mail to the CMS for distribution

  ### Example

  You can create the following rule policy in the **ERMS** context:

  Condition:

  `If E-Mail Sender Equals jdoe@customer.com`

  Action:

  `Push E-Mail to IC WebClient Agent`

- Configure a CMS system ID that points to a CMS system that supports action routing

  Create an entry for the CMS system ID `ERMS_ACTION`. You do this in the **SAP Easy Access** menu under **Service** > **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**.

More Information

[E-Mail Response Management System]

Queue Status

Use

The queue status, located in the bottom right hand corner of the interaction center, allows you to quickly display the channels and queues you are assigned to.

Features

The queue status:

- Displays the current date and time (default display) or queue.

  If queue information is available, then an arrow is shown after the date. You can click on the arrow to show which queue the current incoming contact was sent to.

  Clicking on the arrow again returns the current date and time.

- Displays a tool tip with a quick view of multichannel information.

  The queue status tooltip displays the channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and all queues to which to are currently assigned.

  The queue status tool tip is only available if the connection to the relevant system is currently available.

- Displays the Agent Dashboard.

  If you click on the queue status, the agent dashboard is displayed.

Agent Dashboard

Use

You can use the agent dashboard to view information relevant to your current interaction center (IC) session. It is useful when you need more information about what is displayed in the queue status. You can access the agent dashboard by clicking the queue status at the bottom right corner of the browser window.

Features

The agent dashboard displays the following information:

- Your user ID and business role
- Channels and queues

  You see all channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and the queues to which you are assigned.

  This information is only available if:

  - There is a connection to a communication management software system, and
  - The communication management software passes queue and channel information to the interaction center.

- System information

  System information is used primarily by system administrators for troubleshooting. It includes:

  - Information on the domain in which the servers are operating (for example, wdf.sap.corp)
  - Name and description of the communication management software being used

Specifics for Interaction Center: Toolbar

Use

The toolbar provides an interface between your communication management software (CMS) and the Interaction Center (IC) WebClient. The toolbar contains pushbuttons and icons for communication functions that apply to the various communication channels.

You can modify the content of the toolbar to better meet your business requirements. For example, you can specify that different sets of pushbuttons are available in the toolbar depending on the connection of an agent to the CMS and the status of the agent's interaction with a contact.

Integration

The following processes are examples of how you can use the pushbuttons in the Interaction Center toolbar:

- Communication process

  For a phone call, you start the process by choosing the pushbutton **Dial** or **Accept**, and you end the process by choosing the pushbutton **Hang Up**.

- Interaction process

  Includes account maintenance and the interaction record. Additionally, business processes such as sales and service can also be handled in the interaction center. You start the process by confirming the account and you end the process by choosing the pushbutton **End**.

- Communication process and interaction process

  The two processes can take place simultaneously and do not necessarily start and end in the same order, for example in the following cases:

  - Multiple phone calls

    You can dial and end multiple phone calls during the same interaction. This occurs, for example, when a phone line is disconnected during a customer interaction and needs to be dialed again.

  - Multiple interactions

    Multiple interactions can happen during a single phone call. If the agent receives and accepts the next call whilst wrapping up the previous interaction, they can end the first interaction by choosing the **End** pushbutton despite already being in the next active phone call.

Prerequisites

- Your system administrator has set up the CMS and ensured that the user names there match those in the IC. For more information, see [Specifics for Interaction Center: Communication Management Software] and the documentation for the CMS you are using.

- You have made the settings for the toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

Features

You use the toolbar to work in different channels. The system determines the pushbuttons that appear in the toolbar, depending on the following:

- The communication channel in which you are currently working

- Customizing settings

Example

Pushbuttons in the IC toolbar

Channel	Name	Definition
Universal (available for all channels)	Accept	Accepts the contact. In telephony, Accept is similar to picking up the receiver.
	Reject	Sends the contact back to its original queue.
	Transfer or Blind Transfer (in the telephony toolbar)	In telephony, puts caller on hold while you transfer the call to another agent's extension without first consulting with that agent.
	End	Ends the interaction (indicating that the agent has finished wrap up activities), saves all objects created or changed during the interaction, and clears customer information from the screen and from the business data context.
	Hang Up	Similar to hanging up the receiver. Ends a phone call that was started with Dial or Accept.
	Clear Interaction	Discards the interaction record and resets the account identification. **i Note** Discarding other transactions such as sales or service transactions is not possible in this way. Depending on the process status, this may be done by choosing Cancel if it is part of the transaction toolbar. If you want to clear the entire context of the session including the business data context, you have to choose the Clear Interaction pushbutton followed by the End pushbutton.
	Reset CTI	Resets the context area of the browser session to synchronize it with the communication session state. The different states can for instance be caused by network issues.
	Dial Pad	Displays a dial pad that looks like the numbers on a telephone, which you can use to make a call.
Telephony	Hold	Puts the active call on hold.

Channel	Name	Definition
	Toggle	Switches between a call on hold and an active call, automatically placing the active call on hold and vice versa.
	Retrieve	Takes a call off hold and makes it active once again.
	Consult	Puts customer call on hold while you call another agent. After you are finished consulting with the other agent, the agent call is dropped and the customer call is active once again.
	Warm Transfer	Puts caller on hold while you consult with another agent before transferring the call to the agent's extension.
	Conference	Puts customer on hold while you call another agent. Then you activate both calls at the same time so that all three parties can participate in a discussion.
	Wrap Up	Indicates to the system that you need time to wrap up the call. Wrap up time ends when you choose the End pushbutton.
E-Mail	Pushbuttons specific to the e-mail function, such as Reply, New, or Forward, appear on the e-mail screen dynamically, as and when they are required.	Represent standard e-mail functions.
Chat	Leave	Finishes the chat.

Work Modes

Name	Use
Ready	When you choose Ready, the CMS system sends incoming contacts to you.
Not Ready	When you choose Not Ready, the CMS system does not send incoming contacts to you.

Scratch Pad

A temporary workspace in the interaction center that you use as an electronic notepad to store miscellaneous information during an interaction.

Customers usually provide a lot of information during the first few minutes of conversation, before you have a chance to access the relevant transaction. Most agents have separate notepads to take notes and then enter the data into the system later.

The scratch pad replaces the need to write manual notes. It allows you to electronically store information so it can be copied into specific functions that require this data.

The scratch pad allows you to focus on recording and listening to what the customer is relaying. Customers do not have to hold their conversation until you follow through the sequence of the interaction, gather the preliminary information, such as contact information, and then navigate to the appropriate area in the IC WebClient to record the information. This also saves customers from having to provide information more than once.

The scratch pad is located in the top left corner of the IC, under the title bar. It is available throughout the interaction, and can be copied into different IC functions, such as the interaction record or knowledge search by choosing **Import Scratch Pad**. All of the data on the scratch pad is imported, but you can still edit, add, or delete it, if necessary.

Data remains on the scratch pad for the entire interaction until you either select **End** from the toolbar, or select **Clear**. Closing the scratch pad does not delete the contents.

Alerts and Messages

Use

These functions display texts on your page to alert you about items or situations that require your immediate attention. There are three types:

- Alerts

  Alerts are predefined by system administrators or Interaction Center (IC) managers to be triggered by certain events in the system.

- Messages

  There are three types of messages: informational, warning, and error.

Features

- Placement on page

  Alerts appear in the alerts area and system messages appear in the system messages area.

- Alerts
  - If more than two alerts appear, an arrow appears, which allows you to page through alerts in pairs.
  - All alerts disappear when you choose **End**.

Example

Alerts

Marcus is a customer service representative at an interaction center. He accepts a call from Ms. Anna Glenn and confirms her identity in the system. The following alert appears:

Service order 00225

Marcus asks Ms. Glenn if she is calling regarding this order. She says that she would like to know the status of the service order. Marcus simply chooses the link to the service order and can inform Ms. Glenn of its current status.

Creating Alerts

Use

You can create alerts in that use system information to provide agents with information they may need during customer interactions.

Note

No preconfigured alerts are delivered. Interaction center managers define alerts in the Alert Editor and assign them through rule policies. Alerts are created independent of any rule policy context.

Alerts can only be triggered in the Intent-Driven Interaction context. This context allows you to automate specific interaction center functions, for example, triggering and terminating alerts, via business rules. Interaction center alerts are not relevant for other rule policy contexts.

Prerequisites

- You have completed the Customizing for Service under Interaction Center WebClient > Additional Functions > Intent-Driven Interactions > Service Manager > Configure Alerts.
- You have completed the Customizing for Service under Interaction Center WebClient > Additional Functions > Intent-Driven Interaction > Define Intent-Driven Interaction Profiles.
- If you want to user user-triggered alerts, you have selected this alert type in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Communication Processing Profiles.

Features

The following types of alerts are available in the IC:

- Real-time alerts

  With these alerts, the system continually polls the server to check for alerts.

- User-triggered alerts

  With these alerts, the system waits until the user triggers an event to perform a server round-trip to check for alerts.

  This type of alert allows you to show alerts without using server polling and can be used with asynchronous processing.

When creating alerts, you can select a theme that affects the appearance of alerts by changing, for example, the font style, font color, and the alert icon. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Themes must be defined in Customizing before they are available for selection when creating an alert. You can define themes in Customizing for Service under Interaction Center WebClient > Additional Functions > Define Themes for Alerts.

Broadcast Messaging

Use

Interaction Center (IC) supervisors can send messages to IC agents in real time using broadcast messaging. They can also create and maintain distribution lists for broadcast messaging and display sent messages. IC agents can receive messages, but cannot send messages.

IC supervisors can access broadcast messaging under Managing Operations > Broadcast Messaging.

Prerequisites

You have made the settings for broadcast messaging in Customizing for **Service** under Interaction Center WebClient > Additional Functions > Define Broadcast Messaging Profiles.

Features

- Distribution lists

  To create or maintain distribution lists, click **Maintain Distribution Lists**. Broadcast messaging profiles contain an organizational ID. When you create distribution lists, the names that appear in the input help for **Add User** come from the organizational unit entered in the selected broadcast messaging profile. You can also copy an existing distribution list, delete a user from a distribution list, or delete a distribution list. The default distribution list is the one that appears first alphabetically.

  To share distribution lists, the owner of the list can flag other users within the list as administrators. As an administrator, you can add and remove users from the list, use the list to send broadcast messages, and you will be able to view any broadcast messages that were sent to this list in the **Sent Log**.

- Themes

  You can define themes for broadcast messaging in Customizing for **Service** under Interaction Center WebClient > Additional Functions > Define Themes for Broadcast Messages.

  Once a theme has been defined in Customizing, it is available for selection when creating a broadcast message. The theme affects the appearance of broadcast messages by changing, for example, the font style and color. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Interaction Record

Use

The interaction record is a screen in the interaction center (IC) that contains data for the current interaction with an identified business partner. IC agents can log current interactions in this screen, and postprocess them if necessary.

Interaction with a customer always includes an interaction record, which can then be linked to another business transaction, such as a service process. The interaction record can, however, also be saved without creating another business transaction.

Note

From a technical point of view, the interaction record is created as a business transaction.

Once an agent has identified an account, the system navigates to the interaction record. In the interaction record, the agent logs the interaction, records central data from the call, and postprocesses the interaction when the call is finished.

To increase system performance at high volume call centers, you can use the lean interaction record. The lean interaction record is a modified version of the full interaction record.

Prerequisites

- You have defined a date profile and assigned this to your transaction type to use interaction time recording in Customizing for **Service** under **Basic Functions** > **Date Management** and **Transactions** > **Basic Settings** > **Define Transaction Types**.
- You have activated the rescheduling view and optionally enabled automatic text entry for call rescheduling in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**.
- **Using Activity Reasons:**
  - You have made the settings for activity reasons in Customizing for **Service** under **Transactions** > **Settings for Activities** > **Define Activity Reasons**.

    Note

    The interaction reason corresponds to the activity reason.
  - You have assigned the **Reason for Activity** subject profile in your transaction type to the **Business activity** transaction type in Customizing at the header level.

    You can make settings for transaction types in Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Transaction Types**.
- **Using Multilevel Categorization:**

  You have made the following settings in Customizing for **Service**:
  - Activate categorization for the interaction record under **Interaction Center WebClient** > **Business Transaction** > **Define Business Transaction Profiles**.
  - Make the necessary settings under **Transactions** > **Basic Settings** > **Define Transaction Types**.
  - Define the subject profile for the category modeler under **Basic Functions** > **Catalogs, Codes and Profiles**.
  - Make the necessary categorization settings for the interaction record under **Interaction Center WebClient** > **Business Transaction** > **Define Categorization Profiles**.

  See Multi-Level Categorization for more information.
- If you want to use the lean interaction record instead of the full interaction record, you have assigned a business transaction profile that contains transaction type 0100 to the business role. You do this in Customizing for **Service** under **UI Framework** > **Business Roles** > **Define Business Role**.

Activity Clipboard

Use

Interaction Center (IC) agents can use this function to view a list of the business objects involved in a current or past interaction. It is located in the Interaction Record or the navigation area.

Note

The activity clipboard in the interaction record visualizes past and current interactions.

The activity clipboard in the navigation area visualizes current interactions.

The objects that were created or changed during those interactions are listed including navigation links.

Integration

This function is integrated with:

- Interaction Record
- Interaction History
- Business data context

  In the Interaction Center (IC), the activity clipboard is one view of the business data context.

Prerequisites

Depending on where you want the activity clipboard to appear, you have to make different settings in Customizing:

Activity Clipboard in the Interaction Record

- You have made the necessary settings in Customizing to define the display of the activity clipboard so that the business partner appears in the activity clipboard list. Your system administrator can either use the default activity clipboard profile, or define which objects are relevant for linking and should appear in the activity clipboard. You do this in Customizing for Service, by choosing Interaction Center WebClient > Basic Functions > Define Activity Clipboard Profiles.
- Both the programming of the business object itself and your Customizing settings must be in place for a business object to appear in the activity clipboard.

Activity Clipboard in the Navigation Area

You have made additional settings in Customizing for Service:

- Define a direct link group of the type EE under UI Framework > Technical Role Definition > Define Navigation Bar Profile > Define Direct Link Groups.
- Assign the direct link group to a navigation bar profile under UI Framework > Technical Role Definition > Define Navigation Bar Profile > Define Navigation Bar Profiles > Assign Direct Link Groups To Nav. Bar Profile.
- Assign the navigation bar profile to your business role profile under UI Framework > Business Roles > Define Business Role.

Features

If a business object is accessed repeatedly, it only appears once on the activity clipboard, in the order in which it was first accessed.

- By default the data is displayed in three columns, however, in Customizing you can choose the number of columns to be displayed with the maximum being five columns. The first column contains the name of the object or activity. Your system administrator can define what information appears in the other columns in Customizing for the activity mentioned above. By default, the Object ID and the Process Type are displayed.
- You can use the links to navigate to objects that appear in the activity clipboard, such as leads or sales tickets.

Account Identification/Account Fact Sheet

Use

IC agents can use this part of the interaction center (IC) to identify accounts in different ways, such as by name, company, or product, and to obtain an overview of an account and contact. If an account does not yet exist in the system, or if it must be updated, IC agents can also create and change accounts from this screen. In addition, they can quick create an account with just the basic data.

Prerequisites

- You have set up one of the following account identification profiles:
  - The standard account identification profile in Customizing for Service under Interaction Center WebClient > Master Data > Define Account Identification Profiles.
  - The profile that enables confirmation of multiple business partners at once in Customizing for Service under Interaction Center WebClient > Master Data > Define Account Identification Profiles for Multiple Business Partners.
- In your account identification profile, you have made settings for the following:
  - Contact attached data (if you want to use this feature)
  - Configurable context area (if you want to use this feature)
  - Embedded object components
  - Search scenario
  - Search results filtered for account roles and contact person roles
  - Default contact type for contact searches using search fields for contact communication details (Telephone, E-Mail, Fax)
  - International address versions (if you want to use this feature)
  - Employee search (if you want to use this feature)
- If you are using contact attached data (CAD), you have mapped the contact attached data from the communication management software to account identification using an XML file.

  You have created a function profile in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Profiles for Contact Attached Data.

  Additionally, if you want to use CAD attributes when creating alerts or rules, you have enabled the relevant CAD attributes for fact gathering. For more information, see Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Enable CAD Attributes for Fact Gathering.
- If you are using international address versions, you must have activated them. You do this in Customizing for Service under SAP NetWeaver > Application Server > Basis Services > Address Management > International Settings > Activate International Address Versions.

  Mapping from logon language to address version is available in the default implementation of Business Add-In `ADDR_LANGU_TO_VERSMAP_LANGUAGE_TO_VERSION`, or an implementation exists that maps the logon language to that version.
- If you are using employee search, you have defined employees as business partners.
- If you are using the standard account identification profile, you have defined any object components that you want to be displayed in the upper right area of the screen in Customizing for Service under Interaction Center WebClient > Customer-Specific System Modifications > Define Object Components.

  If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you define in Customizing for Service under Interaction Center WebClient > Master Data > Define

Account Identification Profiles for Multiple Business Partners.

- You have defined an interaction history profile in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Interaction History Profiles**.
- If you are using the briefcase, you have made the necessary setting in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Briefcase**.
- If you are using the enhanced search, you have made the necessary setting in Customizing for **Service** under **Master Data** > **Business Partner** > **Search** > **Activate Wildcard Search for Accounts and Contacts**.
- If you want to ensure that all rules and events that are triggered when the **Business Partner Confirmed** (`BPConfirmed`) event is raised are also triggered when a call is transferred, you have added the **Business Partner Confirmed for Call Transfer** (`BPConfirmed_Trans`) event to the relevant rule policy in the rule modeler. Once you have done so, an account that is confirmed by an agent is confirmed automatically if the call is transferred to a second agent, and the system does not create a new interaction record when the call is transferred. For more information about creating rule policies, see Rule Modeler.

Features

- Automatic number identification (ANI)

  If your telephony switch system has ANI, the system can identify accounts even before IC agents accept contacts.

- Business context data (BCD)

  When IC agents confirm an account or product, the system displays all relevant information for the account or product in the activity clipboard. This information is also recorded in the BDC so that it can be reused by other functions.

  For more information, see Activity Clipboard.

- Contact Attached Data

  When IC agents confirm an account, the system can incorporate contact attached data from the communication management software (CMS). This information is displayed on the contact attached data screen.

  You can also enable contact attached data attributes for fact gathering so that these attributes can be used when creating alerts or rules. Using these attributes, you can, for example, retrieve a service request number from the CAD attributes of an incoming call and set up a rule that displays an alert providing this information to the IC agent once the account has been confirmed. The IC agent can then click the service request number in the alert to navigate directly to the service order. When using CAD attributes for this purpose, you must ensure that the data that is passed has the same CAD group name and CAD attribute name that have been defined in Customizing.

- Configurable Context Area

  The context area provides background information for the current interaction. If you would like to display different account information in the context area, ask your system administrator to configure the context area accordingly.

- Embedded Object Components

  If you are using the standard account identification profile, you can replace the upper right area of the **Account Identification** screen with your own views that enable you to search for business partner-related business objects. You can define object components so that the agent can search for:

  - Objects based on the confirmed business partner
  - Business partners based on the confirmed objects

  If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you have defined.

- International Address Versions

  If international address versions are activated in your system, you can use the default character set or the international character set to search for business partners (provided a character set is defined for your logon language).

  The system automatically searches in both the default character set and the international character set. For example, in Japan, agents can search for the caller's name using the furigana version of the name. As a result, all names with the same pronunciation are displayed, irrespective of the character set.

  International address versions are available for all countries for which you have maintained an international address version.

- Search Scenario

  You can customize search fields and search views that appear as tabs on the **Account Identification** screen. For each search, you can define the title of the tab and the type of search (for example, business partner or object component).

  You can also define the sequence of relationship types that appear in the search criteria.

- Filter search results for account roles and contact person roles

  You can set the system to filter search results for account searches by account roles and by contact person roles.

- Define default contact type for searches using the contact communication data search criteria

  For contact searches using the fields for contact communication details (**Telephone**, **E-Mail**, **Fax**), you can set a default contact type in Customizing that determines whether the search criteria are used to search for a contact of the type account, contact person, or both. IC agents can change this setting in the WebClient UI.

- Use contact name in account search

  When searching for accounts in the IC, entering data in both the contact name fields and the account fields will generate a combined search result and provide more precise results.

  > **Note**
  >
  > You cannot search for accounts using the account search fields in combination with the search fields for contact communication details (**Telephone**, **E-Mail**, **Fax**). The contact communication details take priority over search criteria entered in the other search fields. If one of the search fields for contact communication details is filled, the system performs a contact search based on the communication details only.

- Employee Search

  You can configure your interaction center to service internal employees. IC agents can use the employee search to look up employee details such as cost center, personnel area, employee group, and employee subgroup. They can also look up telephone numbers and e-mail addresses, or, when they receive a telephone call or e-mail, the system can identify the employee by ANI or from the incoming e-mail, and display the employee's details.

- Multiple Business Partners

  You can configure your interaction center so that IC agents can confirm more than one business partner at once. You can also define a list of partner functions that can be used to confirm business partners and determine the sequence in which partner functions are displayed. You can then map the partner functions of confirmed business partner to the existing partner functions in business transactions and the interaction record.

- Account Fact Sheet

  The account fact sheet gives an overview of the confirmed account. You can personalize views of the fact sheet. For more information, see Fact Sheet.

- Briefcase

You can set up a briefcase that contains a collection of briefing cards that give you an overview of relevant information, such as addresses and contacts. Each briefing card can be assigned to more than one briefcase.

- Account Quick Create

  You can quick create an account from every business object such as an opportunity or an activity. When you save the new account, the data you have entered in the fields for the business object is automatically filled in for the new account.

- Enhanced Search

  You can search for an account and contact with the following search criteria:

  - Region by Country

    The search criterion refines your search for accounts by filtering the correct region of the respective country.

  - E-Mail, Telephone, and Website

    You can enter an asterisk (*) at the beginning or the end of a search string to perform a wildcard search. For e-mail, you can also put the asterisk within a search string.

Chat

Use

Interaction center (IC) agents can use the chat function to process incoming chat messages. The chat function in the IC includes standard chat functions, such as joining chat sessions, sending messages, and transferring chat sessions. Agents can transfer chat sessions to other queues or agents. Agents can also see additional details such as chat subject, location where the customer initiated the chat, and shopping cart items. Further features are described below.

Integration

Chat is integrated with the following IC functions:

- Toolbars

  Agents use the channel-specific toolbar that is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Interactive scripting

  Agents can use scripts to guide them through the chat session. The script is available directly next to the chat text area, so agents can easily and quickly access the information without navigating to a different screen.

- Knowledge search

  Agents can use the knowledge search to find solutions to problems or other issues reported by customers.

- Telephony

  Agents can remain in the chat channel and make a short phone call by using **Dial** in the toolbar.

- Interaction record

  Depending on Customizing settings, the chat transcript can be saved and later accessed in the activity clipboard of the interaction record.

Prerequisites

- The Integrated Communication Interface (ICI) is configured (see Integrated Communication Interface).
- You have defined the chat profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Chat Profiles**.
- You have assigned chat profiles to agents through business roles.
- If you want to use standard responses, you have defined the standard response group in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Standard Response Groups**.
- If you want to add additional knowledge bases for use with chat, you have made the settings for knowledge base implementation in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Knowledge Base Implementation**.
- If you want a list of available queues when transferring chat sessions, Communication Management Software settings are defined.

Features

- Account identification

  The customer search is automatically started when an incoming chat session is routed to an available agent. The customer's e-mail address or other business partner details are used to determine if customer data exists. If the account is confirmed, then chat transcripts can be saved to the confirmed business partner's interaction record. For more information, see Account Identification/Account Fact Sheet.

- Standard responses

  IC agents can insert standard responses into the chat text. They can choose from a predefined list of commonly used responses, or can use the standard response screen to search and preview other responses.

- Knowledge base integration

  You can define which knowledge bases IC agents can access while using the chat function. You can, for example, integrate knowledge bases such as interactive scripts, knowledge articles, mail forms (standard responses), solution database (problems and solutions), or chat transcripts.

- Multiple chat sessions

  IC agents can process multiple chat requests simultaneously to maximize efficiency.

  Note

  For more information about settings for multiple chat, contact your communication management software (CMS) vendor.

  When you use multiple chat, several parallel HTTP sessions are created on the server in Service. This affects the sizing on the server side.

  If you are using multiple chat, the client switch scenario is not supported.

- Web shop integration

  IC also supports chat in Live Web Collaboration in Web Channel. You can integrate IC with your Web shop and with Internet Customer Self-Service (ICSS) so that customer requests for assistance via chat are forwarded directly to the interaction center.

E-Mail

Use

You can use this function to process incoming and outgoing e-mails. This function is integrated with the following functions in the interaction center (IC):

- Toolbar

  IC agents use the channel-specific toolbar, which is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Account identification

  Upon receipt of an e-mail, the customer search is automatically initiated based upon the **From** e-mail address. For more information, see Account Identification/Account Fact Sheet.

- Information search

  Results from the knowledge search can be added to e-mails as attachments or as text in the e-mail message. For more information, see Knowledge Articles.

- Agent inbox

  IC agents can access e-mail through their inbox and process them. For more information, see Agent Inbox.

- Interaction record

  IC agents have access to previously sent or saved e-mails in the confirmed business partner's interaction record. For more information, see Interaction Record.

- Telephony

  IC agents can make a phone call by using the **Dial** function in the application toolbar while remaining in the e-mail channel.

Prerequisites

- You have made settings to display fields (for example, date and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.

- You have made settings for e-mail in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels**. For example, you have made the following settings:

  - You have defined toolbar properties for e-mail in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

  - You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Standard Response Groups**.

  - You have defined outgoing e-mail address groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Outgoing E-Mail Address Groups**.

  - You have defined e-mail profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

Features

- Standard responses

  IC agents can reuse use predefined text. For more information, see Standard Responses.

- Multiple incoming e-mail addresses

  Interaction centers can use more than one incoming e-mail address. Each address is assigned to a group of IC agents. IC agents can be assigned to more than one address.

- Multiple outgoing e-mail addresses

  Interaction centers can also use more than one outgoing e-mail address. IC agents are assigned to one default outgoing e-mail address based upon their e-mail profile as defined in Customizing.

- Unfinished e-mails

  IC agents can process unfinished e-mails if they need to deal with another transaction before completing an e-mail response.

Smart Responses

Use

Smart responses are standard responses with the ability to include additional dynamic information not limited to simple business partner attributes or system information.

Text is populated with business values and inserted into the e-mail content at runtime. This reduces the amount of time agents have to spend searching for and compiling information in a response. Agents are unaware if the response they choose is a standard response or a smart response.

Example

A customer sends an e-mail to the interaction center asking what his account balance is. After the agent confirms the business partner and initiates a reply to the e-mail, the agent selects a standard response. Variables such as account balance and account number are populated with the business partner's values in the response.

Note

Interaction center agents cannot insert smart responses manually into an e-mail using the **Standard Response** dropdown.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function in the role of an interaction center manager under **Knowledge Management** > **Mail Forms**.
- You have defined Multilevel Categorization. (Not relevant for Auto Acknowledgements)
- You have defined the rule policy with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.
- You have defined the data source classes in Customizing for **Service** by choosing **Interaction Center WebClient** > **Customer-Specific System Modifications** > **Smart Responses** > **Define Data Source Classes**.

Features

Like standard responses, you can insert placeholders in the mail form that can be filled in by data related to the e-mail or to the business partner of the e-mail. You can define how the values for business and transactional data are retrieved and placed in mail forms. You can use data from either the fact base attributes or from the business data context (BDC).

Smart responses can be sent automatically to customers with or without agent intervention based upon multilevel categorization. They can be inserted into the e-mail text or attached to the e-mail. For more information, see the actions and parameters table in ERMS Rule Policies for the **Auto Respond** and **Auto Prepare** actions. They also can be used in auto acknowledgements and response proposals.

Standard Responses

Use

IC agents can use standard responses to help automate e-mail processing. Standard responses use predefined text configured from mail forms that are available to IC agents in the e-mail editor. Standard responses provide consistent handling and messaging to customers regardless of which IC agent processes the e-mail.

On the e-mail editor screen, two assignment blocks offer standard responses.

1. The **Standard Responses** assignment block provides a list of standard responses from four different sources:
    - Standard response groups as defined in the **Define Standard Response Groups** Customizing
    - Standard responses are available in the e-mail editor when you select a relevant category
    - Standard responses from a manual search done by the IC agent
    - Standard responses that the IC agent has saved in their favorites.
2. The **Standard Response Tree** assignment block displays standard responses based on the assigned multilevel categorization schema. Standard responses are also available in chat text. For more information about e-mail and chat, see E-Mail and Chat. Multilevel categorization and Customizing settings help determine which standard responses are available.

Prerequisites

- You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Standard Response Groups**.
- You have assigned the standard response groups to e-mail profiles in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles**.
- If you want to use standard responses for chat, you must assign the standard response groups to chat profiles in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Chat Profiles**.
- If you want to determine the default language for your standard response texts, you must make settings in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > BAdIs for Communication Channels > BAdI: Language Determination for Standard Response Texts**.
- If you want to filter the standard responses to limit their use across channels, you must make settings in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > BAdIs for Communication Channels > BAdI: Filter for Standard Response Texts**.
- If you want to use the **Standard Response Tree** assignment block, you must have your e-mail editor profile to the `EMAIL_PROFILE` parameter for the e-mail application area. You can choose any categorization schema without the restriction of using an email profile when you use the `CRM_IC_STD_RESP_SCHEMA` business add-in (BAdI) definition, within enhancement spot `CRM_IC_STD_RESPONSE`. These standard responses will be in addition to those defined in the **Define Standard Response Group** Customizing.

Features

IC agents can do the following:

- Perform a search and preview standard response texts before choosing a standard response
- Choose the language of standard responses after searches
- Sort and filter standard responses
- Manage favorite standard responses
- View the source of standard response proposals (for example, from favorites or searches)

You can use variables in the mail form to insert placeholders for business partners. When IC agents choose the standard response or insert the standard response into an e-mail, the system fills these placeholders with values. For example, you can populate the customer's name in the salutation.

More Information

- Auto Suggest
- Smart Responses
- Actions and Parameters
- ERMS Rule Policies

Fax and Letter

Use

Interaction center (IC) agents can create, edit, and receive fax messages and letters in the interaction center. IC agents can also use an internal HTML editor to process faxes and letters.

Prerequisites

- You have made settings to display fields (for example, subject, date, and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- If you would like the **To** field of fax messages to be filled automatically, you have maintained the business partner (BP) data with a fax number that includes the country code.
- If you want to use business process push (BPP) with fax and letter, you have to specify the communication management software (CMS) connection by executing the program on the **SAP Easy Access** screen, under **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**. For more information about BPP, see Business Process Push.

  ⚠ **Caution**

  You must enter the following values:

  - Under **Comm. Mgmt Software System ID** enter **FAX_LETTER**.

- Under **Connection** enter the RFC connection to your CMS system.

Features

Agents can search for incoming fax messages and letters in the agent inbox and display them. They can also view old fax messages or letters for a confirmed account in the interaction record.

> **i Note**
> Once a fax has been sent or a letter has been finished, it can no longer be edited.

Using the HTML editor, IC agents can:

- Create, send, and print fax messages and letters
- Display information for the current interaction, such as:
  - Reason
  - Priority
  - Status
  - Name
- Add and display attachments
- Change the default header data for fax messages and letters

Activities

Creating Letters

Before IC agents can process letters in the interaction center, you must upload them to the interaction center by doing the following:

1. Scan the letters.
2. Save them to your hard drive.
3. Go to transaction `oawd`.
4. Expand the node and double-click the highlighted line that appears.
5. Select **Drag & Drop**.
6. In the dialog box that appears, drag and drop the relevant files from your local drive.

Agent Inbox

Use

You can use the agent inbox (inbox) to call up e-mails, business transactions, and various other business objects for display or processing. The inbox is a central worklist that the entire team can use to work on incoming objects.

Prerequisites

- You have made the basic settings required to use the inbox. For more information, see Basic Settings for the Agent Inbox.
- If you want to display an alert showing the number of incomplete e-mails linked to a particular account, you have created a rule policy and a rule to set up the alert. The rule policy must include the **Business Partner Confirmed** (`BPConfirmed`) event and the rule must include the **Show Number of Incomplete E-Mails** (`AH_NO_CUST_MAIL`) action.

  For more information about creating rule policies, see Rule Modeler and ERMS Rule Policies.

Features

Business Roles

You can use the inbox in business roles with the following profile types:

- **IC WebClient Business Role**
- **CRM WebClient Business Role**

  > **Note**
  >
  > The following inbox functions are not available for business roles with the profile type **CRM WebClient Business Role**:
  >
  > - Interact
  > - Link
  > - Next item
  > - Displaying e-mails, fax messages, and letters

Inbox Items

You can retrieve, access, and process the following items in the inbox:

- Business transactions in Service

  You can call up business transactions, such as service orders.

- Workflow work items

  You can process SAP workflow work items, including those that you have customized for use in the inbox. These work items can originate from the current system (SAP S/4HANA or SAP S/4HANA Cloud Private Edition) or from another SAP system.

- E-mails, fax messages, and letters
  - Outbound correspondence

    The inbox supports outbound e-mails, fax messages, and letters created in the interaction center (IC) and linked to an interaction record. The inbox also supports outbound e-mails created by the E-Mail Response Management System (ERMS).

  - Inbound e-mails, fax messages, and letters

    E-mails and fax messages that were received via SAPconnect appear as workflow work items. Inbound letters are also available in the inbox as work items.

    A workflow forwards inbound e-mails to the appropriate group of agents.

- Multiple incoming and outgoing e-mail addresses

  Agents can be assigned to various incoming addresses. All items sent to the agent's various addresses appear in the same inbox. Agents can also be assigned to more than one outgoing e-mail address. A default address is proposed, or the agent can select a different address.

- ERMS routing

  Depending on the rule policies you define, you can use a general task for an ERMS decision to allow e-mail routing to all agents or you can specify individual agents or agent groups to allow e-mail routing only to those agents.

- Number of incomplete e-mails linked to confirmed account

  You can set up the inbox so that once an account is confirmed, an alert is displayed that shows the number of incomplete e-mails linked to the confirmed account. IC agents can click the alert to navigate directly to the inbox, where only the incomplete e-mails from the confirmed contact person of the account are displayed. If no contact person has been confirmed, only e-mails from the e-mail address linked with the account are listed in the inbox.

Inbox Search

For information about the inbox search features, see Inbox Search.

Inbox Result List

For information about the inbox result list features, see Inbox Result List.

Basic Settings for the Agent Inbox

Use

System administrators use this process to set up the agent inbox in the Interaction Center (IC) WebClient.

Prerequisites

- The SAPconnect interface is configured.

  For more information about SAPconnect, search for “SAPconnect (BC-SRV-COM)” on the SAP Help Portal at http://help.sap.com.

- E-mail addresses are created.

- If you want to edit scanned letters in your agent inbox, you have defined an ArchiveLink scenario.

  For more information, see the following Customizing activities under SAP NetWeaver > Application Server > Basis Services > ArchiveLink :

  - Basic Customizing > Edit Document Classes and Edit Document Types
  - Customizing Incoming Documents > Workflow Scenarios > Assign Document Type to a Workflow

Process

Step	Description
Create business partner master data for each employee, with an employee role.	Maintain all inbound e-mail addresses that the agent should be responsible for. Assign a user ID.
Maintain settings for recipient distribution	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Maintain Recipient Distribution.
Maintain automatic workflow Customizing	In Customizing for Service, choose Basic Functions > SAP Business Workflow > Maintain Standard Settings for SAP Business Workflow. For more information, see SAP Business Workflow in the Interaction Center.
Activate event linkage for workflow template `WS 14000164` and assign agents for e-mail handling	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Assign Agents for Fax and Letter Processing. For more information, see Assigning Agents for E-Mail Handling.
Forward to employees/groups	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding. Use this transaction to define the content of the dropdown box Forward To in the result list of the agent inbox.
Define quick searches	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. Use this transaction to define the content of the dropdown box Quick Search in the agent inbox search.
Maintain Customizing activities	In Customizing for Service, choose Interaction Center WebClient > Agent Inbox. For more information, review the Implementation Guide (IMG) documentation for each activity.

More Information

Agent Inbox in the Interaction Center WebClient

Assigning Agents for Fax and Letter Processing

Use

System administrators use this procedure to make the following settings for the agent inbox (inbox) in the Interaction Center (IC) WebClient:

- Activate event linking.

Workflow can define how incoming fax messages and letters are routed and processed. Activating event linking activates the workflow you defined and allows it to be used for the agent inbox. You only have to activate event linking once.

- Assign agents for fax and letter processing.

i **Note**

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see ERMS Rule Policies.

Procedure

Activating Event Linking

1. In the **SAP Easy Access** menu, choose **Service** > **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Agent Inbox** > **Assign Agents for Fax and Letter Processing**.
2. In the **Event Linkage** column, choose **Activate event linking**.
3. Under **IC WebClient: Inbound Process AUI** expand the workflow template `WS 14000164` (**IC: Mail Inbox Handling**).
4. Select the icon in the **Activate/deactivate** column.

   Once selected, the icon should display as **Activated**.

Result

Workflow is activated for the agent inbox.

Define Rules for Fax and Letter Processing

Use

System administrators follow this procedure to define routing rules for incoming fax messages and letters in the agent inbox (inbox) in the Interaction Center (IC) WebClient.

The IC WebClient inbox uses workflow to route incoming fax messages and letters to agents.

Additional routing rules use business partner attributes to route e-mails to agents who specialize in particular areas (see Additional Routing Rules for Fax and Letter Processing).

i **Note**

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see ERMS Rule Policies.

Process

As a system administrator, you define routing rules in the following steps:

1. Create rules.
2. Assign rules to agents or grouping of agents.

Creating Rules

1. In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Rules for Fax and Letter Processing.
2. Choose Create responsibility.
3. Enter the following:
    - Object abbreviation
    - Description
    - Start date
    - End date
4. Choose Continue.
5. Enter a priority.
6. Enter values for each business partner field that you want to use to further define routing.

   To enter more than one range for the same field, select the field and choose New row.

   Enter * (asterisk) for the fields that you do not want to define.
7. Save your data.

Assigning Rules to Agents

1. On the Responsibilities for Rule screen, select the rule that you created.
2. Choose Insert agent assignment.
3. Select an object type.
4. Search for and select one or more objects of the type that you specified in the previous step.
5. Choose Create.

If you want to delete an assignment, select a rule on the Responsibilities for Rule screen and choose Delete agent assignment.

Example

You want to use the business partner language Spanish as an additional routing field. You make the following settings:

Name	From	To
Address	*	<no entry>
Country	*	<no entry>
Language	ES	<no entry>
Name	*	<no entry>
Region	*	<no entry>
Postal code	*	<no entry>

Additional E-Mail Routing Rules

Use

The Interaction Center (IC) WebClient agent inbox uses workflow to route incoming mails to agents. In the standard workflow, after a customer sends an incoming mail to the interaction center, the system retrieves the address to which the customer sent the mail. It determines which agents are assigned for processing incoming mails for that address. This determination is based on the e-mail addresses entered in the employee's business partner master data.

In addition, you can use certain business partner data to define which agents or groups of agents should process the mails (for example, to route e-mails to agents who specialize in particular areas). The routing rules can be assigned to object types such as users, positions, and organizational units.

Prerequisites

Routing rules for e-mails have been defined and assigned to agents or groups or agents (see Defining Routing Rules for E-Mails).

Features

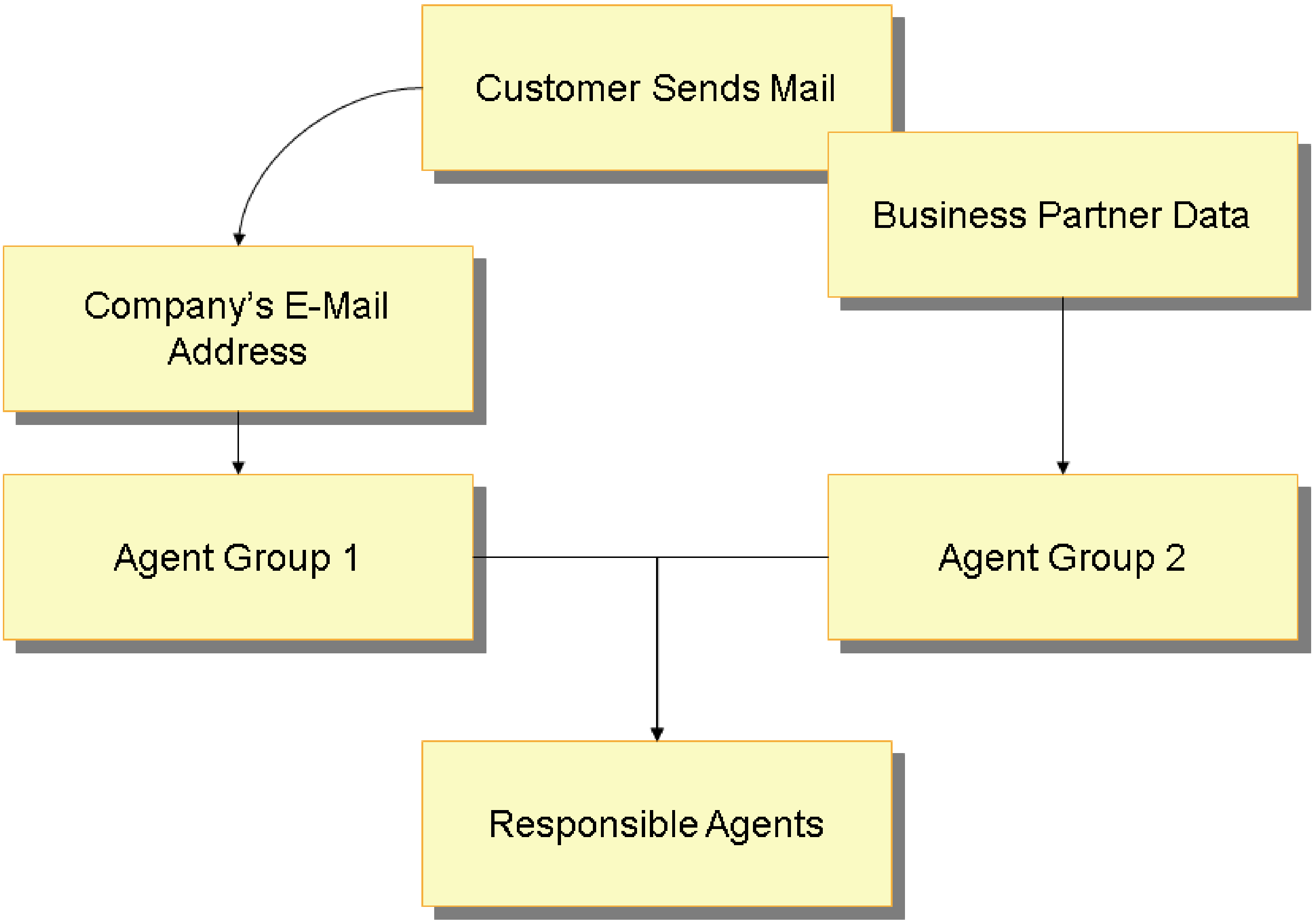
The following business partner data can be used to define how mails are routed:

- Address
- Country
- Language
- Name
- Region
- Postal code

To use other fields, you need to create a new rule. From the **SAP Easy Access** menu, choose **Architecture and Technology** > **ABAP Workbench** > **Development** > **SAP Business Workflow** > **Definition Tools** > **Rules for Agent Assignment.** > **Create**. The **Responsibilities** tab allows the creation of rule definitions and assignments to agents.

Activities

The figure below shows how the system determines which agents process incoming mails and is followed by an explanation:



1. With additional routing rules, the system tries to determine the business partner by the customer's e-mail address. If the business partner is determined, attributes from the business partner master data can be used to further define which agents process the mail.

   Note

   If the business partner cannot be determined, the additional routing rules are not used. In this case, agents who are assigned to process mails for that incoming address or number receive the mail (see [Defining Routing Rules for E-Mails]).

2. The system checks which agents are assigned to the incoming mail address (**Agent Group 1**), as specified in the inbound distribution.

3. The system checks which agents are assigned to process the specific business partner attributes (**Agent Group 2**) if the business partner address or number is found. This depends upon the defined rules and which agents are assigned to them.

4. The routing depends on how many agent groups exist:

   - Two agents groups

     The mail is routed only to agents who are in both of the groups (**Agent Group 1** and **Agent Group 2**). If no agents are found in both groups (agents in **Agent Group 1** and **Agent Group 2**, but none belonging to both groups) the mail is routed to all agents in each group.

   - One agent group

     If no agents are in **Agent Group 1**, the mails are routed to agents in **Agent Group 2**. Alternatively, if there are no agents in **Agent Group 2**, the mails are routed to agents in **Agent Group 1**.

Example

Routing works as follows when two agent groups exist:

1. Mr. Smith sends an e-mail to service@company.com.

   Julie and David (Agent Group 1) are assigned to process mails sent to this e-mail address.

2. The system identifies Mr. Smith's business partner data based upon his outgoing e-mail address.

3. The system determines that an additional rule can be used based upon Mr. Smith's country (US).

   Michael and Julie (Agent Group 2) are assigned to an e-mail rule for US customers.

4. The system determines that only Julie should receive the incoming mail because she is the only agent in both of the agent groups.

   If Julie was not in both of the groups, then all of the agents, David, Michael, and Julie, would receive the incoming mail.

Define Recipient Profile for Forwarding

Use

Use this procedure to define to which person(s) you may forward items for further processing from the agent inbox in the Interaction Center (IC) WebClient. Inbox items may be forwarded to an individual employee or a group.

Prerequisites

Business partners specified as recipients must be defined.

Procedure

In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding.

Quick Searches

Use

You can define searches that can be used as options for finding business objects in the agent inbox (inbox).

Prerequisites

Features

You can define quick searches in the SAP Easy Access menu, under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches.

SAP delivers one Quick Search (0001) that is defined to select My Open Items sorted first by priority, then by main category. You can define additional quick searches based on your own selection criteria.

You can use an asterisk (*) as a wildcard to specify the title of a business object that has been derived from the business transaction.

You can also define a default quick search for each business role in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Note that the inbox supports the following time periods for quick searches: creation period, change period, and due period.

Example

You can define quick searches to retrieve item groups such as:

- All overdue items
- All items for which you are responsible
- All high-priority items

Inbox Search

Use

The agent inbox (inbox) search provides a broad range of search attributes and search features that you can use to tailor the inbox search according to your business requirements.

Prerequisites

You have set up and configured the inbox. For more information, see Agent Inbox in the Interaction Center WebClient.

Features

Search Attributes

For information about the available search attributes, see Search Attributes.

Search View: Standard and Advanced

The standard search provides a fixed arrangement of search fields that the user cannot change.

The advanced search provides selectable search attributes, the multiple use of search attributes, and the use of saved searches.

You can specify whether you want to use the standard search or the advanced search. You do this in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Lean Search

During a lean search, the system determines only the required data and does not store any data. It only determines the object attributes that are relevant for the search result list. This enables a faster search and faster processing times when working with the result list. The remaining attributes are determined when the object is used, for example, when the object is opened in the display mode or in the edit mode.

Predefined Searches

Quick Searches

You can create predefined search queries that appear as selection criteria in the **Quick Search** field. For example, you can define a search that agents can use to retrieve all high-priority items. For more information, see Quick Searches.

Quick searches are assigned to business roles through the inbox profile.

Saved Searches

The saved searches function is only available in the advanced search. Saved searches are user-defined.

Search Attribute Validity Check

The following attribute validity check prevents unclear search results:

If a main category cannot be combined with a selected search attribute, the main category is excluded from the search. For example, a search involving the main category **E-Mail** in combination with the **Checklist ID** search attribute is not valid, and the search is not performed.

This system behavior also applies if only one main category has been selected or if no main category has been selected. In the latter situation, the validity check is performed for all main categories that are defined for the corresponding inbox profile in Customizing, and the search is performed for all main categories that have passed the validity check.

Search Traceability

A search processing log consolidates various detailed search messages, such as which search attribute is not supported by which main category, in a structured way. The processing log can be accessed using a corresponding button in the result toolbar.

In addition to the entries in the processing log, the following general messages are displayed on the UI, as appropriate:

- Search has been only partially performed
- Search could not be performed

This enables the user to understand the search behavior and to tailor future searches accordingly, or to contact the system administrator, as required.

Predefined Inbox Search Tables

You can use predefined inbox search tables that contain the inbox search attributes and inbox result attributes that are relevant to specific inbox item types. Predefined inbox search tables are available for the following inbox item types:

- **Inbox Item: OneOrder**
- **Inbox Item: Workitem - Inbound Email, Fax and Letter**
- **Inbox Item: Outbound Correspondence**

For information about the prerequisites and features, see Predefined Inbox Search Tables.

Search Attributes

Use

You can use various search attributes and search attribute combinations to locate the required agent inbox (inbox) items.

Prerequisites

For information about the necessary settings for the inbox search, see Inbox Search.

Features

Available Search Attributes

Area	Search Attribute	Function
Predefined searches	Quick Search	Predefined search criteria (for example, My Open Items) You can configure these searches on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. You must assign the relevant quick searches to the corresponding inbox profile, and you can define a default quick search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Saved Searches	The saved searches function is available in the advanced inbox search.
Inbox item	Main Category	Inbox item type as defined in the following Customizing activities in Customizing for Service under Interaction Center WebClient > Agent Inbox: • Inbox Search Definitions > Define Item Types for Searches • Define Inbox Profiles
	Object ID	ID of the inbox item
	Description	Description of the inbox item
Partner information	Created By	User who created the inbox item
	Account	Account for which the inbox item has been created
	Assigned To	Assigned agent or agent group To use the Assigned To search attribute to search for business transactions and cases, the relevant user or group must be maintained as a business partner in the system. The business transaction and case search is based on the business partner in combination with the inbox partner mapping (inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The search for work items (inbound e-mails, fax messages, and letters), outbound correspondence (outbound e-mails, fax messages, and letters), social media posts, and workflow work items is based on the assigned users. Assigned To: Me The search returns the inbox items that are assigned to the current user. The search returns the following results: • Business transactions and cases to which the current user is assigned as the employee responsible • Work items for which the current user is the employee responsible (assigned agent)

Area	Search Attribute	Function
		• Work items that are available for processing by the user and have not yet been picked by another user • Outbound correspondence that was last changed by the user • Social media posts for which the current user is the employee responsible (assigned agent) Assigned To: My Groups The search returns the inbox items that are assigned to the selected groups. The search returns the following results: • Business transactions and cases to which the selected groups are assigned as the groups responsible • Work items to which the users of the selected groups are assigned as the processor • Work items that are available for processing by the users of the selected groups • Outbound correspondence that was last changed by users of the selected groups • Social media posts to which the users of the selected groups are assigned as the processor Assigned To: <no entry> For work items, the Assigned To: My Groups search is performed. Note that the search for workflow work items always returns the workflow work items assigned to the current user. Therefore, the Assigned To search attribute is not relevant for workflow work items.
	Employee Responsible	Assigned employee responsible The specifics for the Assigned To: Me search attribute also apply to the Employee Responsible search attribute. The Employee Responsible search attribute cannot be combined with the search attribute Assigned To.
Time	Creation Period Change Period Due Period	Predefined periods of time for searching inbox items (for example, Last 3 days) In the standard search, you can use the respective From/To fields to enter a user-defined date or date range. You can define the time periods in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Time Periods for Searches. You can define which time periods are available for the respective search attributes in the inbox search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Created On Changed On Due On	Date or date range These search attributes are only available in the advanced search.
Importance	Status	Inbox status, such as Open, In Process, Completed

Area	Search Attribute	Function
		You can define the inbox statuses in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Inbox Status. You must map the system statuses and user statuses of the inbox item types to the inbox statuses. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Status to Inbox Status.
	Priority	Inbox priority, such as High, Medium, Low You must map the inbox item type priorities to the inbox priorities. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Priorities to Inbox Priorities.
Multilevel categorization	Category ID	ID of category used in multilevel categorization For more information, see Multilevel Categorization.
Checklist	Checklist ID	Not applicable
E-mail addresses	Sender Address Recipient Address	E-mail addresses that can be used to search for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) If you are not using the predefined inbox search table for work items, entries in the Sender Address field are restricted to 20 characters (use an asterisk in case of longer e-mail addresses). • Outbound correspondence (Outbound e-mails, fax messages, and letters)
Sorting	Sort By	Sort attributes by which the search result is sorted A maximum of two sort attributes can be used for each search.
Generic functions	Clear	Clears the search attribute values
	Reset	Sets the inbox search criteria and the inbox result list back to the initial state: • The Quick Search field is cleared. • The search criteria is set back to the predefined combination of search fields (relevant to the advanced search only), and the field values are cleared. • The Maximum Number of Results field is set back to the default. • The result list is cleared, and the sorting and the filtering are set back to the initial state. Note that personalization settings for the result list and the selected display type (table or tree) are not set back.
	Maximum Number of Results	Maximum number of search results You can define the maximum number of search results to improve the search performance, for example. You can define the default values and range of user entries for the Maximum Number of Results field. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.

Search Attributes not Enabled for Multiple Use:

- Category ID
- Assigned To: Me and Assigned To: My Groups cannot be combined with other Assigned To search criteria. You cannot for example perform a search using Assigned To: Me in combination with Assigned To: My Groups.
- Sort By

Predefined Inbox Search Tables

Use

The standard system contains predefined inbox search tables, which contain the inbox search attributes and inbox result attributes relevant for a specific inbox item type. You can improve the search performance by using predefined inbox search tables. For each of the following inbox item types, a separate table is available:

- Inbox Item: Workitem - Inbound Email, Fax and Letter

  Inbox search table for work items (`CRMD_AUI_WI_IDX`)

- Inbox Item: Outbound Correspondence

  Inbox search table for outbound correspondence (`CRMD_AUI_OUTCORR`)

  i Note

  For outbound correspondence, the predefined inbox search table is mandatory for both of the abovementioned inbox item types.

- Inbox Item: Inbox One Order item

  For this inbox items type a predefined search table is not required because the search is optimized already. However, in case of Customizing changes, this inbox item type must be updated. For more information, see the *Features* section.

Prerequisites

General

- You have defined the business transactions and work items that you want to use in the inbox in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches.
- You have assigned partner functions for use in the inbox in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Business Transactions to Responsible Employees and Groups.
- You have defined the inbox status mapping in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Status to Inbox Status.

Inbox Search Table for Work Items

- You have made the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Assign Standard Tasks to Communication Methods.
- You have made the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles > Work Item Table.

Inbox Search Table for Outbound Correspondence

- If you want to adjust the data record of an outbound correspondence item before it is stored in the inbox search table for outbound correspondence, you make the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Business Add-Ins > BAdI: Outbound Correspondence Database Update.

Features

Data Load

To use the predefined inbox search tables, you first need to upload the data for the relevant main categories. You do this in the SAP GUI, on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Process Data for Inbox-Specific Searches (transaction `CRM_IC_INDEX_LOAD`).

Only the main categories that are mapped to the corresponding inbox item types in Customizing activity Define Item Types for Searches are available in the program Process Data for Inbox-Specific Searches.

Note

If you change any of the Customizing settings detailed in the Prerequisites section, you must update the corresponding inbox search table accordingly.

In this case, you must also update the entries for the inbox item type Inbox One Order Item to fill various header fields in the corresponding business transaction header table, for example `CRMS4D_SERV_H` for service orders.

Process Data for Inbox-Specific Searches: “Inbox Item Type”

- Choose the inbox item type `ONEORDER` Inbox One Order Item to update the following header fields in the corresponding business transaction header table, for example `CRMS4D_SERV_H` (only in the case of Customizing changes):
  - INBOX_STATUS
  - INBOX_ASSIGNED_TO
  - INBOX_GROUP_ASSIGNED
  - INBOX_ACCOUNT
  - INBOX_CONTACT_PERSON
- Choose the inbox item type `WORKITEM` Inbox Workitem to process the data for the inbox search table for work items. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.
- Choose the inbox item type `OUTCORRESP` Outbound Correspondence Item to process the data for the inbox search table for outbound correspondence. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.

Process Data for Inbox-Specific Searches: Displayed Figures

- Source Entries (Not available for outbound correspondence)

  The number of entries in the tables, which the system uses to update the corresponding predefined inbox search table

- Target Entries

  The number of entries in the predefined inbox search table that is being updated

Process Data for Inbox-Specific Searches: Processing Options

- Update

  Loads the data for the selected main categories from the source tables to the corresponding inbox search table

  Note that for the selected main categories, all previous entries in the corresponding inbox search table are removed and new entries are provided.

- Remove

  Removes the selected main categories from the corresponding inbox search table

- Calculate

  Updates the displayed figures

- Reset Status

  Resets the status to Not Started

  You can use the Reset Status function if you have canceled a job and the status of the corresponding main category is still In Process.

The functions above are also available if multiple main categories are selected, unless one of the main categories cannot be processed due to the status.

Process Data for Inbox-Specific Searches: Availability of the “Update” and “Remove” Functions

The following table provides an overview of the availability of the Update and Remove functions and the corresponding status:

Context	Status of Corresponding Main Category	Update	Remove
One of the selected main categories is no longer supported according to the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches.	Not Supported	Not enabled	Enabled
One of the selected main categories is already included in another data load.	In Process	Not enabled	Not enabled
You have canceled a job and reset the status to Not Started using the Reset Status function.	Not Started	Enabled	Enabled
The initial data load for one of the selected main categories has not yet been performed.	Not Started	Enabled	Enabled
One of the selected main categories has already been removed from the inbox search table.	Not Started	Enabled	Enabled

Context	Status of Corresponding Main Category	Update	Remove
The data for one of the selected main categories has been uploaded to the inbox search table.	Completed	Enabled	Enabled

Delta Update

The automatic delta update is limited to the main categories that you have uploaded, or are in the process of uploading, to the relevant search tables:

- Inbox search table for work items

  When a work item is saved

- Outbound correspondence

  - Outbound e-mails

    When an e-mail is saved, sent, or deleted

  - Outbound faxes and outbound letters

    When a fax or letter is saved or deleted and the session is ended

- For outbound correspondence, you are not required to upload the data for the relevant main categories to enable the delta update.

- The system status is used to calculate whether a business transaction is overdue.

Specifics for the Inbox Search Table for Outbound Correspondence

The system fills the following fields from the interaction record using your settings in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Assign Partner Functions to Business Transactions**:

- `ACCOUNT`
- `CONTACT`

If more than one interaction record is assigned to an outbound correspondence item, the interaction record that the fields are supplied from (during data load) is undefined.

During data load, the system loads the outbound correspondence items linked to an interaction record during an interaction center session or during ERMS processing. For more information, see Interaction Record.

⚠ Caution

If you delete outbound correspondence items from the inbox search table, you can use the **Process Data for Inbox-Specific Searches** transaction to restore items that are linked to an interaction record. However, you cannot use this transaction to restore items that are not linked to an interaction record. For example, you cannot restore outbound e-mails sent by ERMS when they were not created with an interaction record.

Inbox Result List

Use

The agent inbox (inbox) result list displays the inbox search results in table or tree format and provides the functions required to process the inbox items.

Prerequisites

- You have set up and configured the inbox. For more information, see Agent Inbox in the Interaction Center WebClient.
- If applicable, you have configured the result toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Toolbar Profiles**.

Features

Result List in Tree Format or Table Format

You can display the result list in tree format or table format:

- The tree format provides context information, such as related business transactions.
- The table format provides better performance, sorting and filtering functions, and the use of table charts.

  You can define whether the result list is displayed in tree format or in table format as a default. You do this in Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

  The user can switch between the tree format and table format in the inbox, as required.

Maximum Number of Hits

You can define the maximum number of search results.

For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Define Inbox Profiles**.

Advanced Warning for Due Date

You can define an advanced warning for each inbox item type, with reference to the due date.

For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Agent Inbox** > **Inbox Search Definitions** > **Define Item Types for Searches**.

Configurable Result Toolbar

You can configure the result toolbar according to your business needs for interaction center (IC) business roles (business role profile type **IC WebClient Business Role**) and for other core business roles in Service:

- Result toolbar buttons

  You can define the following for each button:

  - Description
  - Label
  - Tooltip
  - Icon

- Availability of button for use in IC business roles, in other business roles for core or both (providing the button function is available in the corresponding business role)

  You can also add user-defined buttons.

- Result toolbar profiles

  You can define toolbar profiles for IC business roles and for other core business roles. You can define the following for each profile:

  - Which buttons are available
  - Button order
  - Button grouping

    You can group buttons using separators.

  - Whether the label, icon, or both are displayed for the button

  If you do not define toolbar profiles for your business roles, the system uses default profiles.

Available Toolbar Buttons and Corresponding Functions:

Button Description	Button Function
Complete	The statuses of the selected inbox items are set to **Completed**. The **Complete** button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) • Workflow work items Whether a workflow work item can be completed depends on the corresponding workflow task. • Outbound text messages
Delete	The selected inbox items are deleted. The **Delete** button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters)
Display	The system navigates to the details page for the inbox item in the display mode: • No interaction record is created. This also applies if the user switches from the display to the edit mode from the object. (For workflow work items and work items (e-mails, fax messages, letters), switching from the display mode to the edit mode is not supported.) • The account is not confirmed and not published in the context area. When switching to the edit mode from the object, the system defines the user who is currently logged on as the employee responsible (according to the partner mapping in the inbox Customizing activity **Map Business Transactions to Responsible Employees and Groups**) and confirms the business partner. The **Display** button is not supported for work items (inbound e-mails, fax messages, and letters) for business roles with the profile type **WebClient Business Role**.
Edit	The following steps are performed:

Button Description	Button Function
	• The system navigates to the details page for the selected inbox item in the edit mode. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). • No interaction record is created. • The business partner is confirmed and published in the context area. If other business transactions are opened during the editing process, always the business partner of the business transaction that was most recently opened is confirmed and published in the context area. • The business transaction is linked to the activity clipboard. • All subsequently opened business transactions are also linked to the activity clipboard, unless the user chooses the End button. Choosing the End button clears the activity clipboard. • If the user chooses Interact, all business transactions that were linked to the activity clipboard during the editing process are linked to the interaction record. • Editing during an interaction If the user edits other business transactions during an interaction, the business partners for the other business transactions are not confirmed or published in the context area. The Edit button is only supported for business transactions.
Forward	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile. If the recipient profile contains more than 10 entries, the 10th item of the button dropdown menu is a More entry that launches an input help. The input help contains all remaining recipients that you have defined in the recipient profile for forwarding. The forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Forward (Extended)	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the following entries (maximum of 10 entries): • The recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile • The More entry that launches an input help The input help contains the remaining recipients that you have defined in the recipient profile for forwarding. • The Other Employees entry that launches the employee search • The Other Service Teams entry that launches the business partner search The extended forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Interact	The following steps are performed by the system:

Button Description	Button Function
	• The system navigates to the details page of the inbox item. • The account and contact are confirmed and published in the context area. • An interaction record is created. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). • In the case of work items (e-mails, fax messages, letters) and workflow work items, the system navigates to the account identification where a preview of the work item or workflow work item is displayed. The Interact button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Interact button is not supported for business roles with the profile type WebClient Business Role.
Link	If the user selects a second inbox item during an interaction and chooses the Link button, the second inbox item is linked to the current interaction record. The Link button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Link button is not supported for business roles with the profile type WebClient Business Role.
Next Item	The system navigates to the next inbox item that is available for processing and locks the item for other users. Once the user has chosen the Next Item button, the button is deactivated until the user chooses the End button. If the Next Item button is used for outbound correspondence (outbound e-mails, fax messages, and letters), the system navigates to the next outbound correspondence item that is in process and to which the current user is assigned as the employee responsible. In this case, the outbound correspondence item is not locked for other users. The Next Item button is not supported for business roles with the profile type WebClient Business Role.
Preview (Toggle)	Displays a preview of the currently selected e-mail, fax message, and letter The preview is displayed below the table chart area. The preview does not provide follow-up buttons, such as Reply and Forward.
Reference	Sets a reference to the currently selected inbox item If the user chooses the Reference button, a dropdown menu is displayed that contains the relevant entries from the activity clipboard. If the user chooses an object from the dropdown menu, the currently selected inbox item is referenced from the object (the inbox item is added to the object's business context). Note that the Reference button only works if the `BUSINESS_CONTEXT` function profile is assigned to the business role. In Customizing for the business context, the following is defined: • The inbox item types to which a reference can be added Only these inbox item types are available in the dropdown menu of the Reference button.

Button Description	Button Function
	• Which inbox item types can be added as a reference For more information, see Customizing for **Service** under **Transactions** > **Settings for Service Requests** > **Define Settings for Business Context**. The **Reference** function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type **WebClient Business Role**.
Reference (Extended)	Unlike the dropdown menu for the **Reference** function, the dropdown menu for the **Reference (Extended)** function contains entries from the **Recent Items**. The last entry in the dropdown menu is the **Other Transactions** entry that launches a transaction search. Apart from this, the **Reference** and **Reference (Extended)** functions are identical. The **Reference (Extended)** function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type **IC WebClient Business Role**.
Refresh	The system performs the most recent search again, based on the selected search criteria or based on a quick search. The **Refresh** button enables the user to conduct the most recent search again without expanding the **Search Criteria** view and choosing the **Search** button, or selecting the quick search again.
Reserve	The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity **Map Business Transactions to Responsible Employees and Groups**). The **Reserve** button is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation	Resets the reservation of the inbox item and removes the employee responsible Only the employee responsible assigned to the inbox item can reset the reservation of the inbox item. The **Reset Reservation** function is not supported for the following inbox items: • SAP ERP business transactions • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation (Extended)	Resets the reservation of inbox items on behalf of other users The **Reset Reservation (Extended)** function is not supported for oubound correspondence (outbound e-mails, fax messages, letters, and text messages).
Separator	The separator is used to group toolbar buttons in the result toolbar profile.
Search Processing Log	Provides access to the search processing log. For more information, see Inbox Search (**Search Traceability** section).
Show Table View (Toggle)	Displays the result list in table format
Show Tree View (Toggle)	Displays the result list in tree format

Multiple Selection of Inbox Items

The following functions are available when multiple inbox items are selected (unless the function is generally not available for one of the selected inbox items):

- Delete
- Reserve
- Reset Reservation
- Complete
- Link
- Forward
- Reference

The following functions are not available when multiple inbox items are selected:

- Edit
- Display
- Interact
- Preview On

Business Transaction Routing

Use

You can use this function to forward business transactions and cases in the interaction center (IC) manually or automatically.

Prerequisites

You have made the necessary settings for automatic routing in the BSP application rule editor.

For more information, see Customizing for Service under Interaction Center WebClient > Business Transaction > Define Automatic Routing.

Features

You can forward business transactions and cases to agent groups as well as to individual agents. Both manual and automatic routing are available:

- Manual routing

  You can manually enter a responsible employee or a responsible group as the business partner.

  The transaction or case can be selected in the agent inbox of the assigned responsible employee as an open item for processing.

  - Assigning a responsible employee

    You can either enter an employee directly or using a search.

    If you start a search and a responsible group is assigned, the system searches for agents in the responsible group. If there is no responsible group assigned, the system takes all agents into account during the search.

  - Assignment of a responsible group

You can enter the responsible group directly or using a search. If you start a search and a responsible employee is assigned, the system searches for all organizational units to which this employee is assigned in the organizational model. If no responsible employee is assigned, the system searches for all organizational units that are assigned to the business role. The organizational units are displayed in the search results according to the structural authorization of the agent.

- Automatic routing

  You trigger automatic routing by choosing **Escalate**.

  You can include up to five of the following attributes in the routing:

  - Priority
  - Status
  - Confirmed product
  - Confirmed customer
  - Categorization
  - Transaction type
  - Business role

  You can also push business transactions and cases to external routers or communication management software (CMS) for distribution to IC agents based on their real-time availability. For more information, see Business Process Push.

E-Mail Response Management System

Use

E-Mail Response Management System (ERMS) is a tool for managing large amounts of incoming e-mail. Instead of routing all incoming e-mails into one queue, ERMS provides services for automatically processing and organizing incoming e-mail. There are several automated activities that reduce the need for manual intervention by interaction center agents, thus substantially increasing efficiency and processing accuracy, by helping agents to process e-mail in less time and allowing them to provide the same quality response regardless of agent skill level. ERMS provides tools for agents to efficiently and consistently respond to messages, and for managers to administer, monitor, and report on the whole e-mail process.

SAP delivers preconfigured ERMS settings and functions that you can extend using custom coding.

Integration

- Organizational Management

  Defines how e-mails are routed and to what groups or organizations and how they are going to handle and process e-mails. You can route e-mails to an organizational unit or responsible group, a position, or even an individual user. The e-mails are routed to the agents' inboxes where they are processed.

- SAP Business Workflow

  Enables automated processing and routing of e-mails. For more information, see Workflow for ERMS.

- SAPconnect

  Receives inbound and sends outbound e-mails. Once e-mails are received, SAPconnect creates SAP office documents, which are then processed by SAP Business Workflow.

- Multilevel Categorization
- Machine Learning

  You can connect ERMS with **SAP Service Ticket Intelligence** to automatically categorize incoming e-mails.
- E-Mail Response Management System Analytics

  Allows you to generate reports and analyze your ERMS processing on historical data. You can analyze historical ERMS e-mail volume, handling time, response time and service level.

Features

Manager Features and Tools for Productivity

The WebClient UI is available for modeling rules and categories, and for monitoring and viewing analytics.

- Rule Policies
- Category Modeler
- ERMS Simulator
- E-Mail Workbench

Agent Features and Tools for Productivity

- Agent Inbox in the Interaction Center WebClient

  Central location in which agents receive and respond to e-mails. E-mails are integrated with other communication channels.
- E-Mail Threading
- Automatic Business Object Creation

  Interaction records, along with appropriate interaction reasons, and service tickets can be created and associated to incoming e-mails automatically. For more information, see the **Create Interaction Record** and **Create Service Ticket** actions detailed in the actions and parameters table in ERMS Rule Policies.

Automated Features

- ERMS Responses
- E-Mail Filtering

  Certain incoming e-mails that are not relevant for agent processing, such as out-of-office replies, spam, or bounced e-mails, can be filtered or deleted based upon the e-mail subject or content.
- Rules

  These identify and determine the automated processing of e-mails.
- E-Mail Routing

  Incoming e-mails can be routed to particular queues or agent groups that are most capable of responding to the issue in the e-mail. For example, you can route e-mails that contain foreign languages to queues in which agents are fluent in that language.
- Content Analysis and Fact Gathering

The system analyzes attributes and content regarding each incoming e-mail and stores it in a fact base, enabling the use of the data for making decisions in respect to categorizing, routing, and automatic responses.

- E-Mail Escalation

  E-mails can be escalated if they are not processed within a specific period of time. For more information, see Defining E-Mail Escalation.

- Web Forms

  In addition to accepting incoming e-mails, it is also possible to accept information created from a web form on a website.

- Support of Multiple Incoming and Outgoing E-Mail Addresses

  You can define multiple incoming e-mail addresses. You can also define multiple outgoing addresses that agents can change when they are responding to e-mails.

More Information

Setting Up ERMS

ERMS Rule Policies

Use

You can create rule policies in the e-mail response management system (ERMS) context to handle and evaluate incoming e-mails.

Rules are evaluated every time an incoming e-mail is received if the policy in which they belong is invoked during rule processing.

To create ERMS rules and policies, you use objects configured in the ERMS repository.

Prerequisites

- You have defined attributes, actions, and action parameters in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

- You have created authorization groups in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

- You have assigned the authorizations groups to authorization object `CRM_ERMS_P` in **Role Maintenance** (transaction `PFCG`).

- You have assigned the authorization groups to policies in the WebClient UI in Service. You can see (and in some cases select) authorization groups in the WebClient UI under **Rule Policy Details**, after you select a policy (the top node of the policy structure). The possible values are the entries you maintained in the Customizing activity above.

Activities

- Select attributes

  Most attributes are self explanatory. Only those needing further explanations are listed below.

Attribute	Use
ERMS: Content Analysis Accuracy	A value between 0 and 1 describing the accuracy of the content analysis result. Accuracy = 1 indicates the result is highly focused. All matching categories form a single path in the category hierarchy. Accuracy = 0 indicates the result is contradictory. At least two matching categories point to different paths in the category hierarchy.
ERMS: Best Minus Next Best Score	Distance in score between the category with highest score and the category with second highest score. Each matching category has a score which is a value between 0 and 1 describing how good the content-queries of the category match the given textual content.
ERMS: Number of Top Categories	Number of matching categories that all have the highest score.
ERMS: Best Category Score	Highest score among all matching categories.
ERMS: Top Scoring Category	Matching category with the highest score.
Escalation Date	Date when the escalation of an e-mail occurs. The date is calculated based upon either rule evaluation, Customizing, or service level agreements (SLA) at the time the e-mail arrives.
Escalation Time	Time when the escalation of an e-mail occurs. The time is calculated based upon either rule evaluation, Customizing, or SLA at the time the e-mail arrives.
Escalation Type	Depends on the property METHOD of the service UT_ESCALDETERM. If the METHOD is set to 1, the Escalation Typeis SLA. If the METHOD is set to 2, the Escalation Type is determined as the shortest time out of one of the following: rule evaluation, Customizing, or SLA.
Response Time Duration (Hours)	Used for escalations. Represents the initial response time in hours.

- Set actions and parameters

  This covers the standard delivered actions and their corresponding parameters.

Action	Use	Parameters
Add Attribute Value	Adds a value to a multi-value attribute	Attribute The attribute that a value is added to. Value The value that is added.

<table>
<tr><th>Action</th><th>Use</th><th>Parameters</th></tr>
<tr><td>Auto Prepare</td><td> Same as Auto Respond, except the response is not sent automatically. The agent can review the content first, make any necessary changes, and then send the response. This action is triggered when an agent selects Reply to an incoming e-mail. </td><td> Mail Form Name of the mail form that determines the content of the outgoing e-mail. Outgoing E-Mail Address Address used for the outgoing e-mail's sender address. If you do not use this parameter, the incoming e-mail address that the e-mail was originally sent to, will be used. Category Schema Schema to determine which Top Scoring Category is used. Inline Determines whether standard responses and solutions tied to the category are inserted directly into the e-mail content instead of as an attachment. If both are inserted into the e-mail content, then the standard responses are shown first in the order they are attached to the category. The solutions are then added, also in the order they are attached to the category. </td></tr>
<tr><td>Auto Respond</td><td> Sends an automated response with proposed answers and/or solutions, to an incoming e-mail, without agent intervention. The solutions are inserted into the e-mail as attachments and are determined from the Top Scoring Category of the incoming e-mail. Not to be confused with the Send Auto Acknowledgement action. </td><td> Category Schema Mail Form Outgoing E-Mail Address See descriptions above from Auto Prepare. Create Interaction Record Creates an interaction record, after the response is sent, which includes both the incoming and the outgoing e-mails. Route To (On Exception) Organizational unit that the incoming e-mail is sent to if no solutions are attached to the Top Scoring Category. Inline See description above from Auto Prepare. </td></tr>
<tr><td>Create Interaction Record</td><td> Creates an interaction record in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to interaction record, so the original e-mail can be viewed at a later time, if necessary. </td><td> Description Description saved in the interaction record. </td></tr>
</table>

Action	Use	Parameters
Create Service Ticket	Creates a service ticket in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to service ticket, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the service ticket. You can configure the transaction type (for example, TSRV or TSVO) for this service ticket. You do this in Indirectly Called Services and Properties in Customizing for Service by choosing E-Mail Response Management System > Service Manager > Define Service Manager Profiles.
Delete E-Mail	Deletes an incoming e-mail. Can be combined with the Create Interaction Record action, so the interaction record shows an e-mail was received and was also deleted.	(No parameters)
Forward E-Mail		Forward To Forward From
Invoke Policy	Temporarily suspends the processing of the current policy and continues with the processing of the specified policy. After the specified policy is finished processing, the action stops processing the original policy, unless the processing was stopped previously by the Stop Further Rule Processing action. This action can be nested, that is a policy that is invoked may invoke other policies. If a policy has multiple variants, then the active variant is invoked.	Policy Name of the rule policy that is invoked
Push E-Mail	Allows e-mail routing through the communication management software instead of directly going to an inbox. E-mails that are pushed to agents using this action are also routed to the Agent Inbox by the ERMS default routing, unless another routing destination is specifically maintained. Therefore, we recommend directly specifying a Route E-Mail action after the Push E-Mail action, allowing all pushed e-mails to be monitored by a supervisor or special organizational unit, and avoiding incorrect routing of push e-mails to the default routing that are already completed.	Organizational Object
Route E-Mail	Directs the incoming e-mail to an agent or group of agents. The e-mail is then accessible from the agent inbox.	Organizational Object You can enter an agent's system user name or an organizational unit. If you enter an organizational

Action	Use	Parameters
		unit, then the e-mail can be processed by any agent who belongs to that organizational unit.
Route to Case Processor	Directs the incoming e-mail to the agent who is assigned as the case Processor in the Case Overview, only if the incoming e-mail is related to an existing case and the case number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no case associated to the incoming e-mail or if there is no processor assigned to the case.
Route to Service Ticket Responsible	Directs the incoming e-mail to the agent who is assigned as the Employee Responsible in the Service Header, only if the incoming e-mail is related to an service ticket and the service ticket number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no service ticket number associated to the incoming e-mail or if there is no employee responsible assigned to the service ticket.
Send Auto Acknowledgement	Sends an automatic notification that the e-mail was received.	Mail Form Outgoing E-Mail Address See descriptions from Auto Prepare. Create Interaction Record Creates an interaction record to record that an acknowledgement was sent after receiving the e-mail. (The outgoing e-mail acknowledgement text is not tied to the interaction record.)
Set Attribute Value	Allows you to set an attribute value in the fact base if certain conditions are met. This attribute can then be used in rule evaluation or passed on to actions.	Attribute Value
Show Number of Incomplete E-Mails	Searches for incomplete e-mails linked to an identified account.	(No parameters)
Stop Further Rule Processing		(No parameters)

- Set rule policy authorizations

  You can define your own groupings for authorization, for example, by department or by organization.

Authorization Field	Description	Values
ACTVT	Activity	The following values affect authorizations for policies: 01: Create or Generate 02: Change 03: Display 06: Delete 43: Release(For releasing draft rules)

Authorization Field	Description	Values
ERMS_AUGR	Authorization Group	The values you defined for the authorization groups in Customizing activity **Define Repository**.
ERMS_CTXT	Context	The context in the WebClient UI in Service as defined in Customizing activity **Define Repository**.

More Information

Rule Modeler

Rule Policies

Rules

Conditions

Actions and Parameters

Conflict Resolution

Rule Policy Authorizations

ERMS Simulator

Use

The **ERMS Simulator** allows an interaction center (IC) manager to simulate the results of new or changed rules contained in rule policies that are used for e-mail handling. IC managers also see a detailed view of the resulting actions and categorizations for those e-mails. An IC manager can run the simulation in the background and have the result sent to an e-mail address.

The functions in the **ERMS Simulator** are available in the WebClient UI.

Prerequisites

- You have created rule policies.
- You have set up ERMS and already processed incoming e-mails, using ERMS.

More Information

Setting Up ERMS

E-Mail Workbench

Use

The **E-Mail Workbench** is an overview of e-mail processing in an interaction center, which you can also use as a means to mass process e-mails.

You can access the **E-Mail Workbench** in the WebClient UI by choosing **IC Manager** > **E-Mail Workbench**.

You can perform manual operations for e-mails, such as searching, routing, forwarding, re-categorizing, and deleting e-mails. You can also view the specific rules that were invoked for a particular e-mail and process escalated e-mails.

You can view and in some cases edit the following elements of an e-mail:

- E-mail text
- Attachments
- Rules for processing the e-mail
- Associated categories, that you can add or remove. Automated processing can assign categories to e-mails as determined from Multilevel Categorization. You can change these categories if the assignment is not accurate or complete.
- The Fact Base
- Related e-mails

Prerequisites

- You have set up and actively used ERMS to process incoming e-mails.
- You have set **UT_ERMS_REPLICAT** as your ERMS service manager profile for **Directly Called Services**. You do this in Customizing for **Service** by choosing **E-Mail Response Management System** > **Service Manager** > **Define Service Manager Profiles**.

ERMS Responses

Use

There are several types of E-Mail Response Management System (ERMS) responses you can use to automate e-mail processing. This automated processing saves agents time in compiling and searching for information and ensures consistent handling and messaging to customers. You can use mail forms to configure the e-mail content for the responses.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function under **Knowledge Management** > **Mail Forms**.
- You have defined multilevel categorization. (Not relevant for Auto Acknowledgements)
- You have defined a rule policy with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.

Features

The following is a list of responses you can determine for ERMS:

- Auto Acknowledgement

When an e-mail is received, the system can send an automatic acknowledgment informing the sender that the e-mail was received. If the business partner is identified, you can use business partner attributes, such as first and last name, to personalize the response.

- Auto Response

  The system can send a solution or a response automatically, without intervention from agents, based upon the e-mail content.

- Auto Prepare

  Response proposals are similar to auto responses. However, instead of automatically sending the response, the response is reviewed by agents for accuracy. Agents can send, modify, or delete the proposal.

More Information

ERMS Rule Policies

E-Mail Threading

Use

You can use e-mail threading for service requests and incidents to link e-mails to the associated business transaction, keeping all communication and related e-mails together for easy access to the information, regardless who handles the business transaction. E-mail threading also enables the e-mails to be routed to the person responsible for that business transaction if the customer sends the e-mail back to the interaction center.

Mail forms are used to create a tracking text.

Prerequisites

- You have created tracking text in WebClient UI under **Knowledge Management** > **Mail Forms**. You can add the **Email Response Management** context attribute to the **Text Element**.

  The **Service Request Tracking Text** mail form field is available as a variable for service requests and incident numbers.

  If the variable is not available in the mail form, you have to manually add the `CRMS_ERMS_SCENARIO_FIELDS` structure to the **ERMS** context in Customizing. You do this in Customizing for **Service** by choosing **Basic Functions** > **Communication Channels** > > **Maintain Attribute Contexts for Mail Forms**.

- You have enabled tracking text insertion by maintaining the **Tracking Text Form** field and selecting the **Track Text on Reply** indicator in Customizing for **Service**. You do this by choosing **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

- You have defined rules using the **Route to Service Request Responsible** action. For more information, see the actions and parameters table in ERMS Rule Policies.

Features

- **E-Mail Tracking Thread**

  The threading is enabled by the use of tracking text you define. You use a variable that is filled at runtime with the service request number to insert in the tracking text. This variable allows e-mails to be linked to business transactions.

- **Tracking Text Insertion**

The system inserts the tracking text in the e-mail content when an e-mail is created, either manually or automatically through automated processing, or when the e-mail is replied to. If a service request or incident is automatically created for an incoming e-mail from the auto acknowledgement action, you can link both the incoming e-mail and the outgoing reply to the service request or incident.

- **Thread Detection and Routing**

  When customers reply to e-mails, the system determines whether tracking text is inserted. The incoming e-mail can be associated automatically to the related business transaction. Attributes from these transactions, such as the service request number, can be used during rule evaluation. When you create rule policies, you can create rules that route the e-mail to the agent who is currently assigned to the business transaction.

Example

Example

You create a mail form with the following as the tracking text.

************** DO NOT DELETE **************

{SrvReqNo:[8000006814]}

************** DO NOT DELETE **************

You define a rule, “If Service Request Responsible Is Not Equal To "", Then Route to Service Request Responsible to forward the e-mail”.

More Information

E-Mail Threading Process

E-Mail Threading Process

Use

This process describes e-mail threading using service tickets as the relevant business object.

Prerequisites

You have defined e-mail threading.

Process

1. The customer sends an e-mail.
2. Either an agent sends a manual response or the system sends an automatic response.

   Manual Response

   a. The agent processes the e-mail and creates a service ticket.

   b. The agent replies to the e-mail.

c. The system uses the mail form you entered in the agent's e-mail profile. It inserts the mail form text and tracking text into the outgoing e-mail. The system populates the variable with the actual service ticket number that was just created.

Automatic Response

a. The e-mail response management system processes the e-mail automatically and creates a service ticket, as defined through an action.

b. The system sends a reply to the customer as defined in an action, such as **Send Auto Acknowledgement**.

c. The system uses the mail form that is defined as a parameter in the **Send Auto Acknowledgement** action. The mail form includes the service ticket tracking variable. When the system sends the reply, the variable is populated with the service ticket number that was just created.

3. The customer replies to the e-mail and sends it back to the interaction center.
4. The fact gathering services scan the e-mail and pick up the tracking text and store this in the fact base during automated processing.
5. The e-mail is associated to the service ticket.
6. The e-mail is routed to the agent who is processing the service ticket through a rule that uses the **Route to Service Ticket Responsible** action.

E-Mail Escalation

Use

E-mails can be escalated if they are not processed within a specific period of time, for example, e-mails that are not processed within 24 hours after the initial receipt are escalated. You can define the time periods that determine the escalation and you can be notified and view escalated e-mails as soon as the escalation occurs. From reviewing escalated e-mails, you can reroute the e-mail or decide on another action to speed up processing.

Integration

The SAP Business Workflow is used for the escalation process. In Workflow for ERMS, the escalation times you define are used as the **Latest End** date and time.

Prerequisites

You have defined e-mail escalations.

Features

Escalation Definition

Escalated e-mails are determined when defined initial response times are not met. The initial response time is the time between an e-mail is received in the system until the time that an agent processes the e-mail.

Initial response time is calculated by one of the following:

- Service Level Agreements (SLA)

You can set a deadline for escalation by using a service contract. The system uses the service contract that is assigned to the business partner and reads the duration time from the response profile associated with the service contract.

- Rules and actions

  In the WebClient user interface, you create a rule using the **Set Attribute Value** action with the value **Response Time Duration** to determine at what point an e-mail is considered escalated. You can also use some fact base attributes as additional criteria, for example, if a customer is a high priority and valued customer with a specific system status, and sends an e-mail with a request, but the e-mail is not processed within a specific number of hours as determined in the rule, the e-mail is escalated.

- E-Mail Addresses

  You can determine an escalation time based upon the receiving e-mail address.

Escalation Determination

You can choose which method is used to determine the initial response time. You can either use the:

- Service Level Agreement
- Combination of SLA, rules and actions, and e-mail addresses

  The system uses the definition with the shortest initial response time out of all three methods.

Escalation Process

See E-Mail Escalation Process.

Escalation Notification

You are notified of a new escalated e-mail by receiving an e-mail. Mail forms are used to create the notification text.

Escalation Monitoring

After receiving the escalation notification, you use the overview in the E-Mail Workbench to monitor the number of escalated e-mails per organizational unit and navigate directly to the individual e-mail for further action.

E-Mail Escalation Process

Use

This process describes the e-mail escalation process.

Prerequisites

You have defined e-mail escalations.

Process

1. An e-mail is received in the interaction center (IC).
2. Automated processing begins.
3. The initial response time (IRT) is determined by the escalation definition.

4. If the IRT limit is reached, the e-mail becomes escalated and the workflow escalation process starts.

5. The notification e-mail is sent to the IC manager.

6. The manager accesses the E-Mail Workbench, views the situation and takes further actions with the escalated e-mail.

More Information

E-Mail Escalation

Defining E-Mail Escalation

Prerequisites

You have authorization to read mail forms.

Context

You use this procedure to define and setup E-Mail Escalation.

Procedure

1. Define the method used for determining initial response times.

   - Service Level Agreements

     a. Run transaction `CRMD_SERV_SLA`.

     b. Maintain a response profile and assign a response time. For **Name of Duration**, select **Duration Until First Reaction**. The response profile is associated to a service contract. For more information, see Service Level Agreements (SLA).

   - Rules and Actions

     Define a rule in the WebClient user interface and use the **Set Attribute Value** action with the **Response Time Duration** attribute.

   - E-Mail Addresses

     Define the escalation time period for each receiving e-mail address in Customizing for **Service** by choosing **Interaction Center WebClient** > **Agent Inbox** > **Settings for Asynchronous Inbound Processing** > **Define Receiving E-Mail/Fax Settings**.

2. Assign the method used for determining initial response times and assign the fact gathering service.

   a. In Customizing for **Service** choose **E-Mail Response Management System** > **Service Manager** > **Define Service Manager Profiles**.

   b. Select your service manager profile.

   c. Under **Directly Called Services**, find the Service ID `RE_RULE_EXEC` and add the following services underneath to ensure that they are executed in the correct order with the attribute **Invocation Order**. If you do not use Rules and Actions to determine your initial response times, the order of the services could be invoked before `RE_RULE_EXEC`:

      i. Under **Directly Called Services**, add the Service ID `FG_ESCALTIME` if it is missing. You can maintain the property values using the F1 field help.

ii. Under Directly Called Services, add or select Service ID UT_ESCALDETERM and choose Properties.

iii. In the Property ID field add or select METHOD, and in the Property Value field enter the value. Refer to the F1 field help for e-mail escalation values.

Note

Do not include the service UT_SEND_ESCL as part of your DEFAULT service manager profile.

3. Customizing of the notification mail.

   a. In the WebClient UI, define the mail form used for notifying managers of the e-mail escalation. This is done in the interaction center (IC) manager role under Knowledge Management > Mail Forms. Insert escalation specific placeholders by clicking Attributes in the mail form and adding those specific escalation fields (for example, escalation date, time and type, sender's name and e-mail address and the e-mail subject).

      If these escalation objects are not available, ensure that object CRMS_ERMS_ESC_FIELDS has been added and configured to the correct context in the mail form. In Customizing for Service choose Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms.

   b. Assign the mail form and the receiver's and sender's e-mail information to the service manager profile.

      i. In Customizing choose Service > E-Mail Response Management System > Service Manager > Define Service Manager Profiles.

      ii. Select service manager profile ERMS_ESCALMSG and maintain the property values of the service UT_SEND_ESCL under Directly Called Services. Refer to the F1 field help for e-mail escalation values.

   c. In Customizing choose Service > E-Mail Response Management System > Service Manager > Assign Service Manager Profiles. Assign the service manager profile ERMS_ESCALMSG to the object ObjectID 15107965 without entering the Address/Number.

   d. In transaction SWU3 under Maintain Runtime Environment, maintain the Schedule Background Job for Workflow Deadline Monitoring and specify the interval until next deadline check.

Automated Processes and Architecture

Rule Processing Services

Use

Rule processing, or rule invocation, services is one of the four types of services found in the E-Mail Response Management (ERMS) service manager. There are standard delivered rule processing services available. For more information, see Customizing documentation for Service > > E-Mail Response Management System > Service Manager.

Features

Rule processing services evaluate a set of predefined rules and determine a set of actions that need to be executed during rule processing.

Rules consist of conditions and actions that are triggered if the conditions evaluate to true. The rule processing services support rules and nested rules. Nested rules are only processed if the condition in the higher-level rule is true.

Rule processing starts with the first rule defined within a given policy and continues sequentially with all subsequent rules in exactly the order defined in the rule modeler. All actions that are triggered during rule evaluation are collected and passed back to the service manager. For each of these actions, the service manager determines the appropriate action handler and calls these handlers accordingly.

Special rule processing

The **Stop Further Rule Processing** action can immediately interrupt rule processing. If this action is triggered, only that actions that have executed to that point are passed to the service manager.

The **Invoke Policy** action is executed by the service manager after the current policy is completely processed. It is not executed immediately.

Content Analysis

Use

This function identifies the categories that best match specific content and makes automatic category suggestions.

Content analysis is used in the E-Mail Response Management System (ERMS) to analyze the content of inbound e-mails. It uses predefined search queries that are assigned to categorization schemas, to determine suitable categories.

Integration

Function	Description
Category Modeler	You define content queries for content analysis in the category modeler. If necessary, you can reset the content analysis in the page area **Application Areas** for each application area. You do this by choosing **Reset Content Analysis**.
E-Mail Response Management System (ERMS) - automated processing	ERMS manages a high volume of inbound e-mails. This system partially automates routing and responding to e-mails. The automated processing of e-mails in the ERMS uses content analysis to evaluate conditions assigned to categories. The following ERMS components relate to content analysis: • Fact gathering • Rule operators **Matches Category** and **Rule Attributes**
Interaction Center	Content analysis provides suggested categories based on the content of e-mails received in the interaction center and selected for processing in the agent inbox. These categories are then displayed in the relevant fields of the following IC-specific business transactions: • Service order • Service request • Interaction record

Prerequisites

You have made content analysis settings in Customizing for **Service** by choosing **Cross-Application Components** > **Multilevel Categorization** > **Define Application Areas for Categorization**.

Features

- Automated categorization

  Category suggestions are made automatically using content queries.

- Language detection

  Content analysis also supports automatic language detection for textual content.

More Information

E-Mail Response Management System

Fact Base

Use

The fact base is an E-Mail Response Management System (ERMS) component that serves as a data container for any information that is related to an inbound e-mail and can be relevant during rule evaluation.

Integration

Fact gathering services gather data relevant to the inbound e-mail and add the data to the fact base.

The rule engine runtime retrieves data from the fact base and uses it for rule evaluation and processing.

Features

The fact base stores information that is relevant for exactly one instance of an inbound e-mail. Such information may include e-mail content, business partner information, and the incoming e-mail address to which the e-mail was originally sent.

The fact base is organized as an XML document. Its structure is flexible to include any tree structure and content, depending upon your business requirements.

Updates and read access to the fact base occurs through XSL transformations (XSLT) and XPath queries. These standardized methods are platform- and programming-language independent. This provides maximum flexibility if you need to extend the fact base, for example, for data residing in legacy systems.

Overview Report

This report provides a real-time snapshot of the activity and current status of the E-Mail Response Management System (ERMS). It gives you an overview of the volume of incoming e-mails, which you can display by organizational unit, e-mail address, or Web form ID.

To use this monitoring function, you must be actively using ERMS. To use this function, you must be running Service, and you must activate the necessary BSP service.

Performance Report

You can use this report to monitor the overall effectiveness of the E-Mail Response Management System (ERMS) in your interaction center. This report provides a real time overview of key performance indicators such as volume, average handling time, average response time, and service level. You can sort this information by organizational unit or by e-mail address.

To use this monitoring function, you must be actively using ERMS.

Workflow for ERMS

Use

This workflow is used for the E-Mail Response Management System (ERMS) in the Interaction Center.

ERMS allows automated processing of inbound e-mails received via SAPconnect. E-mails can be automatically responded to or routed to an agent group for manual processing. An inbound communication triggers the workflow template `WS00200001`, which controls the work item processing and fulfills several tasks.

The following system tasks are the most important:

- Determining the responsible agent group and routing the work item to that group
- Deleting and stopping further manual processing

For a complete list of tasks, see the workflow template `WS00200001` in the workflow definition tools.

Prerequisites

ERMS configuration activities are completed.

Process

1. The incoming e-mail is received.
2. Task TS00207915 **ERMS Rule Execution** invokes the service manager.

   The service manager provides capabilities (necessary to process e-mails) such as fact gathering, content analysis, and rule processing.
3. Depending on the outcome of the service manager invocation, the system continues processing the work item by one of the following steps:
   - Delete

     You can determine when the system deletes an item by using a rule, for example, in cases of junk e-mail (spam). Task TS66107918 **ERMS Delete E-Mail** removes the office document. The workflow is ended.
   - Do not process further

     You define when processing is stopped in either a rule or in one of the services in the service manager.

- Route

  Task TS00207914 **ERMS Decision** forwards the work item to an agent or agent group as selected by the processor determination.

The processor determination decides which agents are responsible for a work item. Task TS00207914 determines which agents or agent groups are eligible for processing e-mails. The actual determination of which agent or agent group receives the work item is defined in rules.

Technical Execution

The following technical information is useful if you wish to customize your own workflow:

- Workflow template

  The actual procedure is implemented in the workflow template WS00200001 **(ERMS1).**

- Triggering events for the workflow template

  The event **MailReceived** of Business Object Repository (BOR) object `ERMSSUPRT2` is maintained as a triggering event in the template WS00200001.

- Processing methods

  The following table explains how methods of BOR object `ERMSSUPRT2` are used in the corresponding tasks of the workflow template WS00200001:

Method	**Use**
`RECEIVE`	Receives the mail from SAPconnect
`RULEEXECUTION`	Raises the event **MailReceived** and triggers workflow

Setting Up ERMS

Use

This procedure describes the basic settings needed for [E-Mail Response Management System](#).

Prerequisites

Your IT administrator has created e-mail addresses for receiving incoming e-mails.

Procedure

Mandatory and Recommended Steps

Step	**Description**
Create an organizational model.	In Customizing for **Service** under **Master Data** > **Organizational Management** > **Organizational Model** > **Create Organizational Model**.
Maintain workflow standard settings.	In Customizing for **Service** under **Basic Functions** > > **SAP Business Workflow** > **Maintain Standard Settings for SAP Business Workflow**.

Step	Description
Define receiving e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Define Receiving E-Mail Addresses/Fax Numbers.
Assign receiving e-mail addresses to ERMS workflow.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Recipient Distribution.
Assign agents for e-mail handling.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling. Choose Assign Agents. Under workflow ERMS 1 either: • Use a general task for ERMS decision to allow e-mail routing to all agents, depending on the rules, or • Specify individual agents or agent groups to allow e-mail routing only to those agents, depending on the rules.
Activate event linking. Incoming e-mails can use workflow to define how mails are routed and how they are handled. Activating event linking activates the workflow you defined and allows it to be used for ERMS.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling. Choose Activate Event Linking. For New Recipient, choose possible entries and select SAP Object Instance. Use business object repository (BOR) Object Type ERMSSUPRT2 (ERMS Support 2). When this BOR object type is invoked, it triggers workflow WS00200001 (ERMS1). For more information, see Workflow for ERMS.
Define outgoing e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Sender E-Mail Addresses.
Define service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Service Manager Profiles. Choose Default and double click Directly Called Services to set the DEF_ROUTING for Invocation Order 50 to an organizational unit.
Assign service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Assign Service Manager Profiles.
Create ERMS policies and rules	In the Interaction Center, choose Process Modeling > Rule Policies. Choose E-Mail Response Management System as the Context.
Assign standard tasks to communication methods	In Customizing for Service under Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Assign Standard Tasks to Communication Methods. Choose INT as the communication method for Object ID 207914.
Define e-mail templates and e-mail forms.	In the Interaction Center, choose Knowledge Management > Mail Forms.

Optional Steps/Extensions of Default Customizing

Step	Description
Define additional services. You can define services that are available in the service manager. This activity determines how all necessary services for evaluating policies are invoked. You should only	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Services.

Step	Description
execute this activity if you want to change the default settings or create new services.	
Extend the ERMS repository.	In Customizing for Service under E-Mail Response Management System > Define Repository.
Define additional e-mail profiles.	In Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles.
Define catalogs, codes, and subject profiles. Catalogs are resources which help ensure the uniform use of terms	In Customizing for Service under Basic Functions > Catalogs, Codes and Profiles.
Define additional categorization profiles. This influences which categorization schema(s) are displayed in the corresponding business transactions, and the behavior of the auto suggest function.	In Customizing for Service under Interaction Center WebClient > Business Transaction > Define Categorization Profiles.
Define reporting for e-mail statuses and events.	In Customizing for Service under E-Mail Response Management System > Reporting.
Delete reporting data.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Utilities > Delete Reporting Data.

Integrated Communication Interface

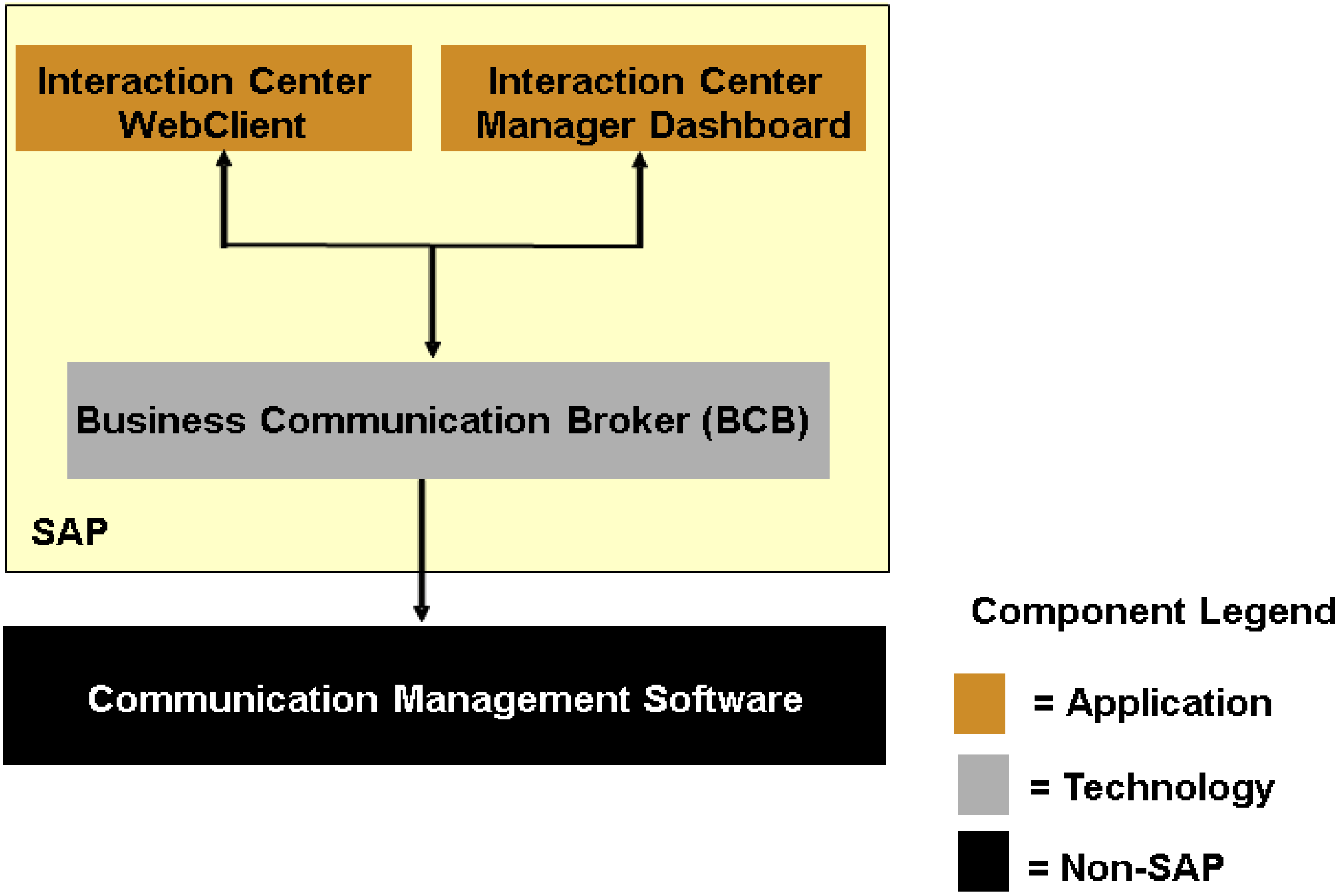
Use

The Integrated Communication Interface (ICI) supports the integration of non-SAP communication products with SAP components that use communication services. The ICI is intended in particular for scenarios involving multiple communication channels such as telephony, e-mail, and chat, but it can also be used in single-channel scenarios. It enables activities such as an agent accepting an inbound phone call or deflecting an inbound chat request, or a supervisor monitoring a queue. The ICI is only one of several components required to implement communication scenarios of this kind.

Note

An example scenario is the integration of communication management software systems with the interaction center (IC), specifically the IC WebClient and IC manager dashboard. For illustration purposes, this example is used throughout this documentation.

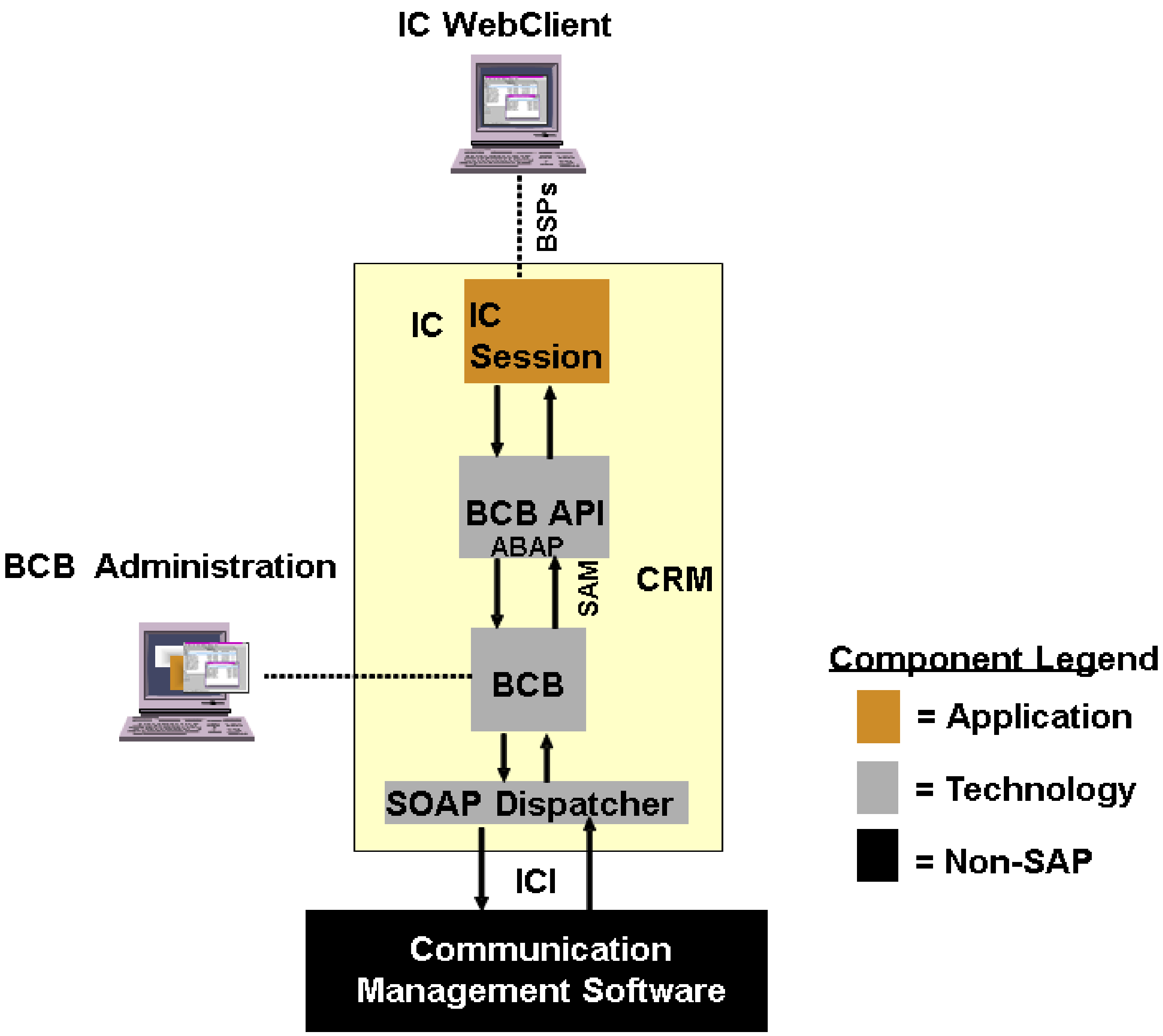
The graphic below represents a simplified view an example scenario without detailing the interfaces. The scenario is made up of functionality from three distinct sources – an application component, a technology component, and an external (non-SAP) component.



Integration

The ICI is based on Simple Object Access Protocol (SOAP) and Extensible Markup Language (XML); they are non-proprietary, Web-oriented, open interface technologies. HTTP is used at the transport level.

The system architecture required to connect the interaction center with external communication management software using the ICI is made up of several components. The system architecture of the interaction center example scenario in a live environment is shown below:



Live System Architecture

BCB Components

The BCB, BCB API, and SOAP dispatcher components enable the CMS. The messages passed between the application component and the external component are passed via the ICI.

> **i Note**
>
> CMS Customizing for IC WebClient is done using transaction `CRMM_BCB_ADM`.

Application-Specific Components

The application-specific components are linked to the ICI-specific components via the Business Communication Broker Application Programming Interface (BCB API). This is the API for application customizing.

In the interaction center example scenario, the application consists of the IC WebClient, which provides an ABAP-based user interface for interaction center agents.

The IC server sends actions to the communication management software via the ICI-specific components and receives events back from the communication management software also via the ICI-specific components. The actions might include agent logon, initiate outbound phone call, or remove chat line from chat session. The events might include new inbound phone call, or new posting in chat session.

External Components

These are non-SAP communication systems. In this scenario, the external product is the communication management software, which is a multichannel communication product. This is responsible for merging requests from different media, routing requests to specific agents or agent groups, queuing requests, signaling inbound requests to the IC, and so on. The external products communicate with SAP via the ICI using SOAP and XML.

Certification

A certification process is implemented to verify the connectivity of external communication products with SAP using the ICI. An external product does not have to support the ICI in full. Partial certification is possible to enable the connection of single-channel communication products, such as telephony products.

Features

Many application components use external communication services. Integrating an external communication product with an application component using the ICI can provide:

- Integration of multiple communication channels such as phone, e-mail, and chat
- Active reporting of new, changed, or completed phone calls, e-mails, or chat sessions (known generically as items)
- Monitoring and statistics

Note

Multichannel management (that is, the synchronization and serialization of incoming communications) is not part of this SAP functionality. The ICI is a server-to-server interface. It does not support direct communication between the SAP client and the external communication product.

Working with WebClient UI

WebClient UI is a UI technology used in Service that delivers a harmonized user interface. Learn how the UI is structured, how to navigate the apps and search the data. Get an overview of the WebClient UI Framework in general, including the framework architecture, tools, generic UI functions, and integration with other UI technologies.

[Getting Started with the WebClient UI](#)
[WebClient UI Framework](#)

WebClient UI Framework

The WebClient UI framework defines the software architecture of the WebClient UI. With the WebClient UI, SAP delivers a harmonized online user interface. The WebClient UI is designed for the business user and presents a role-based workspace that provides an easy-to-use navigation and user interface.

Integration with WalkMe

Use

WebClient UI applications offer integration with WalkMe when applications are launched within SAP Fiori Launchpad.

Features

In the SAP Fiori Launchpad integrated mode, you can launch WalkMe Plugin and use it with WebClient UI applications.

Prerequisites

The configuration for embedding WalkMe is perfomed in Customizing. For the WalkMe integration, a script must be loaded from WalkMe. The path to this script is maintained in the backend.

In Customizing, navigate to **ABAP Platform** > **UI Technologies** > **SAP Fiori** > **SAP Fiori Launchpad Settings** > **Configure Backend URL for WalkMe Integration**.

Here, you can open the customizing activity in the backend by following the path provided. For each sub-node of the customizing activity, you can check the documentation available there and follow the steps described to perform the required activities.

For more information about the WalkMe Integration, see Integration with WalkMe.

Integration with UI Theme Designer

Use

You can use the UI theme designer to create your own themes to adapt the visual appearance of the applications.

In standalone mode, you can use special WebClient UI previews in the theme designer.

If you are using FLP integrated mode, you can create your custom themes in theme designer for Fiori apps. The same themes will apply to the WebClient UI apps in the FLP integrated mode.

Features

In the standalone mode, you can use the following WebClient UI previews in the theme designer.

- **Calendar**
- **Calendar Monthly View**
- **Calendar Weekly View**
- **Guided Activity**
- **Home New Design**
- **L-shape New Design**
- **Overview**
- **Overview Tile**
- **Search**
- **Workcenter**

Procedure

1. To create a new theme, you can use transaction /n/UI5/THEME_DESIGNER to launch the UI theme designer
2. Select the Create a New Theme button
3. Choose the base theme that you want to use from the list and enter the required information
4. Select WebClient UI Control Previews from the Test Suites section

   Note

   The WebClient UI Control Previews is only available for supported themes

5. Make your changes based on the different levels of theming within the tool, then save and build your theme.

In the standalone mode, your custom theme is available in Personalization > Personalize Layout in the WebClient UI.

In the FLP integrated mode, your custom theme is available in FLP user settings.

More Information

To transport your custom theme from the test or quality system to the production system, you can use transaction /n/UI5/THEME_TOOL.

For more information about the theme designer, see SAP Note 1852400.

Integration with Microsoft Teams

Use

WebClient UI Applications offer Share Menu in the object page and advance search page toolbar in Fiori launchpad integrated mode to enable the share features. With this feature, users can share the app with other users.

Features

In the FLP integrated mode, you can share the object page or advance search page with other users. You can share via:

- E-mail
- Microsoft Teams - As Chat, As Tab (if Microsoft Teams integration is enabled)
- Save as Tile

The Share functionality is not available for newly created objects which are not yet persisted, so the button is not visible.

Prerequisites

To use Share with Microsoft Teams option, you must have completed the set up described here: Integration with Microsoft Teams

Procedure

The share feature is available in the header toolbar within object pages and advance search pages.

Users can click on the **Share** button and choose an option to share via **E-Mail** or **Microsoft Teams** – **As Chat**, **As Tab** as well as **Save as Tile** in the Fiori launchpad.

- **Send Email**

  When a user chooses this option, the link to the page opens in the default email client that is configured in the system. When a user chooses the link, the application page opens in the same state in which it was shared.

- **Microsoft Teams**

  Users can use Microsoft Teams:

  - **As Chat**

    Collaborate and quickly resolve issues, if necessary, simply by using Microsoft Teams **As Chat** to share a direct link with co-workers. You can provide them with access to a specific state of an application, for example, so they can easily process any requests you have or tasks that come up.

  - **As Tab**

    Work efficiently with a group of co-workers on specific content that you share in the form of a Microsoft Teams tab. Use the Microsoft Teams environment to work on the same context of an SAP Fiori application, and use the tab conversation option in parallel to share ideas, discuss, and collaborate.

  - **Save as Tile**

    When a user chooses this option, the corresponding **Save as Tile** dialog is displayed. When a user chooses the tile, the application page opens with which the tile was created. Application developers can customize the title and the sub-title of the tile.

More Information

For more information, see Integration with Microsoft Teams.

UI Integration with Attachment Service

Use

The Attachment Service is offered as a central unified solution for uploading attachments across applications. This feature allows you to embed reuse components in the application UI.

WebClient UI offers a reusable component that can be consumed by any WebClient UI application in SAP S/4HANA. SAP applications and customer built applications can use this feature to enable attachment capabilities. The data management and context is handled by the application consuming this utility.

Attachments are stored via Document Management (DMS) or Generic Object Services (GOS).

With this feature you can upload, download, rename and delete attachments. You can add URLs as attachments. The check-in check-out features are not supported.

Prerequisites

All of the customizing and backend implementations required in Customizing of the Attachment Service are completed.

The wrapping object page must have a valid business object layer (BOL) entity that you can get from the parent window through the standard method `IF_BSP_WD_HISTORY_STATE_DESCR~GET_MAIN_ENTITY`.

Procedure

1. In the component where you've defined your overview page, add the **ComponentUsage** `WCF_GC_ATTACH`.
    - Provide the usage **ID** (for example `ATTACHMENTS`)
    - Enter the **Used Component**: `WCF_GC_ATTACH`
    - Enter the **Interface View**: `WCF_GC_ATTACH/MainWindow`
    - The **InboundPlugs** and **OutboundPlugs** aren't required.
2. Add the usage that you created in step 1 to the overview page view area (runtime repository) and adjust the page configuration.
3. Locate your overview page and if you haven't already, implement the `DO_VIEW_INT_ON_ACTIVATION` method. Follow these steps to set the values of the object type and to resolve the object key.
    - You retrieve the attachments node or create it if it doesn't already exist. Use the same usage ID that you used in step `1`
    - The node object provides attributes `OBJECTTYPE` and `OBJECTKEY` to set the object type and object key respectively. These attributes are required to identify the attachment. In some scenarios the final key isn't available, for example when a business object is being created. Applications can use temporal keys (attribute `OBJECTKEYDRAFT`) and be updated with the actual key later on. The values of these attributes can be one of the following:
        a. Constant value
        b. Binding path (for example: `//STRUCT.ATTRIBUTE_NAME`)
        c. BOL path (use prefix `bpath` for example `bpath:./@FLDATE`)

Sample Code

```
    DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us

IF lr_attach_node IS NOT BOUND.
  " Set the properties for service contract attachment navigation
  cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name   =
                                                            RECEIVING rv_node         =
ENDIF.

lr_attach_node->set_property( EXPORTING
                          iv_attr_name = 'OBJECTKEYDRAFT'
                          iv_value     = 'bpath:./@CRM_GUID' ).

lr_attach_node->set_property( EXPORTING
                         iv_attr_name = 'OBJECTKEY'
                         iv_value     = 'bpath:./BTOrderHeader/@OBJECT_ID' ).

lr_attach_node->set_property( EXPORTING
              iv_attr_name = 'OBJECTTYPE'
              iv_value     = 'bpath:./BTOrderHeader/@OBJECT_TYPE' ).
```

Note

The `DO_VIEW_INIT_ON_ACTIVATION` method is executed each time you activate the view (for example, the navigation to the view). As long as you used a binding or BOL path, the actual value is resolved automatically in each roundtrip as these are dynamic. If the main entity isn't available, the application must calculate the values. The `DO_PREPARE_OUTPUT` method is more appropriate than `DO_VIEW_INIT_ON_ACTIVATION` method in this case.

4. To activate the document type selection for DMS repository, the application must set the attribute `SETDOCTYPE` in the method `DO_VIEW_INIT_ON_ACTIVATION.`

Sample Code

```
DATA lr_oc_node1 TYPE REF TO CL_WCF_ATTACHMENTS_NODE.

    lr_oc_node1 ?= CL_WCF_ATTACHMENTS_SERVICE=>get_instance( )->get_node( iv_usage_name = '
    IF lr_oc_node1 IS NOT BOUND.
      lr_oc_node1 ?= cl_wcf_output_control_service=>get_instance( )->create_node(
                                                                      iv_usage_name = 'CUBTAtt
                                                                      class_name    = 'CL_WCF_
    ENDIF.
    CHECK lr_oc_node1 IS BOUND.

   lr_oc_node1->set_property( EXPORTING
                              iv_attr_name = 'setDocType'
                              iv_value     = 'true' ).
```

If attachments are uploaded from DMS source, you can select the document type from a dropdown list when you are in edit mode. The document type is linked to the uploaded attachment. Attachments that are saved show the **document info record**.

5. To enable Harmonized Document Management, set additional attributes in step 3 in the overview page.

To use HDM for attachments at header level, the application developer must pass SAPOBJECTYPE corresponding to the object type used in HDM Customizing. The OBJECTYPE should be initial.

Sample Code

```
        DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us

    IF lr_attach_node IS NOT BOUND.
      " Set the properties for service contract attachment navigation
      cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name   =
                                                                RECEIVING rv_node         =
    ENDIF.

    lr_attach_node->set_property( EXPORTING
                              iv_attr_name = 'OBJECTKEYDRAFT'
                              iv_value     = 'bpath:./@CRM_GUID' ).

    lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEY'
                             iv_value     = 'bpath:./BTOrderHeader/@OBJECT_ID' ).

    lr_attach_node->set_property( EXPORTING
                 iv_attr_name = 'OBJECTTYPE'
                 iv_value     = ' ' ).


    lr_attach_node->set_property( EXPORTING
                 iv_attr_name = 'SAPOBJECTTYPE'
                 iv_value     =  'MaintenanceServiceOrder' ).
```

To enable HDM for attachments on item level, the application developer must pass both SAPOBJECTTYPE and SAPOBJECTNODETYPE in the item overview page.

Sample Code

```
    DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us

IF lr_attach_node IS NOT BOUND.
  " Set the properties for service contract attachment navigation
  cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name   =
                                                            RECEIVING rv_node         =
ENDIF.

lr_attach_node->set_property( EXPORTING
                          iv_attr_name = 'OBJECTKEYDRAFT'
                          iv_value     = 'bpath:./@GUID' ).

lr_attach_node->set_property( EXPORTING
                         iv_attr_name = 'OBJECTKEY'
                         iv_value     = 'bpath:./@HEADER ' ).

lr_attach_node->set_property( EXPORTING
              iv_attr_name = 'OBJECTTYPE'
              iv_value     = ' ' ).


lr_attach_node->set_property( EXPORTING
              iv_attr_name = 'SAPOBJECTTYPE'
              iv_value     =  'MaintenanceServiceOrder' ).

lr_attach_node->set_property( EXPORTING
              iv_attr_name = 'SAPOBJECTNODETYPE '
              iv_value     =  'MaintenanceServiceOrderItem' ).
```

Transactional Handling

The save and cancel are synchronized automatically with the main BOL entity commit and rollback. This is achieved as the WebClient UI attachment component flags the BOL entity as modified by raising event `if_bol_comm~object_modified` when there's an update in the attachments, for example when you upload or delete an attachment. The WebClient UI attachment component delegates the save or cancel to the `CL_ODATA_CV_ATTACHMENT_API` API.

Note

The automatic synchronization of the save and cancel relies on the entity transaction context. If the commit is executed using the global BOL transaction context, the synchronization doesn't work. This is because the global transaction context doesn’t raise event `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_REQUESTED`, which is required to define the type of action (for example commit or revert). The information provided by the `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_REQUESTED` event is used in the `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_FINISHED` event to proceed with the actual saving and canceling of attachments. The application implementation must call the `API CL_ODATA_CV_ATTACHMENT_API` API manually at the time of the save or cancel.

Draft Handling

The WebClient UI attachment service component supports the draft feature. The attachments become persistent when saved and they're deleted on cancel.

This functionality is based on reuse libraries in the front-end system. In the hub scenario, where the front-end and the back-end systems are separate, the call to the reuse library from the back-end to the front-end server may be blocked due to **Cross-Origin Resource Sharing** (**CORS**) settings. If this is the case, you need to maintain the **CORS** setting in the front-end server to allow request from the back-end server in the transaction UCONCOCKPIT.

More Information

- For more information about Attachment Service, see Attachment Service.
- For more information about DMS, go to SAP Help Portal, and see under **SAP S/4HANA** > **Product Assistance** > **Cross Components** > **SAP Enterprise Content Management** > **Document Management**.
- For more information about GOS, see Generic Object Services.
- For more information about Attachment Service data protection, see the Attachment Service section at SAP Help Portal, and search for the S/4HANA product page. On the **Implement** tab, choose **Security Guide** > **SAP S/4HANA Enterprise Business Applications** > **R&D / Engineering** > **Product Lifecycle Management** > **Attachment Service**.
- For more information about HDM, see https://help.sap.com/viewer/DRAFT/36802406aebb4b96b1598246e1d316ee/2021.000/en-US/bc275e43f1b043cc8f97710d5de64e97.html

UI Integration With Output Control

Use

SAP S/4HANA output control is the central entry point in SAP S/4HANA to output functions such as printing, sending e-mails, or maintaining form templates for the output. WebClient UI offers a reusable component that can be consumed by any WebClient UI application in SAP S/4HANA for UI Integration with Output Control.

SAP and customer-built applications can use this feature to display the output control section in WebClient UI object pages. The data management and data context are handled by the application consuming this component.

Prerequisites

To set up SAP S/4HANA output control, you must complete the tasks listed in Setup of SAP S/4HANA Output Control.

Procedure

1. In the component where you've defined your overview page, create the component usage with the following:

   → **Tip**

   Record the **ID** for the next steps. It will be referred to as `Usage ID`.

   - **ID**: usage ID of your choice.
   - **Used Component**: **WCF_GC_OUTPUTC**.
   - **Interface View**: **WCF_GC_OUTPUTC/MainWindow**.

- InboundPlugs/OutboundPlugs: not required.

2. In the `WD_USAGE_INITIALIZE` method of your component controller, set the binding with the context node from which the values for object type and object key can be resolved.

   - `iv_controller_type`: `cl_bsp_wd_controller=>co_type_component` OR `cl_bsp_wd_controller=>co_type_custom`
   - `iv_name`: required if you use `cl_bsp_wd_controller=>co_type_custom`
   - `iv_target_node_name`: your context node
   - `iv_node_2_bind`: `BINDINGCONTEXT`

Code Syntax

```
WHEN 'OutputControl'.
     iv_usage->bind_context_node(
       EXPORTING
         iv_controller_type  = cl_bsp_wd_controller=>co_type_component
         iv_target_node_name = 'BTADMINH'
         iv_node_2_bind      = 'BINDINGCONTEXT'
     ).
```

To activate the edit mode, link the Output Control view to the parent view group context. In this way if the overview page is in edit mode the Output Control view will follow: cl_wcf_output_control_service=>get_instance( )->set_usg_view_gp_ctxt_to_parent(iv_usage).

Code Syntax

```
WHEN 'CUCRMS4OutputControl'.
     iv_usage->bind_context_node(
       EXPORTING
         iv_controller_type  = cl_bsp_wd_controller=>co_type_component
         iv_target_node_name = 'BTADMINH'
         iv_node_2_bind      = 'BINDINGCONTEXT'
     ).
    cl_wcf_output_control_service=>get_instance( )->set_usg_view_gp_ctxt_to_parent( iv_usa
```

Note

Partial edit for Output Control is not supported.

3. If your bound context nodes are not used to resolve the current entity at runtime, locate your overview page and implement the method `DO_VIEW_INIT_ON_ACTIVATION.` To set the attributes from which values of object type and object key can be resolved, complete the following:

   a. Retrieve the output control node or create it if it does not exist. Use the same usage ID that you used in step 1.

   b. The node object provides attributes `OBJECTTYPE` and `OBJECTKEY` to set the object type and object key respectively. These attributes are required to identify the output documents. The values of these attributes can be a constant value, a binding path (for example: `//STRUCT.ATTRIBUTE_NAME`), or a BOL path. If the value of the attribute is a BOL path, you must use the prefix `bpath` (for example: `bpath:./@FLDATE`)

Code Syntax

```
"ouput control parameters
    DATA lr_oc_node TYPE REF TO cl_wcf_output_control_node.

    lr_oc_node ?= cl_wcf_output_control_service=>get_instance( )->get_node( iv_usage_name =
    IF lr_oc_node IS NOT BOUND.
      lr_oc_node ?= cl_wcf_output_control_service=>get_instance( )->create_node(
                                      iv_usage_name = 'OutputControl'
                                      class_name    = 'CL_WCF_OUTPUT_CONTROL_NODE' ).
    ENDIF.
    CHECK lr_oc_node IS BOUND.

    "set object type
    lr_oc_node->set_property( iv_attr_name = 'OBJECTTYPE' iv_value = 'SERVICE_CONTRACT' ).

    "set object key
    lr_oc_node->set_property( iv_attr_name = 'OBJECTKEY' iv_value = 'bpath:./@OBJECT_ID' ).
```

Note

The WebClient UI uses the bound context node to resolve the current entity at runtime and these paths to resolve the values for the object type and object key. In this case, you do not need to implement the method `DO_VIEW_INIT_ON_ACTIVATION.` You can set the binding parameters in method `WD_USAGE_INITIALIZE`.

4. The applications using the output control can activate the actions **Delete** and **Determine Items**. The application must set the previous actions to **true**.

   To activate the actions **Delete** and **Determine Items**, the following properties must be set in the node `lr_oc_node` in the method `DO_VIEW_INIT_ON_ACTIVATION` as described in the step 3.

Code Syntax

```
* Activation of Delete and Determine items Actions
  lr_oc_node->set_property( EXPORTING
                              iv_attr_name = 'setHideDeleteAction'
                              iv_value     = 'true' ).

  lr_oc_node->set_property( EXPORTING
                              iv_attr_name = 'setDeterminationMergeOption'
                              iv_value     = 'true' ).
```

   If the business scenario requires to hide the **Send Output** button, for example in edit mode, set the attribute `setHideIssueOutputAction` with value `true` in method `DO_VIEW_INIT_ON_ACTIVATION`.

Code Syntax

```
lr_oc_node->set_property( EXPORTING
                              iv_attr_name = 'setHideIssueOutputAction'
                              iv_value     = 'true' ).
```

5. Next, you add the usage that you created in step 1 to the overview page view area (runtime repository) and adjust the page configuration.

   This functionality is based on reuse libraries in the frontend system. In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing** (**CORS**) settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction `UCONCOCKPIT`.

More Information

- For more information, see SAP S/4HANA Output Control.
- For more information about Data Protection and Privacy in Output Control, see Blocking of Receiver Data.
- For more information about UCON Cockpit, see Using the UCON Cockpit.

UI Integration With Situation Handling

Use

Situation Handling is a concept for bringing business issues to the attention of specific user groups. It helps the user to recognize, understand, and resolve the situation by gathering all relevant information and recommending solution proposals. This feature enables you to integrate and use Situation Handling service with WebClient UI object pages in SAP S/4HANA. The users can view situation message strip in the object page and take necessary actions such as dismiss the situation. You can use this feature when you're using WebClient UI applications in SAP Fiori launchpad integrated mode and standalone mode.

Prerequisites

You must have completed Situation Handling framework set up and configuration.

Procedure

The situations are linked to your application's object page document through the anchor object. The anchor object can be identified with the following two fields:

- `anchorObjectKey`: Must be bound to the unique ID of the business object displayed by the object page.
- `anchorObjectType`: The name of your **Anchor Object Key** as specified in the situation. The **Anchor Object Key** and type can be found in the Monitor Situations app.

Go to your overview page and call method `SET_ANCHOR_OBJECT` of class `CL_WCF_SITUATION_HANDLING` in the method `DO_PREPARE_OUTPUT`. Pass the parameters `iv_anchor_object_key` and `iv_anchor_object_type`.

Code Syntax

```
*Situation integration
    TRY.
        lv_objkey = me->typed_context->btadminh->collection_wrapper->get_current( )->get_propert
```

```
        CHECK lv_objkey IS NOT INITIAL.

        cl_wcf_situation_handling=>set_anchor_object(
          EXPORTING
            iv_anchor_object_key  = lv_objkey
            iv_anchor_object_type = 'ServiceContractUUID' ).
      CATCH cx_root.
    ENDTRY.
```

This functionality is based on reuse libraries in the frontend system. In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing** (**CORS**) settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction `UCONCOCKPIT`.

In addition, for the hub scenarios you must activate the OData service **C_SITNINSTANCETP_CDS** with transaction **/IWFND/MAINT_SERVICE**, **Activate and Maintain Services** on the backend server and the frontend server. For more information on OData service activation, see SAP Help Portal at http://help.sap.com —> **SAP Gateway Foundation (SAP_GWFND)** —> **SAP Gateway Foundation Developer Guide** —> **OData Channel** —> **Basic Features** —> **Service Life-Cycle** —> **Activate and Maintain Services**.

More Information

- For more information, see Situation Handling.
- For more information about UCON Cockpit, see Using the UCON Cockpit.

Situation Indicator and Popover in Tables

Use

The situation indicator and popover are displayed in the table when there is a situation in an item or object. User can see the details of the situation in popover and can navigate to the object.

Prerequisites

You have a column in the table that contains a number of situations.

Procedure

To display the alert icon, implement the P-getter of the situation attribute in the table context node.

Pass the `iv_property` to static method `cl_wcf_situation_handling=>handle_table_p_getter`.

Code Syntax

```
METHOD get_p_sitn_num_of_instances.
```

```
    rv_value = cl_wcf_situation_handling=>handle_table_p_getter( iv_property = iv_property ).

  ENDMETHOD.
```

To display the button and popover, you must enhance the above code by passing the anchor object key and anchor object type to get the details of the situation.

Code Syntax

```
 METHOD get_p_sitn_num_of_instances.

rv_value = cl_wcf_situation_handling=>handle_table_p_getter(
      EXPORTING
        iv_anchor_object_key = lv_anchor_object_key "Situation object key
        iv_anchor_object_type = 'ServiceContractItemUUID'  "Situation anchor object type
        iv_property = iv_property
        iv_display_mode = iv_display_mode
    ).

ENDMETHOD.
```

With the **Show Details** button, the navigation is triggered via SAP Fiori launchpad. In some scenarios, if you want to navigate to a WebClient UI application instead of SAP Fiori launchpad navigation, a mapping can be implemented from intent to outbound plug. This can be done in the `do_prepare_output method` of the implementation class.

Code Syntax

```
 cl_wcf_situation_handling=>map_ibn_to_wcf_op(
      EXPORTING
        iv_sitnsemanticobject = 'ServiceContract'
        iv_sitnsemanticobjectaction = 'display'
        iv_outbound_plug = 'TO_ITEM_DISPLAY'
    ).
```

Search Functions

You can use the central search, the advanced search, and the simple search to search for business objects in the WebClient UI. You can use the search page wizard to create search pages.

Central Search

Use

You can use the central search to search for business objects the WebClient UI. The central search fields are displayed when you choose the search icon. You can always access the search fields with this icon, regardless of the business object that you are processing.

In the search menu you can choose between a search for all business objects, saved searches, or a search for individual business objects.

Features

The option **Saved Searches** is the default search option for the central search. Below it in the menu you see **All Objects** and then follows a hierarchical list of business objects, in alphabetical order. When you have selected a saved search, the search is automatically performed.

Saved Searches

When you select **Saved Searches**, the search menu contains all your saved searches, and the field label changes to **Saved Searches**.

> **i Note**
>
> The WebClient UI Framework displays the data that the application has set in the search functionality. There are no controls in place to ensure that this data does not contain sensitive personal information. You can create, modify, and delete saved searches. Use these functionalities to ensure that the pre-filled search fields in Saved Searches are up-to-date and follow data protection regulations.

All Objects (Cross-Object Search)

When you select **All Objects**, the field label changes to **Search for All Objects**. You can now search across all business objects that belong to your business role. This search function behind this search technique is “simple search” (see [Simple Search]).

> **i Note**
>
> The **All Objects** entry only appears if more than one searchable business object has been registered in the central search Customizing.

Business Objects (Object-Specific Search)

When you position the cursor over a business object in the menu, a cascading menu with the search attributes assigned to that business object opens.

Search Using Business Object Attributes

After you have selected a search attribute, you can enter the search value in the input field. The field label changes accordingly, for example, to **Accounts/Category**.

The search function behind these search techniques is “advanced search” (see [Advanced Search]).

Free Text Search

You can perform a free text search for a business object by selecting **Free Text**. If **Free Text** is the only search attribute of a business object, it is omitted.

This search function behind this search technique is “simple search” (see [Simple Search]).

> **i Note**

To improve the performance when you start searching for a business object, the table for the search objects displayed in the search menu is cached in a server cookie. This cache depends on the business role and language. Therefore, Customizing changes that have been made for the central search might not be immediately visible in the WebClient UI.

More Information

Configuring the Central Search

Configuring the Central Search

Making Business Objects Available in the Central Search

For every business object that is to be supported by the central search, make the necessary settings in Customizing for **UI Framework** under **Technical Role Definition** > **Define Central Search**.

Making Business Objects Invisible in the Central Search

If you want to make business objects invisible at business role level in the central search, make the necessary settings in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**.

Defining Search Criteria

If you want to include or exclude search criteria from the search page of a business object, you can activate or deactivate the corresponding indicator in the UI configuration of the selected application component and view. For more information, see Search Pages.

Example

To configure the Account search criteria in the central search of your business role, start the UI configuration tool and select application component `BP_HEAD_SEARCH` and view **MainSearch**. Activate the indicator **Use in Central Search**for the search criteria that you want to be displayed in the central search.

Adjusting Entries in the Central Search Menu

You can change entries in the central search menu in the Online Text Repository (OTR). If you want to change these entries, start transaction `SOTR_EDIT` for the OTR. Choose the correct language, search for the OTR text by entering the OTR ID in the **Alias** field, and change the SAP standard text. The following entries are available in the central search:

OTR ID	SAP Standard Text	Description
`CRM_CENTRAL_SEARCH/SEARCH_FOR_ALL_OBJECTS`	Search for All Objects	Field label that appears when the user clicks **All Objects** in the search menu
`CRM_CENTRAL_SEARCH/SEARCH_FOR_OBJECT`	Search for &1	Field label that appears when the user selects a business object in the search menu; &1 is replaced by the selected business object.
`CRM_CENTRAL_SEARCH/SEARCH_IN_ALL_FIELDS`	Free Text	Entry in the search menu that allows the user to enter free text in the

OTR ID	SAP Standard Text	Description
		search field; this entry is used by the simple search.
CRM_CENTRAL_SEARCH/ALL_OBJECTS	All Objects	Entry in the search menu that starts the search across all business objects
CRM_CENTRAL_SEARCH/SEARCH_FOR_OBJECT_BY	Search for &1 by &2	Field label that appears when the user selects a business object and a search attribute in the search menu; &1 is replaced by the business object and &2 is replaced by the search attribute.

Advanced Search

Use

You can use the advanced search in the WebClient UI to search for business objects.

Features

On the search page, you can define the search criteria and start the search. You have a variety of search criteria and search operators with which you can refine your search to find precisely the data you are looking for. In the result list, you can see the data that matches your search criteria.

You can show and hide the search criteria, by clicking the corresponding hyperlinks. You can add or remove search rows, by choosing the icons next to the search row. You can enter the maximum number of business objects that should be in the result list of this search.

You can save your current search, which can be made available as saved search in the central search.

You can define default search criteria in the personalization of an advanced search page. For more information, see Advanced Search Personalization.

To adapt the result list to your personal requirements, you can make the necessary entries in the personalization dialog box.

Search Criteria Displayed as Read-Only

To indicate that certain search criteria that have an impact on the search result may not be changed, certain search criteria can be displayed as read-only. This cannot be modified by users. Plus or minus buttons to add or delete a row in the search criteria are not available.

Note

In the SAP standard system, the search criteria are not displayed as read-only. This feature is used only in special cases.

Mandatory Use of the Input Help

To ensure that certain search fields are only filled by using the input help, the use of the input help in these fields is mandatory. In this case, the look and feel of the search field is as if it is read-only, and users cannot manually place the cursor focus on the search field.

If users click the search field or the input help icon, the input help is automatically started. If users press ENTER in the search field, the input help is also started, instead of executing the advanced search. When users tab to the search field, the focus lies on the search field and the input help icon, because they are one entity. In this case, the input help can only be started by pressing ENTER. If the input help is closed and the focus is still on the input help icon, pressing ENTER restarts the input help, instead of executing the advanced search.

Note

In the SAP standard system, the use of the input help is not mandatory. This feature is used only in special cases.

Example

You are logged on to the system as sales professional: You start the account search page or the contact search page and select search attribute **Marketing Attribute**. You can now use the input help in the search field.

You start the account search page and select search attribute **Account Classification**. You can now use the input help in the search field.

Pasting of Multiple Values as Search Criteria

You can copy a list of values from another application, such as spreadsheet or text processing application, and paste them as search criteria in an advanced search by pressing CTRL + V. The values are handled as separate search criteria.

In the source application, each value should be in a new row separated by tab or new line. Non-adjacent row selection support depends on the source. After pasting the search criteria into the search, they are shown as separate search criteria in individual rows.

You make the settings for this function in Customizing for **UI Framework** under **Technical Role Definition** > **Define Parameters**. By default, the system limits the number of rows that can be pasted at once to 200 to avoid accidental pasting of an overly large amount of data. You can override this default maximum number by adding parameter `SEARCH_MAX_ROW_PASTE` in function profile `PARAMETERS` of the relevant business role.

Groups of Rows

For usability reasons, when a large number of search criteria rows have been created by pasting multiple search criteria, successive rows that have the same attribute and the same operator are grouped. These rows are collapsed and appear with an arrow icon pointing to the right of the root row. A row group that results from pasting is by default collapsed and shown in display mode. To edit the rows that were created, you have to expand the row group. With **-** (**Remove Line**), you can delete all rows of the group. With **+** (**Copy Line**), you can expand the group and add a new row at the bottom of the expanded group. By default, the rows are grouped if there are more than two rows of search criteria that have the same attribute and operator.

More Information

Central Search

Table Personalization

Advanced Search Pages with Fiori Layout

Relative Dates

A relative date is a date or date range that is relative to the current date such as yesterday, last week, or next month. You can use relative dates for date type criteria on advanced search pages.

Use

On advanced search pages, you can use relative dates for date type criteria. A relative date is a date or date range relative to the current date such as yesterday, last week, last year, next month or during the next three quarters.

Prerequisites

- You or your Administrator has defined relative dates and associated them to **Dynamic Queries** using the **Define Relative Dates** Customizing activity available from **UI Framework** > **Generic Interaction Layer/Object Layer** > **Component Specific Settings**.
- You or your Administrator has created relative dates for the default handler using the Maintain BRFplus Settings for the **Relative Dates Default Handler** Customizing available from **UI Framework** > **Generic Interaction Layer/Object Layer** > **Component Specific Settings**.
- You or your Administrator may have used the **BAdI : Relative Dates Handler** in Customizing available from **UI Framework** > **Generic Interaction Layer/Object Layer** > **Component Specific Settings** > **Business Add-Ins (BAdIs)** to create a component handler to define relative dates.

Features

If your system is customized to use relative dates, the relative date section of the date picker will only be visible when the search criteria operator on the advanced search page is set to **is**.

- You can either choose the relative date value from the date picker or enter it in the **Date** field.
- Since relative dates supports type ahead functionality, suggestions will be provided while you are typing within the **Date** field.
- You can enter the key (for example: `WCF_TODAY`) or the value (for example: `Today`) in the **Date** field. If you have used the key, the value will be displayed after the round trip is completed. Note that the key and value are not case sensitive.
- You can enter a specific date in the **Date** field.

Advanced Search Pages with Fiori Layout

Use

The WebClient UI advanced search pages have a user experience consistent with the Fiori layout. The concept of search variants is used to access saved searches. The search criteria view uses tokens. You can pin and unpin as well as expand and collapse the header content.

Features

- **Search Variants**: Search variants allow the user to select, create, update, and delete variants for filter settings and control parameters such as layout, table column visibility, sorting, and filtering.
- **Search Criteria**: In the search criteria section, you can use the input fields and operators to create tokens as your search criteria. The classic search criteria and maximum number number of hits are available under the **Adapt Filters** dialog.

- **Pin and Unpin**: The pin and unpin button is directly below the header, which is used to pin and unpin the header content. Using the pin feature keeps the header expanded and pinned to the top while you scroll through the page content. This mode remains fixed until you use the pin button again or trigger a snap on click.
- **Expand and Collapse**: The expand and collapse button is located directly below the header, which is used to expand and collapse the header content. When the pin is active, the expand and collapse header button is still available and overrules the pin. Alternatively, you can also expand or collapse the header by selecting the header title bar.
- **App State**: In the Fiori launchpad integrated mode, the search criteria and values are saved in the Fiori launchpad app state and can be shared with other users using the URL or the **Share Menu** in the advanced seach toolbar.

Activities

- **Search Variants**: When you activate this feature, it will appear on the advanced search pages. You can use the variant list to swap between your search variants, previously known as saved searches. When you save a new variant, the search fields, values, and result list layout will be saved. When you change your search criteria, an asterisk (*) will appear on your variant name, and you can re-save it, or save it as a new one.
- **Manage Variants**: In this dialog, you can do the following actions:
    - Favorite

      Favorite variants will appear in the variant list on search pages. If you un-favorite a variant, it will be removed from this list, but it will still be available in other areas, like saved searches. New variants are automatically favorited.
    - Rename

      You can change the name of the variant in the **View** column.
    - Apply Automatically

      When this is selected, the search will execute automatically when you select the variant.
    - Default

      You may select a variant to be your default, which will be the one loaded when you open this search page.
    - Delete

      You can delete variants. The **Standard** view cannot be deleted.
- **Search Criteria Input**: In the search criteria, you can select or type values to the search criteria. After you tab out or press enter, your search criteria will be turned into a token. This token will display your search values with the operator. You can easliy click on the **X** of the token to remove it from your search. Choose **Adapt Filters** to see a more detailed view of the search criteria. Choose **Go** to execute your search.
- **Search Criteria Adapt Filters**: In this dialog, you can do the following:
    - View the classic search view with the displayed operators.
    - Add new search criteria items using the dropdown list box.
    - Add multiple rows of the same search criteria with the **Copy Line** button.
    - Remove criteria with the **Remove Line** button.
    - Clear all the values with the **Clear** button.
    - Change the maximum number of results shown in the search results list.
- **Search Criteria Tokens**: You may specify operators in the input fields as follows:

Operator	Pattern
is / equal to	`=[value]` (default when no operator is typed)
contains	`*[value]*`
between	`[value1]`... `[value2]`
starts with	`[value]*`
ends with	`*[value]`
less than	`<[value]`
less than or equal to	`<=[value]`
greater than	`>[value]`
greater than or equal to	`>=[value]`

Activation

To enable or disable advanced search pages with Fiori layout, use the function profile parameter `SEARCH_LAYOUT= 'FIORI'` (enabled) or `'CLASSIC'` (disabled).

By default, the advanced search pages with Fiori layout are enabled.

Limitations

The search criteria view is only available in advanced search pages. It is not available in dialog.

Public variants are not supported.

Simple Search

Use

You can use the simple search to search for business objects in the WebClient UI using keywords, IDs, or free text, which you enter in a single search field.

Integration

The simple search can be integrated in the UI as follows:

- In the central search (see Central Search).
- In the search pages for the business objects.

  On search pages, users can switch easily between the simple search and the advanced search.

To use the simple search in the search pages, applications must have the corresponding business function active.

Prerequisites

To use simple search you need to configure Enterprise Search (see Enterprise Search Integration).

To make the simple search available in the **central search** of the WebClient UI, you have made the following settings in Customizing for UI Framework:

- You have registered your business objects for the central search under Technical Role Definition > Define Central Search.
- For the relevant roles, you have activated your business objects for the central search under Business Roles > Define Business Role.

To make the simple search available in the **master data search pages**, you make settings under UI Framework > Enterprise Search Integration > Define Availability of Simple Search in Business Object Search. For information about assigning the settings to a business role, see the documentation for the Customizing activity.

Features

Free Text Search

You can enter keywords, IDs, or free text in the easily accessible search field, and search across all business objects that belong to a business role. The search is executed in all attributes that belong to a business object.

Search for IDs

If you search for an ID, a wildcard character asterisk (*) is automatically added as a prefix to the ID. This applies to the free text search and the custom filter in result lists on the search result page. The asterisk is necessary because all IDs are automatically saved with leading zeros in the enterprise search index. For example, if you enter `2349` in the search field, business objects with ID `00002349` are displayed in the result lists. From more information about use cases for simple search, see your application specific documentation.

Activities

Using Simple Search in the Central Search

To start the simple search, you select All Objects in the search menu of the central search. Or you select the search attribute Free Text of an individual business object in the search menu and enter free text in the search field. The results list is a simple list without further functions.

Using Simple Search in the Master Data Search Pages

To start the simple search on a master data search page, either the field is visible by default or you toggle from the advanced search, depending on Customizing. The results list offers the same functions as are available as for the advanced search results list for the relevant object.

Enterprise Search Integration

Use

The WebClient UI has been integrated with Enterprise Search (ABAP). The main goal of the Enterprise Search integration is to provide the WebClient UI with a search function based on keywords and free text. This means that you can enter keywords, IDs, or free text in the search field, and search across all business objects that belong to a business role. This search function is named simple search.

Integration

The simple search is based on Enterprise Search (ABAP) or Embedded Search, and uses the complex functions of this search engine, such as full indexing or incremental indexing. Embedded Search is the Enterprise Search component in the Application Server (AS) ABAP.

Features

Enabled Business Objects

You can find the business objects that have been enabled for Enterprise Search (ABAP) in your business role in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**. Select your business role and choose **Adjust Central Search Objects**. All business objects with object action **Product Enterprise Search** are enabled and delivered in the standard system. They are available in the central search. For more information about the central search, see Central Search.

More Information

For more information about Enterprise Search (ABAP), see SAP Help Portal **S/4HANA** > **Enterprise Seach** > **Enterprise Search**.

Enterprise Search Modeling Workbench

Use

You can use the ES modeling workbench to connect business objects to Enterprise Search (ABAP). You can use the workbench to model templates for business objects and technical objects. These templates are used by Enterprise Search (ABAP) to search for business objects.

Integration

You can find the ES modeling workbench in SAP Easy Access Menu for **CRM ES Modeling Workbench** at **Service** > **Architecture and Technology** > **Enterprise Search Integration** > **Tools**.

Prerequisites

Before you start working with the ES modeling workbench, you need to make the necessary settings in Customizing for **UI Framework** under **Enterprise Search Integration**.

Features

Technical objects are mainly templates based on text tables, which are used to extract and index the language-dependent text maintained in these tables. If the user enters, for example, a keyword or free text in the search field, the technical objects are searched for this keyword or the free text.

The structure of the templates in the ES modeling workbench corresponds to a tree, which is based on the Business Object Layer (BOL) Model, associated to the BOL root object as defined in the template header. This tree structure is used to maintain these templates in dedicated detail screens on node, relation, and attribute level.

In templates delivered by SAP, the search fields are a subset of the search attributes that are defined in the advanced search of a business object. You can find all search attributes of a business object, in the corresponding advanced search object in the BOL Model Browser (transaction `GENIL_MODEL_BROWSER`).

Foreign Relation Support

You can model foreign relations to templates in the ES modeling workbench. For every foreign relation, you can define an individual set of search attributes and search result attributes from the referenced foreign template. Search criteria and search result fields can be adapted to specific search use cases.

Complete Attribute Overview

You can display all selected attributes of a business object template in the ES modeling workbench. This information is helpful when you are modeling a business object template, and want to see a list of all attributes that are relevant for the search. You can export this list using different file formats, for example, a spreadsheet.

Activities

Specifying the Search Model

Before you can start modeling your business objects and technical objects in the ES modeling workbench, you can specify the search model. This means that you can define the attributes that are used in the search request and in the result list of a business object.

Modeling the Search Template

You start the modeling process in the ES modeling workbench by creating a new template set, which corresponds to a component set in the BOL model browser. A template set contains two types of templates, which are business templates and technical templates.

In the modeling process you maintain the BOL nodes, relationships, and the relevant fields of your business object. At attribute level, you can maintain the following information:

- Key indicator

  Specifies fields that can be used as key fields in the relations of a business object

- Search criteria

  Specifies attributes that are used when users search for a business object

- Result

  Specifies attributes that are available in the search result of a business object

- Select

  Specifies fields that are extracted for Enterprise Search

- Text

  Specifies fields that contain human-readable text

- Language code

  Specifies fields that are language-dependent

- Template ID

Specifies the template ID

Transferring the Search Template to Enterprise Search (ABAP)

After you have modeled your business object templates in this workbench, you need to transfer them to Enterprise Search (ABAP).

Working With Templates

Use

When you are working with the ES Modeling Workbench, you can either copy existing SAP templates to your namespace and then adapt these templates to your needs, or you can create new templates.

Procedure

Copying Existing Templates

If you copy existing SAP templates, proceed as follows:

1. Copy the SAP template to your namespace.
2. Make the necessary changes.

   Adjust the required search request attributes and the required result list attributes.
3. Save your template.
4. Check if the template is consistent.
5. Transfer your search template to Enterprise Search (ABAP).

Creating New Templates

If you create new templates, proceed as follows:

1. Create a new template.
2. Maintain your search model.

   Define the required search request attributes and the required result list attributes in the workbench.
3. Save your template.
4. Check if the template is consistent.
5. Transfer your search template to Enterprise Search (ABAP).

Configuration and Administration Tools

Use

The following configuration and administration tools are available:

Tools

- Enterprise Search Modeling Workbench
- Enterprise Search Administration Cockpit

Utilities

- Structure Generation for Search Queries
- Client Copy Utility of Templates
- Transport Utility of Templates
- Model Transfer Utility of Templates

Report

- Report to synchronize Enterprise Search templates and templates

You can find the above tools, utilities, and report on the **SAP Easy Access** screen, under **Service** > **Architecture and Technology** > **Enterprise Search Integration**.

Prerequisites

If you want to use the Enterprise Search Administration Cockpit, you need the authorization role `SAP_ESH_CR_ADMIN`, which is necessary for administration and monitoring in this tool.

If you want to define technical and application-specific Customizing settings, you need authorization role `SAP_WCF_ES`. This role contains authorization object `S_WCF_ES`, which controls access to Enterprise Search.

For more information about authorization roles, see Customizing for **UI Framework** under **Business Roles** > **Define Authorization Role**.

More Information

Topic	Source of Information
Tools, utilities, and template synchronization report	Selection screens of each individual tool, utility, and report, under **Program Documentation**.
Enterprise Search Modeling Workbench	Enterprise Search Modeling Workbench
Enterprise Search (ABAP)	For more information about Enterprise Search (ABAP), see SAP Help Portal **S/4HANA** > **Enterprise Search** > **Enterprise Search**.

Search Page Wizard

Use

You can use the search page wizard to create search pages for the WebClient UI. You can create more than one search page for a UI component.

The search page wizard is integrated into the UI component workbench and guides the user through a series of steps.

After the wizard has finished the generation, you need to configure which fields are to be displayed in the search query views and search result views in the WebClient UI.

Features

The search page wizard generates a search query view, a search result view, and a view set. These views are based on a common Business Server Page (BSP) layout.

The search page wizard offers the following functions:

- View set
    - Hiding of search query options
- Search query view
    - Search button
    - Clear button
    - Saved searches (optional)
- Search result view
    - Selection mode: single selection, and multiselection (optional)
    - Navigation links for fields in the result list (optional)
    - Buttons (optional)

Business Roles

Use

You can use business roles to work with content from different applications. You can combine this content according to your personal requirements.

Features

Business Roles in the WebClient UI

You can perform the following activities in the business role application:

- Search for business roles, create new business roles or copy existing ones, and delete business roles
- Start transaction PFCG to assign authorization roles (PFCG roles) to business roles by clicking **Authorization Roles**
- Choose or create a transport request, after you have changed or deleted an existing business role, or created a new business role
- Personalize the overview page, including the assignment blocks of your business role, by clicking the **Personalize** icon
- Export the content of the individual assignment blocks to Microsoft Excel

- Translate the business role description, and titles of direct link groups, work centers, and logical links into other languages by clicking **Translate**
- Assign icons to work centers and to logical links in direct link groups

Where Do I Customize Business Roles?

Use

You can use the business role Customizing to tailor the navigation frame of your business roles to your personal requirements.

Prerequisites

To start the business role Customizing in the WebClient, you need to include a link in the navigation bar of your business role. You have assigned the logical link `CT-BR-SR` to the work center link group `CT-ADM-SR`, in Customizing for **UI Framework** under **Technical Role Definition** > **Define Navigation Bar Profile**.

Features

You can customize your business roles in SAP GUI and in the WebClient UI.

Customizing in SAP GUI

You access the business role Customizing in SAP GUI, by choosing **UI Framework** > **Technical Role Definition** and **UI Framework** > **Business Roles**.

Customizing in the WebClient UI

You access the business role Customizing in the WebClient by clicking a link in the navigation bar of your business role. The business role application consists of a search page and an overview page. The overview page contains the header block **Business Role Details**, with the **General Data** and **Profiles** areas. In the **General Data** area you find general information, such as business role ID and description, role configuration key, and the assigned authorization role (PFCG role). In the **Profiles** area, you see all profiles that are assigned to your business role, such as navigation bar profile, layout profile, and technical profile. Under the header block you can see the following assignment blocks on the overview page:

- Navigation Bar
- Direct Link Groups
- Keyboard Shortcuts
- Central Search
- Function Profiles

In the header block of your business role you can change the business role details. In the assignment block you can customize the side panel, assign function profiles, and adjust central search objects.

More Information

For more information about the central search, see Central Search.

How Do I Customize Business Roles?

Use

The following document describes the business role Customizing in the WebClient.

Prerequisites

Before you start your Customizing activities, you have created a navigation bar profile that you want to use in your business role. The navigation bar profile contains a collection of work centers, direct link groups, logical links, and central search objects. It can be assigned to a business role. You have created the navigation bar profile for your business role in Customizing for **UI Framework**, by choosing **Technical Role Definition** > **Define Navigation Bar Profile**.

Activities

You can perform the following Customizing activities in the business role application in the WebClient UI.

Maintaining Business Roles

The following functions are available for maintaining business roles in the WebClient:

Searching for Business Roles

After you have started the business role Customizing you navigate to the business role search page. Here you can enter your search criteria and start the search. All business roles that correspond to the search criteria are displayed in the result list.

Displaying Business Roles

You can select a business role in the result list by clicking the technical name in the **Business Role** column. You navigate to the overview page of your selected business role. Here you can see the most important information about your business role, at a glance.

Changing Business Roles

You can change the business role details in the header block by clicking the **Edit** icon. You can change the information in every assignment block by clicking **Edit List**. You can use the **Clean Up** button to delete rows in which the indicator **Deleted** is set. This means that the corresponding rows are not part of the underlying navigation bar profile any longer, and that these entries are no longer relevant for this business role. You can use the **Translate** button to translate titles into other languages, for example work center titles in the navigation bar. Save your changes after you have finished changing your business role.

Copying Business Roles

You can copy a business role by clicking the **Copy** icon either on the search page at the top of the result list or on the overview page. You are asked to confirm that you want to copy all dependent entries of the current business role to the new business role.

Creating New Business Roles

You can create a new business role by clicking **New** either on the search page at the top of the result list, or on the overview page. You navigate to a page that contains the **Business Role Details** header block in which you need to enter business role ID, authorization role ID, and navigation bar profile, which are mandatory fields. The content in the assignment blocks depends on the navigation bar profile that you have assigned to your business role in the header block. Save your new business role.

Deleting Business Roles

You can delete an existing business role by clicking the **Delete** icon either on the search page at the top of the result list, or on the overview page. You can delete more than one business role at once on the overview page.

Adjusting Work Centers and Logical Links

In the assignment block **Navigation Bar**, you can see the work centers that are displayed in the navigation bar of your business role. The work centers are displayed at the first level and the corresponding logical links that belong to a work center are displayed at the second level, in the assignment block. The work centers are displayed in the order defined by the navigation bar profile.

Adjusting Work Centers

You can set the indicator **Visible in Menu** to active, if a work center is to be displayed in the navigation bar of your business role. Additionally, you can change the work center title in the **Title** field. You can change the order of the work centers by using **Up** and **Down**.

Adjusting Logical Links

You can set the indicator **Visible in Work Center** to active, if a logical link is to be displayed on the corresponding work center page. Additionally, you can change the logical link title and the order of logical links on a work center page.

Defining the Sorting of Logical Links

The sorting of logical links on work center pages is in alphabetical order. If you want to change the sorting, you can do so in Customizing for **UI Framework** under **Technical Role Definition** > **Define Parameters**.

Enter a profile, for example `PARAMETERS`, and assign the parameter `WC_LINK_ORDER`, which defines the sorting of logical links on work center pages, to that profile. Enter a value of your choice for the parameter `WC_LINK_ORDER`. For more information about the available values, see the above-mentioned Customizing. Finally, you need to assign the profile, for example `PARAMETERS`, to your business role in the assignment block **Function Profiles**.

Adjusting Direct Link Groups and Logical Links

In the assignment block **Direct Link Groups** you can see the direct link groups that are displayed in the navigation bar of your business role. The direct link groups are displayed at the first level and the logical links that belong to a direct link group are displayed at the second level, in the assignment block. The direct link groups are displayed in the order defined by the navigation bar profile.

Adjusting Direct Link Groups

You can set the indicator **Visible** to active, if a direct link group is to be displayed in the navigation bar of your business role. Additionally, you can change the direct link group title in the **Title** field. You can change the order of direct link groups by using **Up** and **Down**, and assign icons that are displayed in front of the link group title in the navigation bar of your business role.

Adjusting Logical Links

You can set the indicator **Visible** to active, if a logical link is to be displayed in the corresponding direct link group. Additionally, you can change the logical link title and the order of the logical links in a work center. You can assign icons that are displayed in front of the logical link in the navigation bar of your business role.

Defining Keyboard Shortcuts

In the assignment block **Keyboard Shortcuts**, you can define shortcuts for navigation operations and focus operations. Navigation operations navigate to a view or component, using a logical link. Focus operations set the focus to an area on the user interface.

In SAP GUI you define keyboard shortcuts in Customizing for **Customer Relationship Management** under **Business Roles > Define Business Role > Define Keyboard Shortcuts**.

Making Business Objects Invisible in the Central Search

In the assignment block **Central Search** you can make business objects invisible in the central search.

You can make business objects invisible in the central search in Customizing for **UI Framework** under **Business Roles > Define Business Role > Adjust Central Search Objects**.

Assigning Function Profiles

In the assignment block **Function Profiles** you can assign function profiles to your business role.

You can assign function profiles to your business role, in Customizing for **UI Framework** under **Business Roles > Define Business Role > Assign Function Profiles**.

Defining Function Profiles

You can find all available function profiles that you can assign to your business role, in Customizing for **UI Framework** under **Technical Role Definition > Define Function Profile**.

More Information

For more information about the central search, see Central Search.

Defining the Organizational Assignment

Use

Before you can start working with your business role, you need to assign your business role to a node, either an organizational unit or a position, in the organizational model. All users who are assigned to that node can access the assigned business role.

Activities

You can perform both assignments in the business role Customizing in SAP GUI or in the WebClient UI.

Assigning a Business Role to an Organizational Unit

You assign your business role to an organizational unit or a user in the organizational model, which is available in transaction `PPOMA_CRM`. The user needs to be assigned to the same organizational unit.

- In the WebClient UI, you can start the organizational model in the navigation bar of your business role via the logical link `MD-ORG-SR`.
- In SAP GUI, you define the organizational assignment in Customizing for **UI Framework**, by choosing **Business Roles > Define Organizational Assignment**.

Assigning Authorization Roles

Use

Before you can start working with your business role, you need to assign an authorization role (PFCG role) to it. If you are using SAP Fiori launchpad integrated mode, see Integration with SAP Fiori Launchpad.

Prerequisites

To maintain authorization roles (PFCG roles) in the WebClient UI, you have set up the transaction launcher Customizing for the transaction `PFCG`. Start transaction `CRMS_IC_CROSS_SYS` in Customizing and create a new entry:

- Mapped logical system: `OWNLOGSYS`
- Logical system: <*logical system*>
- ITS client: <*ITS client*>
- ITS URL: `/sap/bc/gui/sap/its/webgui/!?~transaction=IC_LTX&~okcode=ICEXECUTE"http://<server>:<port>/sap/bc/gui/sap/its/webgui/!?~transaction=IC_LTX&~okcode=ICEXECUTE`

Note

<*server*> means the SAP server.

<*port*> means the port that is used for the HTTP protocol.

Both settings can be reviewed in transaction `SMICM` under **Goto** > **Services**.

For more information, see SAP Note 888931.

Activities

Assigning an Authorization Role to a Business Role

The authorization role contains a certain authorization profile (PFCG profile) that defines the necessary authorizations. You assign this authorization role to your business role. You maintain authorizations in transaction `PFCG`.

Authorization roles have been enhanced with framework-specific authorizations that need to be added to existing authorization roles. Authorizations that you need to run the framework are part of the authorization role `SAP_CRM_UIU_FRAMEWORK`.

- In the WebClient UI, you can start the role maintenance in the business role overview page by clicking **Authorization Roles**.
- In SAP GUI, you define authorization roles for business roles in Customizing for **UI Framework** under **Business Roles** > **Define Authorization Role**.

Integration with SAP Fiori Launchpad

In addition to the standalone mode, WebClient UI Framework offers SAP Fiori launchpad integrated mode. In the SAP Fiori launchpad, you can create business roles and directly access WebClient UI applications in the integrated mode. All the WebClient UI applications needed for your business role are available in the SAP Fiori launchpad, which is the single point of entry.

Integration

Use

In the SAP Fiori launchpad integrated mode, you can directly access WebClient UI applications using tiles in SAP Fiori launchpad. You can also create front-end business roles.

Features

- SAP Fiori launchpad integrated mode can be used for applications that do not have dependencies on the L-shape of the classic WebClient UI. The L-shape is not displayed in this mode. Only the work area is displayed within the SAP Fiori launchpad shell.
- WebClient UI applications are embedded in the SAP Fiori launchpad shell.
- The saved searches and WebClient UI settings are available in the dashboard, which can be accessed with the dashboard icon in the system toolbar.
- The role configuration key **<DEFAULT>** is always used in SAP Fiori launchpad integrated mode.
- The back-end business role and related customizations are not applicable.
- **Help** link is available from the shell toolbar.

The following features are only supported in the standalone mode and not in the SAP Fiori launchpad integrated mode:

- **Apps** menu
- Navigation history and forward button
- Quick **Create** menu
- Direct link group
- **Recent Items**
- **Home** page
- **System News**
- Business role text
- Role selection
- System header

Procedure

Use SAP tools to set up SAP Fiori launchpad content as described in SAP Note 3170196.

The WebClient application ID is the **Target ID** defined in Customizing for **UI Framework** > **Technical Role Definition** > **Define Work Area Component Repository**. The target ID uniquely identifies the component, window, and inbound plug for the navigation. The target ID corresponds to the unique combination of UI object type and object action.

When you assign catalogs with WCF Apps to the PFCG Role, the UIU services corresponding to the target IDs are automatically assigned to the catalog in the PFCG Role menu.

In addition to the target IDs in the catalog, the framework services required for the WCF Apps are added.

The PFCG integration for the WCF Apps is provided in SAP Note 2817664.

More Information

For more information, see Setting Up Launchpad Content.

Function Profiles

Use

With WebClient UI apps in SAP Fiori launchpad integrated mode, you can use the function profiles for different processes offered with function profiles. When using SAP Fiori launchpad integrated mode, you have a backend authorization role, which contains the catalogs for WebClient UI applications to navigate to certain target IDs.

Procedure

In Customizing **UI Framework** > **Technical Role Definition** > **Define Fiori Launchpad Catalogs**,

- Assign the function profiles to SAP Fiori launchpad catalogs.
- Create entries for the WebClient UI function profiles.
- Define the function profiles and values for SAP Fiori launchpad integrated mode for each catalog.
- Assign PFCG roles in the backend, which contains the catalogs.

Note

If the user is assigned multiple catalogs, in the runtime, all the function profiles are loaded. A warning message is displayed if there are conflicting function profiles and the system uses the first value found.

Outbound Navigation

Use

With outbound mapping, you can navigate from WebClient UI applications directly to other apps available in SAP Fiori launchpad. These apps can be SAP Fiori apps, as well as applications based on Web Dynpro and SAP GUI for HTML technology. You must have access to these other apps from your SAP Fiori launchpad for the outbound mapping to work.

The internal navigation within WebClient UI works without any customizing, provided you have access to the applications.

Procedure

To use the outbound navigation, make entries in Customizing **UI Framework** > **Technical Role Definition** > **Define FLP Navigation Parameter Mapping**.

WebClient UI Framework Dashboard

Use

With the WebClient UI framework dashboard you can use features and settings specific to WebClient UI. The main settings like the theme, and the language are available in the SAP Fiori launchpad user settings.

The dashboard contains the following features:

- General Settings: Personalize your general settings specific to WebClient UI.
- Performance Settings: Manage performance settings specific to WebClient UI.
- Shortcuts: Personalize keyboard shortcuts specific to WebClient UI.
- Favorites: Access and maintain WebClient UI favorites.
- Tags: Access and maintain tag clouds.
- Central Sharing Tools: Share and display the objects that have been shared with you.
- Saved Search: To execute and maintain the saved searches.

Procedure

To access the WebClient UI dashboard, use the My Dashboard icon from the system toolbar of the WebClient UI page. You can personalize the content of the WebClient UI dashboard.

The WebClient UI dashboard features are available through function profiles. Link the required function profiles to the catalogs assigned to the user via the PFCG role.

- Function profile for Central Sharing Tool: `CENTRAL_SHARING_TOOL`
- Parameters for Favorites: `FAVORITES_ALLOWED`, `FAVORITES_COMMUNITY`
- Parameters for Tags: `TAG_CLOUDS_ALLOWED`, `TAG_CLOUDS_COMMUNITY`

Fact Sheet

Use

The fact sheet gives you an immediate and compressed overview of a particular business object.

The data displayed in the fact sheet can vary greatly. For this reason, you can configure the fact sheet and determine its layout yourself.

Features

- Format

  The information contained in the fact sheet is displayed in assignment blocks that group together certain types of information.

- Availability

  The fact sheet is available online. You can display the fact sheet and make changes to it such as showing table columns, if you have made the columns personalizable in Customizing. You can also print the fact sheet with your browser's print function.

- Configuration

You can determine which views you want to display in the fact sheet. You can choose between forms, tables, or views with graphics.

- Customizing

  You can determine the way the fact sheet is laid out, provided that you have chosen the tile layout. With this layout, you can determine the number of rows and columns in which the tiles are arranged.

Example

The Account fact sheet is available in Account Management on the WebClient UI. On the overview page of the individual accounts, you can click **Fact Sheet** and receive the most important information about this account online.

This information can come from multiple sources, for example master data, statistical data, and operation data. It enables you to quickly and easily see the most essential details about your key customers and partners from the various business transactions.

For example, you want to visit one of your customers. You open the fact sheet about this customer and receive all the important information about this customer at a glance.

Configuring the Fact Sheet

Use

You can adjust the fact sheet to the special requirements of your company.

Prerequisites

- You can define the page layout of the fact sheet in Customizing for **UI Framework** under **UI Framework Definition** > **Fact Sheet** > **Maintain Fact Sheet**. These settings are only valid for the page layout with tiles.
- You can change the title of the fact sheet and define placeholders by using the Business Add-In (BAdI) `BSP_DLC_FS_BADI` that belongs to the enhancement spot `BSP_DLC_FS`.

Features

- You can choose between different page layouts.

  You have the choice between a single column page layout and a page layout that consists of multiple tiles.

- You can choose the available views and assign them to the fact sheet.

  You can use the same view multiple times. For example, you can assign a view to a report from SAP NetWeaver Business Intelligence (BI) as well and then rename it.

  Example

  You have created an account fact sheet that you want to use in a certain role. This account fact sheet includes the view **Opportunities** which is also displayed in the account fact sheet used in another role.

- You can choose between various load options for the fact sheet views.

Activities

Starting the Application Component

Enter transaction BSP_WD_CMPWB and the application component BSP_DLC_FS. In the Browser Component Structure under Views, select the view BSP_DLC_FS/factsheet and click Configuration.

You can start the fact sheet configuration also in WebClient UI via hyperlink in the navigation bar. To make the fact sheet configuration accessible in your system administrator role in the WebClient UI, you have made the necessary settings in Customizing for UI Framework, by choosing Technical Role Definition > Define Navigation Bar Profile.

Using the Role Configuration Key

Choose an existing role configuration key or use the default one. If you want to create a new configuration, you can choose between a single column layout and a layout that consists of tiles.

Choosing and Assigning Fact Sheet Views

- Single column layout

  Use the UP ARROW and DOWN ARROW keys to move views between the Available Fact Sheet Views and Assigned Fact Sheet Views areas.

  Assign a load option to each view in the Assigned Fact Sheet Views area.

- Page layout with tiles

  Select a tile and use the arrow key to move the views that you want to display in the fact sheet from the Available Fact Sheet Views area to the Assigned Fact Sheet Views area.

Changing the Fact Sheet Title

If you want to change the title of the fact sheet, click Properties. You can use the default title or enter your own title in the Config Title field.

You can make further changes to the title and other fact sheet attributes by using a Business Add In (BAdI). Make these changes in the BAdI Builder transaction (SE18) and the enhancement spot BSP_DLC_FS. There is the default and example implementation BSP_DLC_FS_IMPL and the enhancement implementation BP_FACTSHEET for the Account fact sheet.

Displaying Technical Settings

Click Show Technical Details if you want to see the technical properties of the available and assigned fact sheet views.

Example

You can configure the Account fact sheet yourself. To do so, enter transaction BSP_WD_CMPWB and open the component BP_FACTSHEET.

Using Attributes for Assignment Blocks

Use

You can define attributes, which you can assign to your own application components.

Prerequisites

You have assigned attribute structures using the Table View Maintenance transaction (SM30) in table BSPC_DL_VIEWS.

Activities

1. Start transaction BSP_WD_CMPWB.
2. Start your application component and the required view.
3. Click Configuration.
4. Select More > Attributes .
5. In the Displayed Assignment Blocks area, you can assign attributes to the views for which you have previously defined attribute structures in Customizing.

Adjusting the Navigation Bar Profile

Use

Create an entry for the fact sheet in the navigation bar profiles that you use in generic outbound plug mapping (OP mapping).

Prerequisites

Create the entry for the fact sheet in Customizing for UI Framework under Technical Role Definition > Define Navigation Bar Profile . Select the required navigation bar profile in the IMG activity and navigate to Define Profile > Define Generic OP Mapping . Create a new entry and enter FACTSHEET as the object type, DISPLAY as the object action and BSP_DLC_FS as the target ID.

Creating and Assigning Views

Use

You can create your own views and assign them to a fact sheet. You can choose between the following two options:

- Option 1:
    - Create your own component
    - Create your views and enter a configuration key
- Option 2:
    - Use the framework enhancement concept
    - Create your views in an existing component, for example BP_FACTSHEET, and enter a configuration key

Activities

In both cases, proceed as follows:

1. Create an inbound plug and assign it to a window.

Using this process, you can assign multiple views to a window simultaneously. Alternatively, you can create a window for each view.

2. To calculate the property that is based on the BOL relation, for example, the business partner ID, write the program code in the inbound plug method. Alternatively, you can use the global data context.

3. Configure the views.

4. Assign the views to an existing fact sheet.

   You make this assignment in Customizing for **UI Framework** under **UI Framework Definition > Fact Sheet > Maintain Fact Sheet**.

5. Configure the fact sheet component (`BSP_DLC_FS`) in transaction `BSP_WD_CMPWB`.

More Information

[How Do I Create Components and Views?](#)

[Framework Enhancement](#)

How Do I Create Components and Views?

Procedure

The following procedure corresponds to option 1, in which you create your own component, create your own views within this component, and define a configuration key.

Creating Components and Views

1. Enter the transaction code `BSP_WD_CMPWB`.

2. Enter a component name and click **Create**.

3. Create a new view with the following parameters:

   - Name
   - BOL entity
   - Model node
   - Model attribute
   - View type: Table

Specifying the Configuration Key

1. Select your view under **Browser Component Structure > Views**.

2. Click **Structure**.

3. Double-click the `*.htm` file under **View Layout**.

   You see the layout of the selected view.

4. Click **Page Attributes**.

You see the attributes that are assigned to the view.

Creating an Inbound Pug

1. Under **Structure** > **Inbound Plugs**, create a new inbound plug with the name `DEFAULT`.
2. Under **Runtime Repository Editor** > **Component** > **Windows**, select the **Main Window** to which you want to assign the inbound plug.
3. In the context menu, choose **Add Plug with Follow-Up Navigation**.

   You see the page **Create Inbound Plug with Forwarding Navigational Link**.
4. Enter the appropriate values.
5. Under **Runtime Repository Editor** > **Component** > **Component Interface** > **Interface View**, add the inbound plug.
6. Under **Browser Component Structure** > **Component** > **Views**, select your view.
7. Double-click your inbound plug under **Structure** > **Inbound Plugs**.
8. In the **Class Builder**, enter the program code to calculate the property that is based on the BOL relation.

Assigning Views to an Existing Fact Sheet

1. Click **Configuration** and configure your views.
2. Assign the views to an existing fact sheet.

   You make this assignment in Customizing for **UI Framework** under **UI Framework Definition** > **Fact Sheet** > **Maintain Fact Sheet**.
3. Add a new entry in Customizing for each view.

Configuring the Fact Sheet

1. Start component `BSP_DLC_FS` in transaction `BSP_WD_CMPWB`.
2. Under **Views**, choose the view `BSP_DLC_FS/factsheet`.
3. Configure the fact sheet under **Configuration**.

More Information

Creating and Assigning Views

Configuring the Fact Sheet

Framework Enhancement

Use

The framework enhancement concept is based on component enhancements, which means that you enhance the function of standard components, standard views, and standard controllers that are delivered by SAP.

First, you verify whether you can use the UI configuration tool, for example, if you want to adjust SAP standard components to your business needs. If you want to make functional changes in a component, you need to use the enhancement concept.

You can also create your own components.

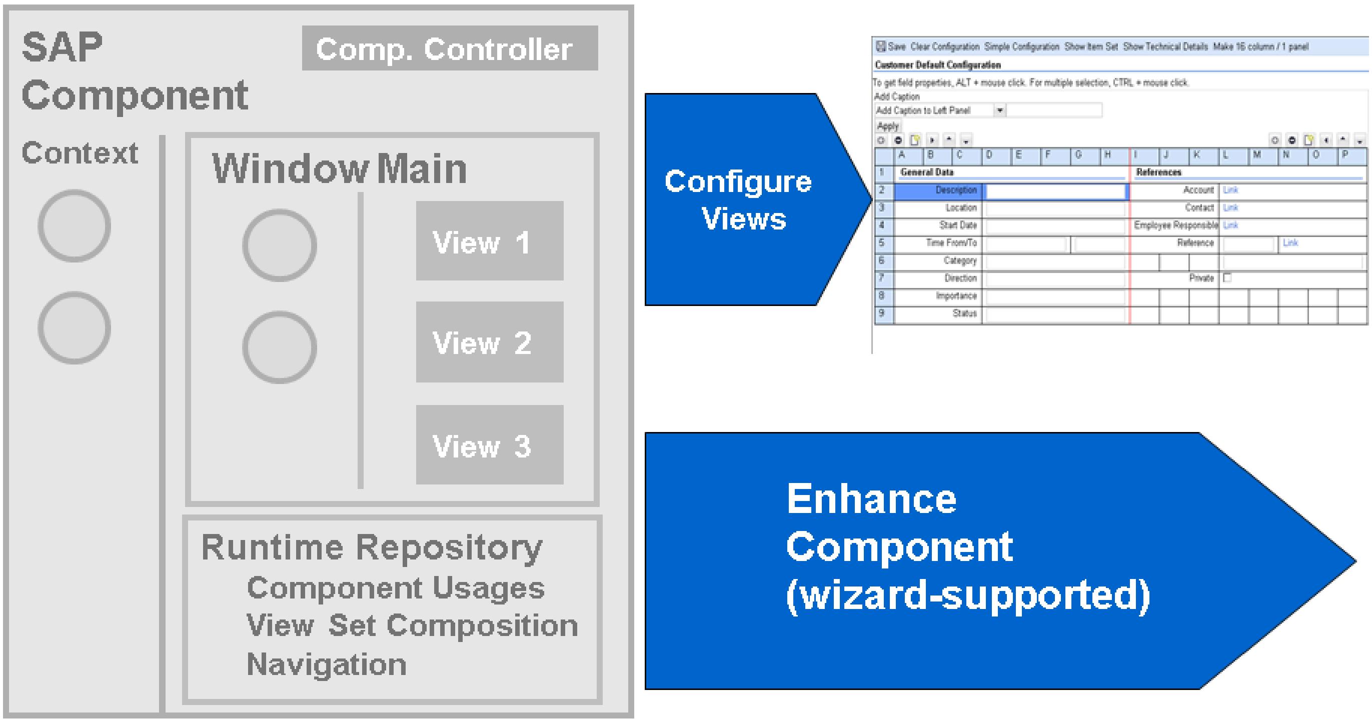
i Note

For more information about the framework enhancement concept, see SAP Note 1122248.

Features

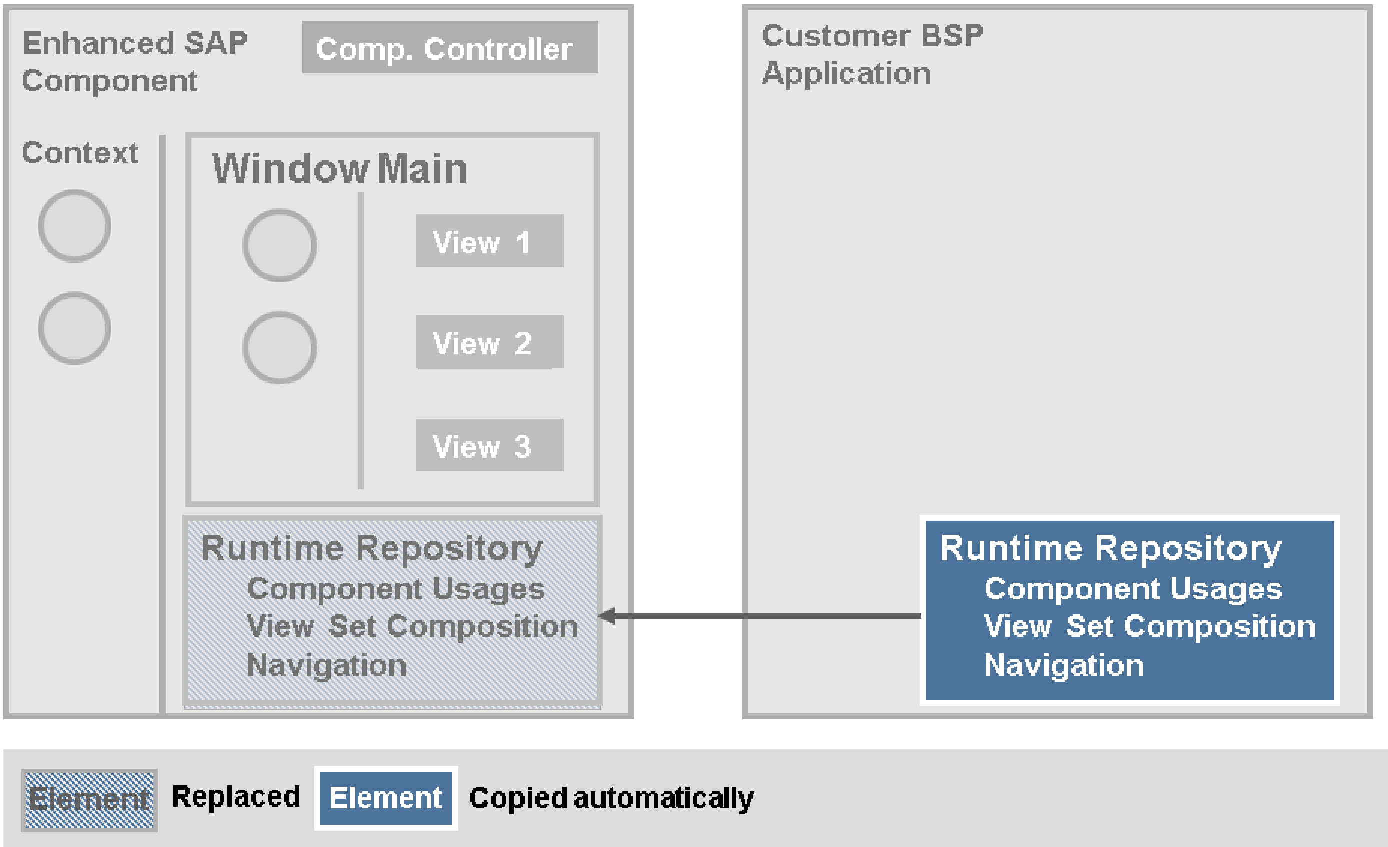
UI Configuration or Component Enhancement?

You can adjust SAP standard components either with the UI configuration tool or with the component enhancement concept. You can use the UI configuration tool, for example, if you want to make specific layout changes in views. If you want to make functional changes in a component, for example, create new events, you need to use the component enhancement concept.



Component Enhancement: Step 1

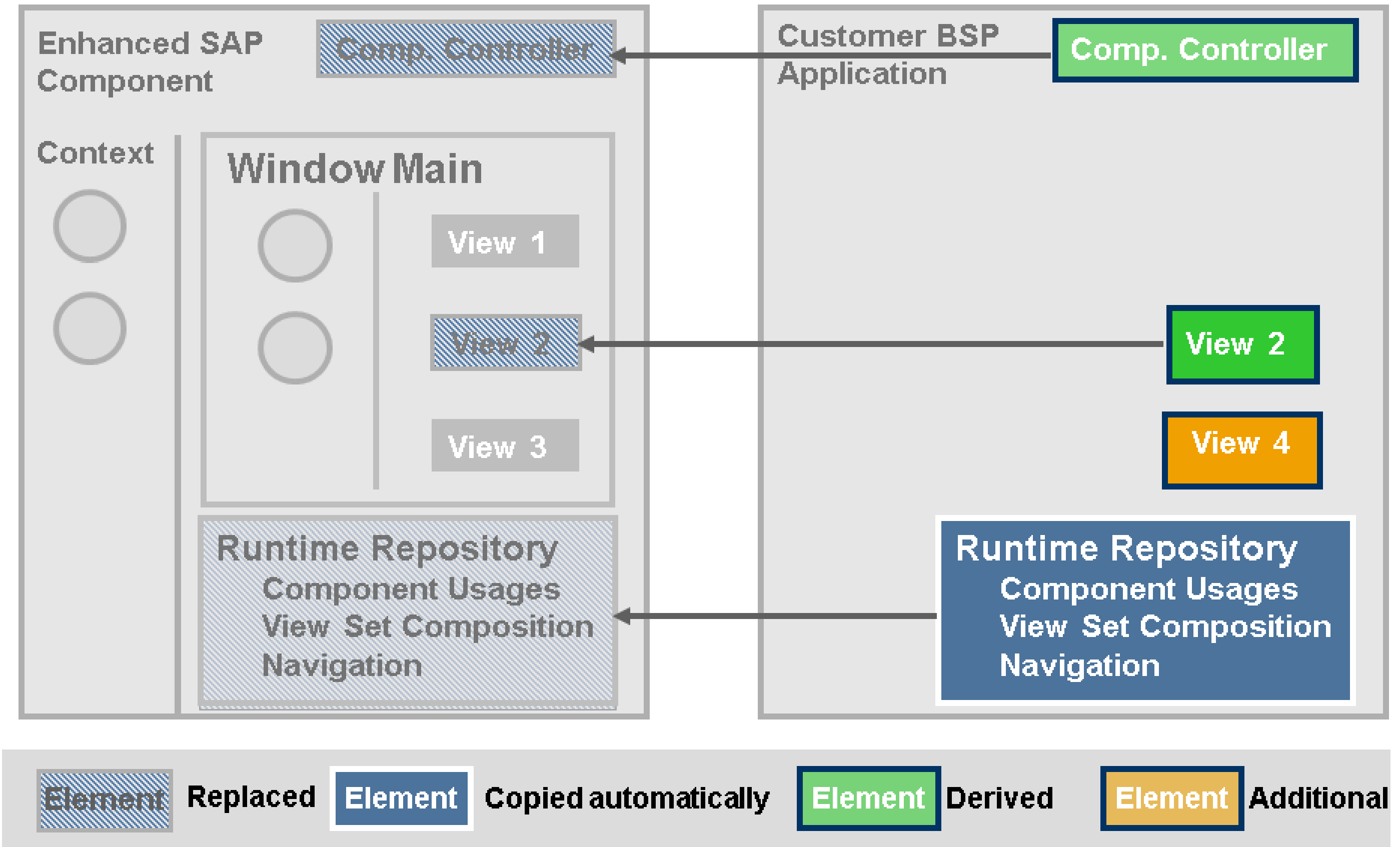
If you use this wizard-supported approach, the system automatically creates a BSP application for the enhanced elements (runtime repository, derived controllers, and so on). It copies the runtime repository and maintains replacement information.



Component Enhancement: Step 2

When a component enhancement has been created, the SAP runtime repository is replaced by a customer version. Now it is possible to enhance views and controllers of the component.

If you need to enhance a view or controller, the wizard automatically guarantees that the required inheritance logic does not need to be changed in your system. The system automatically creates a BSP controller and view. It derives controller and context classes and maintains controller replacement information. Now the SAP version of the view or controller is replaced by the customer version.



Enhancement Set

Use

An enhancement set is used to group several component enhancements together.

Features

- An enhancement set contains the enhancement definition that shows the assigned BSP application and the assigned runtime repository.
- You can define as many enhancement sets as you like, but only one enhancement set is active at runtime.
- Based on the enhancement set, you can create enhancements in the BSP Component Workbench.

⚠ **Caution**

The enhancement set is not yet available as an attribute in the UI configuration tool. Therefore, you can only have one configuration for every enhancement. If a configuration is created for an enhanced object, some objects that are used in this configuration may not be available for the corresponding object that is not enhanced.

Activities

Define Enhancement Set

To define an enhancement set, use transaction SM34. Open the view cluster BSPWDVC_CMP_EXT and create the enhancement set that you can later use in the BSP Component Workbench. You only need to enter a name and a description, because the definition is automatically filled later when you create enhancements in the BSP Component Workbench.

Determine Active Enhancement Set

If the determination of an enhancement set at runtime is not only to be based on the client, but on other criteria, such as the user or the profile, you need to use a Business Add-In (BAdI). You can use the **Object Navigator** (SE80) to open package CRM_BSP_WD. Open the enhancement spot COMPONENT_HANDLING with the BAdI definition COMPONENT_LOADING and the DEFAULT implementation. This implementation uses the customizing entries from BSPWD_EHSET_ASGN to determine the enhancement based on a client. You can also use this BAdI to implement other determination rules for an enhancement set.

Assign Enhancement Set to a Client

To assign an enhancement set to a client, use transaction SM30. Open table BSPWDV_EHSET_ASG to make the necessary assignment. You can only assign one enhancement set to a client.

Customer Namespace

Use

In the SAP standard, the prefix ' Z ' is used for objects that are created in the customer namespace. In the Framework enhancement concept, this has been realized with Business Add-Ins (BAdI). However, you can create and save your enhancements in your own namespace.

Features

- The Business Add-In BSP_WD_APPL_WB with the method GET_CUST_CLASSNAME_PROPOSAL provides you with a proposal for a name for customer-defined classes within the enhancement. The default implementation of this method replaces the first letter in the name with ' Z '. You can create your own implementation for this method to place all new class names in your own namespace / * / if necessary.
- The enhancement spot CLASS_GENERATION with the Business Add-In CLASS_NAME_BUILDER with five methods can be used to generate the class names for customer views and controllers. There is no default implementation. If you implement the BAdI, you get a proposal for a name that you can change.

Activities

Use Your Own Namespace for Enhanced Controllers and Views

If you want to use your own namespace, you can use the **Object Navigator** (SE80) and open the package CRM_BSP_WD_TOOLS. Open the business add-in BSP_WD_APPL_WB. Under **Methods**, you find the method GET_CUST_CLASSNAME_PROPOSAL.

Create Your Own Objects in Your Namespace

If you want to create your own objects in your own namespace, you can use the **Object Navigator** (SE80). Open the package CRM_BSP_WD_GEN_TOOLS. Under **Enhancements** > **Enhancement Spots** you find the BAdI CLASS_GENERATION with the BAdI definition CLASS_NAME_BUILDER and the interface IF_BSP_WD_GEN_TOOLS.

i **Note**

With this BAdI, no default implementation is used.

Create Enhancements

Procedure

If you want to enhance components in the UI Component Workbench, proceed as follows:

1. Start transaction `BSP_WD_CMPWB`.
2. Choose the component that you want to enhance.
3. Choose the **Display** pushbutton.
4. Click the **Enhance Component** pushbutton.
5. Choose the enhancement set.
6. In the **BSP Application for Storing Enhancement Objects** dialog box, choose an existing BSP application or create a new one.

   → **Recommendation**

   We recommend that you create your own BSP application in which you store the enhancements for each component. If you have not yet created enhancements for this component, enter a new name for the BSP application. For more information, see SAP Note [1122248](1122248).
7. Select a view under **Browser Component Structure** > **Component** > **Views**.

   i **Note**

   All gray objects have not yet been enhanced. All black objects have already been enhanced. All blue objects are obsolete, since they are not used in the current release.
8. Choose **Enhance** in the context menu.

   The object is created automatically in the customer namespace.
9. Select a workbench request.
10. Select a package.

    You can create a new package in transaction `SE 80`.
11. Make the necessary enhancements.

    i **Note**

    The different colors of the icons to the left of the methods have the following meanings:

    - Yellow means that the method is inherited
    - Green means that the method is redefined
    - Gray means that the method does not exist

Note

You can delete enhancements by choosing the **Delete Component Enhancement** pushbutton.

Result

In the view cluster `BSPWDVC_CMP_EXT`, which you can open with transaction `SM34`, you can see the substitution.

More Information

Enhancement Set

UI Configuration Determination

Multiple Configurations

Use

The configuration of views, for example field labels, is based on configurations that have been created either with the UI Configuration Tool, in the design layer Customizing, or in the ABAP Dictionary.

Features

Multiple Configurations at Design Time

At design time, you can create multiple configurations using the different keys that are available for views. The available and supported keys are:

- Component (automatically set by the framework)
- View (automatically set by the framework)
- Role configuration key
- Component usage
- Object type
- Subobject type (depending on the object type)

Multiple Configurations at Runtime

At runtime, the multi-configuration is accessed using the following:

- Component (automatically set by the framework)
- View (automatically set by the framework)
- Role configuration key (automatically determined by the business role with which the user has logged on)
- Component usage (automatically set by the framework, as it is derived from the component usage itself)
- UI object type (set by the application)

- Subobject type (set by the application)

Determination of Different Keys

- You can define the role configuration key in Customizing for UI Framework under Technical Role Definition > Define Role Configuration Key.
- You can define the UI object type in Customizing for UI Framework under UI Framework Definition > Define UI Object Types.
- Subobject types are determined via the callback class, assigned to the UI object type in Customizing for UI Framework under UI Framework Definition > Define UI Object Types. The interface is `IF_BSP_DLC_OBJ_TYPE_CALLBACK~GET_OBJECT_SUB_TYPES`.
- UI object type and subobject type can be set by the application in method `DO_CONFIG_METHOD`.

Determination of Assigned Design Objects

For the determination of assigned design objects, the following keys are used:

- Component
- Component usage
- View
- Context node
- UI object type
- Attribute name

More Information

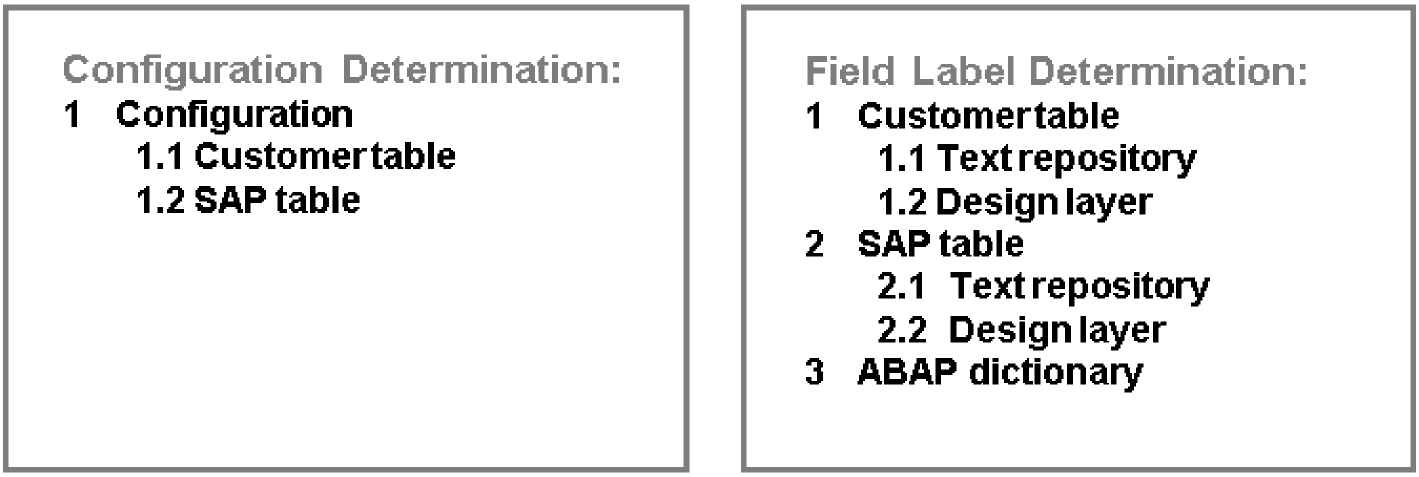
UI Configuration Tool

Design Layer

General Access Sequence

General Access Sequence

The figure below illustrates the general access sequence for configuration determination and field label determination:



There is a special handling for the assignment of block headers in overview pages which shows the technical name of the assignment block if the label does not exist. English values are shown in the case that the assignment block labels do not exist in a requested language.

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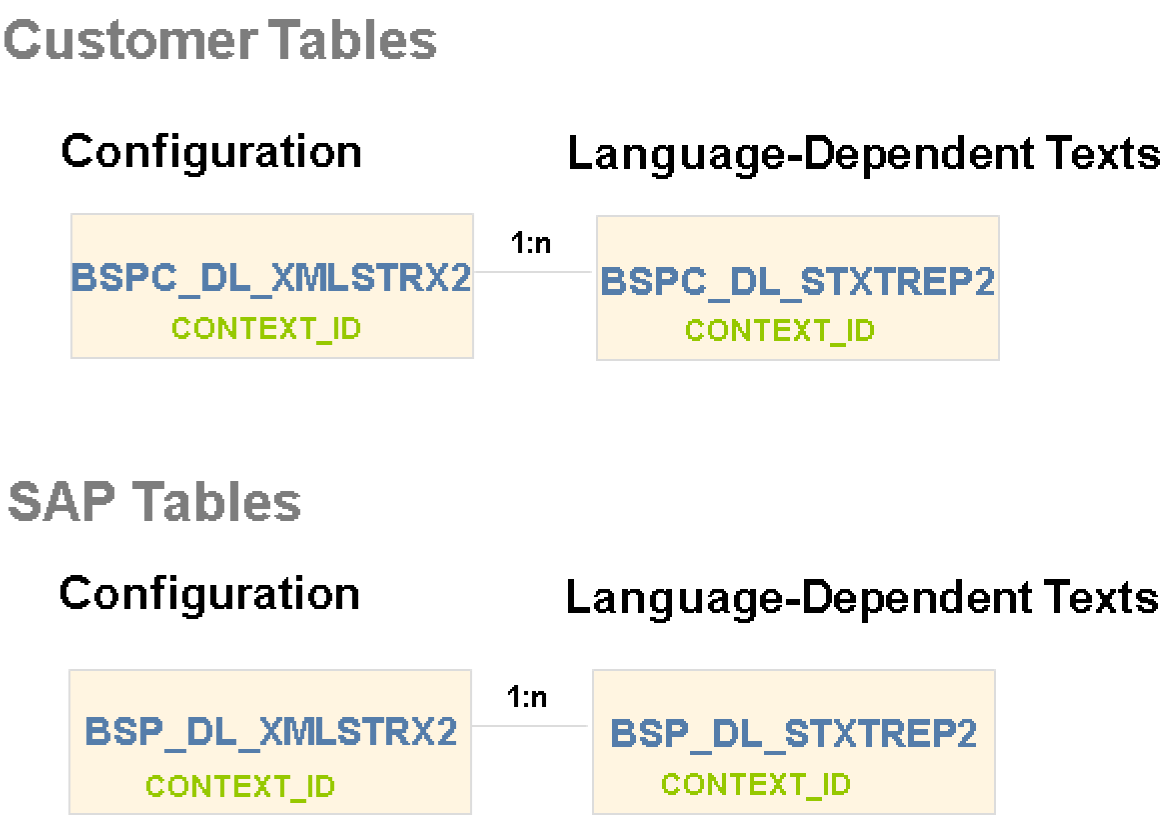
Data Model: UI Configuration/Text Repository

The figure below illustrates the data model of UI configuration and text repository:



Data Model: Design Layer

The figure below illustrates the data model of the design layer:

Access Sequence Logic

Use

The determination takes place in two steps:

- Step 1: A configuration is determined, based on the search key.
- Step 2: Field labels from the text repository are determined, based on the successful key.

Step 1

The table below shows how the configuration is determined, based on the search key. The access sequence for customer tables and SAP tables is identical.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Configuration Table
1	X	X	X	X	Customer Table
2	X	X	X	<default>	Customer Table
3	X	<default>	X	X	Customer Table
4	<default>	X	X	X	Customer Table
5	<default>	<default>	X	X	Customer Table
6	X	<default>	X	<default>	Customer Table
7	<default>	<default>	X	<default>	Customer Table
8	X	X	<default>	<default>	Customer Table
9	<default>	X	<default>	<default>	Customer Table
10	X	<default>	<default>	<default>	Customer Table
11	<default>	<default>	<default>	<default>	Customer Table
12	X	X	X	X	SAP Table
13	X	X	X	<default>	SAP Table
14	X	<default>	X	X	SAP Table
15	<default>	X	X	X	SAP Table
16	<default>	<default>	X	X	SAP Table
17	X	<default>	X	<default>	SAP Table
18	<default>	<default>	X	<default>	SAP Table
19	X	X	<default>	<default>	SAP Table
20	<default>	X	<default>	<default>	SAP Table
21	X	<default>	<default>	<default>	SAP Table
22	<default>	<default>	<default>	<default>	SAP Table

Step 2

The table below shows how the field labels are determined from the text repository, based on the successful key. The search key is used again for the determination of the design object.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	Field Label Saved in
1	<successful key>				X	Customer Table (Text Repository)
2	<default>	<default>	<default>	<default>	X	Customer Table (Text Repository)
3		X	X		X	Customer Table (Design Layer)
4		X	X		<default>	Customer Table (Design Layer)
5		<default>	X		X	Customer Table (Design Layer)
6		<default>	X		<default>	Customer Table (Design Layer)
7		<default>	<default>		X	Customer Table (Design Layer)
8		<default>	<default>		<default>	Customer Table (Design Layer)
9	<successful key>				X	SAP Table (Text Repository)
10	<default>	<default>	<default>	<default>	X	SAP Table (Text Repository)
11		X	X		X	SAP Table (Design Layer)
12		X	X		<default>	SAP Table (Design Layer)
13		<default>	X		X	SAP Table (Design Layer)
14		<default>	X		<default>	SAP Table (Design Layer)
15		<default>	<default>		X	SAP Table (Design Layer)
16		<default>	<default>		<default>	SAP Table (Design Layer)
17					X	ABAP Dictionary

Note

The gray color in the table means that these parameters exist neither in customer tables and/or SAP tables in the design layer, nor in the ABAP Dictionary. When you create a configuration, you can use the above-mentioned parameters. An X means that you have entered a parameter value in the configuration. If you do not use these parameters in the configuration, this is represented by `<default>`.

Features

To determine the field label, the configuration text repository is searched, based on the successful key. If no field label is found and the search is processed in the design layer, the design object determination is started with the search key.

If you use other properties of the design layer Customizing, for example, you have set a field to hidden, the same access sequence logic applies as for the field label determination.

When a design object that has a more specific search key than the related configuration exists, special design layer settings based on this search key are not immediately visible if you are looking at this configuration in the UI Configuration Tool.

Example

Assume that for the account details view (BP_HEAD/AccountDetails), a configuration with the following parameters exists:

- Role configuration key = <DEFAULT>
- Component usage = <DEFAULT>

- Object type = <DEFAULT>
- Subobject type = <DEFAULT>

You can see this configuration in the UI Configuration Tool with the field labels that are based on this configuration, for example **Name 1**.

Assume that a design object with a more specific key combination exists, for example, object type = BP_ACCOUNT, renamed field label **Account** instead of **Name 1**, assigned to the account details view BP_HEAD/AccountDetails. At runtime, the renamed field label from the design layer is displayed. However, if you look at the default configuration in the UI Configuration Tool, you cannot see this specific key combination.

More Information

Design Layer Customizing

UI Configuration Tool

Example: Configuration with a Renamed Field Label

Design Time

The table below shows several configurations that have been entered in the system at design time. In the **Table Type** column, you can see in which layer the configuration exists and in which table (customer table or SAP table) the configuration is saved.

No.	Component	View	Role Configuration Key	Object Type	Subobject Type	Attribute Name	Field Label	Table Type
1	BT116H_SRVO	DetailsEF	<default>	<default>	<default>	DESCRIPTION	<default>	Configuration Customer Table
2	BT116H_SRVO	DetailsEF	<default>	SRVQ	<default>	DESCRIPTION	<default>	Configuration SAP Table
3	BT116H_SRVO	DetailsEF	<default>	SRVT	<default>	DESCRIPTION	<default>	Configuration SAP Table
4	BT116H_SRVO	DetailsEF		SRVT		DESCRIPTION	Information	Design Layer Customer Table
5						DESCRIPTION	Description	ABAP Dictionary

Note

The gray color means that these parameters exist neither in the design layer, nor in the ABAP Dictionary.

Runtime: Configuration Determination

At runtime, the user works with the following component, view, and object type. The configuration is determined using this search key combination, which consists of:

- Component: BT116H_SRVO
- View: DetailsEF

- Object Type: SRVQ

First, the customer tables are checked. According to the general access sequence, the configuration with the default values in the customer configuration is determined by the system.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Configuration Table
1	X	X	X	X	Customer Table
2	X	X	X	<default>	Customer Table
3	X	<default>	X	X	Customer Table
4	<default>	X	X	X	Customer Table
5	<default>	<default>	X	X	Customer Table
6	X	<default>	X	<default>	Customer Table
7	<default>	<default>	X	<default>	Customer Table
8	X	X	<default>	<default>	Customer Table
9	<default>	X	<default>	<default>	Customer Table
10	X	<default>	<default>	<default>	Customer Table
11	<default>	<default>	<default>	<default>	Customer Table

Checked

Determined

Runtime: Determination of the Field Label Renamed in the Design Layer

Based on the successful key, the text repository is searched. If no matching entry was found in the text repository, the design layer is checked, but the search key is used again for the design object determination.

According to the general access sequence, the customer tables are checked first (see first part of the table), before the SAP tables are checked (see second part of the table).

First Part of the Table

Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	UI Configuration Saved in	
X	X	X	X	X	Customer Table (Text Repository)	
X	X	X	<default>	X	Customer Table (Text Repository)	
<default>	X	X	X	X	Customer Table (Text Repository)	
X	<default>	X	X	X	Customer Table (Text Repository)	
<default>	<default>	X	X	X	Customer Table (Text Repository)	
<default>	X	X	<default>	X	Customer Table (Text Repository)	
<default>	<default>	X	<default>	X	Customer Table (Text Repository)	
X	X	<default>	<default>	X	Customer Table (Text Repository)	
X	<default>	<default>	<default>	X	Customer Table (Text Repository)	
<default>	X	<default>	<default>	X	Customer Table (Text Repository)	
<default>	<default>	<default>	<default>	X	Customer Table (Text Repository)	
	X	X		X	Customer Table (Design Layer)	
	X	X		<default>	Customer Table (Design Layer)	Checked
	<default>	X		X	Customer Table (Design Layer)	
	<default>	X		<default>	Customer Table (Design Layer)	
	<default>	<default>		X	Customer Table (Design Layer)	Determined
	<default>	<default>		<default>	Customer Table (Design Layer)	

Second Part of the Table

Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	UI Configuration Saved in
X	X	X	X	X	SAP Table (Text Repository)
X	X	X	<default>	X	SAP Table (Text Repository)
<default>	X	X	X	X	SAP Table (Text Repository)
X	<default>	X	X	X	SAP Table (Text Repository)
<default>	<default>	X	X	X	SAP Table (Text Repository)
<default>	X	X	<default>	X	SAP Table (Text Repository)
<default>	<default>	X	<default>	X	SAP Table (Text Repository)
X	X	<default>	<default>	X	SAP Table (Text Repository)
X	<default>	<default>	<default>	X	SAP Table (Text Repository)
<default>	X	<default>	<default>	X	SAP Table (Text Repository)
<default>	<default>	<default>	<default>	X	SAP Table (Text Repository)
	X	X		X	SAP Table (Design Layer)
	X	X		<default>	SAP Table (Design Layer)
	<default>	X		X	SAP Table (Design Layer)
	<default>	X		<default>	SAP Table (Design Layer)
	<default>	<default>		X	SAP Table (Design Layer)
	<default>	<default>		<default>	SAP Table (Design Layer)
				X	ABAP Dictionary

i Note

The gray color in the table means that these parameters exist neither in customer tables and/or SAP tables in the design layer, nor in the ABAP Dictionary. When you create a configuration, you can use the above-mentioned parameters. An X means that you have entered a parameter value in the configuration. If you do not use these parameters in the configuration, this is represented by `<default>`.

Enhancement Spot

Use

You can change the access sequence logic completely by implementing an enhancement spot. For a given set of four key fields, your implementation selects one of the available configurations.

Assume you want to implement a template approach regarding role-based configurations. The intention is to be able to replace the SAP fallback (role=<DEFAULT>) with the fallback of your customer-specific role. If no customer-specific configuration is available for the current role, you can select a special configuration for a “fallback” role. This means that you do not need to maintain all configurations for all available business roles.

Integration

You can find the enhancement spot in Customizing for **UI Framework** under **UI Framework Definition** > **Business Add-Ins (BAdIs)** > **Define Configuration Access**.

UI Configuration Tool

Use

You can use the UI Configuration Tool to adapt the WebClient UI to your company's specific requirements. You can access the UI Configuration Tool in SAP GUI and in the WebClient UI.

Integration

If you want to change field labels across multiple views, you can use the design layer to consolidate the field changes. You define these settings in Customizing for **UI Framework** under **UI Framework Definition** > **Design Layer**.

Features

The UI Configuration Tool in the WebClient UI enables you to easily adjust pages and views to your requirements.

General Features

The UI Configuration Tool in the WebClient UI offers the following general features:

- The UI configuration is based on an authorization object.

  The UI configuration in SAP GUI is also based on an authorization object.

- The UI configuration is automatically started with the parameters that were found.

  This is true, if you start the UI configuration directly from an application. If you start it from the navigation bar or the work center page, you need to select manually the configuration parameters of the pages or views that you want to change.

- All configuration changes that you have made are automatically visible in the application.
- To save your configuration changes you can create a new transport request or select an existing transport request in the WebClient UI.

UI Configuration Access

You can enter the view configuration and fact sheet configuration in the WebClient UI in the following ways.

Via Logical Links in the Navigation Bar or from a Work Center Page

You can access the view configuration and fact sheet configuration in your system administrator role. They way you access the view configuration and fact sheet configuration depends on your Customizing:

- By using logical links that belong to a direct link group in the navigation bar
- By using logical links that belong to a work center, for example the **Administration** work center

  These logical links are displayed on the second level of the corresponding work centers in the navigation bar of your system administrator role.
- By using logical links on a work center page

  These logical links can be assigned to the **Search** content block on the work center page, for example on the **Administration** work center page.

Via Icons in the Application

When the configuration mode is activated on the central personalization page of the WebClient UI, you see the **Configure Page** icon at the top of **Home** pages, work center pages, and overview pages. If you click the icons, you see the **View Configuration** dialog.

You see the **Show Configurable Areas** icon at the top or in the overflow menu of most pages. If you click the icon, you can see all configurable areas that are available on this page. Every configurable area is surrounded by a frame. If you click a configurable area with a frame you see the view configuration of that specific view. If the configuration mode is activated, the general navigation is deactivated. To deactivate the configuration mode, click outside the configuration frame.

> **Note**
>
> You can use the **Show Configurable Areas** icon to configure only those assignment blocks that are expanded. If they are collapsed, expand them before you click the icon.

Via F2 Help in the Application

To find out the name of an application component or view, click F2. The **Technical Data** dialog with the technical information is started. To access the UI configuration of this view and this application component, click **Configure** at the bottom of the dialog.

Look and Feel of the UI Configuration

View Configuration

When you start the view configuration in the navigation bar you navigate to the search page where you can search for an application component. All views that belong to a certain application component are displayed in the result list. Select a view and click **View Configuration**. You navigate to the standard configuration page of the selected application component and view. All configurations that are available for that specific application component and view are displayed in the **Configurations** block. The current configuration is highlighted in the **Configurations** block. The **View** block contains the fields, assignment blocks, and so on, which are available in that specific view.

If you need more space to display data, you can collapse either the **Configurations** block or the **View** block.

Fact Sheet Configuration

When you start the fact sheet configuration in the navigation or in the BSP Component Workbench you navigate to the search page, where you can search for a fact sheet. All configurations (customer configuration, standard configuration) and the role configuration keys that belong to a fact sheet are displayed in the result list. Select a fact sheet configuration and click **Select Page Type** to define the page type and the page layout of the fact sheet. You can also click **Next** or the hyperlink in the **Role Config. Key** column. Click **Assign Views** to navigate to the next configuration step. In the **Configurations** block, you see the selected fact sheet and its role configuration key. In the **View** block, you can assign fact sheet views to the selected fact sheet and to the tiles. In the **Properties** block you see the different fact sheet titles.

If you need more space to display data, you can hide each of the above-mentioned blocks.

UI Configuration Based on Enhancement Sets

If you select an application component and view in the WebClient UI for which an enhancement set has been previously created in SAP GUI, you can perform the configuration based on the enhanced component.

Configuration Access Sequence

You can change the standard access sequence that is used to determine configurations. The actual configuration that is used for a certain configurable view is determined on the basis of a search key and the available configurations that exist for a specific view. The configuration can either be a standard SAP configuration or a customer-specific configuration.

To use your own access sequence, you need to implement the Business Add-In (BAdI) `BSP_DLC_ACCESS_ENHANCEMENT` in Customizing for **UI Framework** under **UI Framework Definition** > **Business Add-Ins (BAdIs)** > **Define Configuration Access** > **BAdI: Configuration Access Determination**.

More Information

Introduction

Configuring the Fact Sheet

Displaying Technical Information

General Access Sequence

Introduction

Use

The application data is displayed in views. The views can be grouped into view sets. Views and view sets are displayed in windows. SAP delivers configurable and non-configurable views. A view is configurable if a configuration tag is assigned to it. This is the case for many of the views delivered by SAP. Configuration tags enable changes and adaptations to the UI, with the help of the UI configuration.

You can use the UI configuration to do the following:

- Add, remove, and change the position of UI elements
- Rename UI elements

- Add and change captions
- Change field properties (for example, display only or mandatory)
- Define other options (for example, load option for assignment blocks, visible search criteria, and default operators for the search)

View Types

With the UI configuration it is possible to make changes to the layout of a configurable view. Based on the view type, different configuration capabilities are provided. Distinctions are made between the following view types:

- Form view
- Tables and hierarchies
- Overview pages
- Home page and work center pages

SAP Configuration and Customer Configuration

For each configurable view, SAP delivers at least one standard configuration. All configurations delivered by SAP are saved in cross-client SAP tables. They cannot be changed by the customer. If configurations are adjusted or created by the customer, they are saved in client-dependent customer tables. At runtime, an access sequence is used to determine the configuration. The system always checks whether a customer configuration exists and if so, this is displayed. If no customer configuration exists, the SAP configuration is used. If a customer configuration is deleted with the UI configuration, the SAP configuration is determined.

Using Parameters

Additionally, it is possible to have more than one SAP configuration or customer configuration for a view. To implement this, the following parameters are used:

- Role configuration key
- Component usage
- UI object type
- Subobject type

At runtime the correct configuration is determined, based on these parameters. The role configuration key can be defined in Customizing, and assigned to a business role. With this parameter, it is possible to create business-role-dependent configurations. At runtime, this parameter is automatically determined by the business role with which the user is logged on.

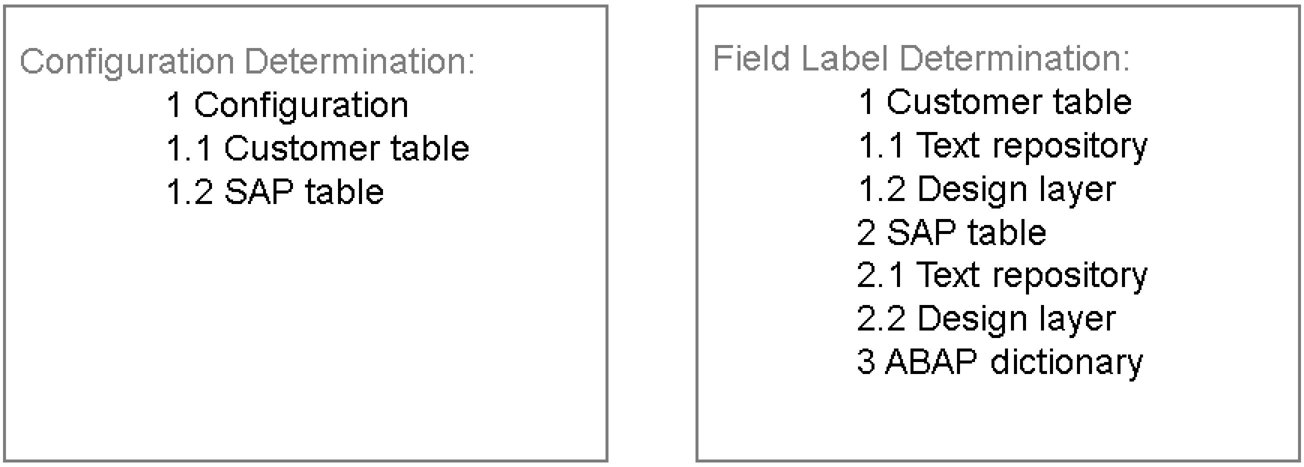
UI objects types have been introduced to define UI objects independently of existing business object layer objects.

Based on the UI object type, configurations can be created. The subobject type is a child object of the object type, for example transaction types, and can also be used. Therefore, by using UI object type and subobject type, you can have different UI configurations for different business processes. At runtime, object type and subobject type are set by the application. Component usage is automatically set by the framework at runtime.

Access Sequence

To determine configuration and labels, the access sequence is executed in two steps. In the first step, a configuration is determined, based on the parameters. The search starts with the definitive parameters (search key) and, if this is not successful, fewer parameters are used step-by-step. The parameters with which a configuration was successfully determined are called successful key. In a second step, the successful key is used to determine the labels from the text repository. If labels are not explicitly set in the configuration, they can be determined from the ABAP Dictionary. In addition to the UI configuration, the design layer Customizing makes it possible to link UI-related settings to the generic design object that spans several views implementing the same business content.

The following picture shows the general access sequence at runtime:



More Information

Design Layer

[UI Configuration Determination]

Starting the UI Configuration Tool

Use

You can access the UI Configuration Tool in SAP GUI and in the WebClient UI.

Prerequisites

- To access the view configuration and the fact sheet configuration in your system administrator role in the WebClient UI, you have made the necessary settings in Customizing for **UI Framework**, by choosing **Technical Role Definition** > **Define Navigation Bar Profile**. The logical links `BSP-DLC-VC` and `BSP-DLC-FC` are assigned to the work center `CT-ADMIN`. You need to ensure that this work center is part of your administrator business role, and that the links have been set to **Visible**.
- To enable the UI configuration of pages and views directly in the WebClient UI, your user needs to be assigned to the authorization object `CRMCONFMOD`, and you must have activated the UI configuration mode on the central personalization page. To activate the UI configuration mode, click **Personalize** in the header area of your business role. Choose **Personalize Settings** on the **Personalization** page. In the **Personalize Settings** dialog, select the **Enable configuration mode** checkbox.

Features

Starting the UI Configuration Tool in SAP GUI

In SAP GUI, the UI Configuration Tool is integrated in the UI Component Workbench. Therefore, you need to start the UI Component Workbench to access the UI configuration. The UI configuration is performed in the UI Configuration Tool, which you can start by using the **Configuration** tab. However, this tab is only activated if a view can be configured.

You can start the UI Component Workbench including the UI configuration in the following ways:

- In Customizing for **UI Framework**, select **UI Framework Definition** › **Access UI Component Workbench**.
- Enter transaction code `BSP_WD_CMPWB` to access the UI Component Workbench.

Starting the UI Configuration Tool from the Navigation Bar

You can start the view configuration and the fact sheet configuration from the second level in your **Administration** work center, in the navigation bar of your administrator business role.

Starting the UI Configuration Tool Directly from the Application

You can start the view configuration and the fact sheet configuration directly from the application in the following ways:

- You click the **Show Configurable Areas** icon at the top of a page, for example on the Account search page.

  The configurable areas are highlighted, and you start the configuration of the related view by clicking one of the configurable areas.

- You click the **Configure Pages** icon at the top of the page, for example on the Account overview page, to configure superior views such as the **Home** page, work center page, and overview page.
- You click the fact sheet configuration and the view configuration links in the navigation bar of your system administrator role.

Activities

In the following text, you will find a description of how to start the UI Configuration Tool integrated in the UI Component Workbench.

1. Enter the transaction code `BSP_WD_CMPWB`.

   You see the page **UI Component Workbench**.

   > **i Note**
   >
   > You can show or hide the field **Enhancement Set** using the icon to the left of the **Component** field. An enhancement set is a parenthesis around multiple enhancements of an application component.

2. Select a component, for example `BP_HEAD_SEARCH`.
3. Click **Display**.

   You see the standard structure view of the chosen UI component.

4. Expand the **Views** node under **Browser Component Structure**.
5. Double-click the view that you want to edit, for example **MainSearch**.
6. Click the **Configuration** tab.

   The view's configuration is displayed.

More Information

Authorization Object

Configuring Views

Configuring Views

You use the UI Configuration Tool to configure views, and pages that consist of different views.

The information that follows describes the configuration of the following views and pages:

- Form views
- Tables and hierarchies
- Search pages
- Overview pages
- Home page and work center pages

Note

When making changes to pages or views, you can choose to discard your saved changes and revert to the default configuration delivered by SAP at any time. To do so, ensure that you have selected the configuration that you want to delete, and click **Delete Configuration**.

Using Drag and Drop

You can use drag and drop in tables to move elements from one table to another, or to change the order of elements within one table. You can use drag and drop in the configuration of form views, where you can move elements from the **Available Fields** table to the grid. If you use drag and drop in tables and grids, a symbol indicates that you are moving an element. The row where you insert the element is indicated.

Copying Configurations

You can copy configurations delivered by SAP and adapt them to your specific needs. When you do so, the configurations are saved as customer configurations. You can also copy field labels of the configuration. Texts that belong to the configuration are copied in all languages, and, depending on the context, may unintentionally overrule other text sources or produce unnecessary duplication. Before you copy configuration texts, take into account the text determination procedure. For more information, see General Access Sequence and Introduction.

Form Views

Use

You can configure form views. In this configuration, the fields are displayed in a grid. You can change the position of fields, and add or remove fields. In the list of available fields, you can add all fields under a node.

Features

You can choose between the following configurations:

- Responsive layout (default)
- Fixed layout
  - Two panels with 8 columns each, 16 in total
  - One panel with 16 columns
  - One panel with 8 columns

More Information

- Responsive Layout
- Fixed Layout

Responsive Layout

Use

You can configure form views. In this configuration, the fields are displayed in a grid. You can change the position of fields, and add or remove fields. In the list of available fields, you can add all fields under a node.

Activities

Start the UI Configuration Tool as described in Starting the UI Configuration Tool.

Select an existing configuration and choose **Display <-> Change**.

You see all the fields that are displayed in the view. Required entry fields have a red asterisk at the end. If the displayed configuration is with a fixed layout, you will see panels. If the displayed configurations is a responsive layout, you will see groups.

Save your entries after you have made all your changes.

Creating Responsive Layouts

If you create a new configuration, it will be responsive by default.

If you use an old configuration with a fixed layout, you can create a responsive layout by choosing **Create Responsive Layout** in the toolbar. The configuration will be saved in the responsive layout and it will be used for the display in the Overview Pages with Fiori Layout.

You can switch back to the fixed layout by choosing **Display Fixed Layout** in the toolbar.

When you create a responsive layout, it will be saved in addition to the fixed layout. Layout changes made in one layout will not be applied to the other layout. When you copy a configuration, both fixed and responsive layouts will be copied.

Deleting Responsive Layouts

You can delete the responsive layout by choosing **Delete Responsive Layout** in the toolbar. When the responsive layout is deleted, you will switch back to the fixed layout.

Adding Groups

The fields in responsive layout are arranged in Groups. You can add new group by choosing the **Create Group** button in the toolbar.

You can select a group by clicking on the group caption. If you select a group and choose the **Create Group** button, the new group is added after the selected group. If no group is selected, the new group is added as the first group in the layout. You can change the group text in field properties.

Deleting Groups

To delete a group and all of the fields contained within that group, you can select the group using the group caption and choose the **Delete** icon from the toolbar.

Adding Fields

Choose **Show Available Fields**to show the field selection with all available fields. In the field selection, click the selector box in front of the field that you want to add to the view. Select the row under which you want to add the new field and choose **Add Field** from the toolbar.

The field is inserted in a new row underneath the row you chose. If you do not choose a row, the field is inserted as the last row in the panel. You can move it to the required row after.

> **i Note**
>
> You can also choose nodes with multiple fields in the field selection, even from different pages, and add all the chosen fields at once.

Removing Fields

You can select one or more fields and remove them by choosing **Remove Field** from the toolbar.

To select multiple fields, hold down CTRL and select the fields. The field or fields are inserted in the list of available fields under the corresponding nodes. The row that the field occupied is not deleted.

Changing the Position of Fields

You can move selected rows up and down by one row using the up and down arrow keys in the toolbar.

You can move selected fields between left and right groups using the left and right arrow keys in the toolbar. If you select a row in the target group and then select the field you want to move, the field will be inserted in the row underneath the row selected in the target group. If you do not select a row in the target group, the field is inserted in the same position in the target group. You can use the up and down arrow keys in the toolbar to move the field to the required row from there.

Adding Blank Rows

Select the row above where you want to insert a blank row. Choose the **Insert Line** icon. The blank row is inserted below the row you selected. If you did not choose a row, the blank row is inserted as the first row in the panel.

> **i Note**
>
> You cannot delete empty rows from a panel. You can only move fields to the new rows.

Resizing Groups

You can select a group using the group caption and choosing the **Resize Group** button to make it 8 or 16 column.

Maintaining Field Properties

When you hold down ALT and select a field in a panel, a column with the field properties can be seen on the right-hand side. Choose **Show Technical Details** at the top of the page so that the field properties are enhanced with technical information. You have the following options:

Activity	Procedure
View technical field information	Check the information in the fields **Technical Field Name**, **Original Label**, **Field Type**, **Length**, **Type**. You cannot change any of these fields. Technical field names are not normally visible to users; they only see the field labels. You can display the technical names if you require support from your hosting provider, or if the support team needs specific field information.
Change the field label	Overwrite the text in the field.
Show or hide the field label	Select or deselect **Show Label**.
Specify if users can, must, or must not enter data in the field	Select **Display Only** if the information in a field is only meant to be viewed. Select **Mandatory** if the user must enter data. Deselect both options if the user may enter data but does not have to. A field cannot be **Mandatory** and **Display Only** at the same time.
Position the field label and the corresponding data area in the panel	Change the entries in the row and column fields as required. **Example** Input field with label **Account**: **Row From**: 3 **Label Col. From**: A **Label Col. To**: C **Field Col. From**: D **Field Col. To**:F The input field and the field label are now in row 3, with the **Account** label occupying columns A through C, and the data area columns D through F. **Note** You cannot change the size of input fields.

To update the column with the changed data, click **Apply**. Then save your entries.

More Information

For more information, see [Fixed Layout](#).

Fixed Layout

Use

You can configure form views. In this configuration, the fields are displayed in a grid. You can change the position of fields, and add or remove fields. In the list of available fields, you can add all fields under a node.

Features

You can choose between the following configurations:

- Fixed layout
  - Two panels with 8 columns each, 16 in total
  - One panel with 16 columns
  - One panel with 8 columns

Activities

Start the UI Configuration Tool as described in Starting the UI Configuration Tool.

Choose an existing configuration and choose **Display <-> Change**.

You see all the fields that are displayed in the view. Required entry fields have a red asterisk at the end. If the view consists of two panels, they are separated in the middle by a red vertical line.

Save your entries after you have made all your changes.

Adding Fields

Choose **Show Available Fields**to show the field selection with all available fields. In the field selection, click the selector box in front of the field that you want to add to the view. In the left or the right panel, click the row under which you want to add the new field. Above the corresponding panel, click the **Add Field** icon.

The field is inserted in a new row underneath the row you chose. If you do not choose a row, the field is inserted as the last row in the panel. You can move it to the required row after.

> **Note**
> You can also choose nodes with multiple fields in the field selection, even from different pages, and add all the chosen fields at once.

Removing Fields

Select a field or fields, and click **Remove Field** above the appropriate panel. To select multiple fields, hold down CTRL and select the field. The field or fields are inserted in the list of available fields under the corresponding nodes. The row that the field occupied is not deleted.

Changing the Position of Fields

You can use the up and down arrow keys above the corresponding panel to move the selected rows up and down by one row.

You can use the left and right arrow keys to move selected fields between the left and right panels or groups. If you select a row beforehand in the other panel or group, the field is inserted in the row beneath it. If you did not previously select a row, the field is inserted as the last row in the panel or group. You can move it to the required row after.

Adding Blank Rows

Select the row above where you want to insert a blank row. Choose the **Insert Line** icon. The blank row is inserted below the row you selected. If you did not choose a row, the blank row is inserted as the first row in the panel.

> **Note**
>
> You cannot delete empty rows from a panel. You can only move fields to the new rows.

Adding Captions

Captions are text-only fields that serve as headings for groups of fields. Choose whether you want to insert the heading in the left or right panel. Enter the text for the heading in the field underneath and choose **Apply**. The caption appears in a new row at the very top of the panel. Use the up and down arrow keys to move the caption to the required row.

Maintaining Field Properties

When you hold down ALT and select a field in a panel, a column with the field properties can be seen on the right-hand side. Choose **Show Technical Details** at the top of the page so that the field properties are enhanced with technical information. You have the following options:

Activity	Procedure
View technical field information	Check the information in the fields **Technical Field Name**, **Original Label**, **Field Type**, **Length**, **Type**. You cannot change any of these fields. Technical field names are not normally visible to users; they only see the field labels. You can display the technical names if you require support from your hosting provider, or if the support team needs specific field information.
Change the field label	Overwrite the text in the field.
Show or hide the field label	Select or deselect **Show Label**.
Specify if users can, must, or must not enter data in the field	Select **Display Only** if the information in a field is only meant to be viewed. Select **Mandatory** if the user must enter data. Deselect both options if the user may enter data but does not have to. A field cannot be **Mandatory** and **Display Only** at the same time.
Position the field label and the corresponding data area in the panel	Change the entries in the row and column fields as required. **Example** Input field with label **Account**: **Row From**: 3 **Label Col. From**: A **Label Col. To**: C **Field Col. From**: D **Field Col. To**:F

Activity	Procedure
	The input field and the field label are now in row 3, with the **Account** label occupying columns A through C, and the data area columns D through F. **i Note** You cannot change the size of input fields.

To update the column with the changed data, click **Apply**. Then save your entries.

Changing Grid Types

You can change the configuration of the grid, to flexibly move fields between the columns. You can choose between various grid configurations. You can, as described in the table, move field labels and the associated data areas between the columns.

> **i Note**
>
> The columns in which you position a field label and its data area cannot be occupied by another field label. Before you move a field label, you must ensure that the columns into which you want to move it are empty.

Creating Separate Configurations

Views on editable overview pages can be used in display and edit mode. In general, both modes have a unique configuration, which means they have the same fields, labels, captions, and so on.

Some views support the creation of separate configurations with different fields, labels, and captions in the display view and edit view of the editable overview page.

If you want to create separate configurations for display mode and edit mode, proceed as follows:

1. Select the required application component and the required view.

   You can use the standard configuration, which is the unique configuration, or copy the default configuration to your role configuration key.

2. Select **Create Separate Config.** in the toolbar.

   You can now see **Display Mode Configuration** and **Edit Mode Configuration** directly below the toolbar.

3. Select **Display Mode Configuration** and create a configuration for this mode.

4. Select **Edit Mode Configuration** and create a configuration for this mode.

5. Save your configurations.

> **i Note**
>
> If a field label is renamed in one of these configurations, this renaming is automatically copied to the other configuration.

Switching Back to Unique Configuration

If you want to return to an editable overview page that looks the same in display mode and edit mode, you can switch back to a unique configuration. In this case, the edit mode is the leading mode, which determines the new unique configuration.

1. Select the configuration with your role configuration key.

2. Select **Switch to Unique Config.** in the toolbar.

3. Save the new unique configuration.

Tables and Hierarchies

Use

You can configure the display and appearance of the columns and rows in tables and hierarchies, for example in result lists on search pages.

Activities

Defining Tables

Once you have chosen and started the required view, you see the list of the available columns on the left, and the list of the displayed columns on the right.

1. In the **Visible Rows Before Paging** field, enter how many rows are displayed in the result list before the paging starts.

   ### Example
   In the result list 150 entries are available. If you enter `100` in the field, paging starts after the defined number of 100 rows. This means that the 150 results are displayed on two pages: 100 results on the first page and 50 results on the second page.

2. In the **Visible Rows Before Scrolling** field, enter how many rows are displayed in the result list before a vertical scroll bar is displayed.

   ### Example
   In the result list 150 entries are available. If you enter `10` in the field, 10 rows are displayed in the result list and scrolling starts after the defined number of 10 rows. This means that a scroll bar is displayed and that you can scroll from the first to the hundredth entry on the first page. For the 101st entry, you enter the second page.

3. You can move fields back and forth between the list of displayed fields and the list of available fields, by using the LEFT ARROW and RIGHT ARROW keys, or drag and drop. You can use drag and drop only on the row selector box, not on the whole row.

   All the fields in the list of displayed fields are available to users, in the result list on the search page.

4. If you want to rename a column, overwrite the text in the field **Column Title** in the list of displayed fields.

5. In the **Column Width** field, enter a whole number between 1 and 100.

   This value specifies how much space the column takes up in the result list on the search page. Columns with no values are equally distributed across the remaining total column width. The following units of measure are supported: `%, px, em, in, cm, mm, pt, pc`.

   ### Example
   Column A: 10 %

   Column B: 10 %

   Columns C and D: No value

Columns C and D occupy 40% of the available column width each.

i Note

Whether a horizontal scroll bar is displayed or not, is based on the following prerequisites:

- If `%` is used to define the column width, no scroll bar is displayed. If the total defined column width exceeds 100 percent, the excess part is cut off.
- If one of the units `%, px, em, in, cm, mm, pt, pc` is used, a horizontal scroll bar is automatically displayed. If `px` is entered in a column, the width of the remaining columns is determined by the ABAP Dictionary.
- If nothing is entered, the percent approach is used. The width is determined by the ABAP Dictionary and converted to percent.
- The mixed use of units, for example, `%` and `px`, is not allowed.

6. In the dropdown box in the **Horizontal Alignment** field, choose the alignment of the text in the field.

   You can choose between **Automatic**, **Left**, **Right**, and **Center**. **Automatic** means that the system determines the appropriate alignment based on the data type. With **Left**, **Right**, and **Center** the automatic alignment is deactivated.

7. If you want to display a column in the table, deselect the **Hidden** checkbox.

   i Note

   If the checkbox **Hidden** is selected, the corresponding column is hidden in the table. However, the user can display hidden columns via personalization of the corresponding table, in the CRM WebClient UI. With **Reset to Default** in the personalization of the table in the CRM WebClient UI, the user can reset the original configuration.

8. Save your entries.

Creating Separate Configurations

Tables and hierarchies can be used in display and edit mode. The way to create separate configurations is the same as for form views.

More Information

[Form Views]

Search Pages

Use

You can configure the search criteria and the result list on search pages.

Features

- Configuration of search criteria

  On the search page, users can search for business objects. To do so, they can choose specific search criteria and search operators, and enter a search term if required. They can:

- Define standard search criteria
- Change search criteria labels
- Change default search operators

- Configuration of the result list

  Users see the result of their search in the result list in the lower half of the search page. They can:

  - Determine which columns are displayed in the result list, and change the column titles
  - Determine which columns are available in the personalization of WebClient UI
  - Specify the order in which the columns appear
  - Set the width and alignment of the individual columns

Activities

Configuring Search Criteria

Once you have chosen and started the required view, you see a list of the available search criteria on the left, and a list of the selected search criteria on the right.

1. Move the search criteria back and forth between the two lists with the LEFT ARROW and RIGHT ARROW keys, or drag and drop. You can use drag and drop only on the row selector box, not on the whole row.

   All the criteria in the list of selected search criteria are available to users, in the corresponding dropdown box on the search page.

2. If you want to rename a search criterion, overwrite the text in the **Label** column in the list of selected search criteria. Your changes are only effective for the selected configuration.

   The field label is the text shown to the user on the search page, either directly in the default search criteria or in the search criteria dropdown box.

3. Select the **Displayed** indicator if the search criterion should be displayed by default.

   In this case the search criterion appears in the default search criteria, which are available to the user on the search page.

4. In the dropdown box in the **Default Operator** column, choose the required search operator.

   Search operators define the process that a user wants to perform to find one or more business objects. Search operators include terms such as **is**, **is empty**, or **is not**.

   The default search operators are preset, which means they are available as soon as the user starts the search page.

5. Select the **Use in Central Search** indicator if the search criterion should be displayed in the central search.

6. Click **Show Technical Details** if you want to display the technical names of all search criteria.

7. Save your entries.

Configuring the Result List

The configuration of the result list is based on the configuration of tables and hierarchies.

More Information

Overview Pages

Use

The editable overview page has been introduced in the WebClient UI Framework for all business objects with overview pages. This concept includes most of the assignment blocks in which information is displayed in form views and tables. The user can now directly change the information on the overview page by clicking **Edit**. In general, editable overview pages look the same in display and edit mode.

You can determine which assignment blocks should be displayed on an overview page. You can define the sequence of the displayed assignment blocks, determine specific characteristics, and change their titles.

Activities

Select the required application component and then expand the **Views** node under **Browser Component Structure**. Choose the required view and click **Configuration**. In the configuration selection, choose the required configuration.

On the right, you see the **Displayed Assignment Blocks** area and on the left, the **Available Assignment Blocks**.

Displaying Assignment Blocks

If an assignment block should be displayed on the overview page at runtime, select it from the list of available assignment blocks and move it to the list of displayed assignment blocks, using the RIGHT ARROW key, or drag and drop.

If you want to remove an assignment block from the overview page, move it from the list of displayed assignment blocks to the list of available assignment blocks, using the LEFT ARROW key, or drag and drop.

Defining and Positioning Assignment Blocks

If you want to move an assignment block up or down within the list of displayed assignment blocks, you can use **Up** or **Down**, or drag and drop in the list of displayed assignment blocks.

You can choose between the load options **Direct**, **Lazy**, or **Hidden**.

- **Direct** means that the assignment block is directly visible and open.
- **Lazy** means that the assignment block is directly visible and collapsed, but you can expand it.
- **Hidden** means that the assignment block is not displayed. However, you can show it by using the personalization option.

Changing Assignment Block Titles

You can change the titles of displayed assignment blocks. Click **Display <-> Change** and enter the required title in the **Title** column. Save your entries.

Showing Technical Details

You can show the **View ID** for the available assignment blocks and the displayed assignment blocks.

Example

You want to add and regroup fields, or add a new caption to the service order details of your service order editable overview page, but only in edit mode. You select application component `BT116H_SRVO` and the `BT116H_SRVO/Details` view, and create two separate configurations for display and edit mode.

Overview Pages with Tile Layout

Use

You can configure overview pages with single column layout or with tile layout.

Overview pages with single column layout consist of one single column. At the top of the page you can see a header block that contains the most important information about the business object. Below the header block you find an undefined number of assignment blocks with information related to the business object.

Overview pages with tile layout are composed of several columns, providing several tiles on one page where header data or information related to the business object is displayed.

i Note

If you switch between both layouts, the information about which views were assigned to the overview page is lost.

If you switch from single column layout to tile layout, you need to ensure that the views that are used on the overview page can be displayed in tiles, because tiles are smaller than a normal page.

Prerequisites

To configure overview pages with tile layout, you need to register these overview pages in Customizing for **UI Framework** under **UI Framework Definition** > **Register Overview Pages for Tile Layout**. If you have registered an overview page, you can switch from single page layout to tile layout in the UI configuration.

You define the layout of overview pages with tiles in Customizing for **UI Framework** under **UI Framework Definition** > **Fact Sheet** > **Maintain Fact Sheet**.

Features

Defining the Tile Layout

For the tile layout definition of overview pages, the tile layout of fact sheets is reused. The following standard layout types are available for pages with tiles:

- 2 x 3 layout, that is, two columns and three rows (six tiles)
- 2 x 2 layout, that is, two columns and two rows (four tiles)
- T-shape (three tiles)

Configuring the Tile Layout

The configuration of overview pages with tile layout in the UI configuration tool has been synchronized with the personalization dialog of standard overview pages. Every tile is displayed as a separate list, and every row in each list is an assignment block represented by a tab link in this tile. The layout of the UI configuration represents the tile layout of the overview page in the WebClient UI.

You can select single assignment blocks in the list of available assignment blocks and add them to a tile, either by clicking the arrow button of the tile or by using drag and drop. You can remove assignment blocks from a tile, either by deleting them in the tile or by using drag and drop to move them back to the list of available assignment blocks. You can move single assignment blocks from one tile to the other either by drag and drop or by selecting them and clicking the arrow button of the target tile.

You can determine the order in which assignment blocks are displayed in a tile by using **Up** and **Down** or by using drag and drop.

You can directly see five assignment blocks in a tile. If more than five assignment blocks are assigned to a tile, you can use a scroll bar to access every assignment block in this tile.

You can assign the attribute “Hidden” to assignment blocks in a tile. If you have selected **hidden** in the UI configuration, these assignment blocks are initially not displayed on the overview page in the WebClient UI. They are available in the personalization dialog of the overview page. To display them on the overview page in the WebClient UI, you can move them from the list of available assignment blocks to the list of displayed assignment blocks.

Design

There are two tile layout designs available:

- Tabs

  You can insert an assignment block within a tile or move it between tiles by using drag and drop or by choosing the **Move** pushbutton on the left of the corresponding tile. You can personalize each tile. The personalization is based on the selected tile layout. For example, the same tile can have a different personalization when different layout types are selected. You can display hidden tabs by choosing the icon in the upper right corner of a tab. This is the standard design.

- Links

  You can enable the design with links in Customizing for **UI Framework** under **UI Framework Definition** > **Technical Role Definition** > **Define Parameters**. To make this design available for a business role, define parameter `TILED_LAYOUT_OLD` with parameter value `TRUE`. This parameter is assigned via parameter profile to the function profile `PARAMETERS`. You need to assign function profile `PARAMETERS` to your business role in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**.

More Information

[Personalization of Overview Pages with Tile Layout]

[Table Personalization]

Button Configuration

The overview page button configuration is used to create, modify, and delete buttons from the overview page.

Use

You can use button configuration as a way to create, modify, and delete buttons from the overview page.

Features

While rendering the overview page, the method `IF_BSP_WD_TOOLBAR_CALLBACK~GET_BUTTONS` from the `CL_BSP_WD_WINDOW` class checks to see if the view controller is implemented. The view controller is implemented when you have defined the CFG_BUTTONS _MODIFY method, which is inherited from the CL_BSP_WD_VIEW_CONTROLLER class. If so, the buttons defined in the BSP configuration are modified according to the logic that you introduced.

You can maintain the following fields for each button:

Field	Description
ID	The ID of the button to be configured.
Text	Use this field to define the text to be displayed on the button. • If you have specified a Type, you can leave this field blank. • You can specify the alias of an OTR text to render the button text. For example, you can define Text as `OTR(CRM_BSP_UI_FRAME_BRC/AUTHORIZATION_ROLES)`.
Tooltip	Use this field to define the tooltip for the button. • If you have specified a Type, you can leave this field blank. • You can specify the alias of an OTR text to render the tooltip text.
Icon Sourse	Use this field to specify the URL of the icon to be displayed. For example, `/SAP/BC/BSP/SAP/thtmlb_styles/sap_skins/serenity/images/edit.gif`.
Type	Use this field to define the Type for a predefined standard button, for example, SAVE or CANCEL. If you have defined a Type, the Text, Tooltip, and Icon Source fields are automatically populated.
Enabled	Use this checkbox to enable or disable the button.
Backend Event	Use this field to defines the name of the backend event, that is triggered when you choose the button.
Frontend Event	Use this field to define the JavaScript event handler, that is executed when you choose the button.

Activities

- To implement button configuration, redefine the method CFG_BUTTONS _MODIFY in the view controller class of the overview page.
- Using the **Button Configuration** assignment block, you specify the maximum number of visible buttons and you can view a table showing the buttons that have been defined.
- You can modify the static configuration depending on runtime information while rendering the overview page. The method CFG_BUTTONS _MODIFY is called only when there are buttons defined in the **Button Configuration** of the overview page in the BSP Workbench.
- When the method CL_BSP_WD_WINDOW-> IF_BSP_WD_TOOLBAR_CALLBACK~GET_BUTTONS is executed, the call to the CFG_BUTTONS _MODIFY method is performed, executing the desired logic for the defined buttons.
  - CFG_BUTTONS _MODIFY: This method has the changing parameter CT_BUTTONS of type CRMT_THTMLB_BUTTON_EXT_T, which is the tabular version of the button configuration details. It contains all information required to render the toolbar. It is possible for you to modify the toolbar data to meet the requirements of the application. This method can be used to activate and deactivate buttons.

- `CFG_NUM_VISIBLE_BUTTONS_MODIFY`: This method has one returning parameter `RV_MAX_BUTTON_NUMBER` of type `I`. The returned value overrides the configured value for **Maximum Number of Buttons Displayed** .
- `CFG_BUTTONS_IS_ACTIVE`: Redefining this method allows you to disable the execution of the logic for button configuration. The default buttons for the toolbar are displayed.

How to Distribute Button Configurations Internally

Consider that the button configuration has different visibility requirements than the implementation of the overview page toolbar in the view controller's `IF_BSP_WD_TOOLBAR_CALLBACK~GET_BUTTONS` function. This means that a configuration must be provided for each config key after switching an overview page to button configuration. If you do not switch the overview page to button configuration, the toolbar will not display any buttons. The `WCF_DISTR_BTN_CONFIG_INTERNAL` report is provided so that you can distribute the button configuration for a specified view without modifying the view configuration.

Overview Pages with Fiori Layout

Use

The dynamic header and layout transformation is introduced in WebClient UI Objects with overview pages. This concept allows configuration of header with facets that can be displayed for the user in display mode or partial edit mode. The overview pages are transformed by default to display flexible responsive layouts.

You can also create a responsive layout in the Configuration Tool.

Fields in header and assignment blocks can be displayed in semantic colors based on business requirements.

Features

- **Header:** Header is displayed below the title in the overview pages. It contains key information about the business object and provides the user with the necessary context.
- **Layout transformation:** The layouts of views are transformed by default to the column design of the object page and to be responsive. The labels are top-aligned, displayed on top of the fields. The captions and empty lines define sections with groups of fields that form a column and can float in the page making it a responsive layout. The floorplans are responsive to screen sizes. Medium screens have a maximum of 2 columns, large screens have a maximum of 3 columns and extra-large screens have a maximum of 6 columns.
- **Semantic colors:** Semantic colors can be used to visualize a meaning or value state in the business context. You can use semantic colors for header content, views, or tables.

Activities

Header Configuration

1. Navigate to the required overview page and choose the **Configure Page** button from the toolbar.
2. Change to edit mode using the **Edit** button in the header.
3. In the **Header Configuration** section, choose the **Add Facet** button and choose a facet type. A corresponding template will be added in the header configuration.
4. Replace the text `ABC/STRUCT.ATRIBUTE1` with `ContextNode/AttributeName`. The binding works only with context nodes that belong to the overview page itself.
5. Add all facets for different attributes as required.

6. Save and close.

Note

Key Value, Form, Image, Progress Indicator and Micro Charts facets are supported. In the Form facet, you can display up to five label-value pairs.

For the Image facet, you need to implement the p-getter of the attribute as below:

Example

Sample Code

```
CASE iv_property.
        WHEN if_bsp_wd_model_setter_getter=>fp_fieldtype.
        rv_value = cl_bsp_dlc_view_descriptor=>field_type_image.
ENDCASE.
```

For the Progress Indicator Facet, you need to implement the p-getter of the attribute as below:

Example

Sample Code

```
CASE iv_property.
        WHEN if_bsp_wd_model_setter_getter=>fp_fieldtype.
        rv_value = cl_bsp_dlc_view_descriptor=>field_type_progressindicator.
ENDCASE.
```

The following types of micro charts can be used for the Micro Chart Facet:

- Column bar chart
- Stacked bar micro chart
- Delta micro chart
- Comparison micro chart

You can change the values in the XML template to your own context node attribute or `ALIAS`. All the bars and columns for the chart are defined as data point elements.

Example

Sample Code

```
<facet>
    <microchart>
    <property name="title" value="ALIAS/LABEL1"  source=" OTR|Binding|Label|String|PGetter"/>
    <property name="subTitle" value="ALIAS/LABEL2"  source=" OTR|Binding|Label|String|PGetter"/>
    <property name="rightBottomLabel" value="//ABC/STRUCT.ATTRIBUTE1"  source=" OTR|Binding|Labe
```

```
      <property name="charttype" value="ColumnMicroChart|StackedBarMicroChart|DeltaMicroChart|Comp
      <data>
      <dataPoint>
      <property name="title" value="ALIAS/LABEL3" source="OTR|Binding|Label|String|PGetter"/>
      <property name="value" value="//ABC/STRUCT.ATTRIBUTE2" source="OTR|Binding|Label|String|PGet
      <property name="displayValue" value="//ABC/STRUCT.ATTRIBUTE2" source="OTR|Binding|Label|Stri
      <property name="color" value="//ABC/STRUCT.ATTRIBUTE2" source="OTR|Binding|Label|String|PGet
      </dataPoint>
      <dataPoint>
      <property name="title" value="ALIAS/LABEL4" source="OTR|Binding|Label|String|PGetter"/>
      <property name="value" value="//ABC/STRUCT.ATTRIBUTE3" source="OTR|Binding|Label|String|PGet
      <property name="displayValue" value="//ABC/STRUCT.ATTRIBUTE3" source="OTR|Binding|Label|Stri
      <property name="color" value="//ABC/STRUCT.ATTRIBUTE3" source="OTR|Binding|Label|String|PGet
      </dataPoint>
      </data>
      </microchart>
   </facet>
```

The following attributes are supported for the chart properties:

Property	Description
`charttype`	The following type of charts are supported: • ColumnBarChart • StackedBarMicroChart • DeltaMicroChart • ComparisonMicroChart
`title`	Title of the chart
`subtitle`	Subtitle of the chart
`footer`	Footer for the chart

`rightBottomLabel`
`leftBottomLabel`
`leftTopLabel`
`rightTopLabel`
	Top and bottom labels are supported for the column chart only

`allowColumnLabels`	Supported only for ColumnBarChart. Decides whether to show the column labels.

The following attributes are supported for the data point properties:

DataPoint Property	Description
`title`	Title of the data point
`value`	Value of the data point
`displayValue`	Display value of the data point

DataPoint Property	Description
`color`	The color of the data point. For more information about color values, see https://sapui5nightly.int.sap.eu2.hana.ondemand.com/#/api/sap.m.ValueCSSColor

The value of the source attribute can be one of the following:

Source	Description	Example

`OTR`	The value is obtained by resolving the `OTR`	`<property name="tit`
		`value="WCF_EPM_DEMO_`
		`source="OTR"/>`

`Binding`	The value is obtained by resolving the binding of the context node attribute	`<property name="valu`
		`value="//ABC/STRUCT`
		`source="Binding"/>`

`Label`	The value is obtained by resolving the label of the context node attribute	`<property name="dis`
		`value="//ABC/STRUCT`
		`source="Label"/>`

`p-getter`	The value is obtained by resolving the p-getter of the context node attribute. This is supported only for the color attribute. You can use the property `if_bsp_wd_model_setter_getter=>fp_microchart_color` to set the color in p-getter.
	Example
	p-getter implementation code
	Sample Code
	`CASE iv_property.`
	`WHEN if_bsp_wd_model_setter_getter=>fp_microchart_color.`
	`rv_value = cl_thtmlb_textview=>c_semanticcolor_positive.`
	`ENDCASE.`
		`<property name="colo`
		`value="//ABC/STRUCT`
		`source="PGetter"/>`

`String`	The value is passed as a constant string.	`<property name="colo`
		`source="String">`
		`<property name="tit`
		`Overview" source="S`

The following properties are supported for each type of chartbar.

ColumnMicroChartData	StackedBarMicroChartBar	DeltaMicroChart	ComparsionMicroChartData
`color`	`valuecolor`	`color`	`color`
`displayvalue`	`displayvalue`	`displayvalue1`	`displayvalue`
`label`		`title1`	`title`
`value`	`value`	`value1`	`value`

If nothing is configured in the header, the header is not displayed.

Semantic Color

To add semantic colors to the fields in the header or other assignment blocks, generate the p-getter for the attribute in the context node and assign the semantic color values as required in the business context.

Example

Sample Code

```
CASE iv_property.
        WHEN if_bsp_wd_model_setter_getter=>fp_semantic_color.
                If delivery_date > current_date.
                rv_value = cl_thtmlb_textview=>c_semanticcolor_positive.
                Else.
                rv_value = cl_thtmlb_textview=>c_semanticcolor_negative.
                Endif.
ENDCASE.
```

The following colors are supported:

- c_semanticcolor_positive
- c_semanticcolor_negative
- c_semanticcolor_critical
- c_semanticcolor_neutral
- c_semanticcolor_information

Layout Transformation

In Fiori launchpad integrated mode, the header and runtime transformation are available by default and this is the only mode available.

In standalone (Compatibility) mode, you can assign the parameter `FIORI_LAYOUT` to the business role to disable the feature and revert to fixed layouts, where available. The responsive layout is displayed when there is no fixed layout.

Home Page and Work Center Pages

Use

You can determine which content blocks should be displayed on the home page and on the work center pages. You can define specific characteristics for the displayed content blocks, determine their position on the page, and change their titles.

You can assign your own views to existing home pages and work center pages without enhancing the corresponding UI components, as long as they are suitable.

Integration

Each SAP business role has its own report page, which the user can launch in the navigation bar. The configuration of reports depends on the SAP business role that you select. You can configure report pages in the same way as the home page and work centers in an SAP business role.

Features

Each SAP business role has one home page, which is automatically launched when the user logs on. The number of work centers depends on the respective business role. This also applies to the appearance and content of the individual work centers. This means that the configuration of the home page and work center pages depends on the respective business role.

Activities

Select the required application component and then expand the **Views** node under **Browser Component Structure**. Choose the required view and click **Configuration**. In the configuration selection, choose the required configuration.

At the top of the home page and the work center page you see the available content blocks. The **Left Visible Column** and **Right Visible Column** are displayed underneath for both pages.

Displaying Content Blocks

If a content block should be displayed on the home page or on the work center page, select it from the list of available content blocks. Then move it to the appropriate column of visible content blocks, using the DOWN ARROW key.

If you want to remove a content block from the home page or work center page, select it in the left or right column of visible content blocks. Then and move it to the list of available content blocks, using the UP ARROW key.

Defining and Positioning Content Blocks

If you want to move a content block from one column of visible content blocks to the other, use the arrow keys between the two columns.

If you want to move a content block up or down within a column, use **Up** and **Down** at the top of the column of visible content blocks.

If a content block in the left column should occupy the entire width of the page, including the right column, select **Whole Width**.

If users should not see a content block directly on a page, but should be able to display the content block via personalization, select **Hidden**.

Note

If you select the **Hidden** checkbox, the corresponding content block is not directly displayed on the home page or work center page. To display the hidden content block on the page, users can select it in the dropdown list box of any content block, when they are in the personalization mode.

Changing Content Block Titles

You can change the titles of the content blocks in the **Left Visible Column** or **Right Visible Column**. Click the **Display <-> Change** icon and enter the required title in the **Title** column. Save your entries.

Showing Technical Details

You can show the **View ID** for the available content blocks and visible content blocks.

Adding Views to Home Pages and Work Center Pages

You can add your own views to existing home pages and work center pages without enhancing the corresponding UI components. You need to register these views in Customizing for **UI Framework**, by choosing **UI Framework Definition** > **Define Additional Views for Home Pages and Work Center Pages**. After you have registered your views, they are available in the UI configuration of the corresponding UI components.

Guided Activities

Use

A guided activity is a task, such as creating a service order, that a user carries out through a sequence of logical steps on a guided activity page.

You can use guided activities to align business processes, for example.

Features

- You can configure an existing guided activity page in the UI Component Workbench. You have the following options:
  - Create guided activity steps (steps)

    For each step, you can create a label.
  - Create substeps

    You can divide complex steps into smaller substeps.
  - Create explanatory texts

    For each step, you can define an explanatory text, providing further information for the business user.
  - Define the step sequence
  - Create multiple UI configurations for guided activity pages and step views
- You can create a guided activity:
  - You can create a guided activity page in the UI Component Workbench.
  - You can control the navigation between the steps.
  - You can enable the sharing of objects from the business object layer (BOL) between the steps.
  - The WebClient UI framework provides the **Previous**, **Next**, and **Finish** buttons for guided activities. You can provide application-specific buttons through the application controller.
- The WebClient UI framework provides a graphical road map on guided activity pages that allows the business users to navigate from step to step. Substeps can be expanded and collapsed. Moreover, the road map provides an overview for the business users as to which step they are currently working on and how many steps are still required.

Example

An example for a guided activity has been provided in UI component `UIF_TBUI_SAMPLE`.

Configurable Data Retrieval

Use

You can use configurable data retrieval to limit the data retrieved in assignment blocks, based on selection parameters defined in the UI configuration tool.

Integration

Configurable data retrieval can be enabled for the following assignment blocks. For assignment blocks, the **Data Retrieval Parameters** area is available in the UI configuration tool. For more information about use cases, see the application specific documentation.

Features

- For assignment blocks that have been set up for configurable data retrieval, you can define which data records are selected and displayed by defining selection parameters as part of the user interface (UI) configuration.

  As a result, only specific data is displayed in the assignment block.

- The system performance can be improved because of a reduced amount of data retrieved from the database. Whether the system performance is improved or not depends on how the configurable data retrieval has been implemented for a certain assignment block.
- You can change the selection parameters at any time, altering the data selection presented to the business user.
- You can set selection parameters for each view usage (business role), because the selection parameters are specified as part of the view configuration.
- Report **Views with Configurable Data Retrieval Option** (`BSP_DLC_CONF_DAT_RETRIEV_USAGE`) provides you with a list of all views that support configurable data retrieval.
- You can use a Business Add-In to define additional data retrieval parameters.

Translation

Use

You can use a dedicated transaction to translate field labels and captions that have been changed.

Example

You have changed field labels and captions on the opportunity overview page (application component `BT111H_OPPT`) and on the opportunity search page (application component `BT111S_OPPT`), in the UI configuration in transaction `BSP_WD_CMPWB`. You want to translate the changes that you have made in the views.

Activities

Finding Out the Context ID

Before you start the translation, you need to find out the context ID in the corresponding configuration.

1. Start the BSP Component Workbench.
2. Select the UI component and the view that you want to translate.
3. Click the Configuration tab, and click Choose Configuration.

   In the Configuration Selection, you can choose a context ID.

i Note

Alternatively, you can find out the context ID in the Technical Data dialog of the WebClient UI. You can start the Technical Data dialog with the F2 key. You can find the context ID in the technical information about the configuration at runtime.

Translating by Using the Context ID

1. Start transaction SE63.

   You see the page Initial Screen: Standard Translation Environment.
2. Enter BSP2 in the transaction code field.

   You see the page ABAP Short Texts: BSP Layout Texts (Client-Dependent).
3. In the field Object Name, enter the context ID, which you have found out beforehand.
4. Choose the source language and target language, English as the source language and German as the target language, for example.
5. Click Edit to translate the object.
6. Save your entries.

Translating by Using the Collection

You can also use the collection for translation purposes. The collection technically corresponds to the package, which you can find out in the Technical Data dialog of the CRM WebClient UI. You can start the Technical Data dialog with the F2 key. The package name is available at the technical information about the UI component.

1. Start transaction SE63.

   You see the page Initial Screen: Standard Translation Environment.
2. Enter BSP2 in the transaction code field.

   You see the page ABAP Short Texts: BSP Layout Texts (Client-Dependent).
3. In the field Object Name, press F4 to start the input help.
4. In the field Collection, enter the collection/package that you found out beforehand.

   You see a list with all context IDs that belong to this collection. Every configuration has its own context ID.
5. Click the RIGHT ARROW key with the quick info text Next Variant to see the corresponding view names.
6. Select the view that you want to translate.
7. Choose the source language and the target language.
8. Click Edit to translate the object.
9. Save your entries.

More Information

Introduction

Starting the UI Configuration Tool

Displaying Technical Information

Translate from Configuration Tool

Use

You can use the Translate from Configuration Tool to translate texts from the text repository.

In the UI Configuration Tool, when edit mode is enabled, you can use the **Translate** button to navigate to the view that you want to translate. Here, the texts that you can edit from the text repository are displayed.

Activities

1. From the UI Configuration Tool, you can access the Translate from Configuration Tool. You can access the UI Configuration Tool from either the UI Component Workbench or the CRM application. For information on accessing the UI Configuration Tool, see Starting the UI Configuration Tool.

2. In edit mode, choose the **Translate** button to see the view that you want to translate. Note that this button may be displayed as a menu option under **More**.

3. Choose the languages where you would like to edit text from the **Languages** drop down list. You can select multiple languages to be displayed for editing at the same time.

4. Enter the revised texts in the appropriate language where you would like to make changes.

> **Note**
>
> You can only use the tool for previously edited texts from the text repository. You cannot edit design layer or DDIC texts through this tool.

Authorization Object

Use

You can use authorization objects to define authorizations concerning the UI configuration.

Features

You can use the authorization object `CRM_CONFIG` to restrict authorization for UI configuration, which is based on component name, BSP view name, UI object type, and role configuration key. You can use activity `02 CHANGE` to create, modify, and delete the authorizations with regard to the UI configuration.

You can use the authorization object `CRMCONFMOD` to start the UI configuration directly from an application in the WebClient UI.

Activities

1. Set up the authorization for your role in transaction Role Maintenance (PFCG).
2. In the Menu tab page, select Transaction.
3. Assign transaction BSP_WD_CMPWB for the BSP WD Component Workbench.

   Note

   To use the BSP WD Component Workbench, you need to assign the authorization objects S_DEVELOP and S_DATASET.

   To enable transport handling, you need to assign the authorization object S_CTS_ADMI.

4. Set up at least display authorization for S_DEVELOP and S_DATASET.
5. In the Authorizations tab page, assign the authorization object for UI configuration CRM_CONFIG to the authorizations in your role.
6. Maintain the values for the following authorization fields:
   - UI Object Type (OBJECT_TYP)
   - Role Configuration Key (ROLE_KEY)
   - Component Name (UI_COMP)
   - BSP View Name (VIEWNAME)
   - Activity (ACTVT)

     Note

     Enter activity type 02 CHANGE, which provides the options Modify, Create, and Delete.

7. Generate the authorization profile and assign the users.

Mash-Ups

Use

You can use mash-ups to combine data from existing applications, and third party data sources and services. You can use mash-ups to retrieve information from different data sources, such as widgets, RSS feeds, or Web services, in an application in the WebClient UI.

Prerequisites

To start the UI configuration of an application, you need to activate the configuration mode in the general settings, on the central personalization page of the WebClient UI.

To enable or disable the UI configuration of mash-ups via the icon Show Configurable Areas in a business role, you need to assign parameter ENABLE_MU_CONF with parameter value ON or OFF via function profile PARAMETERS to your business role. You do this in Customizing for UI Framework under Business Roles > Define Business Role.

To make the necessary Customizing settings for mash-ups, choose Customizing for UI Framework under UI Framework Definition > Global Attribute Tags.

Features

Mash-ups that you add to an application are made available as new assignment blocks on the overview page of the corresponding application in the WebClient UI.

The following mash-up types are available:

- Widget mash-up

  You can enhance an overview page with an assignment block that contains a widget that is, for example, capable of displaying addresses on a map.

- URL mash-up

  You can enhance an overview page with an assignment block that contains a URL that is, for example, capable of searching other software products for all kinds of information.

More Information

Creating Mash-Ups

Creating Mash-Ups

Use

You can create new mash-ups by using the mash-up wizard, which is available in the UI configuration tool. After you have created a new mash-up, it is available in the UI configuration and you can display it on any overview page in the WebClient UI. You can configure the newly created mash-up in the UI configuration, if you have configuration rights.

Procedure

Creating a Mash-Up

To create a new mash-up, proceed as follows:

1. Select the overview page on which you want to create a new mash-up.

2. Click the icon **Configure Page** in the toolbar of the overview page.

3. Choose **Embedded Views** > **Show All** in the **Configurations** block.

4. Click **Add Mash-Up** in the **Embedded Views** block.

   Note

   You can use the following authorization objects by the current business role to control rights for creating and using mash-ups:

   - Authorization object CREATE_MU is used to give rights to create mash-ups.

   - Authorization object TAG_ATB is used to give rights to expose the name and value of a tag attribute that may appear on the URL of the mash-up. If the authorization object check fails on a tag attribute, the name of the attribute and its value at runtime do not appear in the list of available tag attributes at configuration time. Such authorization is given at business role level.

5. Enter the following data in the **Embed Mash-Up** dialog box.
   - Mash-up title
   - Mash-up description

     i Note

     The mash-up ID is automatically created. It corresponds to the view ID, which is needed for the UI configuration of the overview page.

   - Mash-up height

     Optionally, you can explicitly define the height of the mash-up. Otherwise the default height is either the height defined by the source code of the mash-up itself, or a predefined value (300px).

   - Source code

     You can enter the source code, which can either be a URL or a script that defines a widget. To retrieve application data at runtime, you can use attribute tags that you have defined before in Customizing for **UI Framework** under **UI Framework Definition** > **Global Attribute Tags**. Put the cursor in the source code where the attribute tag should be inserted, or select a substring that should be replaced in the source code. Select an attribute tag in the list at the right-hand side of the **Embed Mash-Up** dialog box. Click **Label** or **Value** to insert the required information in the source code.

   - UI object type

     You can filter the tag attribute lists by specifying a UI object type. A search help is provided for this entry.

   - Save your entries, and click **Test Mash-Up** if you want to preview your mash-up.
6. Verify that the new mash-up has been added to the **Embedded Views** list.
7. Return to the UI configuration of the overview page by clicking **Back**.

   i Note

   The mash-up is now available in the list of available assignment blocks in the **View** block. To verify the mash-up ID, click **Show Technical Details**.

8. To display the mash-up on the overview page, move the mash-up from the list of available assignment blocks to the list of displayed assignment blocks.

Modifying a Mash-Up

To modify an existing mash-up, proceed as follows:

1. Select the overview page on which the existing mash-up is available.
2. Click the icon **Configure Page** in the toolbar of the overview page.
3. Choose **Embedded Views** > **Show All** in the **Configurations** block.
4. In the **Embedded Views** block, click **Edit Hierarchy**.
5. Select the mash-up that you want to modify.
6. Click **Edit Details** and modify the mash-up definition in the **Embed Mash-Up** dialog box.
7. Save your entries.

Deleting a Mash-Up

To delete an existing mash-up, proceed as described above at "Modifying a Mash-Up". In the **Embedded Views** block, click **Edit Hierarchy**. Select the mash-up that you want to delete. Click **Remove** to delete the mash-up, and save your entries.

Integration With Custom Fields App

Use

Business Context available in the **Custom Fields** app can be used to extend corresponding WebClient UI application. Key users can add the custom fields to the application views in the WebClient UI configuration tool.

If you add custom fields of type **Association to Business Object**, the fields are displayed as hyperlinks. The link navigates to the business object with the default implementation when the **Semantic Object** is the same as the **Business Object ID** and the **Action** is `maintain`.

Example

Business Object ID: `ZZ_TEAM1`

Semantic Object: `ZZ_TEAM1`

Action: `maintain`

If the **Semantic Object** isn't same as the **Business Object ID**, you can redefine method `HANDLE_ASCBO_EVENT` in the implementation class of your view and adjust the intent values as needed.

In SAP Fiori Launchpad integrated mode, the intent validation is performed for navigation to the business object. If the user cannot navigate due to invalid intent (navigation is not supported), the business object is displayed as text instead of hyperlink.

If you add custom fields of type **Code List based on CDS View** with multiple key fields, all the key fields are added in the context node. You can add one or more of these key fields in the displayed configuration. The F4 dialogue displays all the key fields of the CDS view irrespective of what is added on the UI. Upon selecting an entry from the F4 dialog, all the key fields are updated in the backend to identify a unique record whether or not they are disabled or displayed on the UI.

Features

Once you've published the custom fields in the **Custom Fields** app, the fields are added to the corresponding contexts in the application. The administrator can add fields in the configuration from the list of available fields.

When the context node where the custom fields are added is different from the context node linked to the table configuration, the custom fields added in the **Custom Fields** app aren't available in the table configuration.

The context node with custom fields which is linked to a table or tree view must be set as extensibility related by adding interface `IF_WCF_S4_EXT_CONTEXT_NODE` to the implementation class of the context node. On setting the node as extensibility related, the custom fields are available in the list of the available fields.

The custom fields can also be added to the tree views.

Caution

The management of data in an extension scenario deviates from the management of data in the standard scenarios. You are responsible for ensuring that the data used in an extension scenario is managed in accordance with any applicable legal requirements or business needs, such as data protection legislation or data life-cycle requirements. Please note that the extensibility framework is currently not integrated in the privacy-by-default functionality. Therefore, the extensibility framework is not to be used for the processing of personal data if this processing falls under any applicable data protection legislation.

More Information

- For more information about business contexts, see the extensibility information provided in the documentation for each application.
- For more information, see Custom Fields App and Custom Logic App
- For more information about enhancements, see Create Enhancements

Integration With SAPUI5 Apps via Embedding

Use

WebClient UI offers a reusable component to embed SAPUI5. The component can be consumed by any WebClient UI application in SAP S/4HANA.

Features

The integration of WebClient UI with SAPUI5 supports scenarios with SAPUI5 apps embedded in the WebClient UI work content area in the standalone mode. You can launch embedded SAPUI5 applications via links in the main menu. You can map external navigation from the SAPUI5 App to the Framework. The WebClient UI apps integrate the SAPUI5 application breadcrumbs and navigation history in the header. Only the last state of the SAPUI5 app is recorded.

Prerequisites

The SAPUI5 app must have a component ID and intent defined.

Procedure

You can create links in the navigation bar as follows.

1. Create a parameter handler class for the navigation link. You need this to pass some parameters like the SAPUI5 component ID, the semantic object, and action. This class must implement interface `IF_UI_LINK_PARAMETER_CLASS`. For example, you can see demo class `CL_WCF_UI5_EMBED_PARAM_TEST`.

   The following methods are used:

   - `IF_UI_LINK_PARAMETER_CLASS~GET_AUTHORIZATION` - you can implement authority check, or just return abap_true as the SAPUI5 application has its own authority check. If authority check is performed here, the link in the navigation bar will render only when the user is authorized.
   - `IF_UI_LINK_PARAMETER_CLASS~PROCESS_DATA_COLLECTION` - this is optional. However, it requires at least empty implementation.
   - `IF_UI_LINK_PARAMETER_CLASS~CREATE_PARAMETER_OBJECT` - this should be implemented in order to prepare the navigation descriptor. To do this, create an instance of navigation descriptor of type **cl_wcf_flp_nav_descriptor** that will be used as return value of the method. Pass the following parameters to the constructor.

<table>
<tr><th>Parameter</th><th colspan="2">Description</th><th>Example</th></tr>
<tr><td><code>iv_navigation_object</code></td><td colspan="2">Create an empty instance of <code>if_crm_ui_navigation_object</code> (it is not required to initialize this object).</td><td></td></tr>
<tr><td><code>iv_application_hash</code></td><td colspan="2">The hash part of a shell compliant SAPUI5 URL.

i Note
The server path is determined automatically by the WebClient UI framework by using the method CL_LSAPI_MANAGER=>CREATE_FLP_URL. Use only the application hash.</td><td><code>#Product-display?PV1=PV2&PV4=V5&/display/detail/UU=HH</code></td></tr>
<tr><td rowspan="4"><code>it_parameters</code></td><td colspan="2">List of parameters used for the navigation control. The following two values are supported:</td><td></td></tr>
<tr><td>Value</td><td>Description</td><td></td></tr>
<tr><td><code>sap-ui-app-id</code></td><td>This is mandatory and corresponds to the SAPUI5 component.</td><td><code>sap.ui.demoapps.rta.fiorielemer</code></td></tr>
<tr><td><code>wcf-ui5-mapping</code></td><td>This is optional and it is used to map links from the SAPUI5 application to the WebClient UI framework.

The value should be a JSON string in this format:

<code>{"S01-action1":{"wcf-target-id":"<WCF target 1>"},"S02-action2":{"wcf-target-id":"<WCF target 2>"},...}</code></td><td><code>lv_mapping_json = '{"CADocument create":{"wcf-target-id":"EPMPRODSE"}, "Object2-Action":{"wcf-target-id":"FCC-F OP"}}</code></td></tr>
</table>

i Note

This mapping works when the SAPUI5 application performs navigation to external intent triggering post message `sap.ushell.services.CrossApplicationNavigation.toExternal.` Normally this happens when navigation method `toExternal(oArgs, oComponent?)` is used in the SAPUI5 application. However, it is the most common way of performing external navigation in SAPUI5. Other ways will not be supported. In runtime when the navigation happens, the WCF target ID is used as target, and the SAPUI5 hash is passed as parameter via value node in the parameter `IV_COLLECTION` of the inbound plug. In this way, the WCF application can use and process parameters coming from the SAPUI5 application.

2. Create the navigation link using transaction `CRMC_UI_NBLINKS`. As an example you can see logical link `WCF-UI5RTA` in Customizing under **UI Framework** > **Technical Role Definition** > **Define Navigation Bar Profile**. Use the parameter class

created in step 1 and target ID `WCFUI5EMB`.

3. The link created in step 2 can then be used in your navigation profile and business role.

i Note

The following is a list of limitations:

- Within the SAPUI5 app, internal navigation is fully supported. In the case of cross-application navigation which is not mapped to WebClient UI application, it is forwarded to SAP Fiori launchpad via the URL in a separate browser window.
- Transactional context is not synchronized between WebClient UI and the SAPUI5 app. All communication is done only via parameters and navigation descriptor.
- **Recent Items** is not supported.
- Embedding the SAPUI5 app in assignment blocks is not supported.
- Smart links in SAPUI5 apps are not supported.
- Capture of SAPUI5 events is not supported. For instance, in the SAPUI5 app, if you choose the **SAVE** button this action is not captured.
- SAP Fiori launchpad features in SAPUI5 applications, for example, **Adapt UI**, **Sharing**, **Save as Tile** are not supported in the embedded mode.

Application Enhancement Tool

Use

The Application Enhancement Tool can be used for table extensibility.

- You can add new tables.
- You can add fields to the tables created using the Application Enhancement Tool.
- You can search for table enhancements.
- You can display, create, change, and delete enhancements.

i Note

If you only want to add custom fields to your applications, you can use the **Custom Fields** app and **Custom Logic** app.

⚠ Caution

The management of data in an extension scenario deviates from the management of data in the standard scenarios. You are responsible for ensuring that the data used in an extension scenario is managed in accordance with any applicable legal requirements or business needs, such as data protection legislation or data life cycle requirements. Please note that the extensibility framework is currently not integrated in the privacy-by-default functionality. Therefore, the extensibility framework should not be used for the processing of personal data if this processing falls under any applicable data protection legislation.

Prerequisites

For tables you need to generate a UI component, which hosts the generated views.

If you are not authorized to use the Application Enhancement Tool, you need a certain authorization role.

→ Recommendation

Before you use the Application Enhancement Tool, we recommend that you define the package name and generation prefix. You can make the necessary settings in Customizing for UI Framework under UI Framework Definition > Application Enhancements > Define System Settings. For more information, see the Customizing documentation.

To make saved searches available in the Application Enhancement Tool, you have to make settings in Customizing for UI Framework, under Technical Role Definition > Define Navigation Bar Profile. Under Define Generic Outbound Plug Mappings, add the following entries:

Object Type	Object Action	Target ID
EXT_BO	Search	AXT_SEARCHE
EXT_BO	EXT_BO	AXT_SEARCHG

Features

The Application Enhancement Tool is integrated in the UI Configuration Tool, and can be started in this tool. The tables that you have added to an application are available in the UI configuration of the corresponding UI component and view. You can make these new tables available on the user interface (UI) by adding them to the overview page.

Main Functions

The Application Enhancement Tool offers the function to add tables.

Adding Tables

- Creating tables
- Structuring of deep tables
- Adding fields to new tables
- Assigning a logical key to new fields in a table
- Translating field labels in tables
- Reusing tables in other business objects, if these business objects are based on the same table enhancement place

Business Add-Ins

The following Business Add-Ins are available for enhancements:

Name	Technical Name	Description
BRFplus Application Exit	AXT_BRFPLUS_APPL_EXIT	You can use this BAdI to return the custom BRFplus application exit class.

The following Business Add-Ins are available for table enhancements:

Name	Technical Name	Description
API Exits for Tables	`AXT_RT_TABLES_API`	You can use this BAdI to adjust the behavior of a specific table enhancement based on a database table
Multi APIs	`AXT_RT_TABLES_MULTI`	You can use this BAdI to specify the behavior of all table enhancements that belong to a table enhancement place
Additional Data Source Fields	`AXT_ADD_DATASOURCE_FIELDS`	You can use this BAdI to enrich the extract structure of a table enhancement

More Information

UI Configuration Tool

UI Component Workbench

Adding Tables

Integration With Custom Fields App

Starting the Application Enhancement Tool

Use

There are different ways to start the Application Enhancement Tool in the WebClient UI. Since the Application Enhancement Tool is integrated in the UI Configuration Tool, you start the UI Configuration Tool first, to access the Application Enhancement Tool.

Procedure

Starting the Tool in the UI Configuration of an Application

You can start the Application Enhancement Tool in the UI configuration of an application in the WebClient UI. Therefore, you need to enable the configuration mode in the general settings on the central personalization page of the WebClient UI. After you have enabled the configuration mode, you can choose one of the icons for the page configuration in the header area of the application. You then navigate to the page configuration. To start the Application Enhancement Tool, choose the **Display Enhancements** pushbutton in the **Configurations** block. You can also open the page configuration, where you can configure the views in the current overview page. Here you can either choose the **Display Enhancements** pushbutton or you can directly choose the **Create Table** pushbutton.

Starting the Tool in the Main Menu

You can start the Application Enhancement Tool directly in the main menu of your system administrator role, by choosing **Enhance Applications** in the **Administration** work center. You navigate to the search page where you can search for enhancements.

The logical link to enhance applications (AXT-SEARCH) in the **Administration** work center (CT-ADM) is delivered in the standard system.

Example

You are logged on to the system as a sales professional. Choose **Sales Cycle** > **Sales Orders**. You see the search page where you can search for a sales order. Choose a sales order in the result list and navigate to the sales order overview page, by clicking the sales order ID. Choose the **Show Configurable Areas** icon and start the page configuration of the **Sales Order Details** area. Choose the **Show Enhancements** pushbutton to start the Application Enhancement Tool.

More Information

Starting the UI Configuration Tool

Searching for Enhancements

Context

You can use the Application Enhancement Tool to search for existing enhancements.

Procedure

1. Start the Application Enhancement Tool.

   You can start the Application Enhancement Tool from the main menu of your system administrator role, by clicking **Enhance Applications** in the **Administration** work center. You navigate to the search page.

2. Enter your search criteria.
3. Start the search.

Results

If you enter no search criteria and start the search, the result list displays all objects that have already been enhanced. You can select an existing enhancement in the result list, and navigate to the enhancement overview page where you can see the fields and tables. You can create new enhancements in the result list, by choosing the **New** pushbutton. You then navigate to a list with all objects that can be enhanced. If you select an object that has already been enhanced, you navigate to the overview page where you can see table enhancements that have already been added. If you select an object that has not yet been enhanced, you navigate to the overview page where you can enter new tables in the **Tables** block.

Next Steps

Starting the Application Enhancement Tool

Tables

You can use the Application Enhancement Tool to define your own tables that meet your company's business requirements. Tables that you add to an application are made available as new assignment blocks in the corresponding application. You can use the tool to create, change, and delete custom tables, which do not belong to the standard delivery.

Deep Table

Use

You can use the Application Enhancement Tool (AET) to extend your customer data modeling using deep tables. You can use a deep table to navigate from a table entry to an additional lower-level table. This enables you to organize your customer enhancement data according to a logical and clearly defined structure using tables that can be linked.

i Note

You can structure customer tables as a deep table with any number of levels. However, you should ensure that your specific customer enhancements do not become too complex and can be reused at a later stage.

The following figure shows an example of structuring tables as a deep table with three levels. The customer table at the first level is always assigned as an enhancement to the header data (ORDERADM_H) or item data (ORDERADM_I) for the respective business transaction category. The tables for all subsequent levels reference a higher-level table in each case. This produces the overall structure of a deep table.

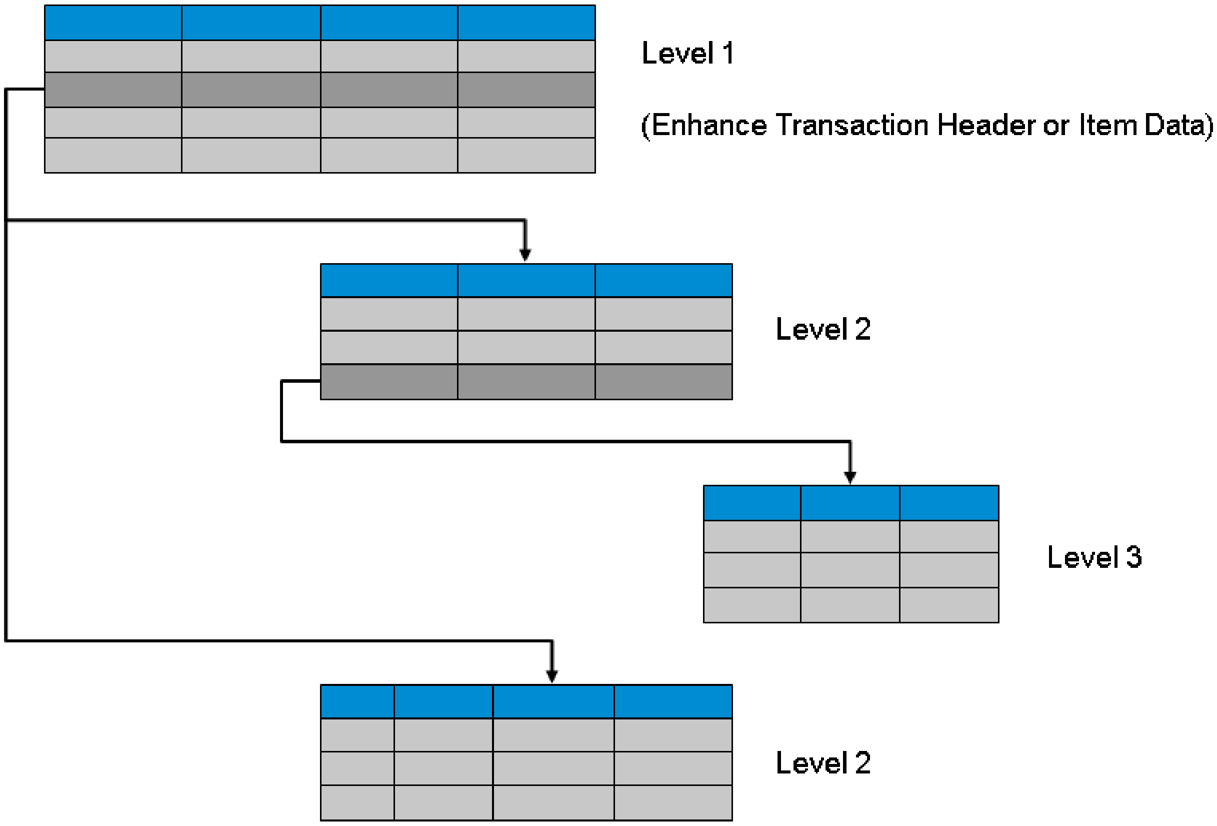


Figure: Structuring of a Customer Enhancement as a Deep Table

Prerequisites

- You have defined a suitable package and generated a UI component for your customer enhancement.
- You can only use deep tables for enhancements to business transaction types.

Features

- **Display and Navigation in the WebClient UI**

  You include the highest table in a deep table in the header or item data as an enhancement to a business transaction type. The table is displayed as a separate assignment block here. The name of the enhancement is used as the title of this assignment block.

Navigation within a deep table for a table entry is possible using the **Details** link in the **Actions** column. When navigating to the next level down, you switch to a lower-level overview page.

Data for the referencing higher-level table entry is displayed as header data on this lower-level overview page. You can return to the higher-level tables using the WebClient UI standard navigation by choosing ✔. You must always save data in the lower-level tables from the highest table (level 1).

> **Example**
>
> In the diagram above, the table (level 1) in the header or item data for a business transaction is integrated as a separate assignment block. The lower-level tables at level 2 are displayed on a joint lower-level overview page. When navigating to the table at level 3, you switch to an additional lower-level overview page.

- **Support During the Search**

  When defining the table fields, you can define in the **Search Relevance** parameter whether a field and its contents are to be made available to the search. The table fields that you define as being relevant here are then available as search criteria.

  > **Note**
  >
  > Note here that a deep table as an item enhancement has a negative influence on the search performance. The search algorithm for deep tables as item enhancements requires the table entries to be found to be provided from the lowest level through the highest level 1.

- **Archiving Deep Tables**

  You can archive your customer data from deep tables using the respective archiving object in the enhanced business transaction.

Activities

Structuring of a Deep Table

if you want to make a customer table you have defined available to be structured in deep tables, you must select the **Allow Sub Tables** checkbox field in the table header data. This checkbox field is displayed on the **Table Details** dialog box in the expert mode.

Tables that you have defined in this way are displayed as higher-level tables that can be selected when you create new tables. When you assign the respective higher-level tables, you define the structuring of several tables as a deep table. The ORDERADM_H and ORDERADM_I objects are provided as higher-level tables as a default setting. You must assign the highest table in a deep table to one of these objects as an enhancement to the header or item data.

For more information about the processing steps for tables in the AET, see Adding Tables.

Adding Tables

Use

You can use the Application Enhancement Tool to add tables that include new fields to an object. You can enable the expert mode to adapt some of the technical settings, for example, data element and technical field names.

Prerequisites

There are applications that require that you activate a business function to make the functions for table enhancements available.

You have enabled the configuration mode in the general settings on the central personalization page of the WebClient UI.

You need to generate a UI component in Customizing for **UI Framework** under **UI Framework Definition** > **Application Enhancements** > **Define System Settings**. You can however create several UI components and set one UI component as default. These UI components are available in the search help of the **Component** field.

Procedure

1. Create a new table

1. Start the WebClient UI and select the application that you want to enhance.
2. Start the page configuration in the application.
3. Choose the **Display Enhancements** pushbutton to display or create a new table.

    Note

    You can also click **Create Table** if you want to directly create a new table. However, you cannot enable or disable the expert mode if you do this.

4. Enable or disable the expert mode.
5. Choose the **New** pushbutton in the **Tables** block.

2. Select an object part

1. Select a valid package. The namespace is automatically derived from the package name.

    Note

    You can select a package only if you have enabled the expert mode beforehand.

2. Select an object part in which the new table is available, for example, header, item, and so on.

3. Enter the details of the new table

You enter the details of the new table in the **Table Details** dialog box. Once you have created a new table in the **Tables** block, the enhancement ID and the table ID are automatically generated.

1. Enter the new table name in the **Description** field.
2. In the **Relationship** field, you can choose one of the following relationships:
    - 1: N Table View

      This relationship specifies that the table contains an unlimited number of entries.
    - 1: 1 Form View

      This relationship specifies that the table contains one entry for every parent object.

      You can make the table fields available in the form view of the parent object.

      Note

      1:1 relationships can be used instead of fields to prevent the main application tables are being overloaded, especially if the fields are rarely used.

3. In the **Component** field, you can choose a UI component.

4. In the **Window** field, you can enter a name for the window.

   The window belongs to the UI component, in which you create the new table. The window name needs to be unique.

> **Note**
>
> If you have enabled the expert mode beforehand, you can change the table ID.

4. Select or deselect the generation checkbox

The **Generate** checkbox indicates that you want to generate a table. The checkbox can be selected or deselected in the **Tables** block or in the **Table Details** dialog box. This checkbox is selected by default for new and changed tables. If you want to exclude a table from generation, you need to deselect the checkbox.

5. Add new fields to the table

You can add fields to the new table. In the **Table Details** block, choose the **Add** pushbutton in the **Table Fields** block. You can select the **Logical Key** checkbox if you want to assign a logical key to the new field in the table. For more information, see Adding Fields to Tables.

In a table, you have to define at least one field that is not a calculated field. For more information, see Adding Calculated Fields to the Tables.

In addition to the field types that are available for field enhancements, you can define fields of field type **Long Text** with sub-type **Text Area**. This field type allows you to enter text of up to 1333 characters. The sub-type renders the field as a text area instead of an input field. This happens if the field is shown in a form view and not in a table. Therefore, you can use it in table enhancements with 1:1 cardinality if there is a form view for the table.

6. Save and generate the new table

For more information, see section “Saving and Generating the Field” at Adding Fields to Tables.

7. Make the new table available on the UI

- You are using the WebClient UI

  After the WebClient UI has been restarted, select the application and start the page configuration of the application, to make the new table available on the user interface. If you have started the Application Enhancement Tool from the UI Configuration Tool, you can return to the UI Configuration Tool, by choosing the **Back** pushbutton, and add the new table to the UI configuration before you restart the WebClient UI.

  > **Note**
  >
  > Tables that you have added to a business object might also appear in the **Available Assignment Blocks** list of a page configuration that is associated with another enhanced business object, if these business objects have the same enhancement place. However, at runtime only the tables created for the rendered enhancement object is displayed. If you want to display these tables at runtime, you can add them as reusable tables.

Adding Fields to Tables

Procedure

1. Create a calculated field

1. Select the Calculated (Read-Only) checkbox. If this checkbox is active, the Calculated Field Value block is shown. The block stays in display mode as long you have not defined the field details.

2. Enter the new field name in the Field Label.

3. In the Search-Relevance field, specify whether the new field is to be used in the search.

   If the field is relevant for the search, it is available in the search criteria and/or the result list. You can select one of the following options:

   a. Not search-relevant

   b. Search criteria

   c. Result list

   d. Search criteria and result list

4. In the Field Type field, select one of the following field types: decimal number, numerical, date, time, indicator, currency, quantity, text with capital letters, text with small and capital letters, or application reference.

   a. If you have selected Currency or Quantity as the field type, an additional reference field for the currency or quantity unit is created, to which the related check table is assigned.

   b. If you have selected Application Reference as the field type, the field is displayed as a hyperlink that allows you to navigate to an object (business object).

   c. If the user defines a field with his or her own data element, the field type is derived from the data element. The Application Enhancement Tool tries to match the derived field type with the field types provided by the tool. If the tool does not find any match, the Manual field type is selected. This option is only available if the expert mode is enabled.

5. In the Field Sub-Type field, select the field sub-type:

   a. If you have selected Application Reference as the field type, select the navigation object, that is, the business object where the navigation leads.

      Note

      Fields of type Application Reference can have different length. The length depends on what was defined by the application.

   b. If you have selected Quantity as a field type, select the dimension. The dimension determines the units of measurement that are made available.

6. In the Render/Validate as field, select a rendering option to determine the field behavior. Depending on the field type, the following options are available:

   a. Show Check Table as DDLB

   b. E-Mail

   c. Calculated Image

   d. Hyperlink

   If you have selected Hyperlink, a hyperlink is added to the field enhancement. This hyperlink can contain a maximum of 60 characters. The field is displayed as a hyperlink in display mode, and as normal text in edit mode.

7. Enter the field length (number of characters).

Field type	Field length (characters)
Uppercase / lowercase text	Maximum: 60 (for calculated fields: 255)
Currency	Maximum: 31 Fixed number of decimals: 2
Field type	Field length (characters)
Quantity	Maximum: 31
Decimal number	Maximum number of decimals: 10
Numerical	Maximum: 32 (for calculated fields: 31)
Date	Fixed length: 8
Time	Fixed length: 6
Indicator	Fixed length: 1

8. Enter the decimals, if you have selected decimal number, currency or quantity as the field type.

9. Define the formula for a calculated field.

   If you have selected the **Calculated (Read-only)** checkbox, define the formula in the **Calculated Field Value** block. For more information, see Adding Calculated Field to the Table.

Adding Calculated Fields to the Tables

Use

You can create a read-only field enhancement associated with a formula expression. The formula expression can be comprised of operations, constants, and application-specific data attributes.

Prerequisites

You define the formula expression in the **Calculated Field Value** block.

In the field details, you have selected the **Calculated (Read-only)** checkbox.

Features

Operations

You can use operations to calculate the value of the field at runtime. You can select the operations displayed in the **Operations** block by clicking them. In addition, you can enter the following operations manually: <, >, =, >=, <=, +, -, *, /.

The operations are provided by Business Rule Framework plus (BRFplus).

Custom Operations

You can also define custom operations. For more information, see the following:

- http://www.sdn.sap.com –>**Articles**–>**Business Rules Management** –>**How to Create Formula Functions**.
- Documentation for Business Add-in (BAdI) **BRFplus Application Exit** (AXT_BRFPLUS_APPL_EXIT)

Global Attribute Tags

You can display global attribute tags (left table) by filtering on the UI object type. You can add them to the formula expression by clicking them. They are inserted at your cursor position. If you want to cut and paste attribute tags, make sure that you cut the complete tag: **<VALUE[...]>**. For information about global attribute tags, see Global Attribute Tag.

You can use the existing attribute tags from the table. In addition, you can create new attribute tags in Customizing for **UI Framework** under **UI Framework Definition** –> **Global Attribute Tags**.

To use an attribute tag in a formula, the following conditions must be met:

- The field type of the attribute tag must fit to the position in the formula.

  You can use attribute tags based on different field types together within a formula. However, make sure that they are appropriate for the operation and the position within the formula.

  ### Example
  You have defined a formula expression that contains the timepoint field type. The result of the calculation can be a Boolean value.

- To create calculated fields as currency and quantity fields, you have defined a reference field in Customizing for **UI Framework** under **UI Framework Definition**--> **Global Attribute Tags** .
- The attribute tag must be accessible from the BOL object where the new calculated field is created.

  User attribute tags and constant attribute tags are always accessible. BOL attribute tags are accessible in the following cases:

  - The BOL entity exists at runtime.

    ### Example
    A BOL attribute tag for contact details of an account that has no contacts is not accessible

  - One of the following applies:
    - The BOL object that is to be enhanced by the calculated field must be in the path of the BOL attribute tag.
    - The path to the attribute tag is unique. That is, there must only be one path (one relation) from the root object to the enhanced BOL object.
    - The BOL attribute tag path has only 1:1 relations.

    Calculated fields are added to a BOL object. The BOL object that is enhanced depends on the enhanced business object part that you have selected. The Application Enhancement Tool (AET) uses only the path of the BOL attribute tag. The path of the BOL attribute tag starts with the root object and follows the relations to an attribute. The AET has to find a path from the enhanced BOL object that contains the calculated field to the attribute of the BOL attribute tag.

- The relation is not an association. Associations cannot be traced backward from child object to parent object.

Using a Calculated Field in Table Enhancements

BOL attribute tags are defined based on an existing BOL object. Since the BOL object needs to be generated with a table enhancement, you have to proceed as follows to use BOL attribute tags:

1. Generate the table enhancement without the calculated field.
2. Create BOL attribute tags for this BOL object.
3. Edit the table enhancement and use the attribute tag in the formula expression.

1:n Relations

If the path of a BOL attribute tag contains at least one 1:n relation, a table of data is returned by the BOL attribute tag. In this case, you have to use the table operations to get only one result value, which can be passed to the calculated field as result of the formula

Length

Global attribute tags, for example <VALUE[BOL:BP_ACCOUNT:ID]>, can have a length of up to 255 characters. They are not passed to BRFplus directly, but converted to BRFplus parameters. The string literals in formula expressions can have a length of up to 60 characters. If you try to define a literal that is longer, for example, if you want to calculate the URL of an image, an error is displayed. You can use two shorter strings instead of one and concatenate them.

Example

You want to calculate the URL of an image, but the following formula fails:

CONCATENATE('http://URL_that_has_more_than_60_characters/', <VALUE[BOL:BP_ACCOUNT:ID]>).

You can concatenate this URL string using two shorter strings:

CONCATENATE('http://first_part_of_the_URL_that_has_more_than_60_characters', CONCATENATE('_second_part_of_the_URL_that_has more_than_60_characters/', <VALUE[BOL:BP_ACCOUNT:ID]>)).

Validation of the Formula

You can choose the **Check** pushbutton to validate the formula expression. If you do not validate the formula, the AET automatically validates the formula when you choose the **Back** pushbutton.

The field types of the AET and BRFplus are slightly different. If you choose the **Check** pushbutton, the calculated field that displays the result of the formula is checked against the field types of BRFplus. For this reason, you may encounter valid formulas that do not work at runtime, since the checks could not be executed at design time

Calculation of the Field Value at Runtime

If the calculation of a field fails at runtime and the field is displayed on the UI, a standard user sees the message **Calculation for field <field name> failed.**

Some fields are not displayed on the current view, but they are calculated, since the calculation is performed when the BOL entity is read.

Example

- Errors from sales order fields are not displayed on the opportunity overview page.

- On the search result page, you do not see the calculated field, but the calculation has happened. Therefore, the message is not displayed on the search result page or overview page.

Technical users can increase their message level by setting the user parameter BSPWD_USER_LEVEL to 6. This displays all messages of the calculation. For example, a calculation message is created if either an attribute tag cannot be resolved at runtime because a relation points to BOL entities that do not exist yet, or if the field types within the calculation do not match

Activities

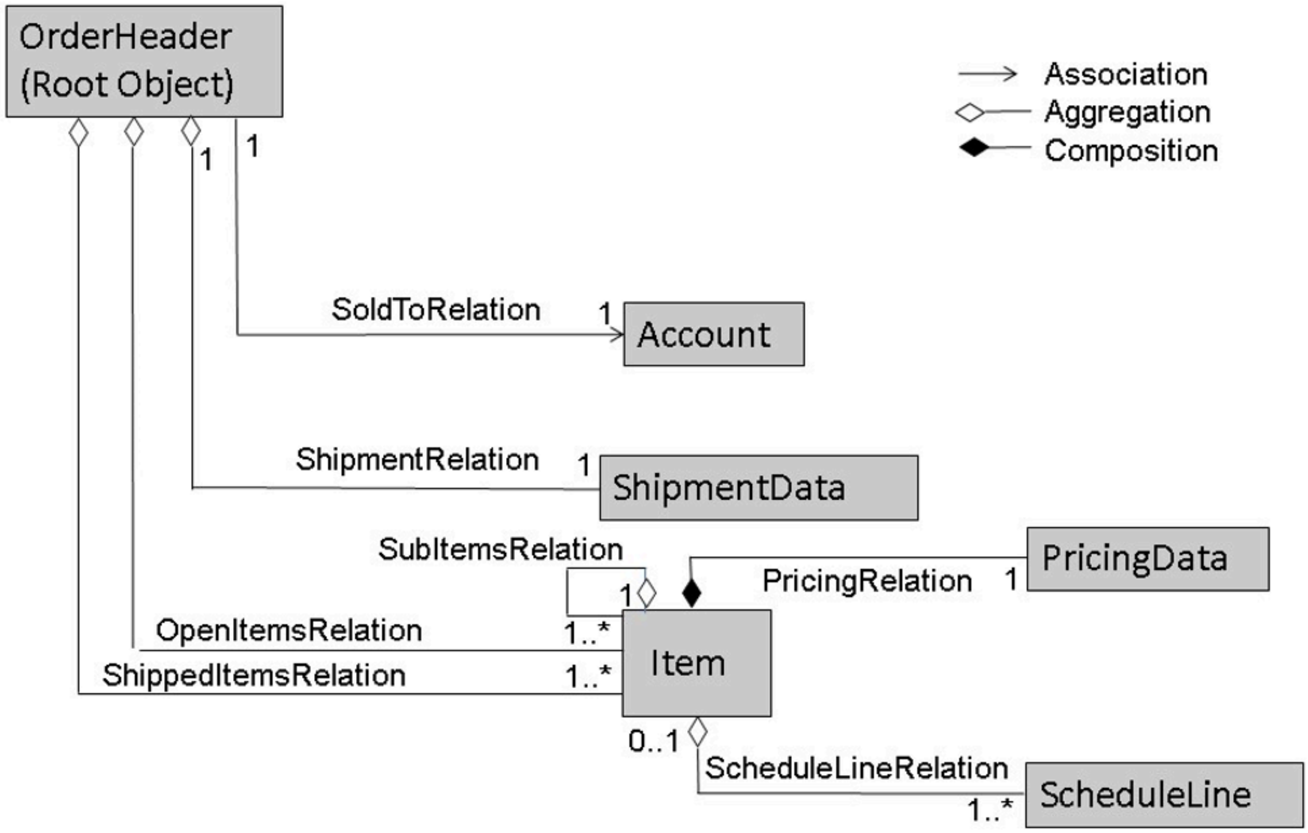
1. Define the field details.
2. Add the operation that is to be used to calculate the field value.
3. Insert one or more global attribute tags as operands.
4. If necessary, create new global attribute tags.

   Note

   May be necessary to generate the enhancement first.

5. Validate the formula.
6. Save and generate the enhancement.

Example



The following table shows whether BOL attribute tags can be used in the calculated fields of the enhanced BOL objects:

BOL Attribute Tag	BOL Object					
	OrderHeader	Account	ShipmentData	Item	PricingData	ScheduleLine
OrderHeader->@HEAD_ID	Yes	No Associations cannot be traced backward	Yes (a unique parent relation exists)	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations
OrderHeader>SoldToRelation/@NAME	Yes	Yes	Yes (a unique parent relation exists)	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations
OrderHeader>ShipmentRelation/ @CAN_BE_SHIP PED	Yes	No Associations cannot be traced backward	Yes	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations	Yes The BOL attribute tag path has only 1:1 relations
OrderHeader>ShippedItemsRelation/ @ITEM_NO	Yes (the result is a table and you have to use the table functions)	No Associations cannot be traced backward	Yes A unique parent relation exists (the result is a table and you have to use the table functions)	Yes	No *	No *
OrderHeader>ShippedItemsRelation/ SubItemsRelations/ @ITEM_NO	Yes (the result is a table and you have to use the table functions)	No Associations cannot be traced backward	Yes A unique parent relation exists (the result is a table and you have to use the table functions)	Yes	No *	No *
OrderHeader>OpenItemsRelations/ PricingRelation/@NET_VALUE	Yes (the result is a table and you have to use the table functions)	No Associations cannot be traced backward	Yes A unique parent relation exists (the result is a table and you have to use the table functions)	Yes	Yes	No *
OrderHeader>OpenItemsRelation/ ScheduleLineRelation/ @SHIPMENT_DATE	Yes	No Associations cannot be traced backward	Yes A unique parent relation exists (the result is a table and you have to use the table functions)	Yes (the result is a table and you have to use the table functions)	No *	Yes

* The root object has more than one path to the enhanced BOL object. The Item has two possible parent objects, that is, OrderHeader and Item itself. Due to the SubItemsRelation, an item is its own parent object.

Changing Existing Tables

Use

You can change existing tables after the initial generation.

Prerequisites

You have enabled the configuration mode in the general settings on the central personalization page of the WebClient UI.

There are applications that require that you activate a business function to make these functions for table enhancements available.

Procedure

Changing the Table

1. Start the WebClient UI and select the application with the existing table.
2. Start the page configuration in the application.
3. Choose the **Show Enhancements** pushbutton.
4. Select one enhanced object, if several enhanced objects are available.
5. Click the hyperlink in the **Enhancement ID** field to enter the table details.

   Note

   You can also choose the **Edit List** pushbutton in the **Tables** block, to change an existing table.

6. Change the table. You can change the table description.
7. Save and generate your data.

Changing the Table Fields

You can also change the fields that belong to this table. However, you cannot change the logical key of the table later. This prevents data inconsistencies.

Deleting Existing Tables

Use

You can delete existing tables that you created using the Application Enhancement Tool.

Prerequisites

You have enabled the configuration mode in the general settings on the central personalization page of the WebClient UI. Before you delete the table enhancement, you need to remove the table from the UI configuration.

> **Caution**
> If you delete an existing table enhancement, the Application Enhancement Tool determines whether an overview page configuration is affected. If this is the case, the tool displays the affected configuration and prevents you from deleting the table enhancement. Before you can delete the table enhancement, you need to remove it from the affected configuration in the UI Configuration Tool.

Procedure

Deleting the Table

1. Start the WebClient UI and select the application in which you want to delete a table.
2. Start the page configuration in the application.
3. Choose the **Display Enhancements** pushbutton.
4. Select one enhanced object, if several enhanced objects are available.
5. Choose the **Edit List** pushbutton in the **Tables** block to change to edit mode.
6. Select a table and choose the **Delete** icon.

   > **Note**
   > If the table is still included in an overview page configuration, you cannot delete the table, and the list of configurations is displayed.

7. Choose the **Save and Generate** pushbutton to delete the table.

Deleting the Table Fields

You can also delete the fields that belong to this table.

Result

The table is removed from all overview page configurations where it was used before.

Adding Reusable Tables

Use

You can reuse existing tables for other enhanced business objects. The tables can be reused in business objects that are based on the same table enhancement place.

Using the Same Enhancement Place

If a reusable table is based on the same enhancement place, no new enhancement is created. The reused enhancement is activated in the business object in which this enhancement is reused. Reusing a table in other enhancement places is not supported.

Prerequisites

You have enabled the configuration mode in the general settings on the central personalization page of the WebClient UI.

Procedure

1. Start the WebClient UI and select the application where you have created a field that you want to reuse.
2. Start the page configuration in the application.
3. Choose the **Display Enhancements** pushbutton.
4. Select one enhanced object, if several enhanced objects are available.

Adding a Reusable Table to the Current Enhanced Business Object

1. Click **Add Reusable Table** in the **Tables** block to check which reusable tables are available.
2. Select a table and add it to the enhanced object.

Adding a Reusable Table to Different Enhanced Business Objects

1. Open the **Table Details** dialog box of the table that you want to reuse.
2. Expand the **Reused in Object** assignment block.
3. Select the indicator **Active** for all object parts where you want to reuse the table.
4. Close the window and choose the **Save and Generate** pushbutton.

Result

You can now add the table to the UI configuration of the overview pages of the new business objects.

Example

You have created a table in the sales order header. You want to add this table to the service order header. Both business objects are based on the same enhancement place ORDER_ADMINH_TABLE. If you reuse this table in the service order header, it is activated for the service order in this enhancement place.

Translation

Use

You can use the Application Enhancement Tool to translate table descriptions, field labels of fields (including fields in tables) and entries in custom dropdown lists.

Prerequisites

You have enabled the configuration mode in the general settings on the central personalization page of the WebClient UI.

Activities

You have started the Application Enhancement Tool in the application in which you want to translate field labels and entries in dropdown lists.

Translating Field Labels

You can enter the edit mode by choosing the **Edit List** pushbutton in the **Tables** block. To enter the **Translation** screen, click **Translate**. You can select additional languages for translation by clicking **System Languages**.

You can also translate in the **Translation** block in the field details and table details, which you can start by clicking the enhancement ID in the **Tables** block.

Translating Entries in Dropdown Lists

You can enter the edit mode by choosing the **Edit List** pushbutton in the **Tables** block. Select an entry in the **Dropdown List Box** column in the **Fields** or **Tables** block. To enter the **Translation** screen, choose the **Translate** pushbutton. If you want to select additional languages for translation, choose the **System Languages** pushbutton.

You can also translate in the **Translation** block in the field details, which you can start by clicking the enhancement ID in the **Fields** block.

More Information

[Starting the Application Enhancement Tool]

Error Handling

Use

You can use the error handling function to find out about errors that occurred during the generation of tables that you created using the Application Enhancement Tool. You can analyze enhancements and see all objects that have been registered for enhancement.

Features

Generation Status

In the **Status** column in the **Tables** block, you can see the current status of every enhancement. The following statuses are available:

- Green

  This color means that the table has successfully been generated.

- Yellow

  This color means that you have entered data, but the table has not yet been generated. This status is also used if you edit a table with a green status. After you have edited the enhancement, it is still inactive. You need to save and generate your data to apply your changes.

- Red

  This color means that the generation of the table has failed. For further information, choose the icon or the hyperlink in the **Enhancement ID** column.

Error Messages

If you choose the status icon of an enhancement, a dialog box opens that shows all error messages that have occurred during the generation. You can use this information to fix the error; then restart the generation of the enhancement.

Analysis of Enhancements

You can use report AXT_EXT_ANALYZE to analyze enhancements. You can start this report via transaction AXTSHOW, or in Customizing for **UI Framework** under **UI Framework Definition** > **Application Enhancements** > **Analyze Enhancements**.

Showing Registry Information

You can use report AXT_REGISTRY_SHOW to show registry information of enhanced business objects. You can start this report via transaction SE38.

More Information

Adding Tables

Notification Center

Notifications for use in the Interaction Center.

Use

The notification center is available in the WebClient UI framework and is used to display the notifications produced by the interaction center. The notification center allows you to view immediate updates on the latest events that are related to your business role.

Features

The notification center exists independently of the application in the work area.

The notification panel shows both, read and unread notifications. It has a badge with the number of notifications that you have received since you last opened the notification panel. High-priority notifications received while the notification center is closed, show in a banner for 5 seconds. The items of the notification list are color-coded according to the level of priority.

The notification center is fully responsive. In minimum width resolution, only one panel, either the side panel, or the notifications panel, is visible at a time. There is a functionality to access truncated text.

In a multisession scenario, the notifications feature can be used seamlessly among all the sessions you have open.

Design Layer

Use

The design layer links UI-related settings to design objects, which span several views that implement the same business content. The UI-related settings are field settings that you can adapt to your requirements.

You can use the design layer to do the following:

- Rename field labels
- Hide fields that are not needed

  You must have selected the option **Hidden** or **Field excluded from field set** in Customizing. This means that the fields are not available in the UI Configuration Tool.
- Assign existing value helps from the ABAP Dictionary without any customer-specific development

  Special getter methods (type V) have been implemented that define the following:

  - Value help
    - Search help
    - Check table
    - Domain values
  - Field type
    - Input field
    - Checkbox
    - Dropdown list box

Note

The checkbox **Value help from ABAP Dictionary** needs to be selected in the design layer Customizing.

Features

The design layer controls field labels in form views, table columns, and in the search criteria of the advanced search. It also controls the field visibility on the UI at runtime and in the available field set in the UI Configuration Tool.

These settings are determined by design objects, which you define in the design layer Customizing.

To reuse the settings of a design object, the reference design object has been introduced.

More Information

UI Configuration Tool

Design Layer Customizing

General Access Sequence

Design Layer Customizing

Use

You can use the design layer Customizing to define design objects and assign these design objects to views.

In the first step, you define design objects and the settings regarding field labels and their visibility in Customizing. In the second step, you assign your design objects to views and their context nodes, in the UI Component Workbench.

In addition, you can define reference design objects. A reference design object inherits its design to design objects that are based on this reference design object.

You can copy existing SAP design objects and SAP reference design objects and adapt them to your requirements. You can also create your own design objects and reference design objects.

You can use the design layer Customizing for form views, tables, and advanced searches.

Activities

Creating Design Objects

You can create your own design objects in Customizing for **UI Framework** under **UI Framework Definition** > **Design Layer** > **Define Design Object**.

The object type and the business object layer (BOL) object in this Customizing activity are used for filtering and to reduce the number of BOL attributes in the list. You can assign the design object to every BOL object and to every BOL attribute that you want.

Copying SAP Design Objects

You can copy existing SAP design objects in Customizing for **UI Framework** under **UI Framework Definition** > **Design Layer** > **Copy SAP Design Object**. You can adapt the copied SAP design objects to your requirements in Customizing for **UI Framework** under **UI Framework Definition** > **Design Layer** > **Define Design Object**.

Assigning Design Objects to Views

You can assign the design object to a context node, or to single attributes under a context node in a selected view. The assignment is only necessary if you have created your own design object and did not copy an SAP design object. Copied SAP design objects have the same name as the original SAP design object. Therefore, an assignment to views is not necessary. The assignment is valid for all configurations of the view.

Note

You cannot assign design objects to tree views (tree context nodes).

1. Enter transaction `BSP_WD_CMPWB` to start the UI Component Workbench.
2. In the UI Component Workbench, select a full context node or an individual context node attribute, to which you want to assign a design object.
3. In the context menu, choose **Assignment to Design Layer** to access the assignment dialog box. In the **Design Layer Assignment** dialog box, you can see which design objects are assigned to the context node. The assignment is defined by the following key fields:
    - Component
    - View
    - Context node
    - Context node attribute

      The standard context node attribute is `<DEFAULT>`; this includes all attributes.
    - Object type

Defines the UI object type used at runtime for the assignment; <DEFAULT> includes all UI object types.

- Component usage

  Defines the component usages used at runtime for the assignment; <DEFAULT> includes all component usages.

- Deviating attribute

  Defines the attribute name of the design object; this field is used only if the attribute name of the context node differs from the attribute name of the design object.

4. Choose Insert Assignment to add a new assignment.
5. Expand the UI object type and select the design object.
6. Choose the UI object type.

   You can choose between the name of the UI object type and <DEFAULT>.

7. Enter the component usage if required.
8. Save your assignment.
9. With Display SAP Assignments and Edit Customer Assignments, you can toggle between both assignments of design objects.

Creating Reference Design Objects

You can create your own reference design objects in Customizing for UI Framework under UI Framework Definition > Design Layer > Reference Design Objects > Define Reference Design Objects.

The object type and the BOL object in this Customizing activity are used for filtering and reducing the number of BOL attributes in the list. You can assign the reference design object to BOL objects and attributes as required.

Copying SAP Reference Design Objects

You can copy existing SAP reference design objects in Customizing for UI Framework under UI Framework Definition > Design Layer > Reference Design Objects > Copy SAP Reference Design Object. You can then adapt the copied SAP reference design objects to your requirements in Customizing for UI Framework under UI Framework Definition > Design Layer > Reference Design Objects > Define Reference Design Objects.

Design Layer Web Tool

The design layer web tool allows you to create, modify, or delete UI design objects using a web-based UI.

Use

You can use the design layer web tool to create, modify, or delete UI design objects.

UI Object Type

The object type screen allows you to create, modify, or delete UI object types using a web-based UI, independent of the BOL.

Use

On the object type screen, you can create, modify, or delete UI object types independently from the existing business object layer (BOL).

Features

Create Your Own UI Object Types

You can create new UI object types in your own namespace, which are client-dependent.

To use the new UI object types, you enhance the component according to the framework enhancement concept or create your own component . You use the `DO_CONFIG_DETERMINATION` method to set the view for the UI object types. Within this method, you can use the `SET_CONFIG_KEYS` method to set the UI object type for the UI or `SET_CONFIG_KEYS_4_CHILDREN` method to set the UI object type for the UI and it's children.

Modify Existing UI Object Types

The standard UI object types are client-independent. When you change a standard UI object type, an entry with the same name is added to the customer version of the table. When there is a standard UI object type and a customer UI object type with the same name, the customer version overrules the standard settings. Your changes are client-dependent.

Generic UI Object Types

A generic UI object type is a UI object type that cannot be set by a UIU component at runtime. You can choose or create any ID for the generic UI object type. However, to avoid an unrestricted proliferation of generic UI object types in the system, you define the generic UI object type ID as the ID of the model component to which the model object is assigned.

Screen Fields

Field	Description
UI Object Type	The current UI object type ID.
Description	You can edit this description of the UI object type.
Callback Class	This field is not available generic object types. You can edit this optional field for specific object types. This class is an implementation of interface `IF_BSP_DLC_OBJ_TYPE_CALLBACK` and is used for the definition of sub object types.
GenIL Component Name	You can edit this optional parameter to define the name of the associated generic interaction layer (genIL) component.
BOL Object Name	You can edit this optional field to define the BOL object associated with the object type. This object must be at the root level.
BOR Object Type	This field is not available for generic object types. You can edit this optional field to define the BOR object type under which this UI object type falls.
Extensible BO	This parameter is not available for generic object types.

Checkbox (Read-Only)	Description
Generic	Selected for generic UI object types. Deselected for specific UI object types.
Standard	Selected for SAP standard UI object types.
Customer	Selected for customer UI object types.

Example

In Service, three UI object types have been created for the complaint business object (BUS2000120):

- Complaint
- In-house repair
- Return.

You can use these UI object types to determine

- UI configuration
- Design layer settings
- Dynamic navigation
- ABAP callback class that is used for the definition of subtypes
- Name of the corresponding genIL component
- Corresponding BOL and BOR objects.

More Information

- Framework Enhancement
- Application Enhancement Tool

Create UI Object Types

Create UI object types using a web-based UI, independent of the BOL.

Use

You can create UI object types with the dialog that results when you

- Choose **UI Object Type** from the **Quick Creates** group link
- Choose **New** from the UI object type search page
- From the **Create** section of the design layer work center, choose **UI Object Type**.

Features

Screen Fields

Field	Description
Generic	If you would like to create a generic UI object type, use the value X for this field. If you would like to create a specific UI object type, leave the value for this field blank.
UI Object Type	The value of this field is the current UI object type.

Button	Description
OK	The OK button validates the input and navigates you to the UI object type page where you can complete the information required for the rest of the attributes.

More Information

UI Object Type

Design Object

The design object screen allows you to create, modify, or delete design objects using a web-based UI, independent of the BOL.

Use

On the design object screen, you can create, modify, or delete design objects independently from the existing business object layer (BOL). A design object is the main technical object in the design layer that can be assigned to context nodes. It controls the appearance of the node attributes in UI configuration and at runtime. There are different kinds of design objects:

Design Objects	Description
Specific design objects without reference	Contains a specific UI object type in its key.
Specific design objects with reference	Contains a specific UI object type in its key with the addition of the reference section in which a generic design object can be selected. A new specific design object inherits the attributes from the referred generic design object.
Preferred design object	If you create many specific design objects that reference a generic design object with the same model object, you choose which of these will be selected using the Preferred Design Object parameter.
Generic design object	You either identify or create a generic UI object type before you can create a generic design object.

Features

Specific and generic UI object types are stored in different tables. The standard specific and generic design objects are stored in the `BSP_DLC_SDOBJ` table whereas any that you create are stored in the `BSP_DLC_DOBJ` table. These tables do not contain any identifier for specific verses generic design objects. Specific and generic design objects are distinguished by the UI object type. Generic design objects reference generic UI object types and specific design objects reference specific UI object types.

Screen Fields

Field	Description
Object Type	In this field, enter the value of the UI object type ID. If the assigned object type is generic, the design object is also generic. If the assigned object type is specific, the design object is also specific.
Design Object	The current design object ID. When creating a new generic design object, you can choose any ID you want, but it is recommended to keep the specific design object ID equal to the referenced generic design object ID.
BOL from ObjTyp	This field is read-only and shows the underlying BOL object associated with the UI object type.
Description	You can edit the description of the design object.
BOL Object Name	You can use this editable field if you want to refer to another BOL object within the same hierarchy of the BOL object from the object type.
Get/Set Handler	You can assign a getter or setter handler object for the design object. These objects are maintained separately.
Hdl. ClassName	The handler class name is automatically populated from the Get/Set Handler.

The following fields are only visible and editable for specific design objects.

Field	Description
Design Object	Referred generic design object.
Generic UI Object Type	Referred generic object type.
Preferred Design Object	When multiple specific design objects reference the same generic design object with the same model object, you select one as the preferred design object.

Checkbox (Read-Only)	Description
Generic	Selected for generic design objects. Deselected for specific design objects.
Standard	Selected for SAP standard design objects.
Customer	Selected for customer design objects.

Create Design Objects

Create design objects using a web-based UI, independent of the BOL.

Use

You can create UI design objects with the dialog that results when you

- Choose **UI Object Type** from the **Quick Creates** group link
- Choose **New** from the UI object type search page
- From the **Create** section of the design layer work center, choose **UI Object Type**.

Features

Screen Fields

Field	Description
Object Type	In this field, enter the value of the UI object type ID. This value could be the ID of any generic or specific UI object type.
Design Object	Enter the new design object ID for this field.

Button	Description
OK	The **OK** button validates the input and navigates you to the design object page where you can complete the information required for the rest of the attributes.

More Information

[Design Object]

Handler Page

The handler page allows you to create, modify, or delete handlers using a web-based UI, independent of the BOL.

Use

On the handler page, you can create, modify, or delete handlers independently from the existing business object layer (BOL). The handler object represents a context node class specialized for the delegation of getter and setter methods. When a specific field of a design object is used in the UI, a centralized method can manage the logic behind that field, regardless of where that field is implemented. These handler objects are not meant to replace getter and setter methods. They are only used when there are no standard or customer implementations of those getters and setters in the UI component.

Features

Screen Fields

Field	Description
Handler ID	The ID of the handler object.
Description	You can edit the description of the handler.

Field	Description
Class Name	The class name of the handler. You derive the class name from `CL_BSP_WD_CONTEXT_NODE_DL`.

Checkbox (Read-Only)	Description
Standard	Selected for SAP standard handlers.
Customer	Selected for customer handlers.
Disabled	Selected for disabled handlers.

Create Get/Set Handler

Create getters and setters from the handler objects using a web-based UI, independent of the BOL.

Use

You can use the create get/set handler dialog to create getters and setters from the handler objects.

Features

Screen Fields

Field	Description
Handler ID	Enter the new handler object ID for this field.

Button	Description
OK	The OK button validates the input and navigates you to the handler page where you can complete the information required for the rest of the attributes.

More Information

Handler Page

Assigning Value Helps

Use

You can use the design layer to assign value helps that already exist in the ABAP Dictionary, without any specific development effort.

Procedure

Creating a Design Object

1. In Customizing for **UI Framework**, choose **UI Framework Definition** > **Design Layer** > **Define Design Object**.
2. Create a design object by clicking **New Entries**.
3. Enter the details of the design object and save your entries.
4. Choose **Design**.
5. Click **New Entries**.
6. Select the checkbox **Value help from ABAP Dictionary**.
7. Choose one of the following field types:
   - **Input field**, if a search help or check table is defined in the ABAP Dictionary
   - **Dropdown List Box**, if fixed domain values are defined in the ABAP Dictionary

Assigning the Design Object to the Views

1. Start transaction **BSP_WD_CMPWB** for the BSP Component Workbench.
2. Assign the design object to the views in which you want to use it.

More Information

Design Layer Customizing

Translation

Use

You can translate field labels that have been created in the design layer by using transaction SE63 and the translation object type DLLC. This object type has been introduced especially for the translation of field labels in the design layer. Your translation is saved in a separate customer table.

Activities

Finding Out the UI Object Type and the Design Object

Before you start the translation, you need to find out the UI object type and the design object.

1. Start WebClient UI and select the business object.
2. Select an input field and press F2 to start the **Technical Data** dialog.

   You can find the UI object type and the design object in the technical information about the design layer.

Note

The label origin must be the design layer. You can find this information in the technical information about the configuration at runtime in the **Technical Data** dialog.

Translating by Using the UI Object Type and the Design Object

1. Start transaction SE63.

You see the page **Initial Screen: Standard Translation Environment**.

2. Enter `DLLC` in the transaction code field.

   You see the page **Translation: ABAP Short Texts: Design Layer Field Labels (Customer)**.

3. In the field **Object Name**, enter the UI object type followed by the design object that you found out beforehand. They need to be separated by an asterisk (*), for example, `BP_ACCOUNT*BP_SEARCH`.

4. Press F4 and then ENTER.

   The system automatically includes the necessary empty spaces between the UI object type and the design object.

5. Choose the source language and target language, **English** as the source language and **German** as the target language, for example.

6. Click **Edit** to translate the object.

7. Save your entries.

Translating by Using the Collection

You can also use the collection for translation purposes. The collection technically corresponds to the package, which you can find out in the **Technical Data** dialog of the WebClient UI. You can start the **Technical Data** dialog with the F2 key. The package name is available in the technical information about the UI component.

1. Start transaction SE63.

   You see the page **Initial Screen: Standard Translation Environment**.

2. Enter `DLLC` in the transaction code field.

   You see the page **Translation: ABAP Short Texts: Design Layer Field Labels (Customer)**.

3. In the field **Object Name**, press F4 to start the input help.

4. In the field **Collection**, enter the collection/package that you found out beforehand.

   You see a list with a combination of UI object types/design objects that belong to this collection.

5. Click the RIGHT ARROW key with the quick info text **Next Variant** to see more technical details.

6. Select the design object that you want to translate.

7. Choose the source language and the target language.

8. Click **Edit** to translate the object.

9. Save your entries.

More Information

Translation

Displaying Technical Information

Runtime Repository Comparison Tool

Use

You can use the Runtime Repository Comparison Tool to compare the runtime repository of an enhanced UI component with the SAP standard runtime repository.

On the comparison result screen, you can manually edit the runtime repository of the enhancement to reflect the changes in the SAP standard runtime repository as needed. Or, you can automatically adapt the enhancement runtime repository to match the SAP standard runtime repository.

When you create a UI component enhancement, the enhancement runtime repository is automatically copied to your namespace. Therefore, changes to the SAP standard runtime repository are not transferred to the enhancement runtime repository when you import a support package or perform an upgrade.

Activities

You can call up the Runtime Repository Comparison Tool as follows:

- In the UI Component Workbench under **Environment** > **Runtime Repository Comparison**. The option **Runtime Repository Comparison** is available only in a component enhancement.
- In Customizing for **UI Framework** under **UI Framework Definition** > **Compare Runtime Repositories**
- As report `BSP_WD_RT_REP_COMPARE`

WebClient UI Framework Check

Use

You can use the WebClient UI Framework Check to analyze the consistency of user interface (UI) repository data and configuration data for the WebClient UI framework. If errors are detected, you can navigate from the error messages to the affected system objects.

For example, you can perform checks after you have upgraded your installation to a higher release.

In the WebClient UI Framework Check, you can select the following check types:

- **Consistency of Enhanced Views**
- **Comparison of Runtime Repositories**
- **Design Layer**
- **UI Configurations**
- **UI Personalization**

Features

- Check types
  - **Consistency of Enhanced Views**

    With this check type, you can analyze the consistency of enhanced views in terms of the corresponding enhancement sets.
  - **Design Layer**

With this check type, you can analyze design object definitions in Customizing.

- **Comparison of Runtime Repositories**

  With this check type, you can compare the runtime repository of the UI components of an enhancement set with the SAP standard runtime repository. For more information, see Runtime Repository Comparison Tool.

  In the check result, you can click the status icon of the messages to display the message long text.

- **UI Configurations**

  This check type supports all types of UI configurations, such as view configuration, overview page configuration, and fact sheet configuration. With this check type, you can analyze whether UI elements that are contained in UI configurations are consistent in terms of the existence and properties of the corresponding context node attributes or runtime repository elements.

- **UI Personalization**

  With this check type, you can analyze the personalization of overview pages, work center pages, home pages, report pages, and tables. Inconsistent UI personalization settings can optionally be adjusted automatically during the check.

i Note

The system performs the checks in the order in which they appear on the UI, and each check builds on the previous check. Therefore, we recommend that you perform the checks in the order in which they appear on the UI, and correct any error messages resulting from a check before executing the next check.

You can improve the system performance by defining the number of background processes that can be used for the checks.

- View selection

  You can use the following parameters to select the views to be checked:

  - Enhancement Set
  - UI Component
  - View
  - Business Role

  You can manually further restrict the list of views to be checked.

i Note

If you specify an enhancement set, the system restricts the selection to the views belonging to the enhancement set and the active SAP standard views according to your selection criteria. In the check result hierarchy, the standard views and enhanced views are displayed separately for each UI component.

If you do not specify an enhancement set, the system selects the views of the SAP standard components according to the following logic: Linked views are not analyzed if they are contained in UI components other than the ones to which the checked views belong.

You can include obsolete views in the check. If you do not select the **Check Obsolete Views** checkbox, only active views are selected, and obsolete views are omitted.

- Display check result history

  You can search for the results of former checks.

- Delete UI Personalization

  With this function, you can delete multiple UI personalization settings according to various selection criteria, such as user, UI component, and view.

Field Label Analysis

Use

You can use this report to determine the layers in which field labels exist, and the field label text available in each layer. You can also get an overview of all fields belonging to one or more UI components, and the origin of different labels.

While the report allows you to display all field labels for one or more UI components, you can determine the origin of a specific field label by pressing F2 at runtime

The field label in the uppermost layer is displayed on the user interface. For more information about the text determination procedure, see General Access Sequence.

Features

- You can click the field label in the design layer, or click a design object to navigate from the field label to the design layer Customizing. You also find this in Customizing for **UI Framework**, under **UI Framework Definition** > **Design Layer**.
- You can delete field labels in the layers.

  ### Example
  There is a redundant field label in the text repository of the UI configuration. To use the field label from the layer below, that is, from the design layer, you would delete the redundant field label.

  Before you delete a field label, you have to lock the view by choosing the **Edit View** pushbutton. This prevents other users from making changes to the view at the same time.
- You can filter by field label origin to display field labels that you want to work on.

Selection

You can select one or more UI components to check all field labels in these components.

Output

In the hierarchy tree you can see the UI components and their views, like in the UI Component Workbench. For each view, you see the SAP and customer configurations, identified with their four configuration keys (**Role Config. Key**, **Component Usage**, **Object Type**, **Object Subtype**). For each configuration, the available fields are displayed.

The available columns include the following:

- **Active Field Label**

  Current field label that is displayed at runtime, also shown in the UI Configuration Tool
- **Label Origin**

  Layer from which the active field label is used
- **Text Rep.** (Text Repository), **Design Layer**, **ABAP Dictionary**

The field label that is available in each layer is displayed in the relevant column.

- **DO: Object Type**, **DO: Design Object**, **Design Obj. Attribute**

  For field labels from the design layer, the object type, the design object, and the design object attribute are displayed.

- **DO: Ref. Dsgn Obj.** (Design Object: Reference Design Object)

  Reference design objects are indicated with an X.

- **DO: No. of Affected Views** (Design Object: Number of Affected Views)

  Number of views that use the relevant design object. You can open a list of the affected views by clicking this number.

Activities

You can access the report as follows:

- In Customizing for **UI Framework** under **UI Framework Definition** > **Field Label Analysis**
- As report `BSP_DLC_FIELD_ANALYSIS`
- By using transaction `WCF_FA`

Transaction Launcher

Use

You can use the transaction launcher to launch transactions to URLs, Business Server Pages (BSPs) of other systems, and Business Object Repository (BOR) transactions.

Note

The transaction launcher is only supported in the standalone mode and not in the SAP Fiori launchpad integrated mode.

Prerequisites

To configure the transaction launcher, you need to understand:

- BOR methods
- BOL browser and BOL objects
- IS-U front office processes

Features

The transaction launcher is configured using the Customizing activities found in **UI Framework** > **Technical Role Definition** > **Transaction Launcher**.

- SAP GUI for HTML transactions via the Internet Transaction Server (ITS) from both systems in SAP ECC and SAP S/4HANA Service
- IS-U front office processes from IS-U systems

- URL based transactions from external applications or SAP systems

Information can be passed to and from the IC WebClient and the launch transaction, depending upon Customizing. For example, the confirmed business partner can be passed to SAP ERP sales order and the created document can be returned.

More Information

For more information about configuring the transaction launcher, see Customizing activity **Configure Transaction Launcher** under **UI Framework > Technical Role Definition > Transaction Launcher**.

Global Attribute Tag

Definition

A global attribute tag is a semantic identifier (ID) that allows you to expose and consequently access application and system data.

You can tag different application and system attributes by assigning a unique, untranslatable ID to them. This ID can be used to reference and retrieve attribute values within the context of a business object (application).

You can tag the following types using global attribute tags:

- Business object layer (BOL) attributes
- Constants (literals)
- User variable attributes

You tag BOL attributes for a specific UI object type. A UI object type corresponds to a business object. You can determine the UI object type assigned to the corresponding application overview page by pressing F2.

You create global attribute tags in Customizing for **UI Framework** under **UI Framework Definition > Global Attribute Tags**.

Use

The following applications use global attribute tags:

- Tag clouds

  For more information, see Tag Clouds.

- Mash-ups

  For more information, see Mash-Ups.

- Calculated fields (created using the Application Enhancement Tool)

  For more information, see Adding Fields to Tables

For most applications, SAP delivers only some examples of BOL attribute tags. You can create more attribute tags according to your business requirements.

Each UI object type is assigned to a BOL root object. When you tag a BOL attribute, you start from that BOL root object. You can navigate through the BOL model using the BOL relationships and tag any attribute that matches your requirements. As a result, you tag a BOL path and give it a semantic ID.

Note

Note the following information about how applications use global attribute tags:

- Tag clouds

  Global attribute tags are used to specify the columns in the tag search result list. For more information, see documentation in Customizing for **UI Framework** under **UI Framework Definition** > **Tag Clouds**.

- Embedding of rapid applications and mash-ups

  If you tagged a BOL attribute path that contains 1:n or 0:n BOL relationships, at runtime, the tag evaluation results in a table with more than one entry. The reason for this is that the system starts evaluating the BOL path from the BOL root object.

  However, when embedding views of rapid applications, you have to map to a single value. In mash-ups, there is no standard way to convert a table to a string. Therefore, in both cases, the system returns only the first entry of the obtained table.

  #### Example

  You tagged an attribute in order items. The BOL path starts from the header (main BOL entity), goes through the 1:n relationship and points to an item attribute. Since an order can have more than one item, the result is a table that contains the attribute values from all the existing items. When parsing such a result, the system uses the first entry from the table and returns the corresponding attribute value.

- Calculated fields

  You create calculated fields at the business object part. The business object part points to a specific BOL object, which is not necessarily a root object.

  #### Example

  You tag an attribute in the items. The BOL path contains a 1:n relationship.

  If you create a calculated field in the header, the BOL path is evaluated starting from the header BOL entity (usually the root object). The behavior is the same when you embed views and mash-ups. The tag value is a table.

  However, you can create a calculated field for an item. The evaluation begins at the corresponding BOL entity, which is the item. The resulting path no longer contains a 1:n relationship. Therefore, at runtime, the attribute tag returns a single value (the attribute value from the current item).

  If an attribute tag returns a table value, you have the following options:

  - You can use table operations like summation, maximum, minimum, and average. These appear in the list of the available operations under **Table Functions**. The result of these functions is a single value, for example, the result of counting the number of existing items.
  - You can use the operations listed under **Functions for First Table Row** to get the first entry from a table.

    #### Example

    Your organization decides to have only one contact for each account. Contacts are connected to the account header using a 1:n relationship. An attribute tag from the contacts returns a table if the attribute tag is used at the account header level. However, in your business processes, there can be only one contact. The **Functions for First Table Row** allow you to retrieve an attribute from your only contact and use it in your calculations.

  For more information, see Adding Calculated Fields to the Tables.

Caution

Global attribute tags are client-dependent. Therefore, you can create tags with the same name but different BOL paths in different clients. Do not define such tags. Some global attribute tag consumers are client-independent, for example, calculated fields created using the Application Enhancement Tool.

In addition, make sure that the tags you created are transported together with the enhancements that use them.

Example

In the `BP_ACCOUNT` UI object type, you tag the `BP_NUMBER` attribute from the BOL object (`BuilHeader`). You call this attribute tag `BP_ID`. You create another attribute tag called `CONTACT_ID` that references the ID of the account's contacts (`BuilHeader/BuilContactPersonRel/@CONP_NUMBER`). The BOL path of `CONTACT_ID` contains a 1:n relationship, namely `BuilContactPersonRel`.

Embedding of Views and Creation of Mash-ups

You can use `BP_ID` when you embed views or create mash-ups. `CONTACT_ID` returns a table. At runtime, the system uses the first entry in the table. Therefore, when embedding views or creating mash-ups, use this tag attribute only if you are interested in the first contact.

Calculated Fields

You create a calculated field in the account header. `BP_ID` returns a single value. `CONTACT_ID` returns a table. You can use the table operations and retrieve information related to the contact's tagged attribute. For example, you can count the number of contacts.

If you create a calculated field in the contacts, you cannot access `BP_ID`, since it belongs to the header.

Example: Enterprise Procurement Model

Definition

The implementation of Enterprise Procurement Model (EPM) in the WebClient UI is an example that shows some of the WebClient UI framework functions. The implementation is based on EPM, which is a reference application provided by SAP NetWeaver.

The implementation of EPM gives you an overview of how a business application is implemented in the WebClient UI framework.

Use

You can use EPM as an example to implement your customer-defined applications.

The following EPM business objects are used in the WebClient UI:

- Purchase order
- Business partner (and address)
- Product
- Organizational unit
- Employee (and address)

The following user interface (UI) functions are available:

- Advanced search page (including saved search)

  For more information, see Advanced Search and Advanced Search Personalization.

- Overview page

  For more information, see Overview Pages.

- Form view

  For more information, see Form Views.

- Table view

- Tree view

- Fact sheet

  For information, see Example: Creating a Fact Sheet.

- Home page

  For information, see Example: Creating a Home Page for the Administrator Business Role.

- Work center

- Guided activity

  For more information, see Guided Activities.

- Mash-up

  Google Maps is integrated as a mash-up on the business partner overview page. For information about mash-ups, see Mash-Ups.

Structure

The EPM implementation in the WebClient UI consists of the following UI components:

- `EPM_DEMO_WCC` for the work center
- `EPM_DEMO_ORDER` for the purchase order
- `EPM_DEMO_PROD` for the product
- `EPM_DEMO_BP` for the business partner
- `EPM_DEMO_ORG` for the organizational model and the employees
- `EPM_DEMO_MAIN` for handling the navigation between the different UI components of EPM outside the WebClient UI
- `EPM_DEMO_FS` for the fact sheet

Integration

The **Administrator** business role includes the **Procurement Demo** work center in the navigation bar. This business role also provides recent items and data loss handling.

Features

The following general functions are available in the work area:

- Page history
- Message handling
- Navigation to the field that issued a message (in business partner, product, and purchase order)

  For more information, see Message Bar.

BOL/genIL Functions

In the generic interaction layer (genIL), an implementation is available for purchase order, product, business partner, employee, and its address.

The genIL implementation uses the genIL handler concept. Each business object is implemented in a separate genIL handler class, which inherits from class `CL_GENIL_NODE_HANDLER_TX`.

The BOL (business object layer) model is implemented in genIL component `EPM` and can be displayed by using the genIL Model Editor (transaction `GENIL_MODEL_EDITOR`). The genIL implementation of the BOL model for EPM consists of the following handler classes:

- `CL_WCF_EPM_PO_IL` for purchase order
- `CL_WCF_EPM_PROD_IL` for product
- `CL_WCF_EPM_BP_IL` for business partner
- `CL_WCF_EPM_ORG_IL` for organization and employees

Example: Creating a Fact Sheet

Use

This procedure describes the creation of fact sheets based on the example implementation of Enterprise Procurement Model (EPM).

A fact sheet is a condensed form of information about a business object taken from several sources, such as master data or statistical data.

Online fact sheets are read-only pages that give an overview of a particular business object, such as a business partner.

i Note

An online fact sheet is different from a print fact sheet. This procedure covers online fact sheets.

The example implementation of EPM provides a business partner fact sheet with fact sheet ID `EPM_BP_FS`. You can display the business partner fact sheet at runtime, in the EPM application, by choosing the **Online Fact Sheet** pushbutton on the overview page of a business partner. For more information about EPM, see Example: Enterprise Procurement Model.

Fact sheets are implemented using a generic user interface (UI) component and view (`BSP_DLC_FS/factsheet`).

Procedure

i Note

For more information about creating fact sheets, see Creating and Assigning Views and How Do I Create Components and Views?.

To create a fact sheet for an application, proceed as follows:

1. In the UI Component Workbench, create the fact sheet views that form the content of the fact sheet assignment blocks.

   The fact sheet view is located in UI component `EPM_DEMO_FS`. View `FsBpPurchaseOrders` is assigned to window `EPM_DEMO_FS`. The view is implemented in a way that it receives the entity of the business partner through the inbound plug.

   The following alternatives are possible (not implemented in EPM):

   - The view receives the entity of the business partner through the inbound plug of the window.
   - You can use a global data context. For more information, see Customizing for **UI Framework** under **Technical Role Definition > Define Global Data Context Parameters**.

2. Define the fact sheet and assign the fact sheet views to it in Customizing for **UI Framework** under **UI Framework Definition > Fact Sheet > Maintain Fact Sheet**. For more information, see the definition of fact sheet `EPM_BP_FS` in view cluster `BSPVC_DLC_FS`, and the assignment of fact sheet view `EPM_DEMO_FS/FsBpPurchaseOrders`.

3. Configure the layout of the UI by using the UI Configuration Tool. You can access the UI Configuration Tool for fact sheets in the following ways:

   - From the **Configuration** tab in the UI Component Workbench for UI component `BSP_DLC_FS/factsheet`
   - From the fact sheet page in the WebClient UI at runtime, by choosing the **Page Configuration** icon
   - In the work area of the WebClient UI (after you have added delivered links to your role)

4. Implement the relevant Business Add-In (BAdI) to obtain the desired fact sheet title. For more information, see the enhancement implementation `EPM_BP of BAdI BSP_DLC_FS_BADI` in enhancement spot `BSP_DLC_FS`. This enhancement implementation allows you to do the following:

   - Create the fact sheet title
   - Define a sample pushbutton that calls a sample dialog box

5. Make the following settings in Customizing for **UI Framework**:

   a. Enable navigation to the fact sheet under **UI Framework Definition > Define UI Object Types**.

      For more information, see view `BSPDLCV_OBJ_TYPE` and UI object type `EPM_FACTSHEET`.

   b. Define the target ID under **Technical Role Definition > Define Work Area Component Repository**.

      For more information, see target ID `EPMFSBP` for UI component `BSP_DLC_FS` and window `BSP_DLC_FS/MainWindow`.

   c. For the relevant navigation bar profile, define the generic outbound plug mapping under **Technical Role Definition > Define Navigation Bar Profile**.

      For more information, see navigation bar profile `EPM_FACTSHEET` with generic outbound plug mapping `EPMFSBP`.

6. Create the code for the navigation to the fact sheet (typically from the overview page). For more information, see the implementation of the business partner overview page:

   - UI component `EPM_DEMO_BP`
   - View `PartnerOverview`

- Event handler EH_ONFACTSHEET

More Information

[Fact Sheet](#)

Situation Templates in Service

Get an overview of situation templates available in Service, together with the apps that display the situations. You can filter them according to various criteria.

> **i Note**
>
> For more information about the Situation Handling framework in SAP S/4HANA, see the Product Assistance for Situation Handling.

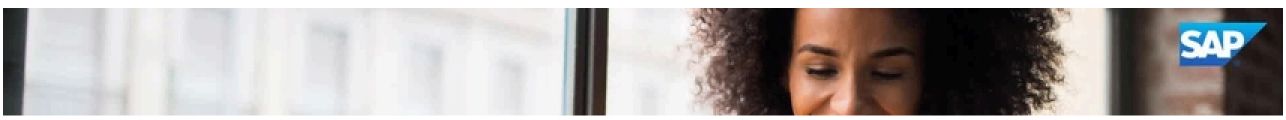
Situation Templates in Service

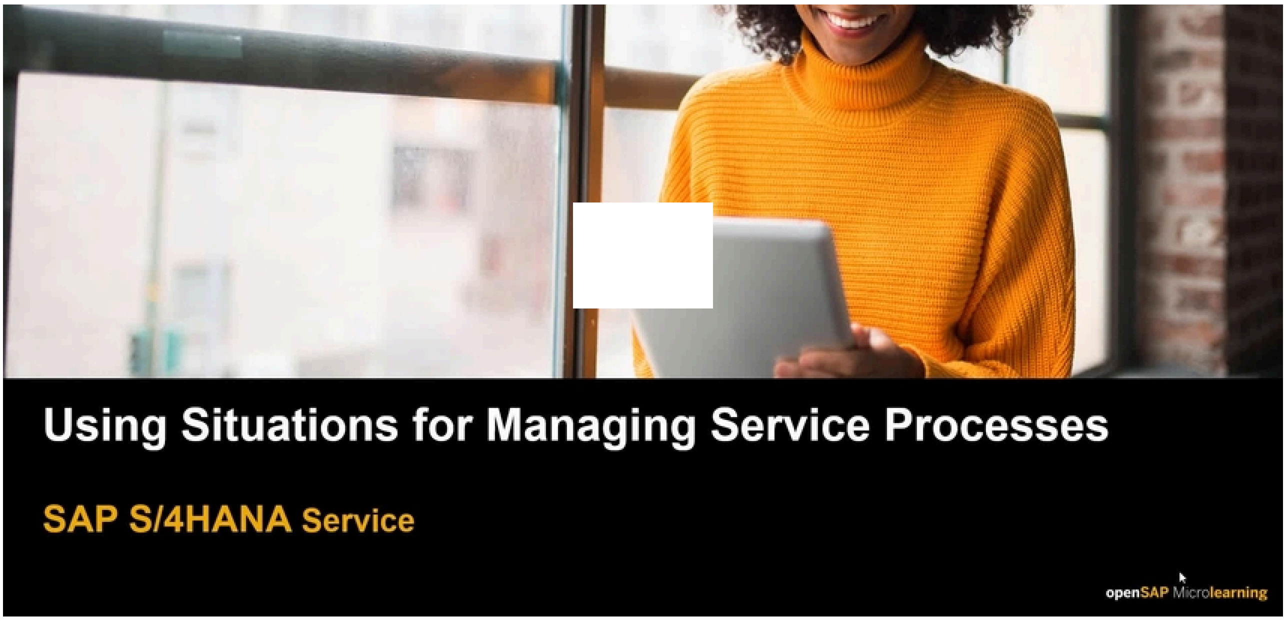
Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	In-House Service	Situation Template: In-House Service Not Confirmed	SRVC_DELAY_IN_IHR_CONFIRMATION	Standard object-based	No	Manage In-House Services	F402!
Service	In-House Service	Situation Template: Processing of In-House Service Overdue	SRVC_DELAY_IN_IHR_PROCESSING	Standard object-based	No	Manage In-House Services	F402!
Service	In-House Service	Situation Template: Follow-Up Step of Precheck Pending	SRVC_IHRITM_PRECHECK_DELAYED	Standard object-based	No	Manage In-House Service Objects	F769!
Service	Service and Maintenance Plan	Service Maint. Plan Mass Schedule Issue	SRVC_MPLAN_MASS_SCHEDULING_ISSUE	Standard object-based	No	Mass Schedule Maintenance Plans - Service	F492!
Service	Service Contract Management	Service Contract Due to Expire Soon	SRVC_EXPIRING_SERVICE_CONTRACT	Standard object-based	No	Service Contracts	TBT11 (Web(UI ID)
Service	Service Contract Management	Service Contract Item Due to Expire Soon	SRVC_EXPIRING_SRVC_CONTR_ITEM	Standard object-based	No	Service Contracts	TBT11 (Web(UI ID)
Service	Service Contract Management	Service Contract Item Not Released	SRVC_CONTRACT_ITEM_RELEASE_DUE	Standard object-based	Yes	Service Contracts	F3763

Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	Service Contract Management	Service Contract with Zero Billing Value	SRVC_CONTR_WITH_EXP_BILL_VALUE	Standard object-based	No	Service Contracts	TBT11 (Web(UI ID)
Service	Service Contract Management	SrvcContr Item Blocked for Billing	SRVC_CONTRACT_ITEM_BILLG_BLOCKED	Standard object-based	Yes	Service Contracts	F3763
Service	Service Contract Management	Price Check for Automatic Renewal	SRVC_CONTR_AUTO_RENEW_PRICE	Standard object-based	No	Service Contracts	F3763
Service	Service Contract Management	Service Contract GTS Blocked/Pending	SRVC_CTR_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Service Contracts	F3763
Service	Subscription Order Management	Subscription Contract Item Due to Expire	SOM_EXPIRING_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Subscription Order Management	Subscription Contract Item Grace Period	SOM_GRACE_PERD_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Subscription Order Management	Subscription Contract Item Distribution	SOM_DISTR_ISSUE_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Service Order Management	Trade Compliance Check Blocked/Pending	SRVC_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Manage Service Orders	TBT11 TBT11
Service	In-House Service	Situation Template: Stock Inconsistency for Service Object	SRVC_IHRITM_PROD_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695
Service	In-House Service	Situation Template: Inconsistency in Logistics Information for Service Object	SRVC_IHRITEM_EQUIP_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695

Using Situations for Managing Service Processes (English Only)

Watch this video at learning.sap.com.





Related Information

[Use Cases for Situation Handling]

Data Management in Service

Data management includes data archiving, blocking, unblocking and deleting personal data, as well as data destruction.

- **Data Archiving in Service**
- **Blocking, Unblocking, and Deletion of Personal Data in Service**

Related Information

[Data Management in SAP S/4HANA]

Data Archiving in Service

Data archiving is used to remove mass data that is no longer required but must be kept in a format that can be analyzed from the database. This section includes archiving objects for Service.

For more information about the archiving objects that are used for leads, opportunities, and activities, see Data Archiving in Sales.

Prerequisites for Archiving Service Transactions

Depending on the Finance integration scenario, you should consider the following when archiving service orders and service contracts:

- If you use item-based accounting, the service orders and service contracts to be archived must be set to **Business Completed**.
- If you use the integration with account assignment manager, the service orders and service contracts to be archived must be set to **Completed**.

Regardless of the Finance integration scenario used, all other transactions, such as service requests, service quotations, and service confirmations, must be set to **Completed** to be archived.

Related Information

Data Archiving
Data Management in Global Trade Management
Item-Based Accounting
Finalizing the Service Order
Finalizing the Service Contract

Archiving Service Contracts Using CRM_SRCONT

You can use archiving object `CRM_SRCONT` to archive service contracts.

Tables

`CRM_SRCONT` archives data from several tables. To check which tables these are, call up transaction `SARA`, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_SRCONT` can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

`CRM_SRCONT` is delivered with the following programs:

Program	Name
Write	`CRM_ARC_SRCONT_SAVE`
Delete	`CRM_ARC_SRCONT_DELETE_2`
Preprocessing	`CRM_ARC_SRCONT_CHECK_2`

When archiving service contracts, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SRCONT

Prerequisites

The following prerequisites must be fulfilled before you can archive a service contract:

- The service contract has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service orders and service confirmations are archivable.
- If item-based accounting has been enabled: The service contract has been set to **Business Completed**.

  For more information about item-based accounting, see Item-Based Accounting.

The following checks have been carried out:

- You've executed an archivability check for service contracts by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service contracts that meet the archivability criteria, receive the status **Archivable**.

- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_SRCONT` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_SRCONT`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance, SC for service contract.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization the rule applies to.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT `)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service contract was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object `CRM_SRCONT` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation of the following BAdIs for `CRM_SRCONT`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)

- **BAdI: Archiving Information of Tables** (`BADI_DTINF_ARCHIVE_DETAILS`)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SRCONT, there is at least one active archive information structure that contains the following selectable field:
    - **Transaction ID** (object_id)

  This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SC that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SRCONT archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service contracts.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Contracts Archived With CRM_SRCONT

You have the following options for displaying archived service contracts:

- From within **Archive Administration** (transaction SARA), you can search for archived service contracts using the read program CRM_ARC_SRCONT_READ
- You can also use the **Archive Information System** (transaction SARI) to display archived service contracts by entering the archiving object CRM_SRCONT and the archive infostructure SAP_CRMS4_SC

In both cases, the system provides a range of criteria that enable you to select the service contracts that you want to display.

Variants for CRM_SRCONT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for `CRM_SRCONT`.

Defining Write Variants

The write variant contains the parameters for the service contracts that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service contracts that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_SRCONT_CHECK_2` determines archivable service contracts.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  i Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Requests Using CRM_INCDNT

You can use archiving object `CRM_INCDNT` to archive service requests.

Tables

`CRM_INCDNT` archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_INCDNT` can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

`CRM_INCDNT` is delivered with the following programs:

Program	Name
Write	`CRM_ARC_SRQM_INCIDENT_SAVE`

Program	Name
Delete	CRM_ARC_SRQM_INCIDENT_DELETE_2
Preprocessing	CRM_ARC_SRQM_INCIDENT_CHECK_2

When archiving service requests, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_INCDNT.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service request:

- The service request has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service orders and service confirmations are archivable.

The following checks have been carried out:

- You've executed an archivability check for service requests by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service requests that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_INCDNT` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_INCDNT`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance, SRVR for a service request.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization the rule applies to.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT `)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service request was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management].

Enablement for Information Retrieval

The ILM object `CRM_INCDNT` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for `CRM_INCDNT`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)
- **BAdI: Archiving Information of Tables** (`BADI_DTINF_ARCHIVE_DETAILS`)

For more information, see SAP Note [2939146].

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction `DTINF_MAINTAIN_PURP`), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction `DTINF_MAINT_PROFILE`) defining that archived personal data must be included in the search.
- Ensure that for archiving object `CRM_INCDNT`, there is at least one active archive information structure that contains the following selectable field:
  - **Transaction ID** (`object_id`)

  This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure `SAP_CRM_INCDNT` that is provided in the standard system.

For more information, see [Information Retrieval Framework (IRF)].

Performing Application-Specific Configuration

Before you can use the `CRM_INCDNT` archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service requests.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Requests Archived With CRM_INCDNT

- From within Archive Administration (transaction SARA), you can search for archived service requests using the read program CRM_ARC_SRQM_INCIDENT_READ
- You can also use the Archive Information System (transaction SARI) to display archived service requests by entering the archiving object CRM_INCDNT and the archive infostructure SAP_CRMS4_SRVR

In both cases, the system provides a range of criteria that enable you to select the service requests that you want to display.

Variants for CRM_INCDNT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_INCDNT.

Defining Write Variants

The write variant contains the parameters for the service requests that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

> **i Note**
>
> Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_SRQM_INCIDENT_CHECK_2` determines archivable service requests.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Orders Using CRM_SERORD

You can use archiving object CRM_SERORD to archive service orders.

Tables

CRM_SERORD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SERORD can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SERORD is delivered with the following programs:

Program	Name
Write	CRM_ARC_SERORD_SAVE
Delete	CRM_ARC_SERORD_DELETE_2
Preprocessing	CRM_ARC_SERORD_CHECK_2

When archiving service orders, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SERORD.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service order:

- The service order has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service confirmations are archivable.
- If item-based accounting has been enabled: The service order has been set to **Business Completed**.

  For more information about item-based accounting, see Item-Based Accounting.

The following checks have been carried out:

- You've executed an archivability check for service orders by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service orders that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SERORD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SERORD, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SRVO for a service order.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service order was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_SERORD assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRM_SERORD:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SERORD, there is at least one active archive information structure that contains the following selectable field:

- Transaction ID (object_id)

These fields are used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRVO that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SERORD archiving object, you must first check the settings in Customizing in **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service orders.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Orders Archived With CRM_SERORD

You have the following options for displaying archived service orders:

- From within **Archive Administration** (transaction SARA), you can search for archived service orders using the read program CRM_ARC_SERORD_READ.
- You can also use the **Archive Information System** (transaction SARI) to display archived service orders by entering the archiving object CRM_SERORD and the archive infostructure .

In both cases, the system provides a range of criteria that enable you to select the service orders that you want to display.

Archiving Service Orders in Hierarchical Settlements

It is possible to archive service orders even when one or more items are part of a financial hierarchy with a service contract and when hierarchical settlement is used. It is, however, required that all service contracts involved are already set to **Business Completed**.

If you revoke a completed service contract, you can also revoke any completed follow-up service orders associated with it, provided they are within the same financial hierarchy. The following applies when doing so:

- Lower levels in the hierarchy cannot have a status that precedes the status of the higher levels.
- If the service contract is archived, the associated service orders and PM orders must also be archived.

- If the service contract is **Business Completed**, the associated service orders and PM orders must be at least **Business Completed** as well. They cannot be reopened for business.

Variants for `CRM_SERORD`

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction `SARA`, you can create variants of the write and preprocessing programs available for `CRM_SERORD`.

Defining Write Variants

The write variant contains the parameters for the service orders that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SERORD_CHECK_2 determines archivable service orders.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving In-House Services Using CRMS4_REPA

You can use archiving object CRMS4_REPA to archive in-house services.

Tables

CRMS4_REPA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRMS4_REPA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRMS4_REPA is delivered with the following programs:

Program	Name
Write	CRMS4_ARC_REPA_SAVE
Delete	CRMS4_ARC_REPA_DELETE
Preprocessing	CRMS4_ARC_REPA_CHECK

When archiving in-house services, you must use these programs or custom variants. For information about creating variants, see Variants for CRMS4_REPA.

Prerequisites

The following prerequisites must be fulfilled before you can archive an in-house service:

- The in-house service has the status **Complete**.
- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for in-house services by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). You can use the program to run several check jobs in parallel, which is especially useful when archiving mass data. During the archivability check, in-house services that meet the archivability criteria, are assigned the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case, parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the ILM object CRMS4_REPA as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

You can select several of the following condition fields and put them into a specific order for the ILM object CRMS4_REPA. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type to which the rule applies, for instance, REPA for in-house service.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization to which the rule applies.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, to which the rule applies.

Condition	Description
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you've selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the in-house service was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRMS4_REPA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data for a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRMS4_REPA:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in the Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRMS4_REPA, there's at least one active archive information structure that contains the following selectable fields:
    - **Contact Person** (CONTACT_PERSON)
    - **Employee Responsible** (PERSON_RESP)
    - **Ship-To Party** (SHIP_TO_PARTY)
    - **Sold-To Party** (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_CRMS4_REP1 that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRMS4_REPA archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity Configure Archiving Preselection, you define parameters for the check and preprocessing programs. In the activity Define Residence Times, you define the residence time for in-house services.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying In-House Services Archived With CRMS4_REPA

You have the following options for displaying archived in-house services:

- From within Archive Administration (transaction SARA), you can search for archived in-house services using the read program CRM_ARC_REPA_READ.
- You can also use the Archive Information System (transaction SARI) to display archived in-house services by entering the archiving object CRMS4_REPA and the archive infostructure SAP_CRMS4_REPA.

In both cases, the system provides a range of criteria that enable you to select the in-house-services that you want to display.

Archiving of Attachments

For more information about archiving attachments using the Document Management Service (DMS), see Archiving Document Info Records Using CV_DVS.

Variants for CRMS4_REPA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRMS4_REPA.

Defining Write Variants

The write variant contains the parameters for the in-house services that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the in-house services that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRMS4_ARC_REPA_CHECK` determines archivable in-house services.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Confirmations Using CRM_SRVCON

You can use archiving object `CRM_SRVCON` to archive service confirmations.

Tables

`CRM_SRVCON` archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_SRVCON` can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

`CRM_SRVCON` is delivered with the following programs:

Program	Name
Write	`CRM_ARC_SRVCON_SAVE`
Delete	`CRM_ARC_SRVCON_DELETE_2`
Preprocessing	`CRM_ARC_SRVCON_CHECK_2`

When archiving service confirmations, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_SRVCON.

Prerequisites

The following prerequisites must be fulfilled before you can archive a service confirmation:

- The service confirmation has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up transactions are archivable.

The following checks have been carried out:

- You've executed an archivability check for service confirmations by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service confirmations that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_SRVCON` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object `CRM_SRVCON`, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (`OBJECT_TYPE`)	Use this field to define the leading business transaction category.
Transaction Type (`PROCESS_TYPE`)	Use this field to define the transaction type the rule applies to, for instance, `SVC1` for a service confirmation.
Sales Organization (`SALES_ORG_SD`)	Use this field to define the sales organization the rule applies to.
Country/Region Key (`BS_COUNTRY_OF_VKORG`)	Use this field to define a country/region.
Organizational Management Object (`BS_CRM_ORGUNIT`)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (`BS_CRM_COUNTRY_OF_ORGUNIT`)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service confirmation was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management].

Enablement for Information Retrieval

The ILM object `CRM_SRVCON` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for `CRM_SRVCON`:

- **BAdI: Tables Used by an ILM Object** (`BADI_DTINF_ILM_OBJ_TABLES`)
- **BAdI: Archiving Information of Tables** (`BADI_DTINF_ARCHIVE_DETAILS`)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SRVCON, there is at least one active archive information structure that contains the following selectable field:
  - **Transaction ID** (object_id)

  This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRVC that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_SRVCON archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service confirmations.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Confirmations Archived With CRM_SRVCON

You have the following options for displaying archived service confirmations:

- From within **Archive Administration** (transaction SARA), you can search for archived service confirmations using the read program CRM_ARC_SERORD_READ
- You can also use the **Archive Information System** (transaction SARI) to display archived service confirmations by entering the archiving object CRM_SRVCON and the archive infostructure SAP_CRMS4_SRVC

In both cases, the system provides a range of criteria that enable you to select the service confirmations that you want to display.

Variants for CRM_SRVCON

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for `CRM_SRVCON`.

Defining Write Variants

The write variant contains the parameters for the service confirmations that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service confirmations that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SRVCON_CHECK_2 determines archivable service confirmations.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **i Note**
  >
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Knowledge Articles Using CRM_KA

You can use archiving object CRM_KA to archive knowledge articles.

Tables

CRM_KA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_KA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_KA is delivered with the following programs:

Program	Name
Write	CRM_ARC_KA_SAVE
Delete	CRM_ARC_KA_DELETE_2
Preprocessing	CRM_ARC_KA_CHECK_2

When archiving knowledge articles, you must use these programs or custom variants. For information on creating variants, see Variants for CRM_KA.

Prerequisites

The following prerequisites must be fulfilled before you can archive a knowledge article:

- The knowledge article has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for knowledge articles by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, knowledge articles that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_KA ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_KA, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (BUKRS)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance KNAR for a knowledge article.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the knowledge article was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_KA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRM_KA:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146 .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes**DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_KA, there is at least one active archive information structure that contains the following selectable field:
  - **Employee Responsible** (PERSON_RESP)

  This field is used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_CRMS4_KNAR that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_KA archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of knowledge articles.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE

Activity	Required Authorization Object
Preprocess	`S_ARCHIVE`

Displaying Knowledge Articles Archived With `CRM_KA`

You have the following options for displaying archived knowledge articles:

- From within **Archive Administration** (transaction SARA), you can search for archived knowledge articles using the read program `CRM_ARC_KA_READ`
- You can also use the **Archive Information System** (transaction `SARI`) to display archived knowledge articles by entering the archiving object `CRM_KA`. You must create an archive infostructure for this purpose. For more information, see [Providing Archive Infostructures for Data Access]

Variants for CRM_KA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for `CRM_KA`.

Defining Write Variants

The write variant contains the parameters for the activities that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CRM_ARC_KA_CHECK_2` determines archivable knowledge articles.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Cases Using SCMG

You can use archiving object `SCMG` to archive cases.

Tables

SCMG archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SCMG can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

SCMG is delivered with the following programs:

Program	Name
Write	CASE_WRITE_PROGRAM
Delete	CASE_DELETE_PROGRAM
Preprocessing	CASE_PRE_PROGRAM

When archiving cases, you must use these programs or custom variants. For information on creating variants, see Variants for SCMG.

Prerequisites

The following prerequisites must be fulfilled before you can archive a case:

- The case has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- All follow-on transactions related to the case must be archivable.

The following checks have been carried out:

- You've executed an archivability check for cases by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, cases that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SCMG ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object SCMG, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Case Type (CASE_TYPE)	Use this field to define the case type.
Area ID (AREA_ID)	Use this field to define the area the rule applies to, for instance case management.
RMS ID (RMS_ID)	Use this field to define the records management system ID. The RMS ID is a logical grouping of all objects that have been created for a specific business unit.
System Status (STATUS)	Use this field to define the case status the rule applies to, for instance, **Completed**.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The available time references are the end date of the case as well as the month, quarter or year of the end date. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object SCMG assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for SCMG:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note 2939146.

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object SCMG, there is at least one active archive information structure that contains the following selectable fields:
  - **Case Key** (CASE_GUID)
  - **Customer** (FIN_KUNNR)

  These fields are used by the IRF to determine which data should be fetched from the archive.

  You can use the archive information structure SAP_FSCM that is provided in the standard system.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the SCMG archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of cases.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Cases Archived With SCMG

You have the following options for displaying archived cases:

- From within **Archive Administration** (transaction SARA), you can search for archived cases using the read program CASE_READ_PROGRAM
- You can also use the **Archive Information System** (transaction SARI) to display archived cases by entering the archiving object SCMG and archive infostructure SAP_CRM_CMG.

In both cases, the system provides a range of criteria that enable you to select the cases that you want to display.

Related Information

BAdis for SCMG

Variants for SCMG

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for SCMG.

Defining Write Variants

The write variant contains the parameters for the cases that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

  You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

  You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

  In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  Note

  Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

  Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program `CASE_PRE_PROGRAM` determines archivable cases.

You can define variants for this program using the following parameters:

- Transaction Type

  You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

  You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

  In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

  Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

  > **Note**
  > Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

  Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

BAdis for SCMG

You can enhance the archiving functions provided for cases using Business Add-Ins (BAdi).

Business Add-In	Use
ARC_SCMG_CHECK	You can use this BAdi to define custom checks that must be met before a case can be archived.
ARC_SCMG_WRITE	You can use this BAdi to write additional data to the archive that isn't directly associated with the case.
ARC_SCMG_DELETE	You can use this BAdi to delete additional data that isn't directly associated with the case.

For more information on the BAdis, see the system help in Customizing under **Service** > **Case Management** > **Extended Customzing** > **Case Archiving** > **BAdis for Case Archiving**.

You can access the BAdIs using transaction SE18.

Archiving of Subscription Product-Specific Data Using SOM_PROD

You can use archiving object SOM_PROD to archive subscription product-specific data.

Tables

SOM_PROD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_PROD may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before subscription product-specific data can be archived:

- The related product master is marked for deletion.
- No drafts exist for the subscription-product specific data.
- The product is not used in any subscription order/contract.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SOM_PROD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- PRODUCT: Product number

The following time references are available:

- CREATION DATE: Date when the product was created
- LAST_CHANGE_DATE: Date when the product was last changed

For more information, search for SAP Information Lifecycle Management in the relevant version of SAP S/4HANA on the SAP Help Portal.

Performing Application-Specific Configuration

There is no specific Customizing for the SOM_PROD archiving object.

Defining Preprocessing Variants

There is no preprocessing program for SOM_PROD.

You can define variants for this program using the following parameters:

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

This program analyzes the subscription product-specific data for archiving and then writes the data to an archive file. The program selects all subscription product-specific data that fits in with the selected products. You can execute the report in either test or production mode. Data, however, is written to the archive file in production mode only.

In the **Comment for Archive Run** field, you can enter your own comment for the archive run.

You can archive subscription product-specific data only if the following prerequisites are met:

- The related product master is marked for deletion.
- No draft exists for the subscription product-specific data.
- The product is not used in any subscription order or subscription contract.

A write variant contains the parameters for the subscription product-specific data that you want to archive.

SAP delivers the following parameters:

- Product

  This is the product number for the related product.

The write program is `CRMS4_SOM_PROD_ARCHIVE_WRITE`.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRMS4_SOM_PROD_ARCHIVE_DELETE for SOM_PROD deletes the subscription product-specific data based on an existing archive. The program first reads the subscription product-specific data belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program `CRMS4_SOM_PROD_ARCHIVE_READ` displays the archived subscription product-specific data. This data must be archived and deleted. The selected product IDs are displayed using ALV. You can navigate to the details of each product by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Subscription product-specific data UUID
- Product ID
- Key of archive file

Defining Postprocessing Variants

There is no postprocessing program for `SOM_PROD`.

Defining Reloading Variants

There is no reloading program from `SOM_PROD`.

Dependencies of `SOM_PROD`

Subscription orders and subscription contracts must be archived before you can archive subscription-product specific data.

The related material master (archiving object MM_MATNR) can only be archived after the subscription-product specific data is archived.

Authorizations for SOM_PROD

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_PROD
ACTVT: 01 - Add or create ARCH_OBJ: SOM_PROD APPLIC: IS	S_ARCHIVE

Enhancements for SOM_PROD

The SOM_PROD archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
ARC_SOM_PROD_CHECK	Archiving subscription product-specific data: Check archivability	Check if an object can be archived
ARC_SOM_PROD_WRITE	Archiving subscription product-specific data: Write additional table	Write additional tables for archiving

Displaying Subscription Product-Specific Data Archived with SOM_PROD

You can use program CRMS4_SOM_PROD_ARCHIVE_READ to display archived and deleted subscription product-specific data.

Archiving of Allowance Definition Groups Using SOM_ADG

You can use archiving object SOM_ADG to archive allowance definition groups.

Tables

SOM_ADG archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_ADG may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before an allowance definition group can be archived:

- Status of the allowance definition group must NOT be **In Preparation**.

  As long as the allowance definition group has this status, it can be deleted directly.

- No draft exists for the allowance definition group.

- The allowance definition group, to be deleted, is not used in any subscription product-specific data.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `SOM_ADG` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Allowance Definition Group ID

The following time references are available:

- CREATION_DATE: Date when the allowance group was created.

- LAST_CHANGE_DATE: Date when the allowance group was last changed.

For more information, see [SAP Information Lifecycle Management].

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds].

Performing Application-Specific Configuration

There is no specific Customizing for the `SOM_ADG` object.

Defining Preprocessing Variants

There is no preprocessing program for `SOM_ADG`.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the allowance definition group that you want to archive. This program covers the following features:

- The analysis of allowance definition groups with respect to archivability.

- Writing allowance definition groups to an archive file based on the analysis in the previous step.

The following database tables are included:

- CRMS4_SOM_ADG
- CRMS4_SOM_ADGT
- CRMS4_SOM_AD

Additionally, change documents related to the archived allowance definition group are also archived.

SAP delivers the following parameters:

- Allowance Definition Group ID

  The write program is `CRMS4_SOM_ADG_ARCHIVE_WRITE`.

Defining Delete Variants

The program CRMS4_SOM_ADG_ARCHIVE_DELETE is used to delete the allowance definition groups based on an existing archive.

The program first reads the allowance definition groups belonging to an archiving run from the archive, and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program `CRMS4_SOM_ADG_ARCHIVE_READ` displays the selected allowance definition groups using ALV. You can navigate to the details of each allowance definition group by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Allowance Definition Group UUID
- Allowance Definition Group ID
- Key for Archive File

Defining Postprocessing Variants

There is no postprocessing program for `SOM_ADG`.

Defining Reloading Variants

There is no reloading program for `SOM_ADG`.

Dependencies of `SOM_ADG`

Authorizations for `SOM_ADG`

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	CRM_ISX_AD

Activity	Required Authorization Object
ACTVT: 01 ARCH_OBJ: SOM_ADG APPLIC: IS	S_ARCHIVE

Enhancements for SOM_ADG

The SOM_ADG archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
ARC_SOM_ADG_CHECK	Archiving Allowance Definition Group: Check Archivability	Check if an object can be archived.
ARC_SOM_ADG_WRITE	Archiving Allowance Definition Group: Write Additional Table	Write additional tables for archiving

Displaying Allowance Definition Groups Archived with SOM_ADG

You can use the program CRMS4_SOM_ADG_ARCHIVE_READ to display archived and deleted allowance definition groups.

Archiving Subscription Orders Using CRM_PRV0

You can use archiving object CRM_PRV0 to archive subscription orders (BUS2000265) in transaction SARA.

Tables

CRM_PRV0 archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRV0 may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a subscription order can be archived:

- The subscription order status must be set to Completed at the header level.
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- If you have set up ILM integration for the CRM_PRV0 ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_PRVO ILM object as part of SAP Information Lifecycle Management. In the transaction IRMPOL, you can create policies for residence time and retention rules, depending on the available policy category. Here, you can also see the available time reference and the existing condition fields. You can decide which of these will be used to define your rule structure. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- OBJECT_TYPE - Business Transaction Category
- PROCESS_TYPE - Business Transaction Type
- BS_CRM_COUNTRY_OF_ORGUNIT - Country of Org Unit

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On

For more information, see SAP Information Lifecycle Management.

Enablement for Information Retrieval

The ILM object CRM_PRVO assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVO:

- BAdI Implementation BADI_CRM_SUBSCR_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRVO_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVO, there is at least one active archive information structure that contains the following selectable fields:
    - Field name (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_PRVO archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide > Service > Transactions > Data Archiving**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVO_CHECK_2 for CRM_PRVO checks the prerequisites discussed above so that subscription orders can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.
- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The write program CRM_ARC_PRVO_SAVE for CRM_PRVO writes the data to the archive file. The program contains the parameters for the subscription orders that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.
- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.
- Changed On

  Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRM_ARC_PRVO_DELETE for CRM_PRVO deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

  Changed On

  Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Read Program Variants

The read program CRM_ARC_PRVO_READ for `CRM_PRVO` reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVO and field catalog SAP_CRMS4_PRVO, you can maintain your own infostructure and field catalogs.

Archiving Subscription Contracts Using CRM_PRVC

You can use archiving object `CRM_PRVC` to archive subscription contracts (BUS2000266) in transaction SARA.

Tables

`CRM_PRVC` archives data from several tables. To check which tables these are, call up transaction `SARA`, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

`CRM_PRVC` may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction `AOBJ`, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction `SARA` enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a `subscription contract` can be archived:

- The subscription contract statuses must be set to Completed and Finished at the header level
- The residence time (number of days after the last change to the business transaction) must have elapsed.

- All linked subscription orders must be archived and deleted from the system.
- The FI-CA provider contracts that correspond to the subscription contracts must be archived and deleted from the system. For more information about archiving a FI-CA provider contract, see Archiving of Provider Contracts (FI_MKKCAVT) on the SAP Help Portal.
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- Sharing Contracts can only be archived when no more linked Shared Contracts exist, i. e. Shared Contracts have to be archived and deleted before the Sharing Contracts they refer to.
- If you have set up ILM integration for the `CRM_PRVC` ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM-PRVC` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- `OBJECT_TYPE` - Business Transaction Category
- `PROCESS_TYPE` - Business Transaction Type
- `BS_CRM_COUNTRY_OF_ORGUNIT` - Country of Org Unit

The following time references are available:

- `LAST_CHANGE_DATE` - Last Changed On

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Enablement for Information Retrieval

The ILM object `CRM_PRVC` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for `CRM_PRVC`:

- BAdI Implementation `BADI_CRM_SUBSCR_ILM_OBJ_TABLES` of BAdI: Tables Used by an ILM Object (`BADI_DTINF_ILM_OBJ_TABLES`)
- BAdI Implementation `SOM_CRM_PRVC_ARCH_DETAILS` of BAdI: Archiving Information of Tables (`BADI_DTINF_ARCHIVE_DETAILS`)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVC, there is at least one active archive information structure that contains the following selectable fields:
  - Field name (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.
- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see [Information Retrieval Framework (IRF)](#).

Performing Application-Specific Configuration

Before you can use the CRM-PRVC archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide** > **Service** > **Transactions** > **Data Archiving**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVC_CHECK_2 for CRM_PRVC checks the prerequisites discussed above so that subscription contracts can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.
- Transaction ID

  Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The write program CRM_ARC_PRVC_SAVE FOR CRM_PRVC writes the data to the archive file. The program contains the parameters for the subscription contracts that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.
- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Changed On

Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRM_ARC_PVC_DELETE for CRM_PRVC deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVC_READ for CRM_PRVC reads data from archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVC and field catalog SAP_CRMS4_PRVC, you can maintain your own infostructure and field catalogs.

Archiving Master Agreements Using CRM_PRVMA

You can use archiving object CRM_PRVMA to archive Master Agreements (BUS2000267) transaction SARA.

Tables

CRM_PRVMA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRVMA may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a `master agreement` can be archived:

- The header status must be set to Completed.

- The residence time (number of days after the last change to the business transaction) must have elapsed.
- All subscription orders, subscription contracts, and solution quotations that refer to the master agreement must be archived.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `CRM_PRVMA` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- `PROCESS_TYPE` - Business Transaction Type

The following time references are available:

- `LAST_CHANGE_DATE` - Last Changed On
- `CONTRACT_END_DATE` - Contract End Date

For more information, see SAP Information Lifecycle Management.

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see Setting Legal Holds.

Enablement for Information Retrieval

The ILM object `CRM_PRVMA` assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for `CRM_PRVC`:

- BAdI Implementation `BADI_CRM_MA_ILM_OBJ_TABLES` of BAdI: Tables Used by an ILM Object (`BADI_DTINF_ILM_OBJ_TABLES`)
- BAdI Implementation `SOM_CRM_PRMA_ARCH_DETAILS` of BAdI: Archiving Information of Tables (`BADI_DTINF_ARCHIVE_DETAILS`)

  This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction `DTINF_MAINTAIN_PURP`).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction `DTINF_MAINT_PROFILE`) defining that archived personal data must be included in the search.

- Ensure that for archiving object CRM_PRVMA, there is at least one active archive information structure that contains the following selectable fields:
    - Field name (SOLD_TO_PARTY)

  These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.

- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Performing Application-Specific Configuration

Before you can use the CRM_PRVMA archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide** > **Service** > **Transactions** > **Data Archiving**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVMA_CHECK for CRM_PRVMA will check the prerequisites discussed above so that BUS2000267 can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

  Transaction type/Process type of a document to be archived.

- Transaction ID

  Object ID of a document to be archived.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the <business objects> that you want to archive.

SAP delivers the following parameters:

The write program CRM_ARC_PRVMA for CRM_PRVMA writes the data to the archive file. The program contains the parameters for the master agreements that you want to archive.

SAP delivers the following parameters:

- Transaction Type

  Transaction type/Process type of a document to be archived.

- Transaction ID

  Object ID of a document to be archived.

- Changed On

Last changes made to the document.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRM_ARC_PRVMA_DELETE_2 for CRM_PRVMA deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVMA_READ for CRM_PRVMA reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVMA and field catalog SAP_CRMS4_PRVMA, you can maintain your own infostructure and field catalogs.

Archiving Sharing Group Using SOM_SHGR

You can use archiving object SOM_SHGR to archive sharing groups.

Tables

SOM_SHGR archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_SHGR may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose **Archiving Classes Used**.

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a sharing groups can be archived:

- No draft exists for the sharing group.
- The sharing group is not used in a subscription order, subscription contract, or a master agreement.

ILM-Related Information for the Archiving Object

You can use this archiving object with the `SOM_SHGR` ILM object as part of SAP Information Lifecycle Management. In transaction `IRMPOL`, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Sharing group ID

The following time references are available:

- CREATION_DATE: Date on which the sharing group was created
- LAST_CHANGE_DATE: Date on which the sharing group was last changed.

For more information, see [SAP Information Lifecycle Management].

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds].

Performing Application-Specific Configuration

There is no specific Customizing for the `SOM_SHGR` archiving object.

Defining Preprocessing Variants

There is no preprocessing program for `SOM_SHGR`.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the sharing object that you want to archive. The program CRMS4_SOM_SHGR_ARCHIVE_WRITE is used to write the archive.

This report carries out the first step in archiving sharing groups and covers the following features:

- Analyzing sharing groups with respect to archivability
- Writing sharing groups to an archive file based on the results of the analysis

The following database tables are included:

- `CRMS4_SOM_SRG`
- `CRMS4_SOM_SRGT`

Additionally, change documents related to the archived sharing group are also archived.

SAP delivers the following parameters:

- Sharing Group ID

The write program is CRMS4_SOM_SHGR_ARCHIVE_WRITE

Defining Delete Variants

The program CRMS4_SOM_SHGR_ARCHIVE_DELETE is used to delete sharing groups based on an existing archive, and is the second step of the archiving process.

The report first reads the sharing groups belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program CRMS4_SOM_SHGR_ARCHIVE_READ is used to display the archived and deleted sharing groups.

You must ensure the following prerequisites are fulfilled before executing the read program:

- Data is archived and deleted

You can select the archived data by using the following selection parameters:

- Sharing Group UUID
- Sharing Group ID
- Key for Archive File

The selected sharing groups are displayed using ALV. You can navigate to the details of each sharing group by double clicking on the relevant record.

Defining Postprocessing Variants

There is no postprocessing program for `SOM_SHGR`.

Defining Reloading Variants

There is no reloading program for `SOM_SHGR`.

Dependencies of `SOM_SHGR`

Subscription orders, subscription contracts, and master agreements using a specific sharing group ID must be archived before you can archive the sharing group.

Authorizations for `SOM_SHGR`

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_SHGR
ACTVT: 01 ARCH_OBJ: SOM_SHGR	S_ARCHIVE

Activity	Required Authorization Object
APPLIC: IS	

Enhancements for SOM_SHGR

The SOM_SHGR archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
ARC_SOM_SHGR_CHECK	Archiving Sharing Group: Check Archivability	Check if an object can be archived
ARC_SOM_SHGR_WRITE	Archiving Sharing Group: Write Additional Table	Write additional tables for archiving

Displaying Sharing Groups Archived with SOM_SHGR

You can use the program CRMS4_SOM_SHGR_ARCHIVE_READ to display the archived and deleted sharing groups.

Blocking, Unblocking, and Deletion of Personal Data in Service

You can use SAP Information Lifecycle Management (SAP ILM) to control the blocking, unblocking, and deletion of business partner, customer, and supplier master data. According to the business partner approach, blocking and deletion of customer and supplier master data is part of the blocking and deletion of the business partner master data. This means that the business partner considers all residence periods of linked customer or supplier master data. Personal data collected in master data can be blocked as soon as the business activities for which this data was collected have been completed and the residence time for this data has elapsed.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, customer and supplier master data must be archived and deleted from the database before the associated business partner master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction IRMPOL).

The following scenarios for blocking business partners are supported:

1. You want to block business partner master data for which the associated customer and supplier data is blocked. The business partner master data satisfies the EoP check criteria and therefore can be blocked.
2. You want to block business partner data for which the associated customer and supplier master data is not blocked. The associated customer and supplier master data satisfies their EoP checks as does the business partner master data. All of the associated customer and supplier master data as well as the business partner master data can be blocked.
3. You want to block business partner master data for which the associated customer and supplier master data is not blocked and where at least one customer or supplier does not meet the EoP check criteria. The business partner master data cannot be blocked until all of the associated customer and supplier master data meet the EoP check criteria and can also be blocked.
4. Independent blocking of the customer and supplier master data is possible. The corresponding roles of the business partner are also blocked.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, associated customer and supplier master data must be archived and deleted from the database before the associated business partner master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction `IRMPOL`).

Specifics Business Partner Master Data

The functions required for blocking and unblocking or viewing blocked business partner master data are provided through the data protection specialist roles:

- `SAP_CA_BP_DP_DISPLAY`
- `SAP_CA_BP_DP_BLOCK`
- `SAP_CA_BP_DP_UNBLOCK`
- `SAP_CA_BP_DP_BLOCK_REQUEST`

Specifics for Customer and Supplier Master Data

The functions required for blocking and unblocking or viewing blocked customer and supplier master data are provided through the data protection specialist roles:

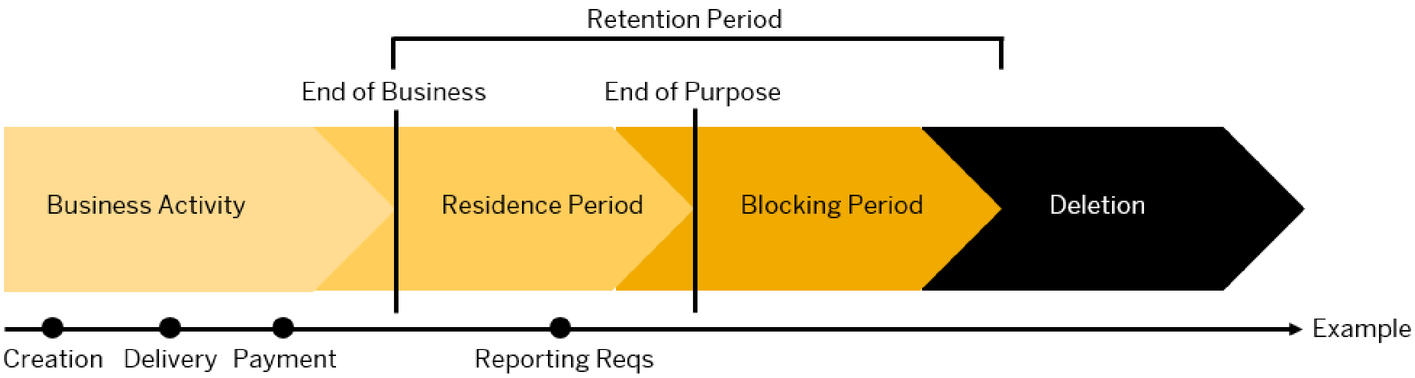
- `SAP_LO_MD_CVP_DISPLAY`
- `SAP_LO_MD_CVP_BLOCK`
- `SAP_LO_MD_CVP_UNBLOCK`

You can copy these template roles to custom roles, adjust them to meet your requirements, and assign them to your users.

Business Partner End of Purpose (EoP) Check in Service

An end of purpose (EoP) check determines whether data is still relevant for business activities based on the retention period defined for the data. The retention period is part of the overall lifecycle of personal data, which consists of the following phases:

- **Business activity:** The relevant data is used in ongoing business, for example contract creation, delivery or payment.
- **Residence period:** The relevant data remains in the database and can be used in subsequent processes related to the original purpose, for example reporting obligations.
- **Blocking period:** The relevant data must be retained for legal reasons. During the blocking period, business users of SAP applications are prevented from displaying and using the data. However, it can be processed, if so required due to mandatory legal provisions.
- **Deletion:** The data is deleted and no longer exists in the database.



Personal Data Lifecycle

Blocking of data impacts system behavior in the following ways:

- **Display:** Only users with specific auditing authorizations can display blocked data.
- **Change:** It is not possible to change a business object that contains blocked data.
- **Create:** It is not possible to create a business object that contains blocked data.
- **Copy or follow-up:** It is not possible to copy a business object or perform follow-up activities for a business object that contains blocked data.
- **Search:** Only user with specific auditing authorizations can search for blocked data or search for a business object using blocked data.

Customizing for the End of Purpose Check

Activity	Customizing Path
Define the settings for authorization management in Customizing for Cross-Application Components under:	Data Protection > Authorization Management
Configure the settings related to the blocking and deletion of business partner master data in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner
Configure the settings related to the blocking and deletion of customer and supplier master data in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Customer Master/Supplier Master Deletion
Define whether you want to use no blocking or dual control for blocking in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner > Set Activation Status of No Blocking/Dual Control for Role Group
Define whether you want to use dual control for unblocking in Customizing for Cross-Application Components under:	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Reasons for Unblocking Business Partner
	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Registered Function Modules for Unblock BP
	Data Protection > Blocking and Unblocking of Data > Business Partner > Define Dual Control Setting for Unblocking
	Data Protection > Blocking and Unblocking of Data > Business Partner > Set Status for Automatic Unblocking
Configure the settings for transactional data specific to service solution in the Customizing for Service under:	Cross-Application Components > Data Protection

For more information about the possible Customizing settings, refer to the system help of the respective activity.

Process Flow

Activity	Additional Information
Enable the end of purpose check function for the central business partner by running transaction BUPA_PRE_EOP	System help for the **Blocking of Business Partner Data** report
Enable the end of purpose check function for the customer and supplier master data by running transaction CVP_PRE_EOP	System help for the **Block Customer and Supplier Master Data** report
Request the unblocking of blocked data	Request Unblocking of Master Data
Unblock master data	Unblocking Master Data
Delete master data	Deletion of Master Data
Display application logs for the business partner, customer, and supplier	Logging

ILM-enabled Objects in Service

Service uses the central business partner CA_BUPA.

Authorizations

Users who have a role based on the SAP_CA_BP_DP_DISPLAY or SAP_LO_MD_CVP_DISPLAY template roles, can display blocked business partner master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Users who have a role based on the SAP_LO_MD_CVP_DISPLAY template role with the B_BUP_PCPT authorization object including the assignment of activity 03, DISPLAY can display blocked customer and supplier master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Business Partner Master Data in Multisystem Landscapes

In a multisystem landscape, the EoP for business partner master data is checked for an application in the local system as well as the remote systems. The EoP check is used to ensure data integrity in case of potential blocking. The EoP determines whether there are any data dependencies, that is, whether the data is still required for business activities. If there is such a dependency, the system does not block the data. If you still want to block the data, the dependent data must be blocked or deleted using archiving and deletion tools. If the EoP criteria are met, the business partner master data is blocked in the local and remote systems.

Blocking Master Data

You can use the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP) to block data related to a business partner including associated customer and supplier master data or the **Block Customer and Supplier Master Data** report (transaction CVP_PRE_EOP) to block data related to customer and supplier master data. These reports determine whether data can be blocked using the End of Purpose (EoP) check. If the data is still in use and business activities have not been completed, the system does not block the data. If business is complete and the residence period has expired, the reports block the business partner, customer or supplier master data as defined in the report parameters. To block business partner master data, any associated customer or supplier master data must first be blocked in your system.

Business Partner Master Data

Business partner master data is blocked completely in one step.

Customer and Supplier Master Data

Data is blocked by the system in the following order:

1. Block contact persons and/or the company codes
2. Block general data

You can perform both steps at the same time. However, performing the first step separately allows the contact person and/or company code data to be blocked and the residence time of the general data to be reset. The general data can then be blocked when the revised residence time is met.

More Information

- Report documentation for the **Blocking of Business Partner Data** report (transaction `BUPA_PRE_EOP`)
- System help for the **Block Customer and Supplier Master Data** report (transaction `CVP_PRE_EOP`)

Request Unblocking of Master Data

You can request the unblocking of business partner, customer or supplier master data using the **Request Unblocking** report (transaction `BUP_REQ_UNBLK`). You may do this because new business starts with a historic business partner, customer or supplier and their master data has not yet been deleted.

More Information

System help for the **Request Unblocking** report (transaction `BUP_REQ_UNBLK`).

Unblocking Master Data

You can use the **Request Unblocking** report (transaction `BUP_REQ_UNBLK`) to unblock business partner data. If the dual control setting for unblocking data has not been set, you can unblock data from your own unblocking request. If your system has dual control for unblocking set-up, another user must process the request to unblock data.

You can unblock customer and supplier master data with the **Unblock Customer and Supplier Master Data** report (transaction `CVP_UNBLOCK_MD`).

Blocked data can only be unblocked, if the master data has not been archived and it is still within the retention period.

Unblocking is performed in the opposite order of the blocking process. That means that business partner master data is unblocked in one step.

Blocked customer or supplier master data that extends across several company codes and has several contact persons, is unblocked as follows:

1. Unblock the general data
2. Unblock the contact persons and the company codes

More Information

- Report documentation for the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP)
- System help for the **Request Unblocking** report (transaction BUP_REQ_UNBLK)
- System help for the **Unblock Customer and Supplier Master Data** report (CVP_UNBLOCK_MD)
- For more information on unblocking, see [Request Unblocking of Master Data]
- For more information on dual control and unblocking, see the product assistance on SAP Help Portal and search for *Unblocking*.

Deletion of Master Data

After the residence period has been completed and data is blocked, it either remains in the database or is stored in an external archive. If the blocked data remains in the database, the data manager can delete the blocked business partner, customer, and supplier master data from the database using the corresponding archiving objects. The business partner master data can be deleted once all associated customer and supplier master data has been deleted from the database.

To destroy data from the database or archives, you can use the **Data Destruction** report (transaction ILM_DESTRUCTION). Subsequently, the data can no longer be retrieved.

Logging

You can use the **Analyze Application Log** report (transaction SLG1) to view the centralized business partner log. To do so, use the BDT_DATAARCHIVING object with the BUPA_EOP subobject.

You can use the **Display Stored Application Logs for Customer and Supplier** report (transaction CVP_DISPLAY_LOG) to display the application log for one or various subobjects including:

Subobject	Subobject Description
CUSTOMER_BLOCKING	Customer blocking
CUSTOMER_UNBLOCKING	Customer unblocking
SUPPLIER_BLOCKING	Supplier blocking
SUPPLIER_UNBLOCKING	Supplier unblocking

This report provides more contextual details at the level of the log items compared to the **Analyze Application Log** report.

More Information

- System help for the **Display Stored Application Logs for Customer and Supplier** report (transaction CVP_DISPLAY_LOG)
- System help for the **Analyze Application Log** report (transaction SLG1)

Blocking and Deleting Personal Data in Subscription Order Management

The processes and functions of Subscription Order Management use personal data of the SAP business partner. If, from the point of view of these processes and functions, there is no longer any reason to store the data (for example, for a possible tax audit), you can block this data from the perspective of Subscription Order Management and ultimately destroy it.

For destroying personal data, use the functions of Information Lifecycle Management (ILM). Using the ILM functions, you first block the personal data of business partners for whom there is no further business. Then you delete this data as soon as it is no longer needed.

All archiving objects in Subscription Order Management support data protection notifications (see **Deletion of Personal Data in SAP BW/4HANA and SAP BW**).

Related Information

Deletion of Personal Data in SAP BW/4HANA and SAP BW
Data Protection

ILM Objects in Subscription Order Management

Use

In Subscription Order Management, the following archiving objects are enabled for use with SAP Information Lifecycle Management (SAP ILM):

ILM-Enabled Archiving Objects

Technical Name	Description	Available Fields of Retention Period	Available Start Times
`SOM_PROD`	Subscription product-specific data	`PRODUCT` - Material ID	`CREATION_DATE` - Created On `LAST_CHANGE_DATE` - Last Changed On
`SOM_ADG`	Allowance definition group	`GROUP_ID` - Allowance definition group ID	`CREATION_DATE` - Created On `LAST_CHANGE_DATE` - Last Changed On
`CRM_PRVO`	Subscription order	`BS_CRM_COUNTRY_OF_ORGUNIT` - Standard field *Country* from Org Unit `OBJECT_TYPE` - Business Trans. Cat. `PROCESS_TYPE` - Business Transaction Type	`LAST_CHANGE_DATE` - Last Changed On
`CRM_PRVC`	Subscription contract	`BS_CRM_COUNTRY_OF_ORGUNIT` - Standard field *Country* from Org Unit `OBJECT_TYPE` - Business Trans. Cat.	`LAST_CHANGE_DATE` - Changed On

Technical Name	Description	Available Fields of Retention Period	Available Start Times
		PROCESS_TYPE - Business Transaction Type	
CRM_PRVMA	Master agreement	OBJECT_TYPE - Business Trans. Cat CRMT_SUBOBJECT_CATEGORY_DB PROCESS_TYPE - Business Transaction Type CRMT_PROCESS_TYPE_DB	CONTRACT_END_DATE - Contract End Date LAST_CHANGE_DATE - Last Changed On
SOM_SHGR	Sharing group ID	GROUP_ID – Sharing Group ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On
CRMS4_SOM_MR_DESTRUCTION	Mass Runs	SERVICEMASSRUNTYPE – Mass Run Type SERVICEMASSRUNSTATUS – Mass Run Status	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On

Prerequisites

To use the blocking and delete functions, you must:

- Set up SAP Information Lifecycle Management. For more information, see SAP Information Lifecycle Management.
- Activate the business function ILM. For more information, see Activating SAP ILM.

ILM Notifications for Data Protection

The archiving objects in Subscription Order Management support data protection notifications. The table below highlights the specific ILM objects and the relevant fields that are included in the notifications.

ILM Object	Source Fields
Subscription Order (CRM_PRVO)	OBJECT_ID OBJTYPE_H
Subscription Contract (CRM_PRVC)	OBJECT_ID OBJTYPE_H NUMBER_INT CI_CONTRACT_ID
Master Agreement (CRM_PRVMA)	OBJECT_ID OBJTYPE_H
Allowance Definition Group (SOM_ADG)	ALLOWANCEDEFINITIONGROUPID

ILM Object	Source Fields
	ALLOWANCEDEFINITIONGROUPUUID
Subscription Product-Specific Data (SOM_PROD)	PRODUCT SUBSCRPNPRODSPECUUID
Sharing Group (SOM_SHGR)	SUBSCRPNCONTRACTSHARINGGROUPID SUBSCRPNCONTRACTSHARINGGROUPUUID

Notification Modes

ILM notification logs the following processes categorized as notification modes:

Notification Mode	Process Step
01	Archiving Write
02	Archiving Delete
03	Data Destruction Run Using Archiving Write Report
04	Data Destruction of Archive Files
05	Destruction Run Using Data Destruction Object
*	All

Authorization Objects

The authorization object S_ILM_NOTI is created with the Display, Delete, and Distribute activities. This object is used for the following authorizations required for performing actions on the notification logs:

- S_ARCHIVE is checked for creation of log entries during archive process.
- S_ARCD_RUN is checked for ILM destruction mode during deletion of archive files.
- S_ARCD_OBJ is checked when the Create API (SO Create SOAP API?) Is called from the destruction object program.
- S_ILM_NOTI with activity = Distribute is required to extract the notification log data.

Veto Check for HR Master Data in Service

Perform a veto check for HR master data in Service.

Use

Service provides a veto check for the deletion of the HR master data. The veto check enables the application to prevent HR master data being deleted for as long as transactions requiring HR master data information exist.

The veto check is relevant to the deletion of personnel numbers. This reference indirectly extends the residence and retention policy of the HR master data. As long as such references exist, HR master data is not being deleted.

Application Name	Function Module Type	Function Module for Check
S4CRM	For Deleting Personnel Numbers	CRMS4_HR_PER_VETO_CHECK

You have to activate all applications that should participate in the veto check for blocking data destruction

If not already available, create an entry for S4CRM in Customizing under Personnel Management > Personnel Administration > Tools > Data Privacy > Data Destruction > Block Data Destruction (Veto) > Activate Applications for Veto Check. To activate the veto checks, set the entry with S4CRM to **Needs to be checked** (**Application Active** checkbox).

If necessary, enter the logical system for S4CRM as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- CRM_SERORD (service orders)
- CRM_SRVCON (service confirmations)
- CRM_SRCONT (service contracts)
- CRMS4_REPA (in-house services)

Veto Check Implementation

Service provides the following functionality for the veto check of the HR master data:

- The application searches for the following data in relation to the HR master data:
  - Service orders, service contracts, and in-house services that do not have the status **Completed**.

    Service confirmations that have the status **Completed** but have not been released for billing.
- In the case of deletion of the personnel numbers, the application returns the personnel number to be deleted.

Based on this information, the function modules can return information at personnel number level about whether the data in the applications is still required. The **Veto** checkbox is selected at personnel number level.

If the **Veto** checkbox is selected, the system prevents both the destruction of infotype and subtype data and the deletion of personnel numbers.

Analysis

In the **ILM Data Destruction** app, once the job that performs the veto check has finished, the results of the check are written to the application log.

If the veto check has failed, the log provides information about any service transactions that triggered the veto and are therefore preventing the deletion of HR master data.

The log also provides the personnel numbers for the affected employees.

Related Information

Veto Check for HR Master Data in Subscription Order Management

Use

Subscription Order Management provides a veto check for the deletion of the HR Master Data. The veto check enables the application to prevent the deletion of the HR Master Data as long as transactions requiring HR Master Data information exist.

The veto check is relevant for the following situations:

- For deletion of employee data. A veto check is required to ensure that the employee data remains accessible for users (for audit and revision reasons, for example) with special authorization to the application and its processes.

These references indirectly extend the residence/retention policy of the HR Master Data. As long as such references exist, HR Master Data will not be deleted.

Application Name	Application Description	Function Module Type	Function Module for Check
Register the application and implementation class in view cluster VC_WFD_EOP_APP. Application Name: S4CRM Application Component: CRM-BF-AR	Subscription Order Management	Remote Enabled Function Module	CRMS4_HR_PER_VETO_CHECK (Veto check for HR Master - S4CRM)

You have to activate all applications that should participate in the veto check for blocking data destruction

For this, if not already available, create an entry for S4CRM in Customizing under **Personnel Management** > **Personnel Administration** > **Tools** > **Data Privacy** > **Data Destruction** > **Block Data Destruction (Veto)** > **Activate Applications for Veto Check**. To activate the veto checks, set the entry with S4CRM to **Needs to be checked** (**Application Active** checkbox).

If needed, enter the logical system for S4CRM as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- HRPA_PERNR (Destruction Object for Personnel Number)

Veto Check Implementation

Subscription Order Management provides the following functionality for the veto check of the HR master data:

- The application searches for open transactions with relation to the personnel number. If open transactions exist, the application will not allow data related to the personnel number to be deleted.

- The end of business is reached when it reaches the end of the retention period.
- The application returns the following information, depending on the scenario:
    - For data destruction, the type of data (infotype/subtype) to be destroyed as well as a list of personnel numbers and the date on which the data for each personnel number is to be destroyed
    - For deletion of personnel numbers, the personnel number to be deleted.

  Based on this information, the function modules can return information on personnel number level about whether the data in the applications is still required. If the veto checkbox is selected, the system prevents the destruction of infotype and subtype data as well as the deletion of personnel numbers.

Analysis

Analysis Question	Answer
What is the purpose of the HR Master Data?	The fields from CRM tables always refer to the Business Partner ID, which is associated with (replicated from) an employee. In Service, it is mandatory to have an associated business partner.
Would a deletion in HR Master Data impact the integrity of your records?	Yes
Describe the impact.	The veto check against the blocking of the HR Master Data person is needed in case a 1Order is not completed as the data is retrieved from HR Employee and not stored in parallel in the BP.

Data Destruction Objects in Subscription Order Management

You use data destruction objects to delete data from the database that is no longer relevant to your business processes, and that you aren’t legally required to keep.

SAP has defined a minimum residence time for some data objects. You can also define your own residence time. If more than one residence time is defined, the longer residence time applies.

You destroy data by using **Schedule Data Destruction Run**. Execute a data destruction run for each data destruction object.

Destroying Mass Runs Using `CRMS4_SOM_MR_DESTRUCTION`

You can use the data destruction object `CRMS4_SOM_MR_DESTRUCTION` for destroying Subscription Order Management mass runs.

For more information, see:

- Processing Audit Areas
- Editing ILM Policies
- Editing Retention Rules

ILM-Based Information for the Data Destruction Object

You create policies and rules for the related ILM object of this data destruction object.

The following fields for the ILM object are defined in the ILM policy and are visible when creating or processing the **ILM Policies**:

- Available **Time References**
  - `CREATION_DATE` - Creation date of the mass run
  - `LAST_CHANGE_DATE` - Last change date of the mass run
- Available **Condition Fields**
  - `SERVICEMASSRUNSTATUS` - Status of the mass run
  - `SERVICEMASSRUNTYPE` - Type of the mass run

Enablement for Information Retrieval

The ILM object `CRMS4_SOM_MR_DESTRUCTION` assigned to this data destruction object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation (`BADI_CRM_MRUN_ILM_OBJ_TABLES`) of BAdI: Tables Used by an ILM Object (`BADI_DTINF_ILM_OBJ_TABLES`) for `CRMS4_SOM_MR_DESTRUCTION`. For more information, see SAP Note 2939146 (S/4HANA).

Before you can start a search for personal data related to this ILM object, you must assign it to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction `DTINF_MAINTAIN_PURP`).

Once the profile has been created, it can be used for data collection using the transaction `DTINF_START_COLL` - Start Data Collection. You can view the collected data using the transaction `DTINF_PROC_COLL` - Process Data Collection Results.

For more information, see Information Retrieval Framework (IRF).

Compatibility Scope

With this compatibility pack, SAP provides a limited use right to SAP S/4HANA customers to run certain classic SAP ERP solutions on their SAP S/4HANA installation. Use the compatibility scope to understand the compatibility packages offered.

For more information about the compatibility scope and the compatibility packages offered for SAP S/4HANA, see SAP Help Portal at http://help.sap.com/s4hana > ***<Choose an on-premise release>*** > **Discover** > **Compatibility Packs**.

i Note

- The features described in this entire compatibility scope are only available in SAP S/4HANA.
- The system behavior described in this entire compatibility scope is only valid in SAP S/4HANA.

Related Information

SAP Note 2269324